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(71)	Applicant: SOCLEAN, INC., PETERBOROUGH, NH (US)	(51)	Int. Cl.
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(72)	Inventors: James KNIGHT, Bedford, NH (US); Kevin Beaulieu, Peterborough, NH (US)	(52)	U.S. Cl.
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- Related U.S. Application Data

(63)

Continuation of application No. 18/623,620, filed on Apr. 1, 2024, now Pat. No. 12,280,170, which is a continuation-in-part of application No. 18/368,598, filed on Sep. 15, 2023, now Pat. No. 12,064,527.

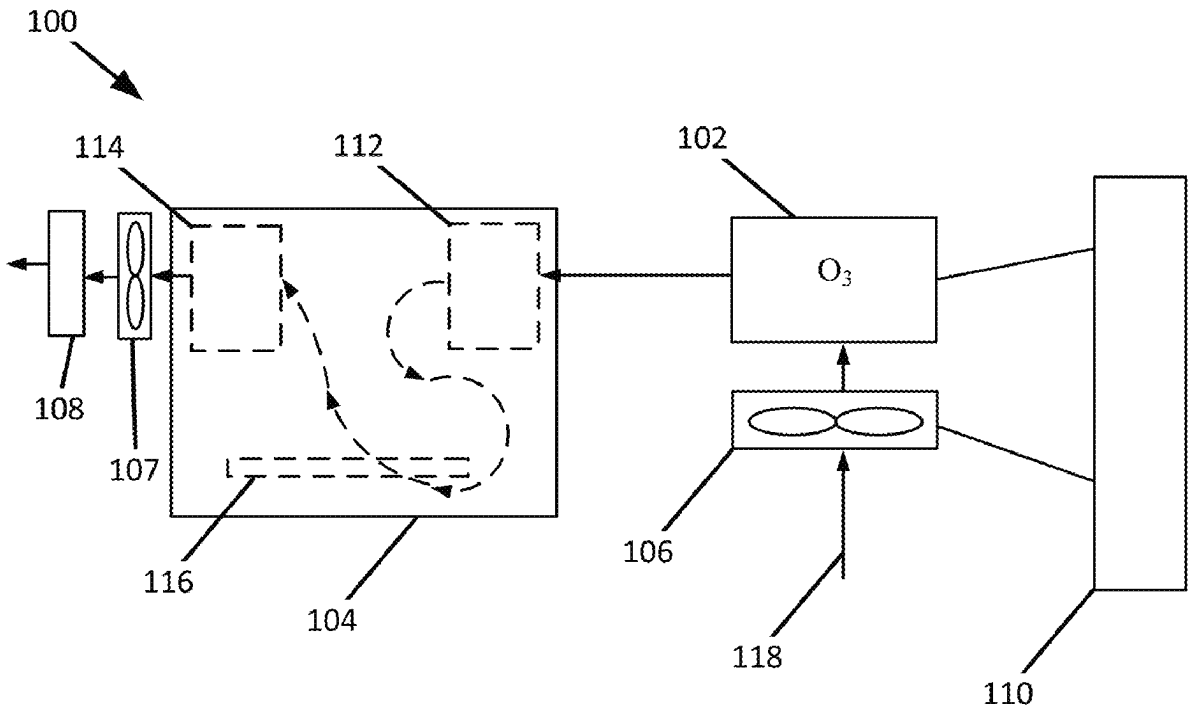
(60)

Provisional application No. 63/458,578, filed on Apr. 11, 2023, provisional application No. 63/458,527, filed on Apr. 11, 2023, provisional application No. 63/375,992, filed on Sep. 16, 2022, provisional application No. 63/458,578, filed on Apr. 11, 2023, provisional application No. 63/458,527, filed on Apr. 11, 2023, provisional application No. 63/609,991, filed on Dec. 14, 2023, provisional application No. 63/613,867, filed on Dec. 22, 2023.

(57)

ABSTRACT

A disinfection device may include a disinfection chamber, a device controller, an ozone generator configured to generate ozonated air, and a humidifier assembly configured to encourage a mixing of the ozonated air with humidity. The humidifier assembly may include a humidification chamber that includes a humidification chamber outlet, a chamber housing, a chamber base coupled to the chamber housing, and chamber electrical connectors electrically coupled to the device controller and a liquid reservoir that includes a liquid container and a reservoir lid removably coupled to the liquid container, the reservoir lid including a piezo-electric atomizer and lid electrical connectors configured to selectively electrically couple with the chamber electrical connectors.



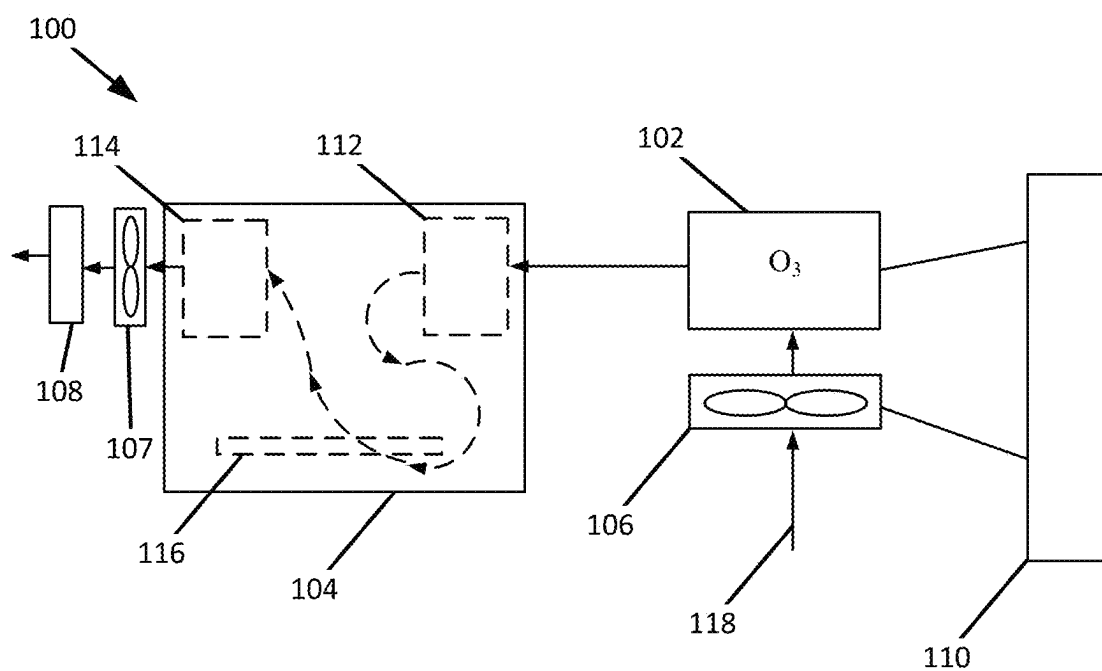


FIG. 1

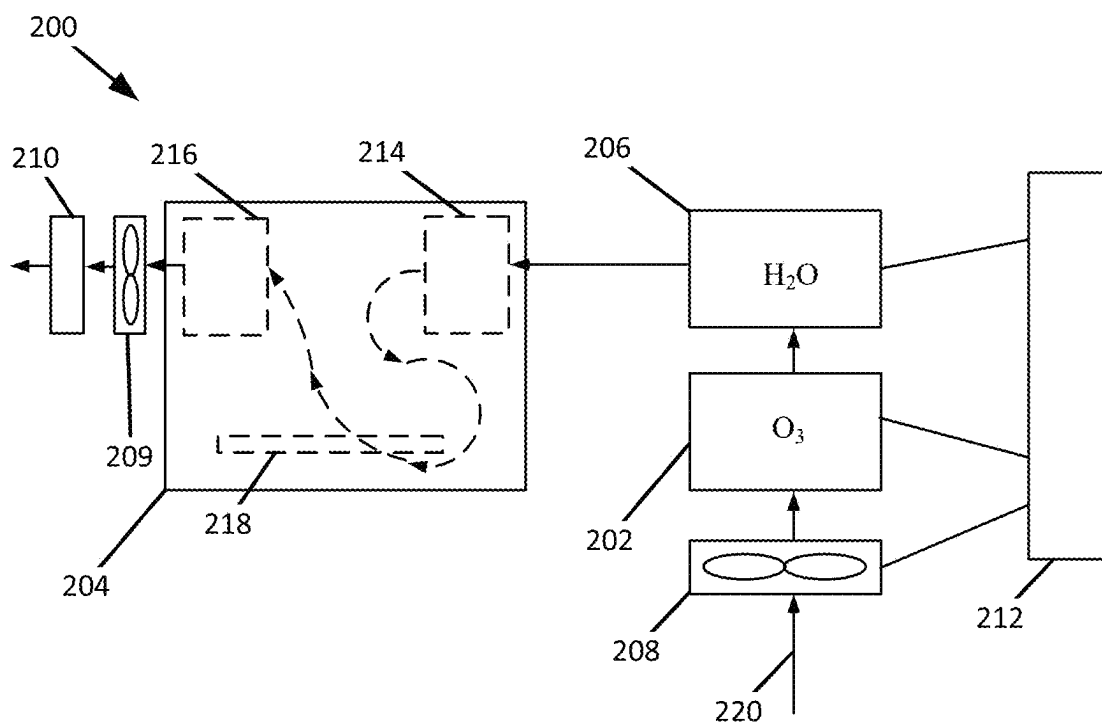
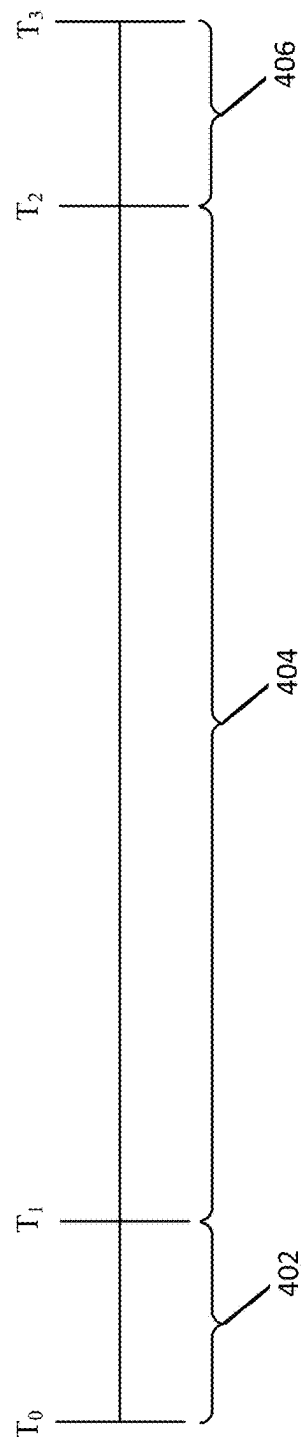
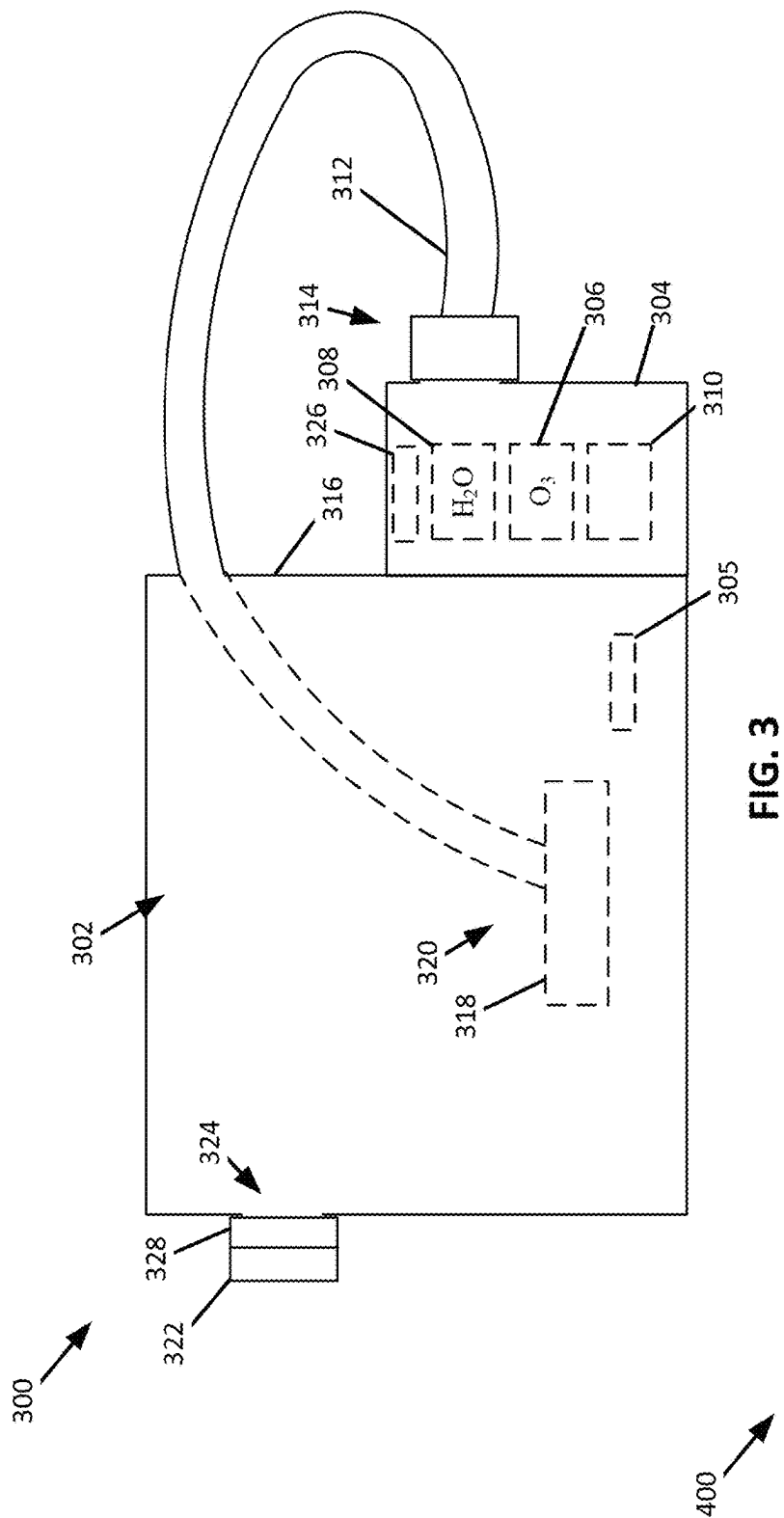


FIG. 2



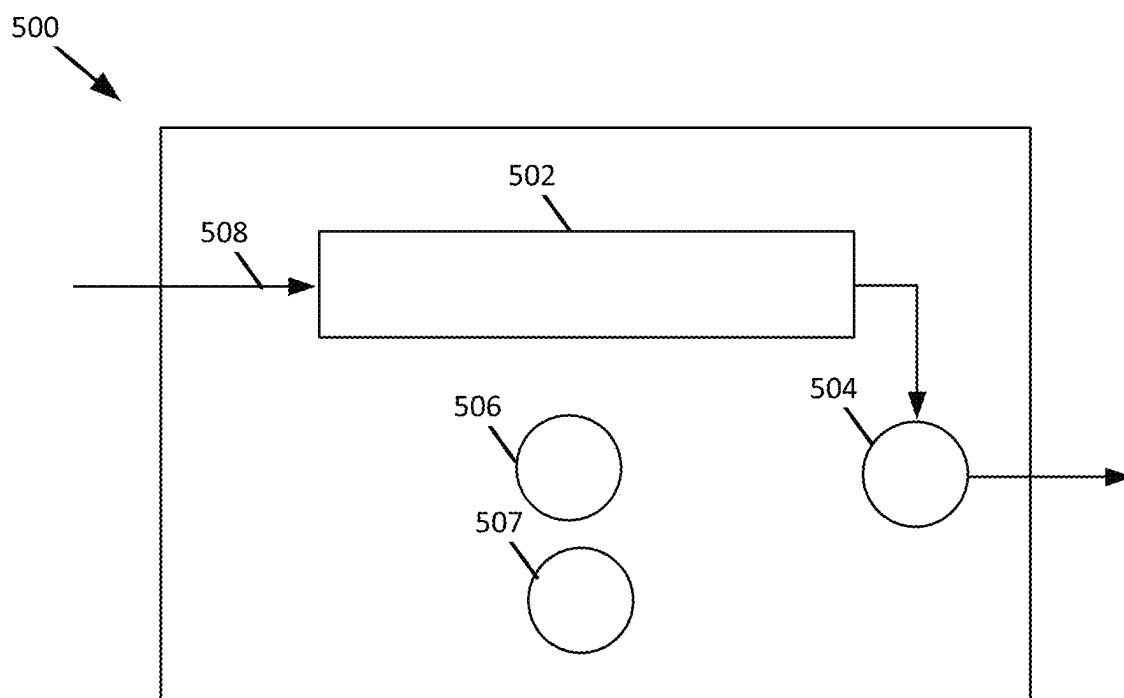


FIG. 5

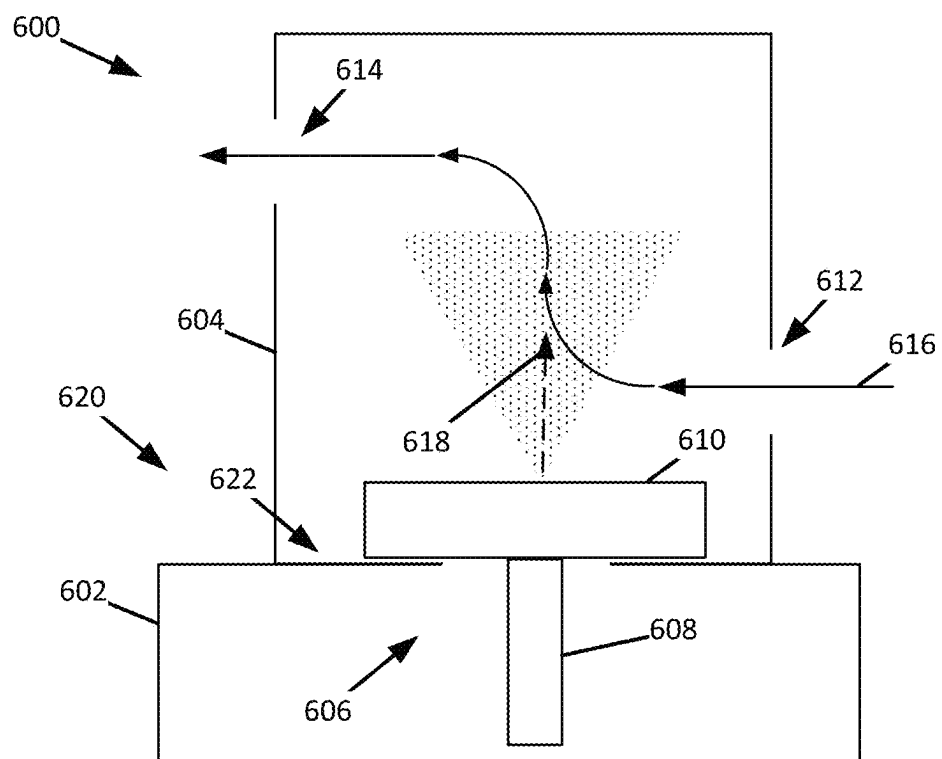


FIG. 6

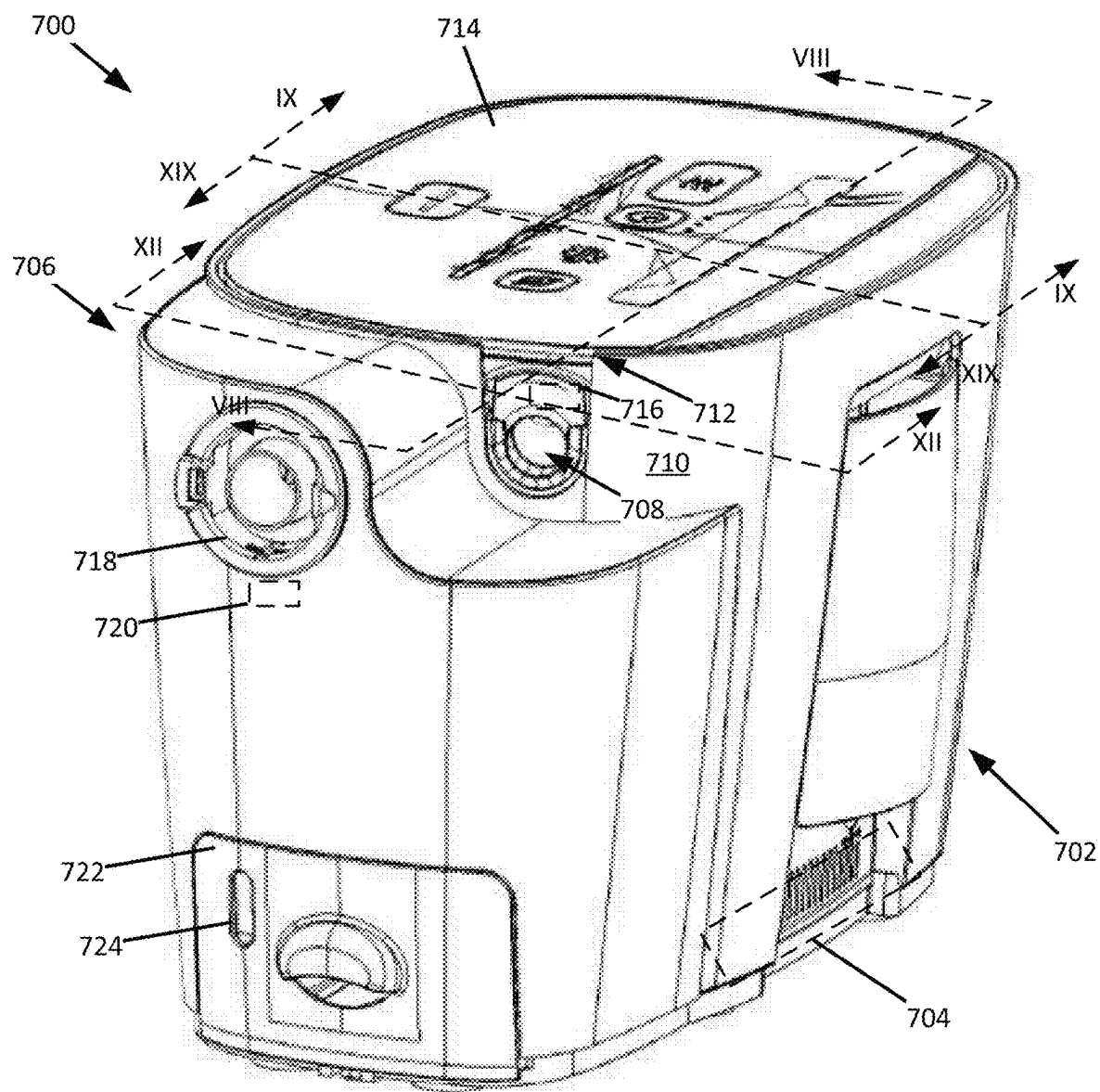


FIG. 7

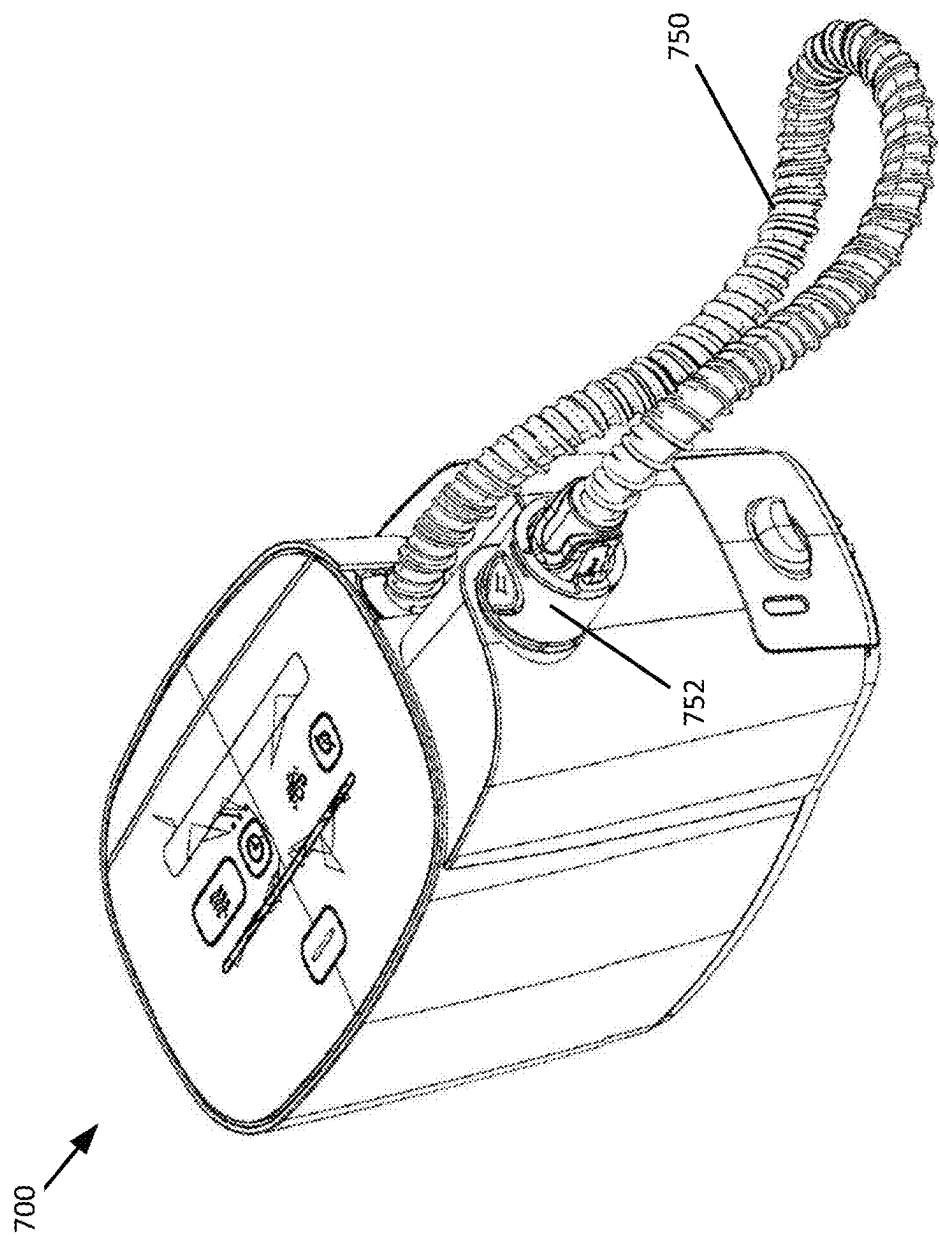


FIG. 7A

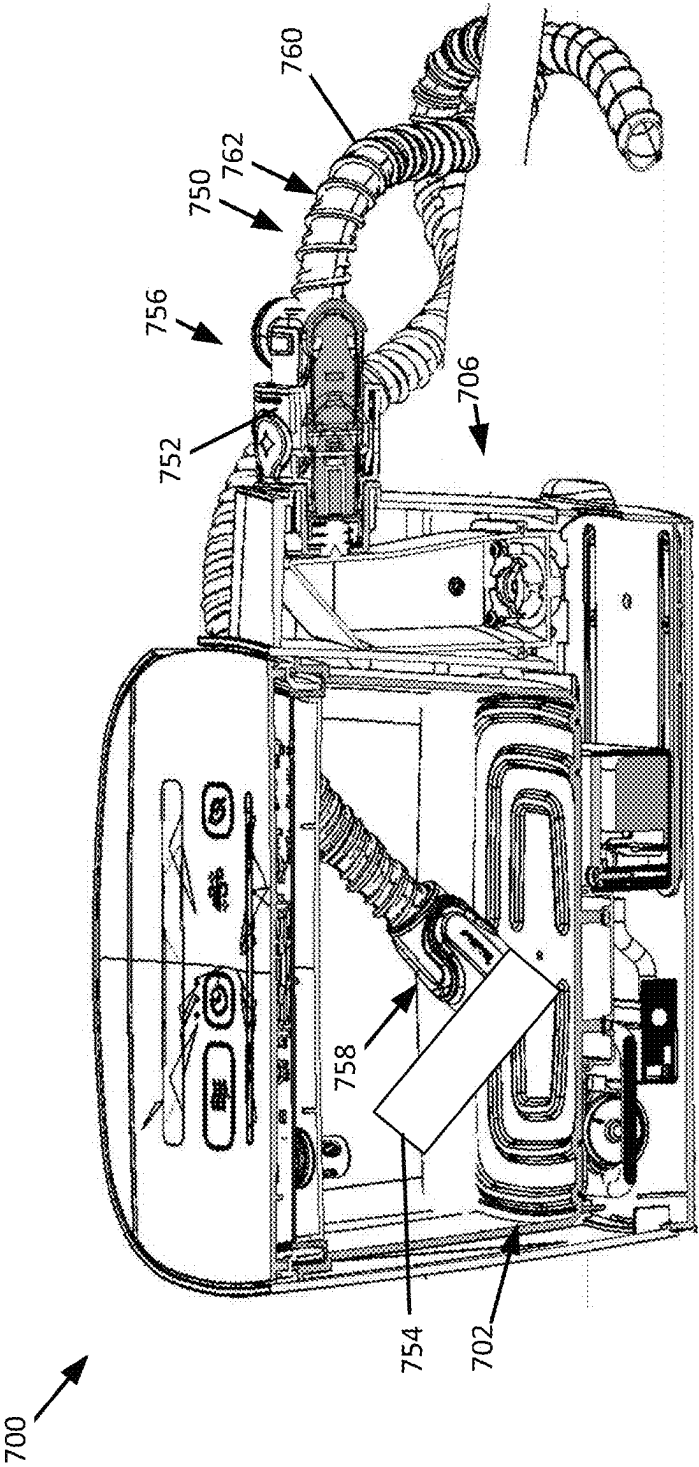


FIG. 7B

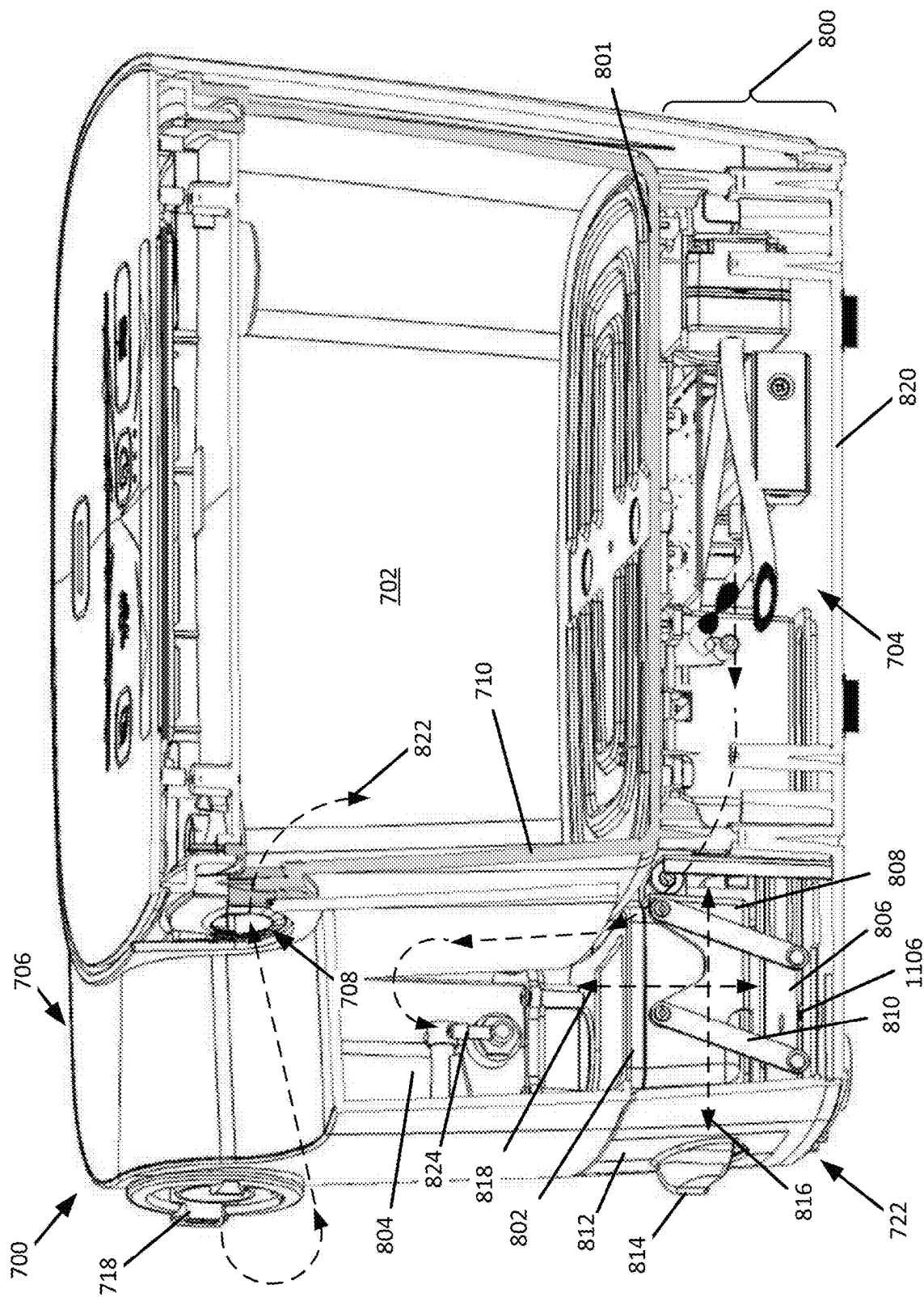


FIG. 8

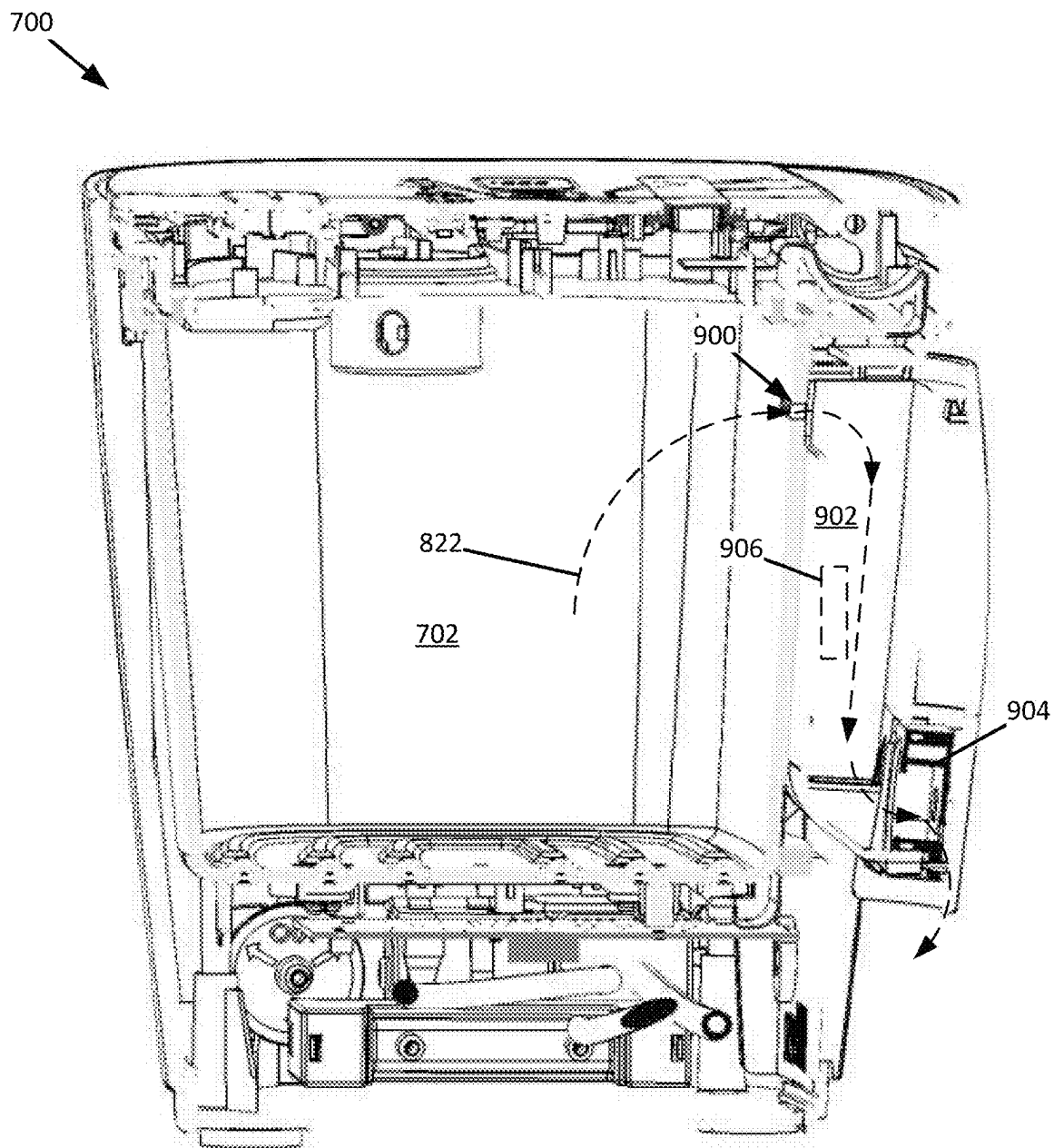


FIG. 9

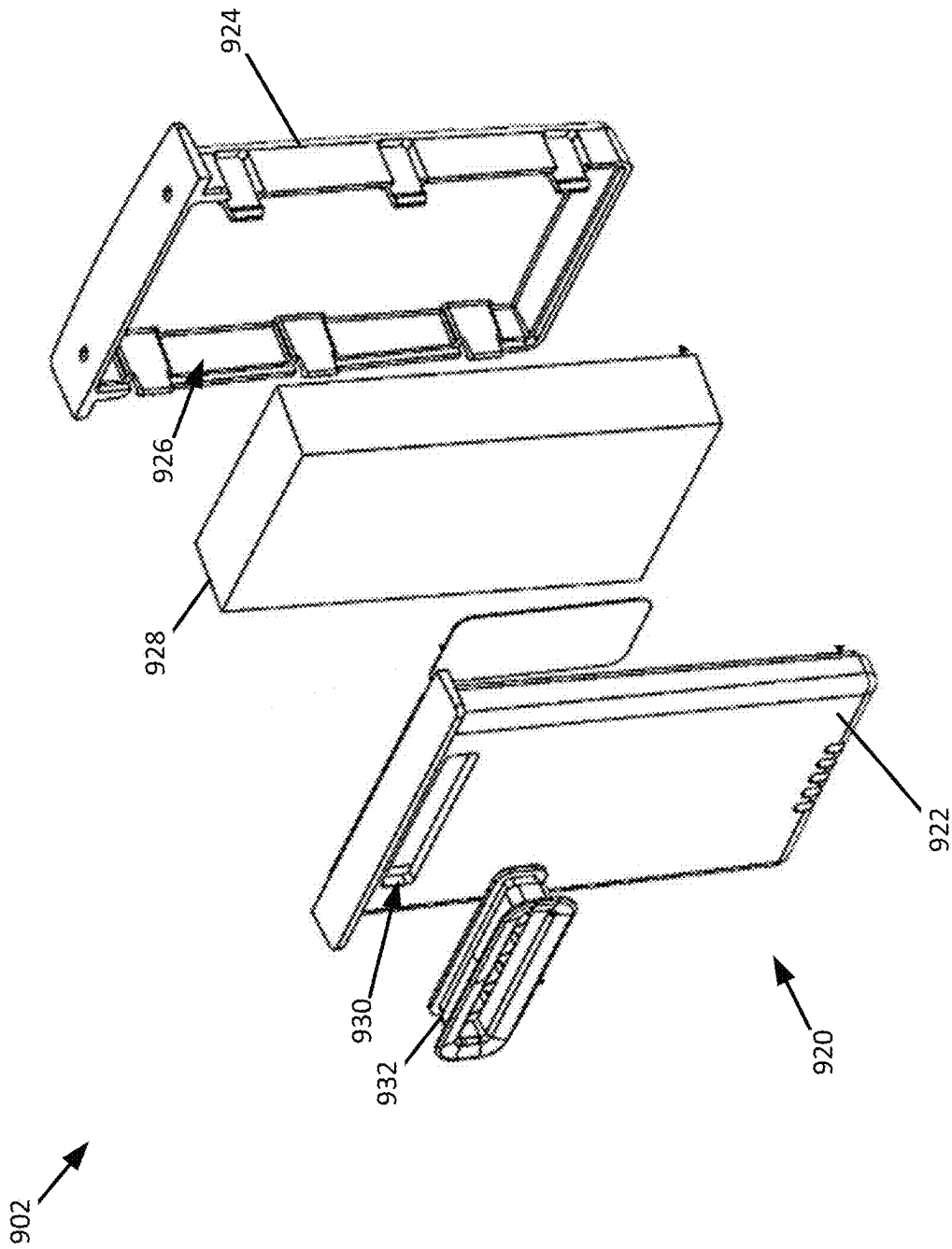


FIG. 9A

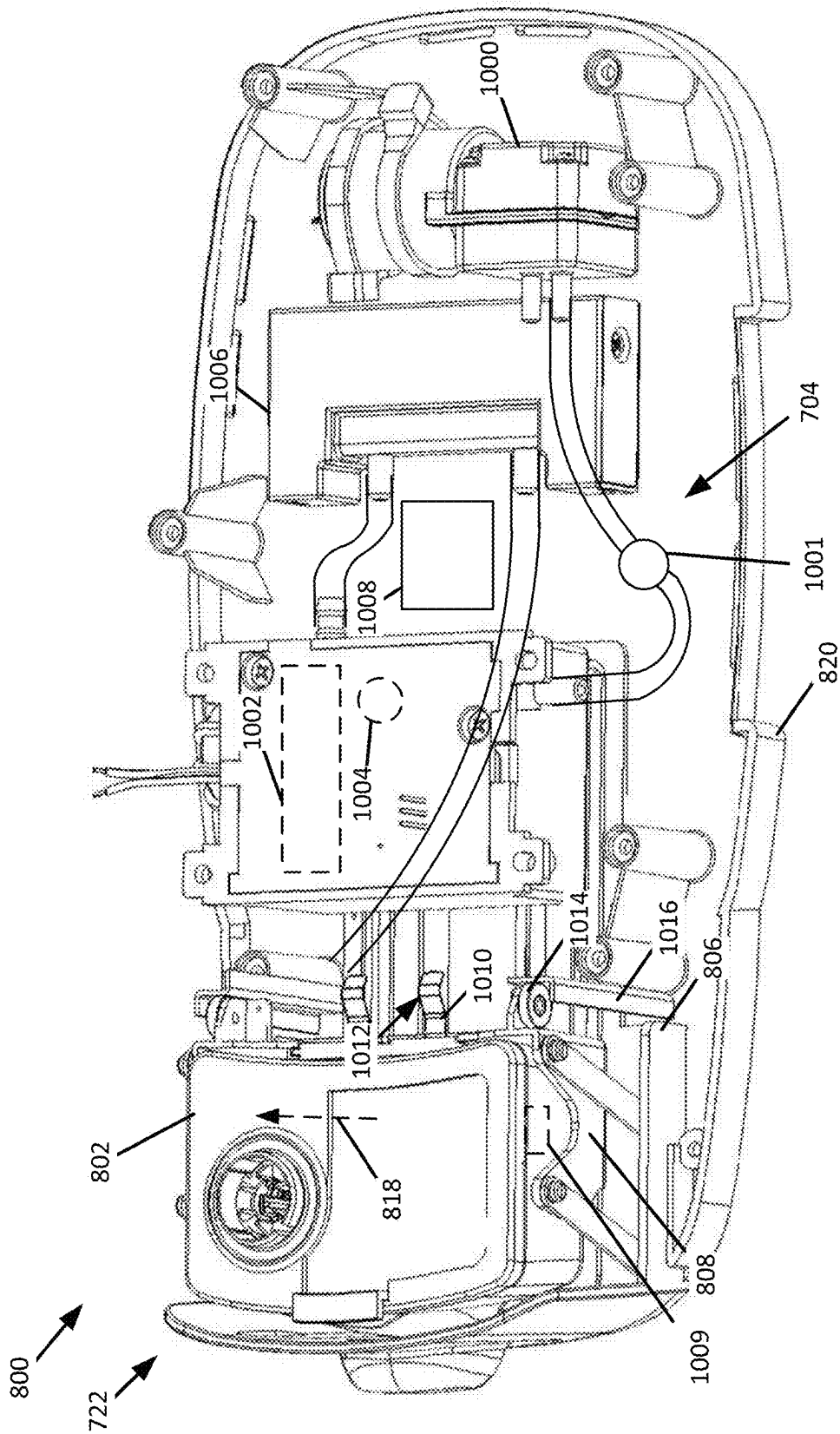


FIG. 10

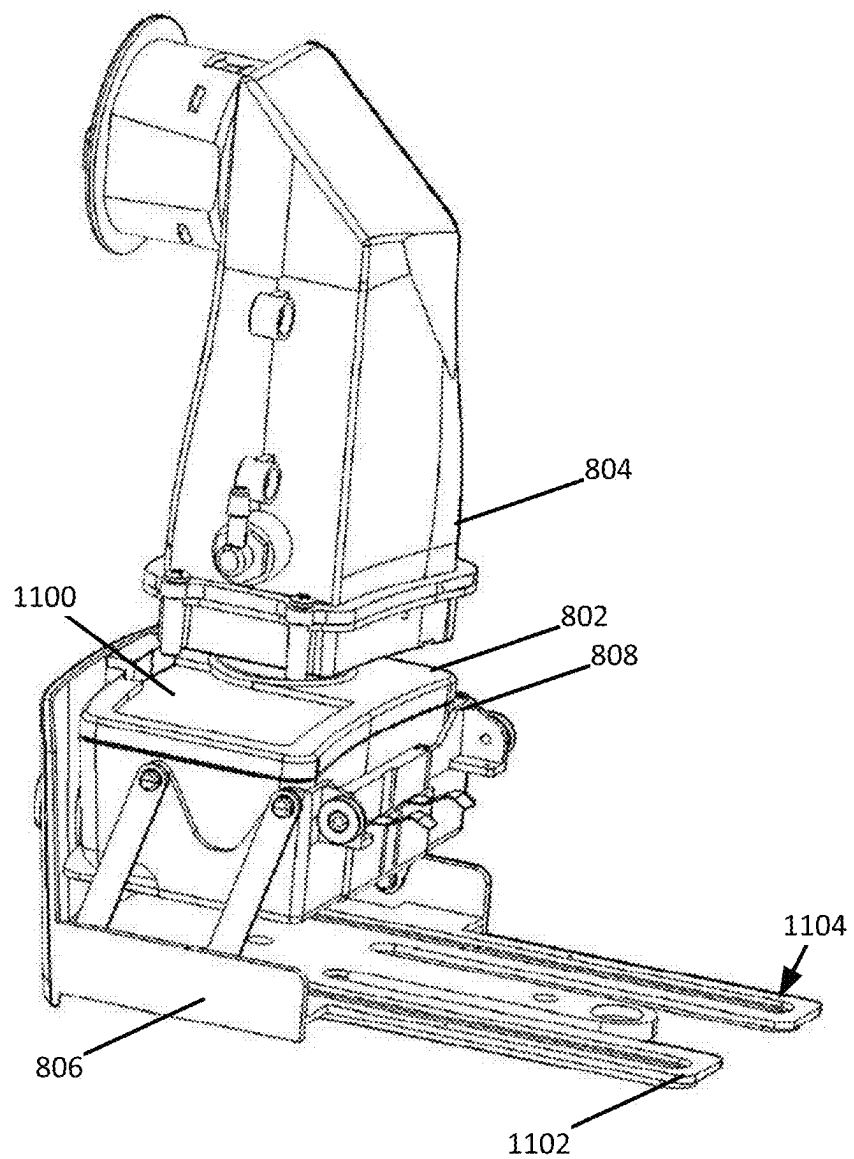
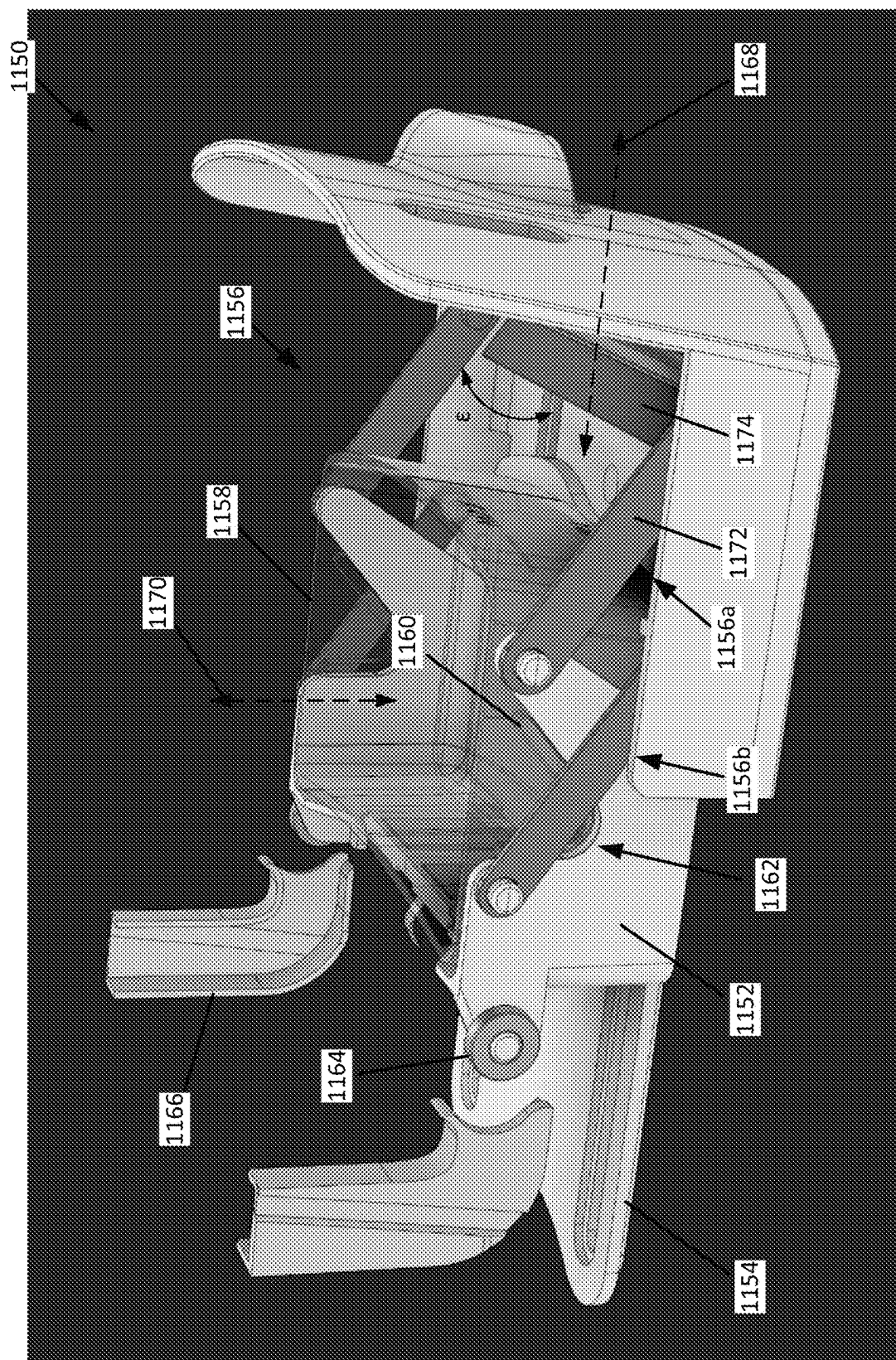
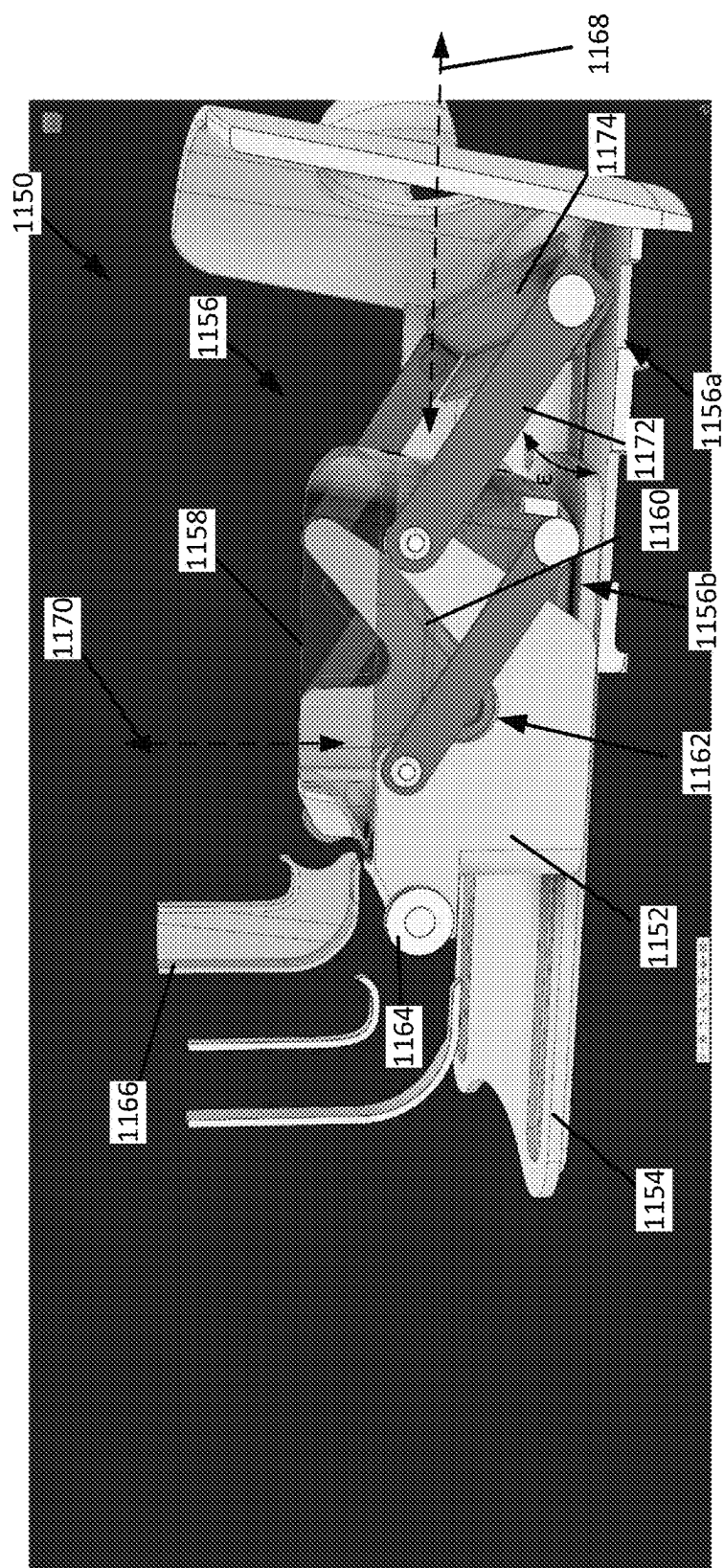


FIG. 11





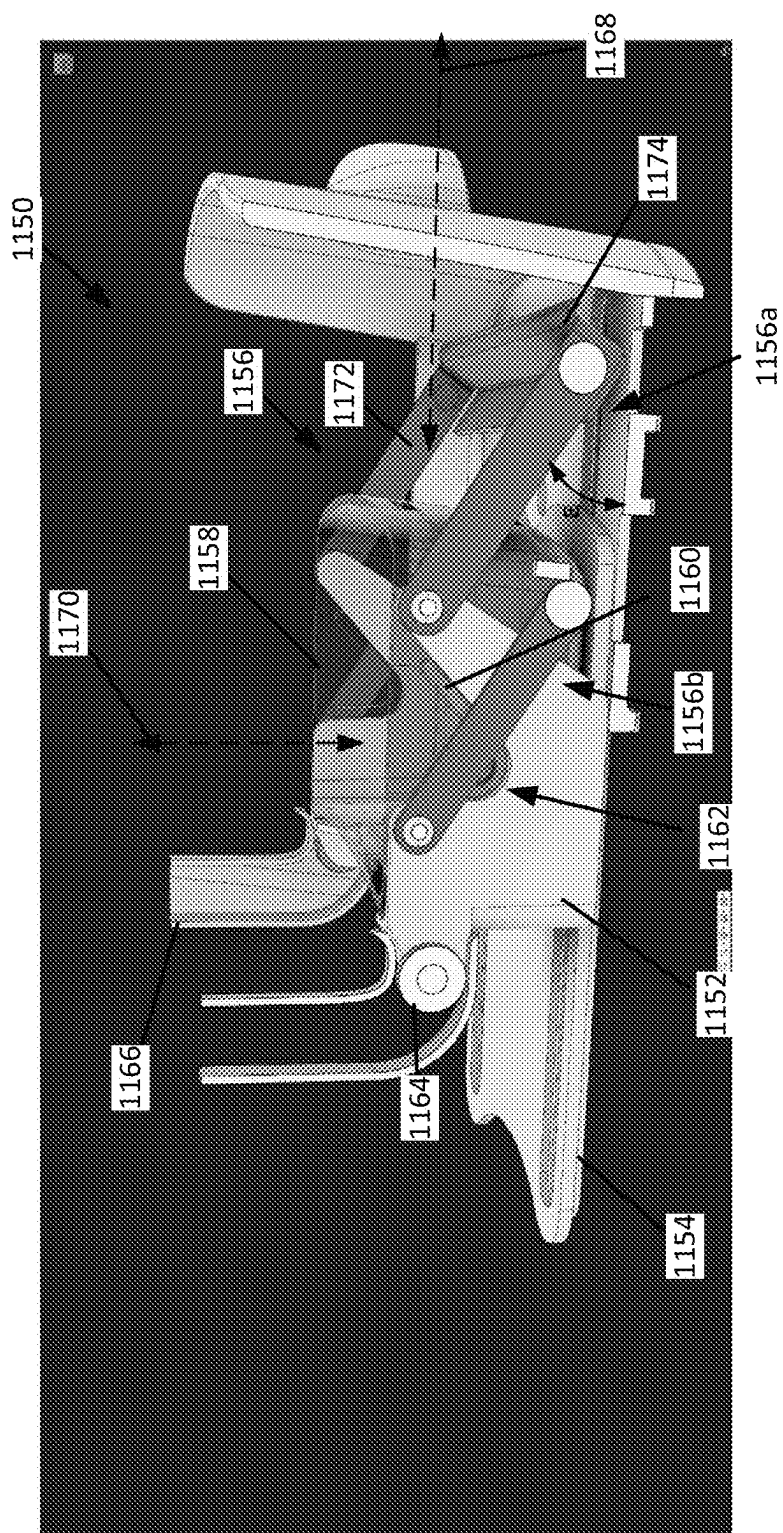


FIG. 11C

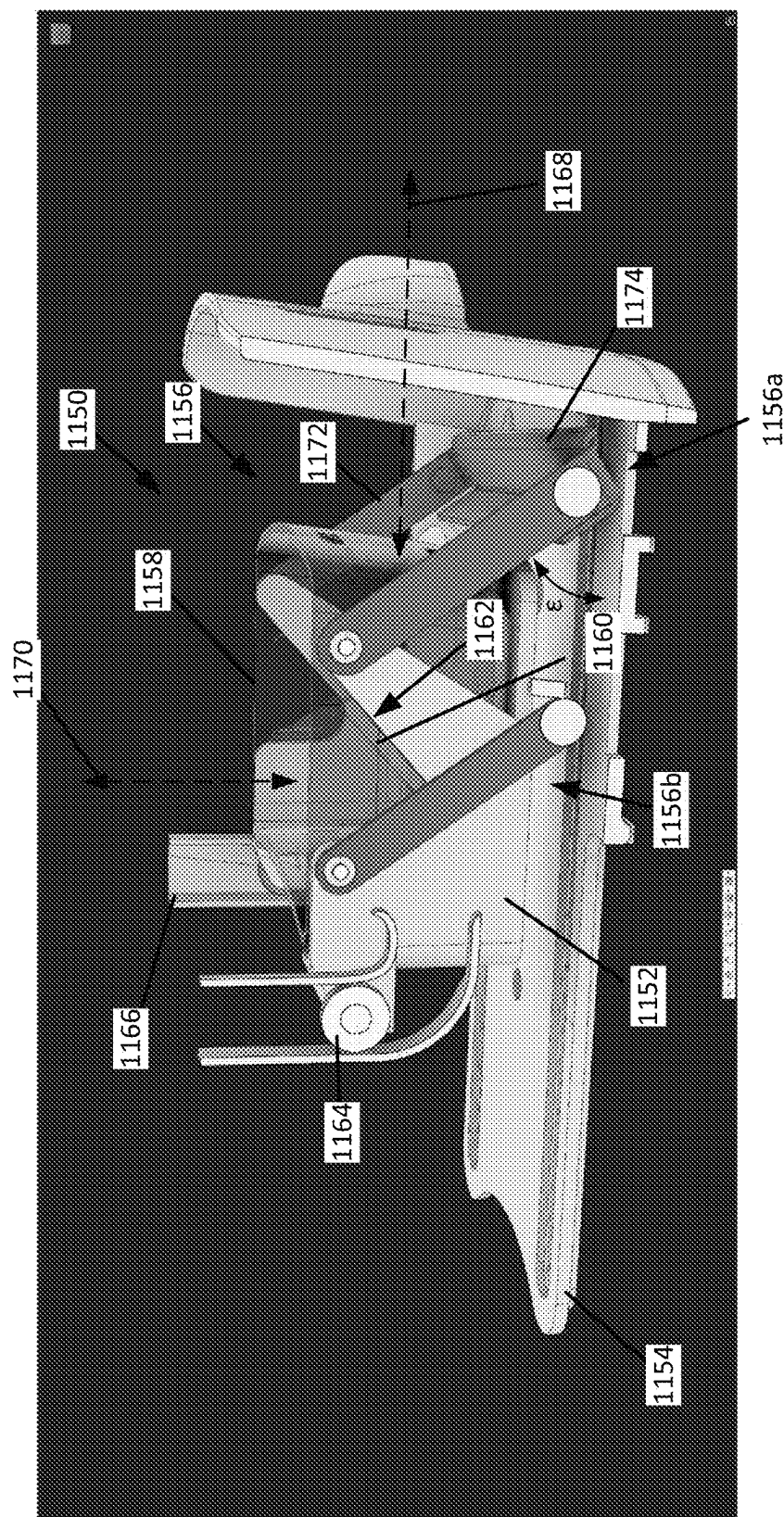


FIG. 11D

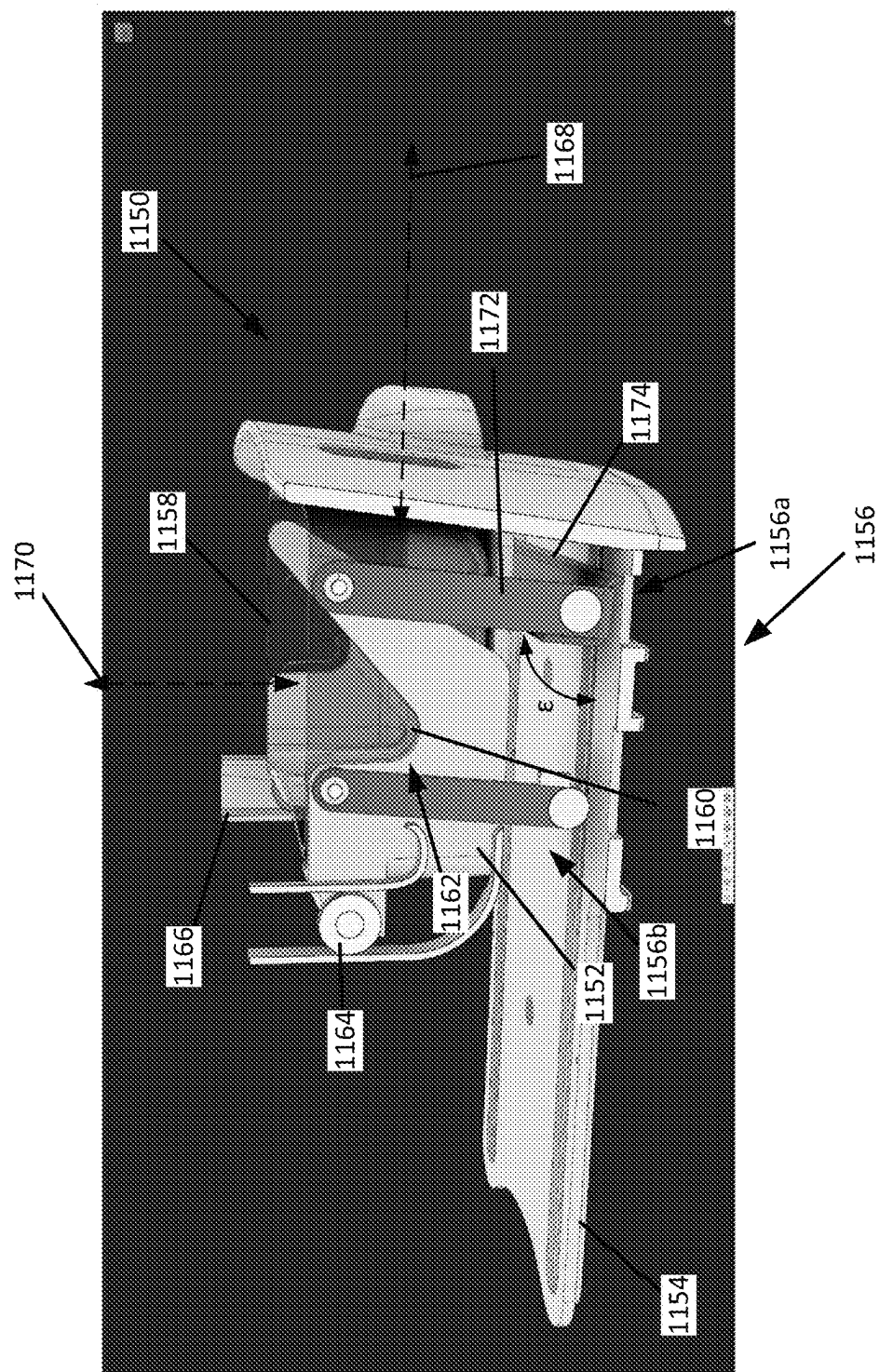


FIG. 11E

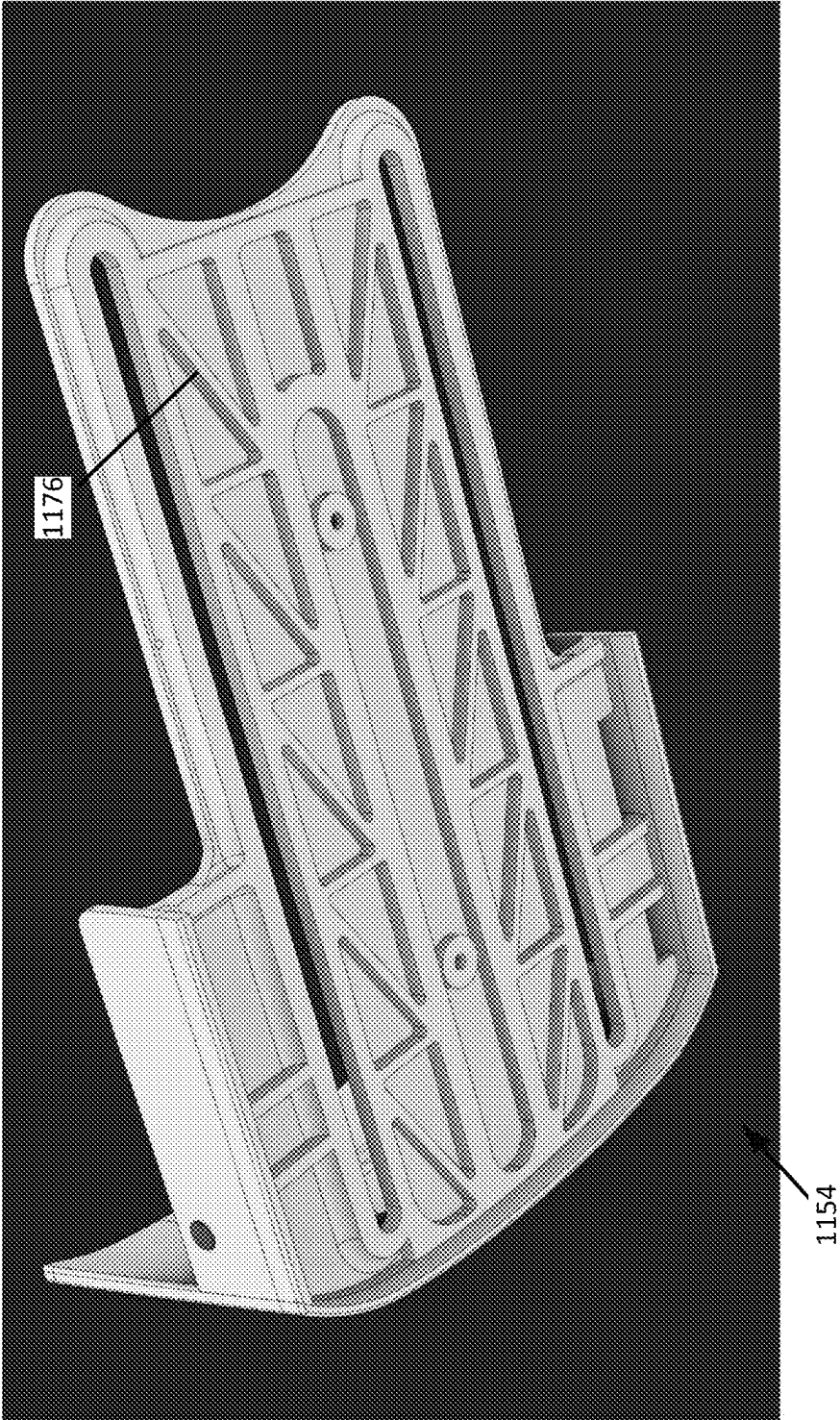


FIG. 11F

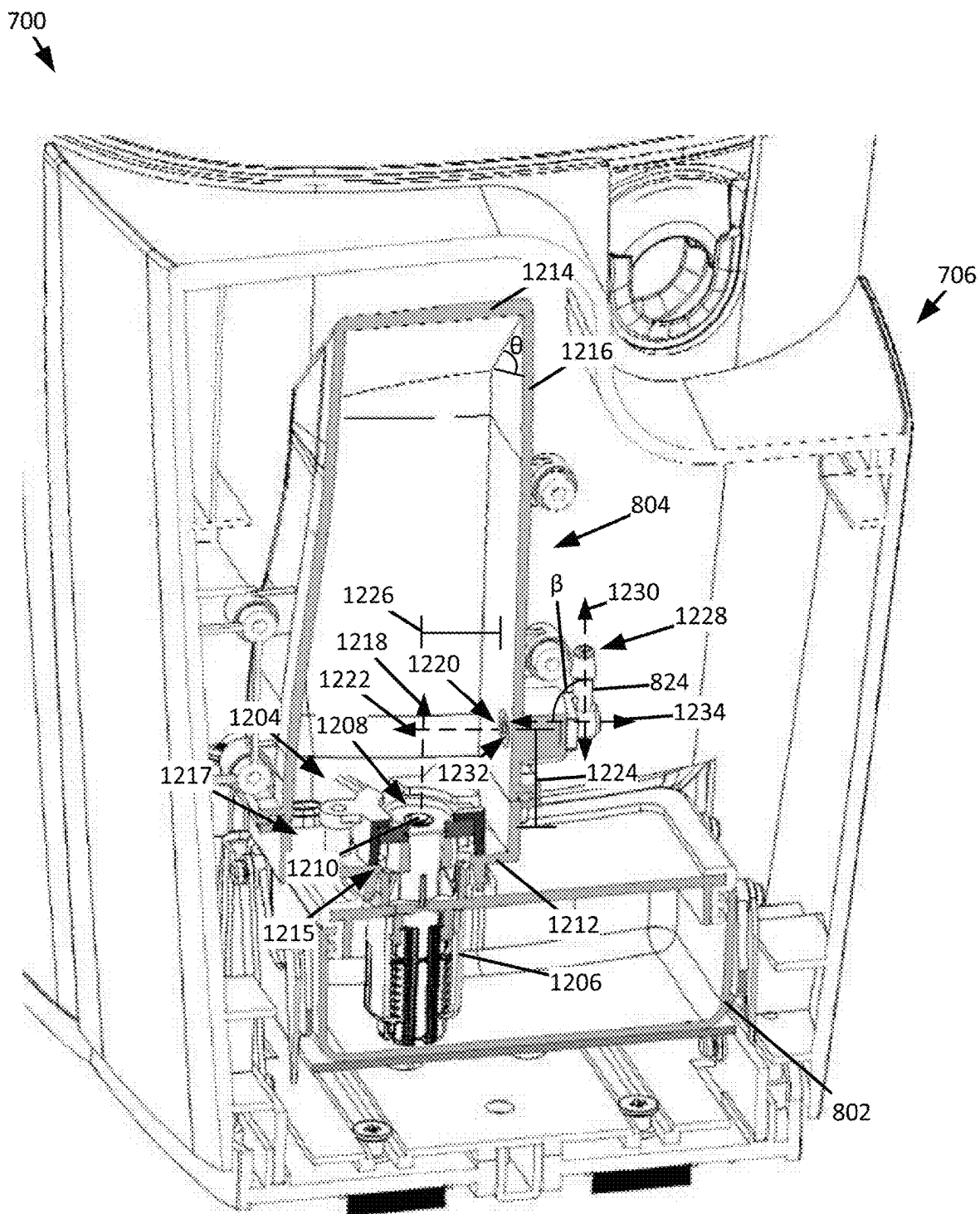


FIG. 12

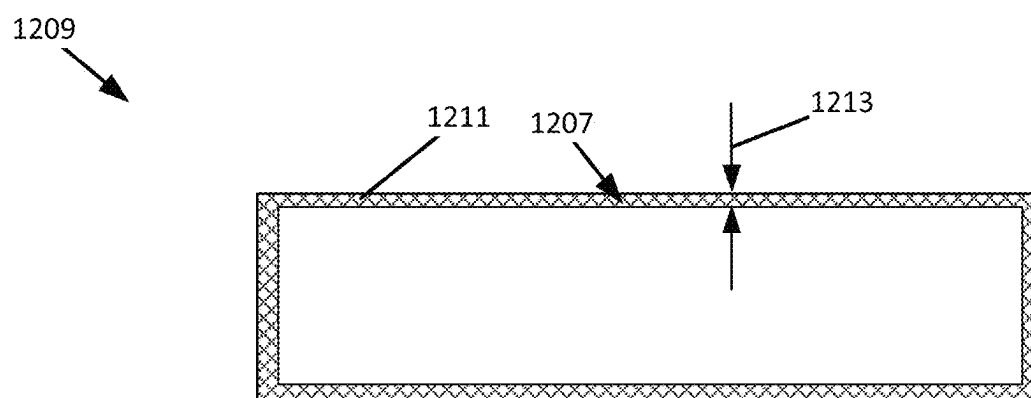


FIG. 12A

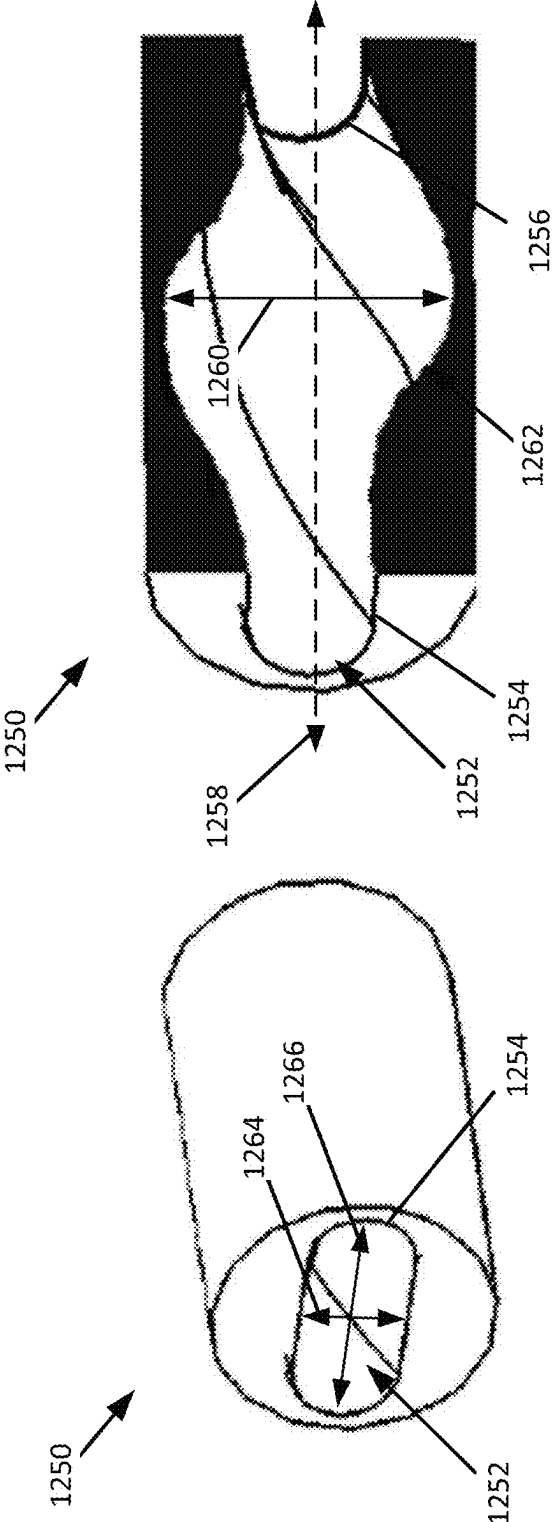


FIG. 12C

FIG. 12B

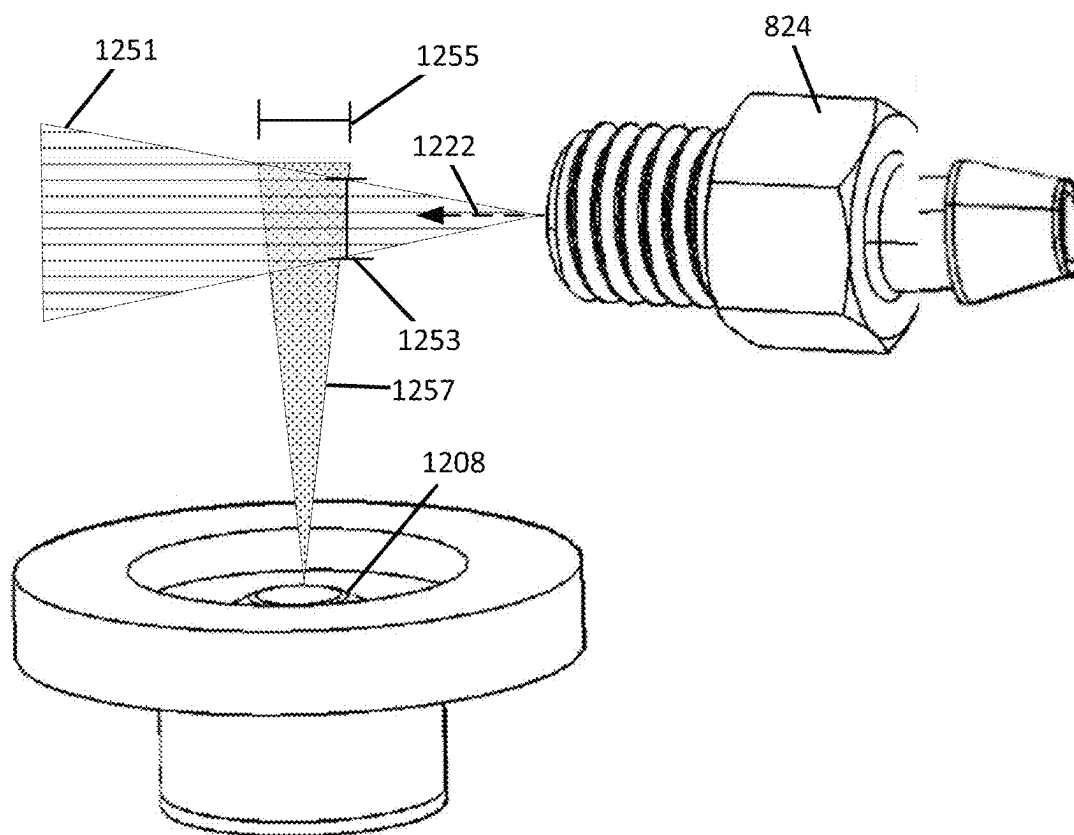


FIG. 12D

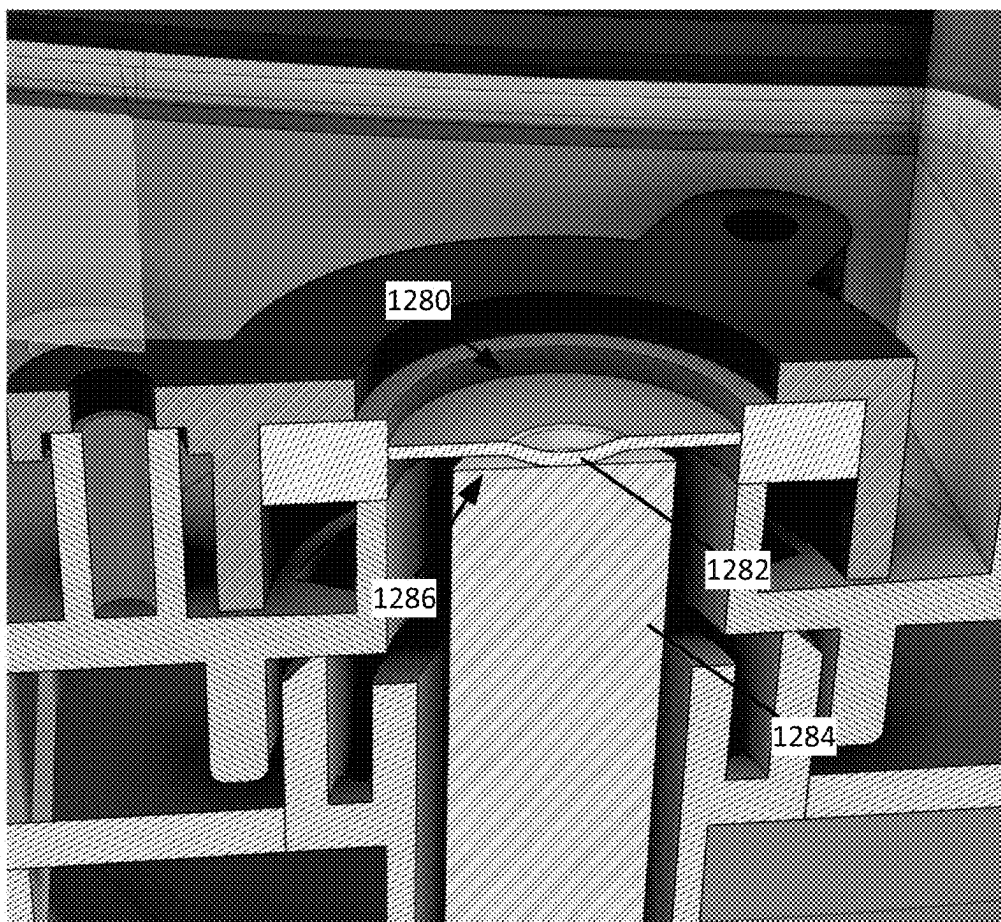


FIG. 12E

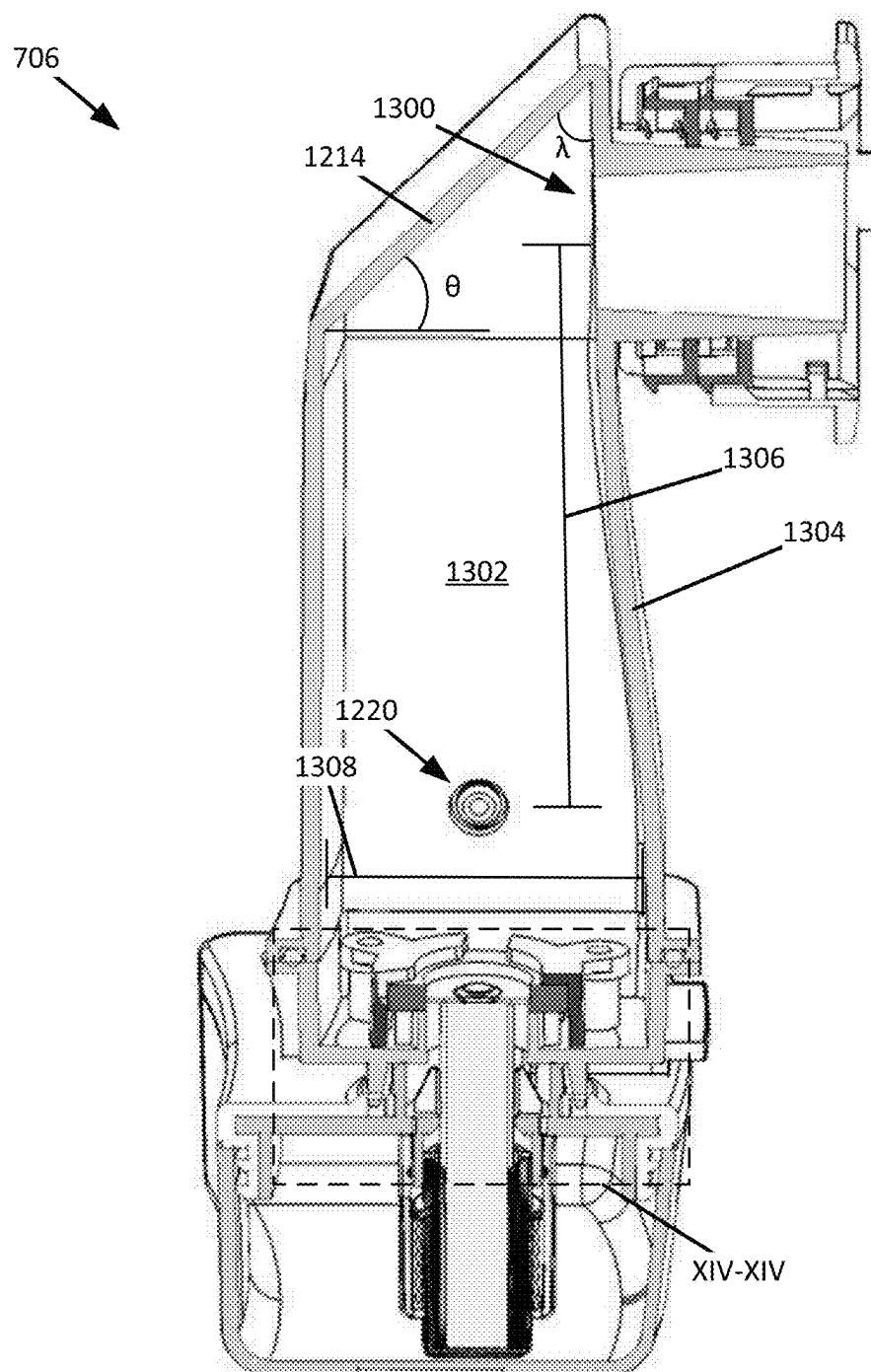


FIG. 13

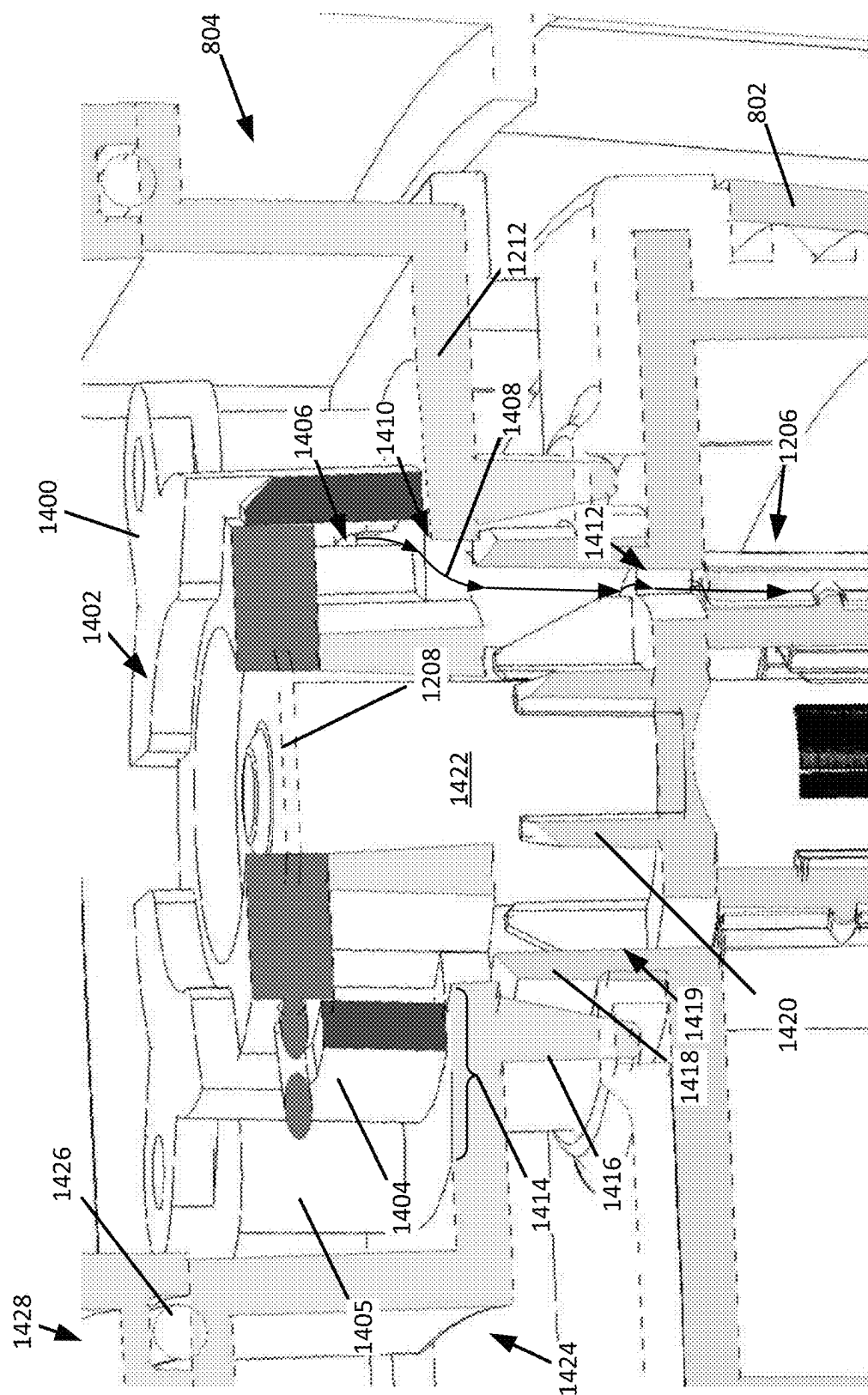


FIG. 14

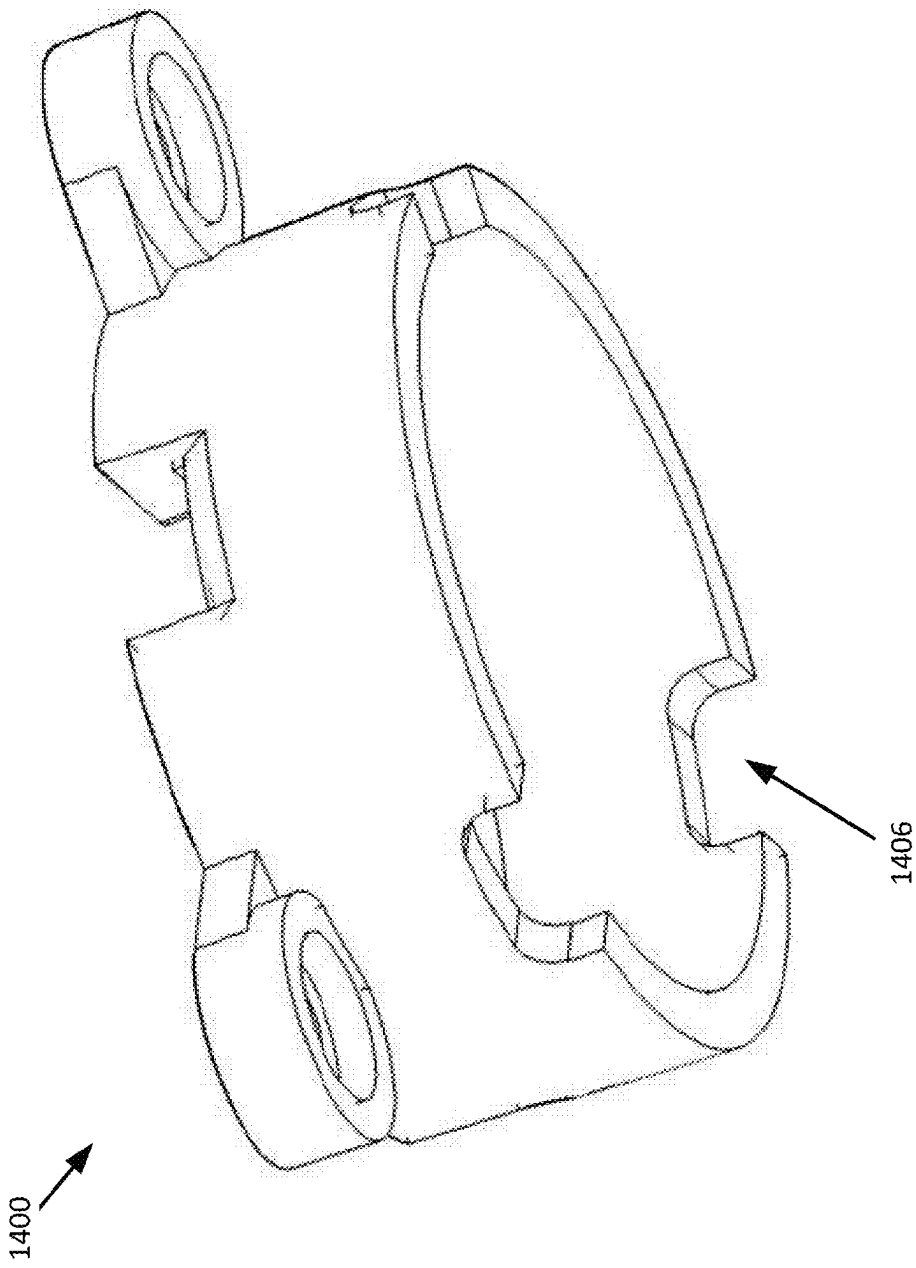
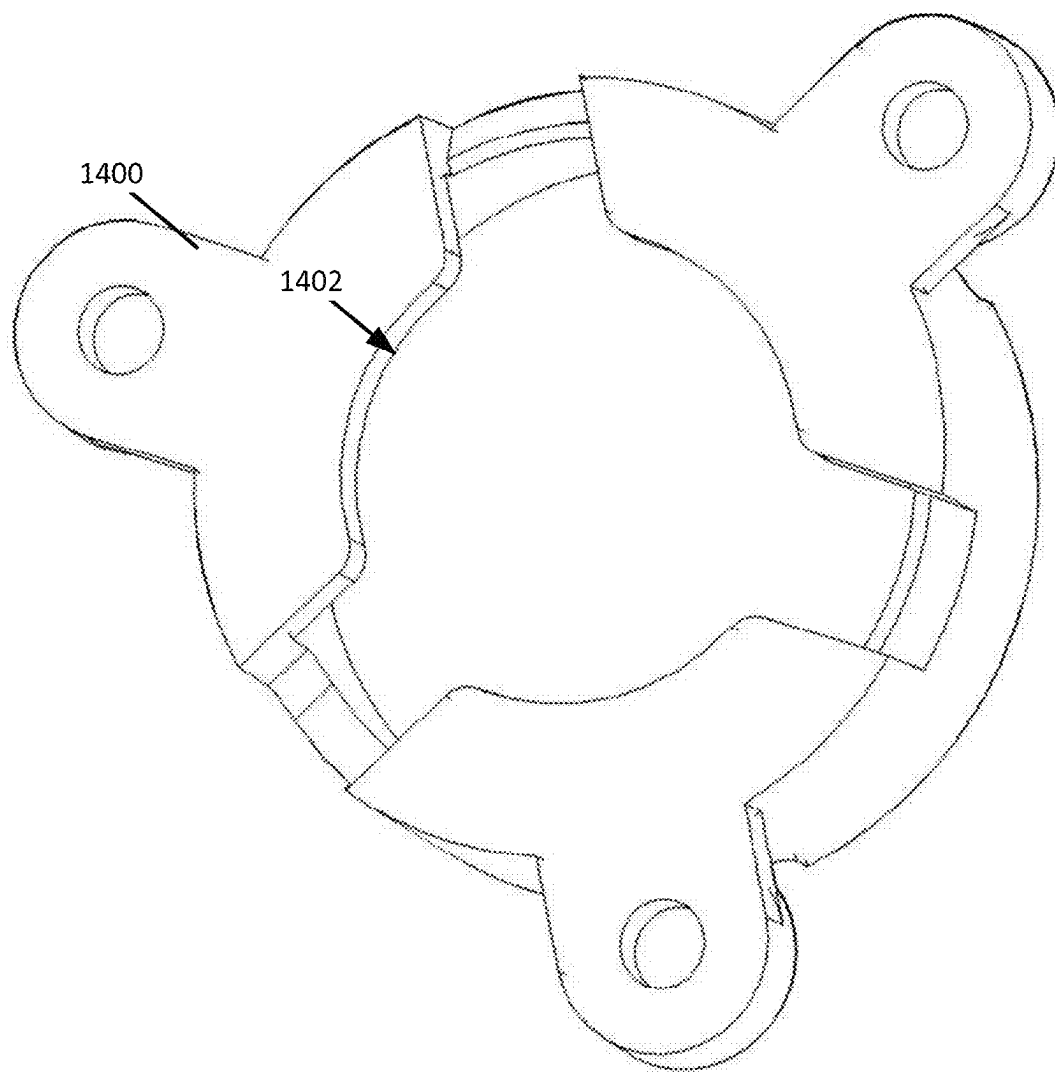


FIG. 14A

**FIG. 14B**

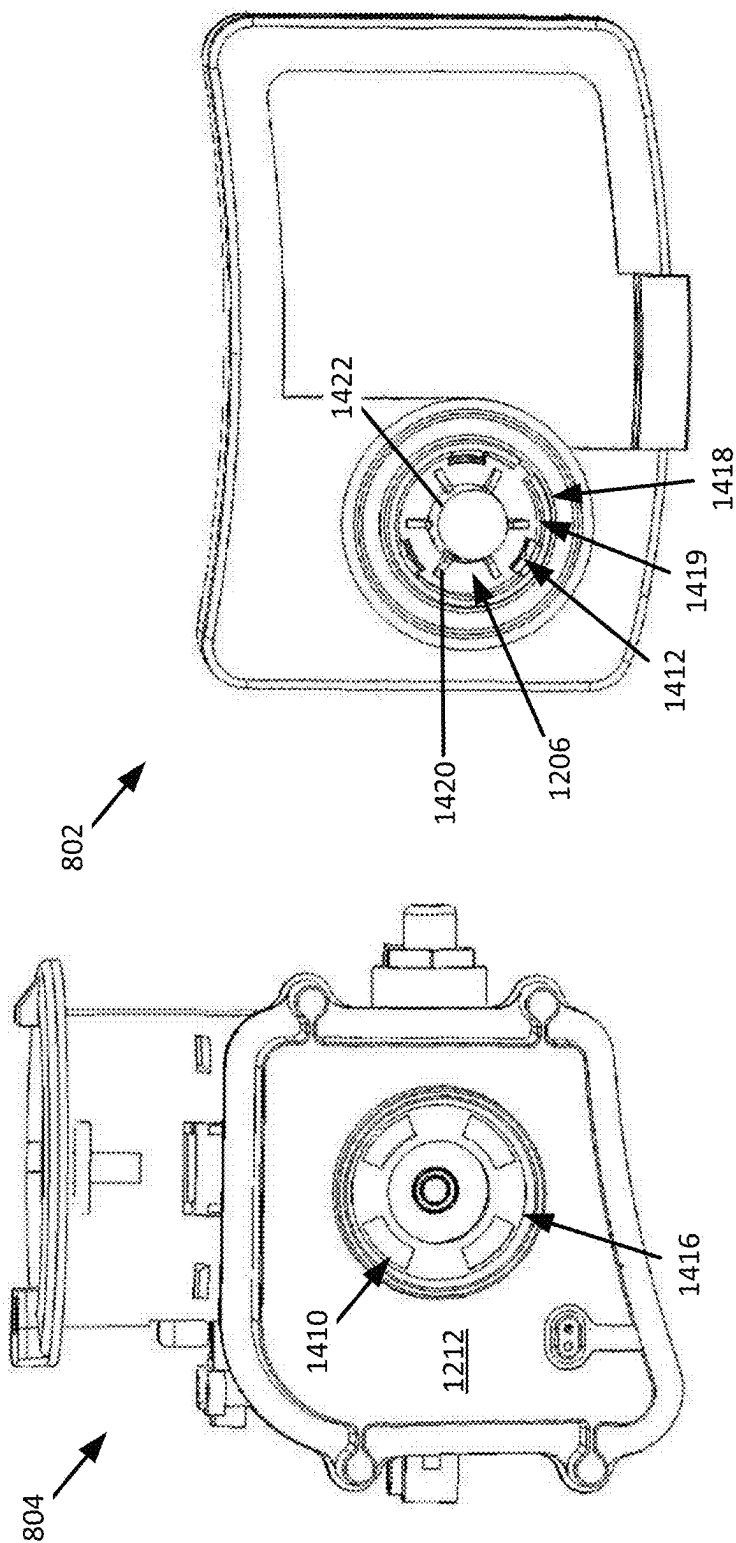


FIG. 15

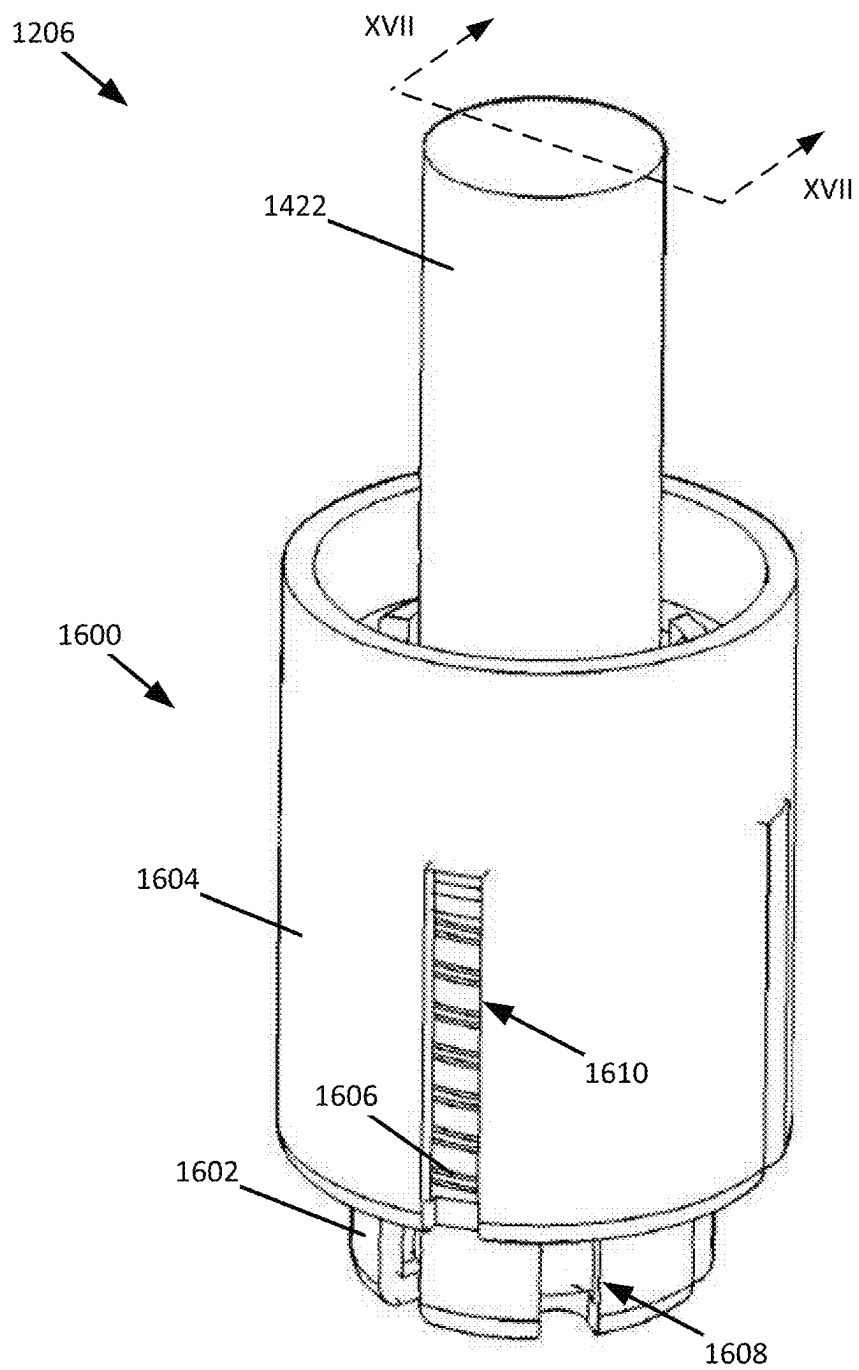


FIG. 16

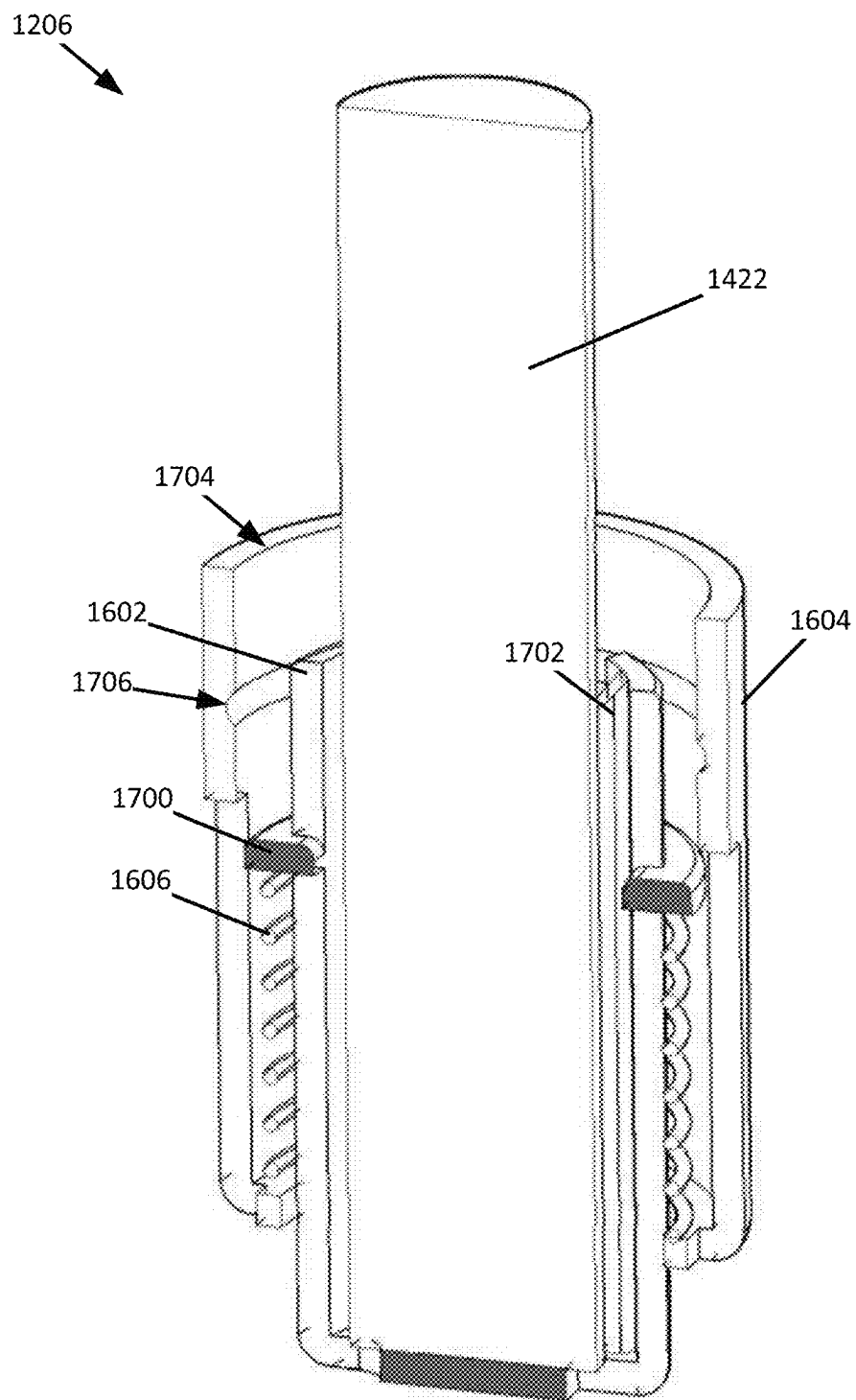


FIG. 17

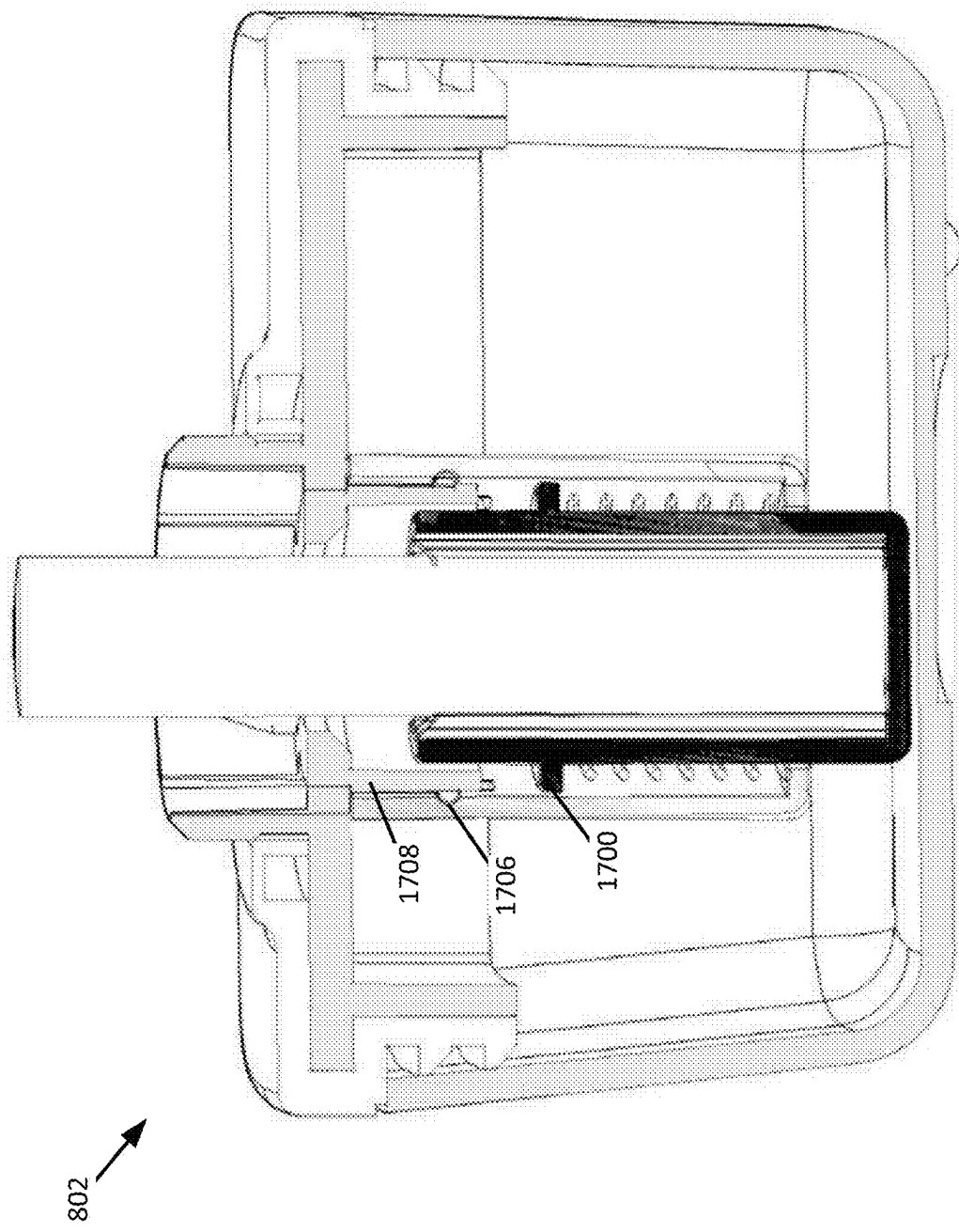


FIG. 18

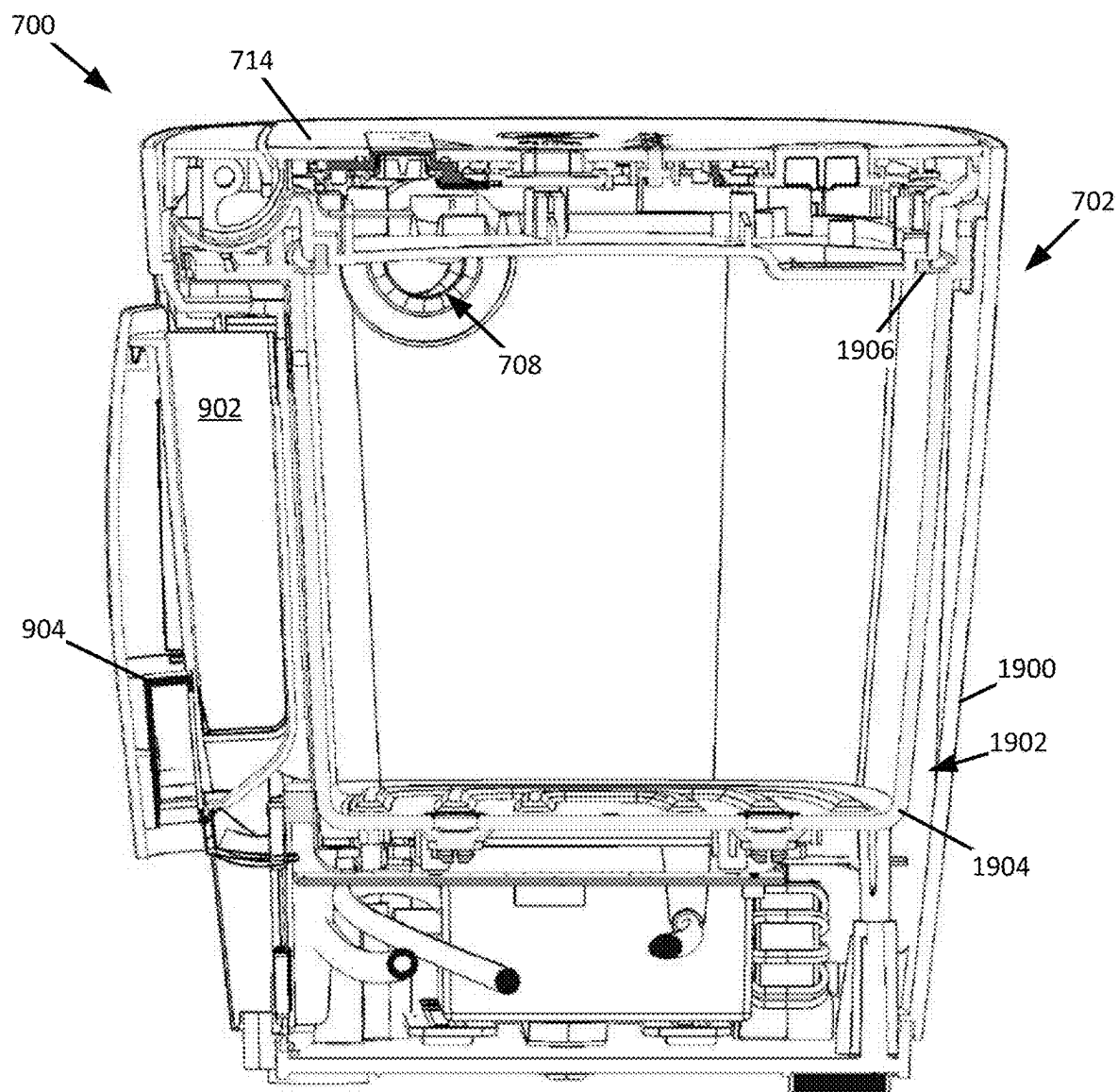


FIG. 19

1920

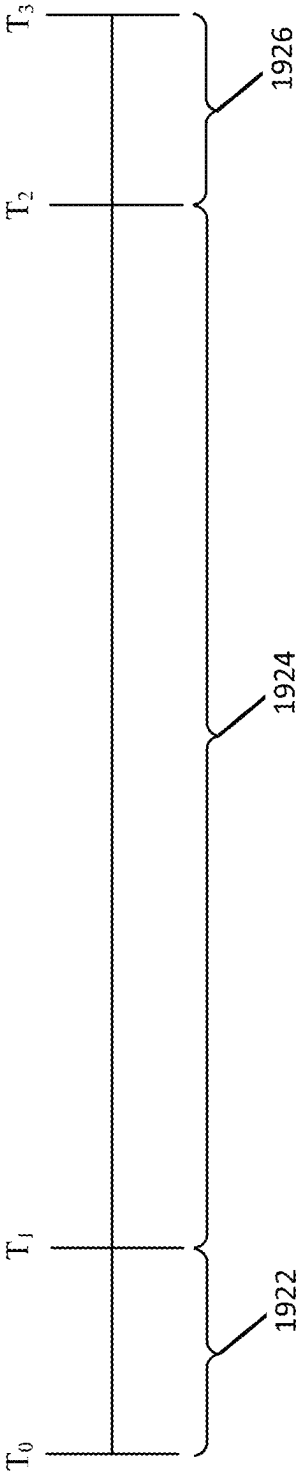


FIG. 19A

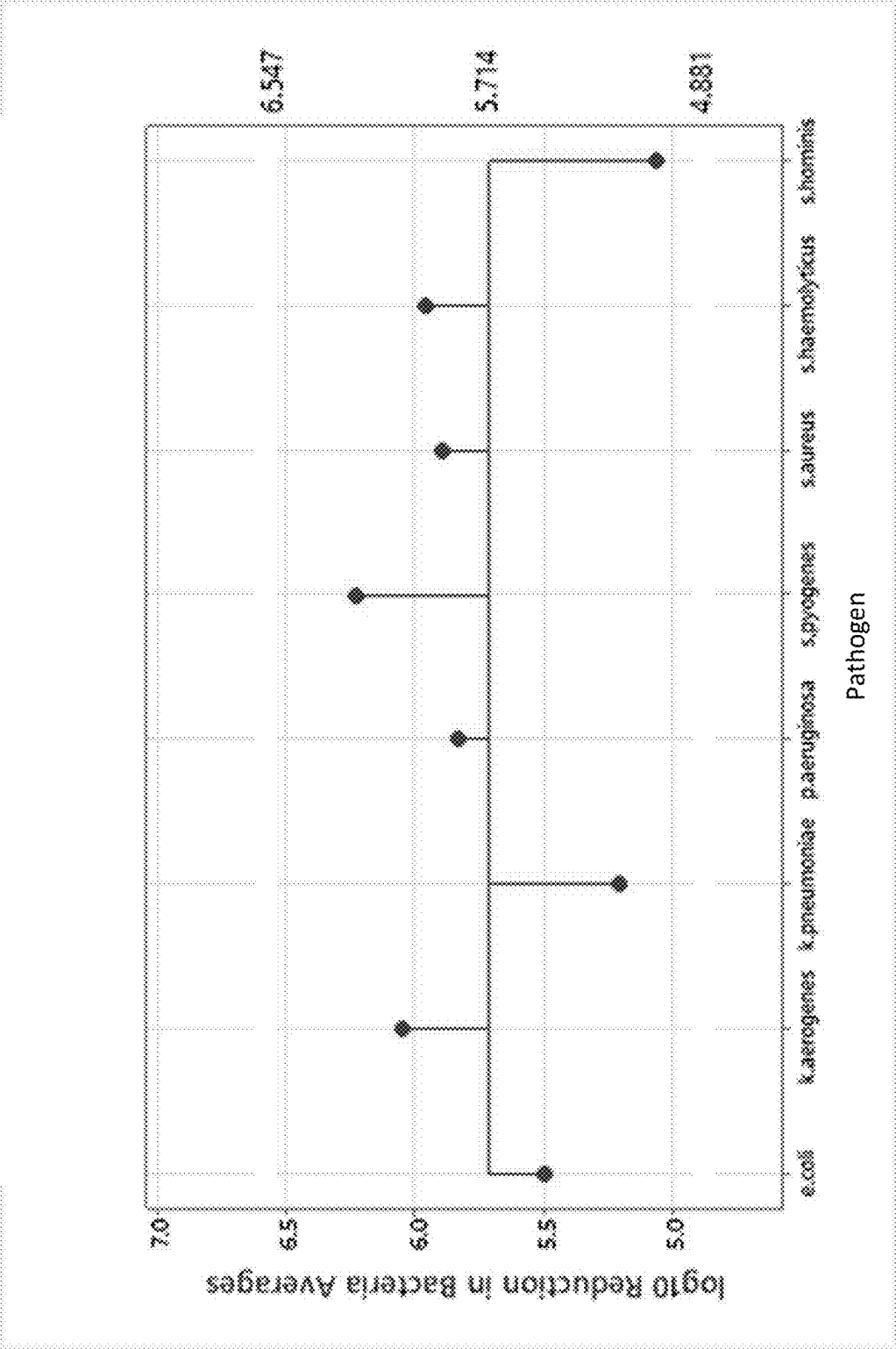


FIG. 19B

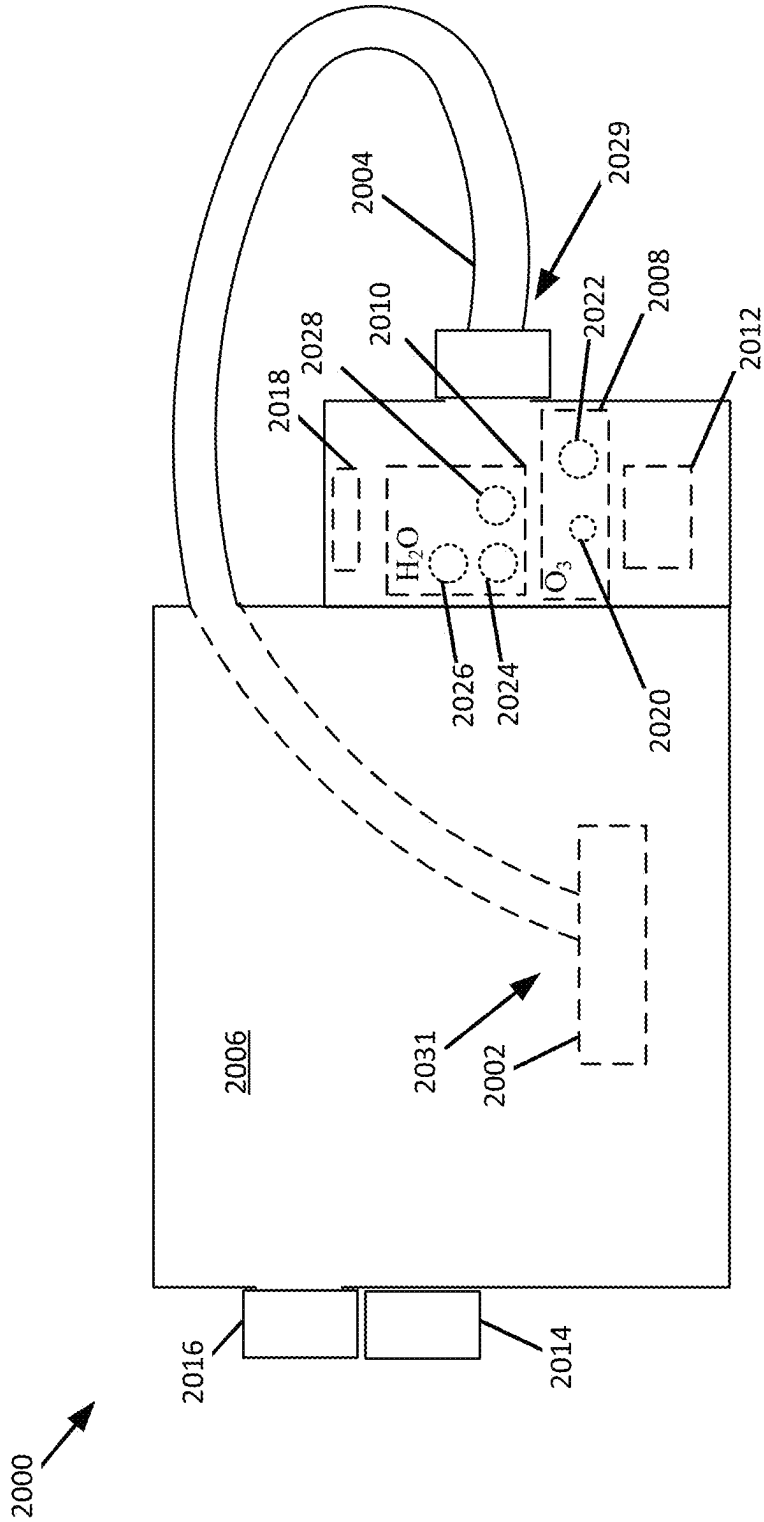


FIG. 20

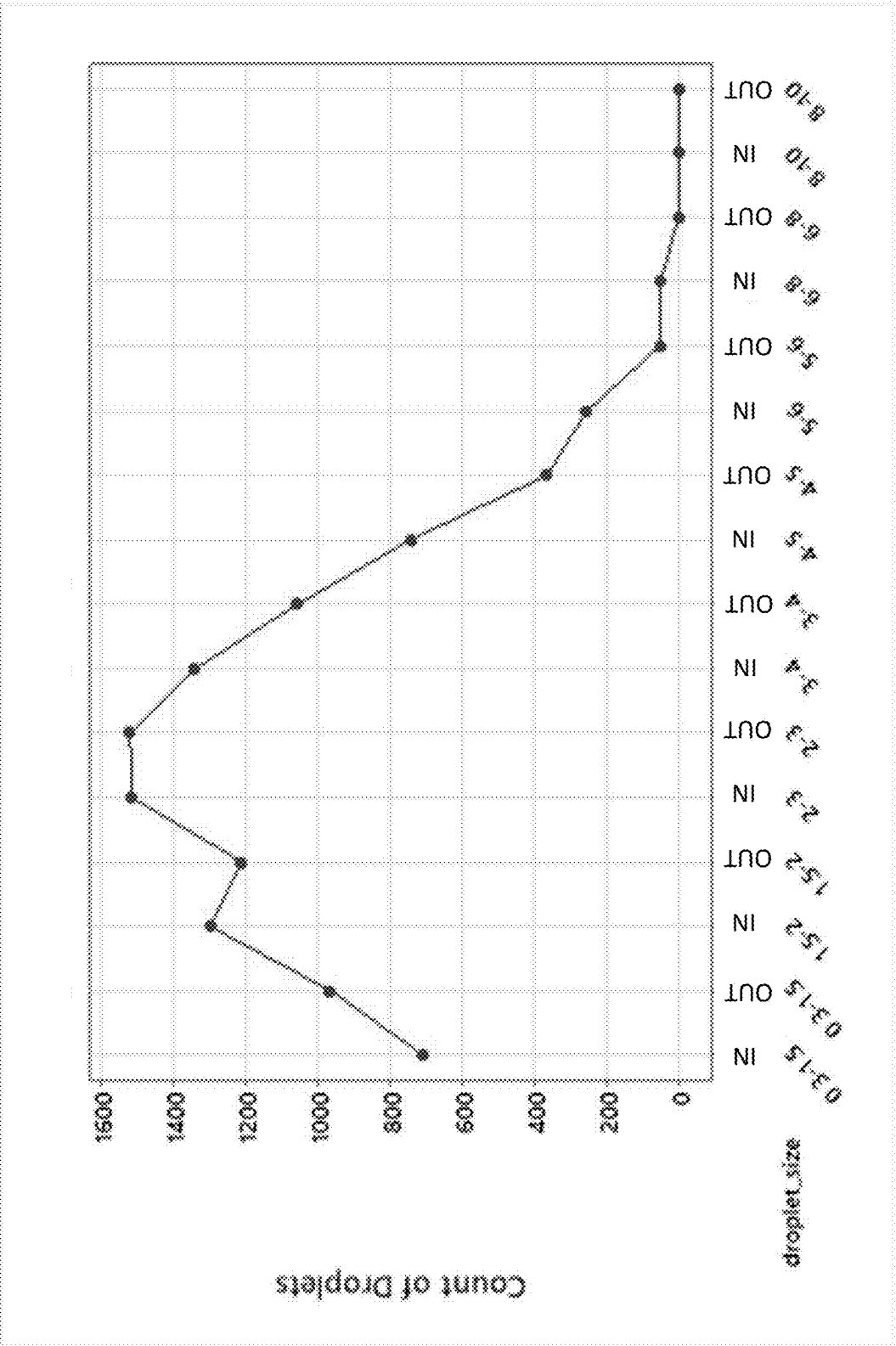


FIG. 20A

2100

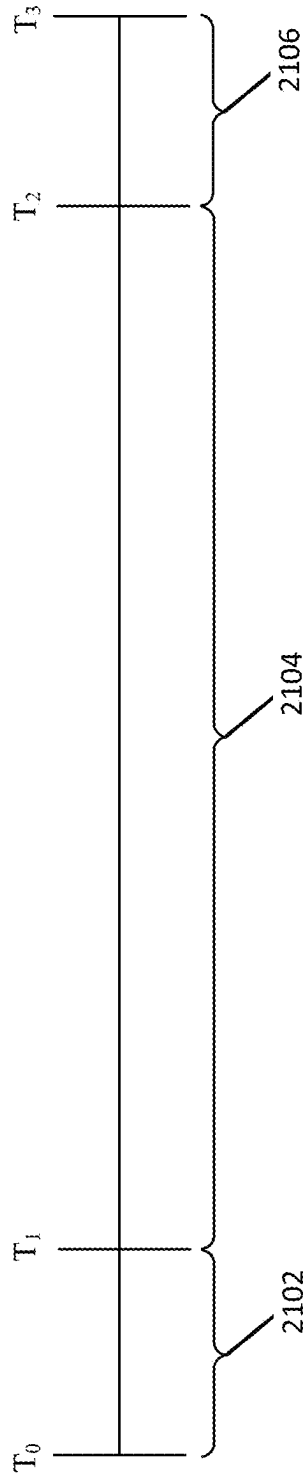


FIG. 21

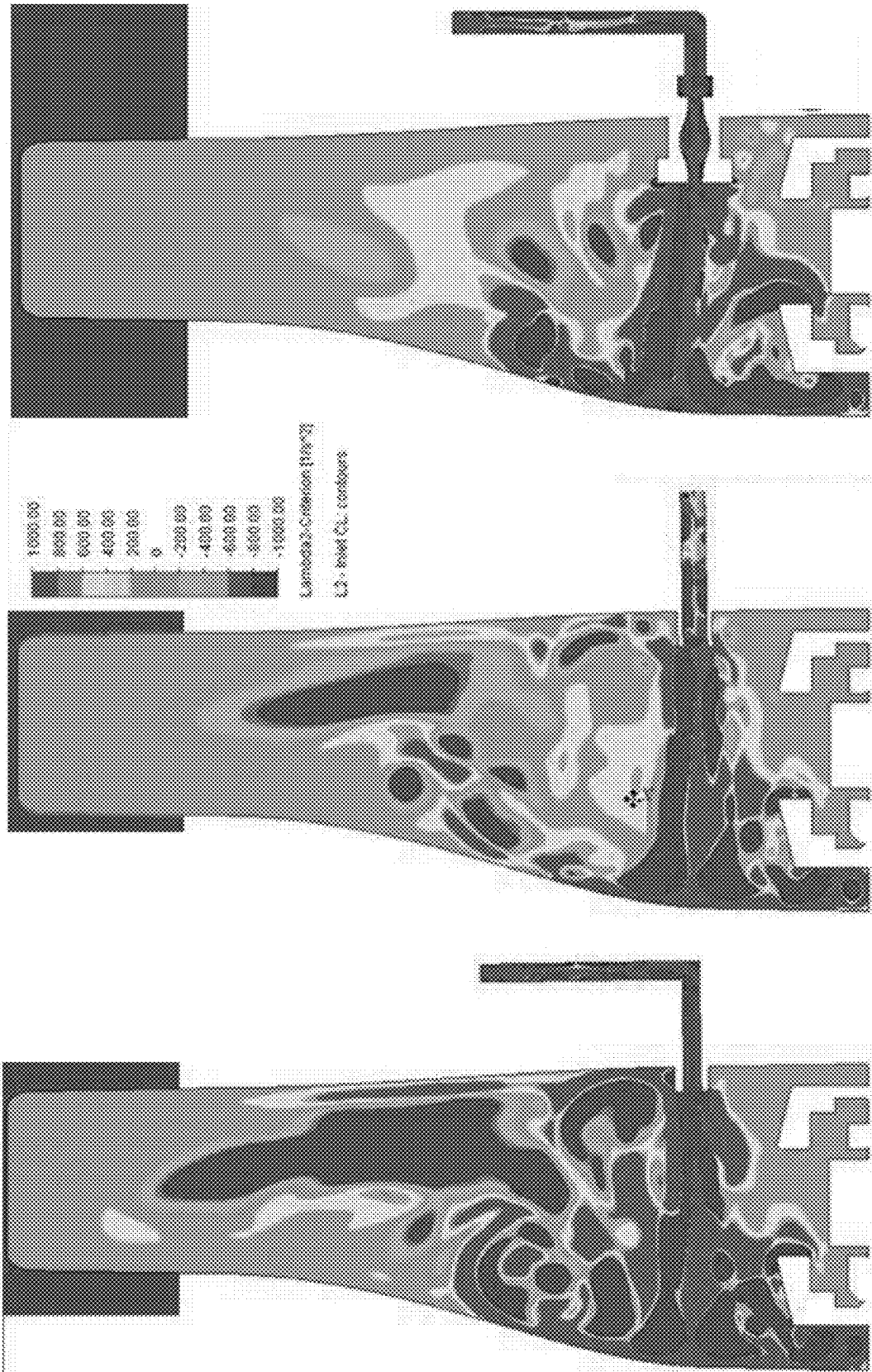


FIG. 24

FIG. 23

FIG. 22

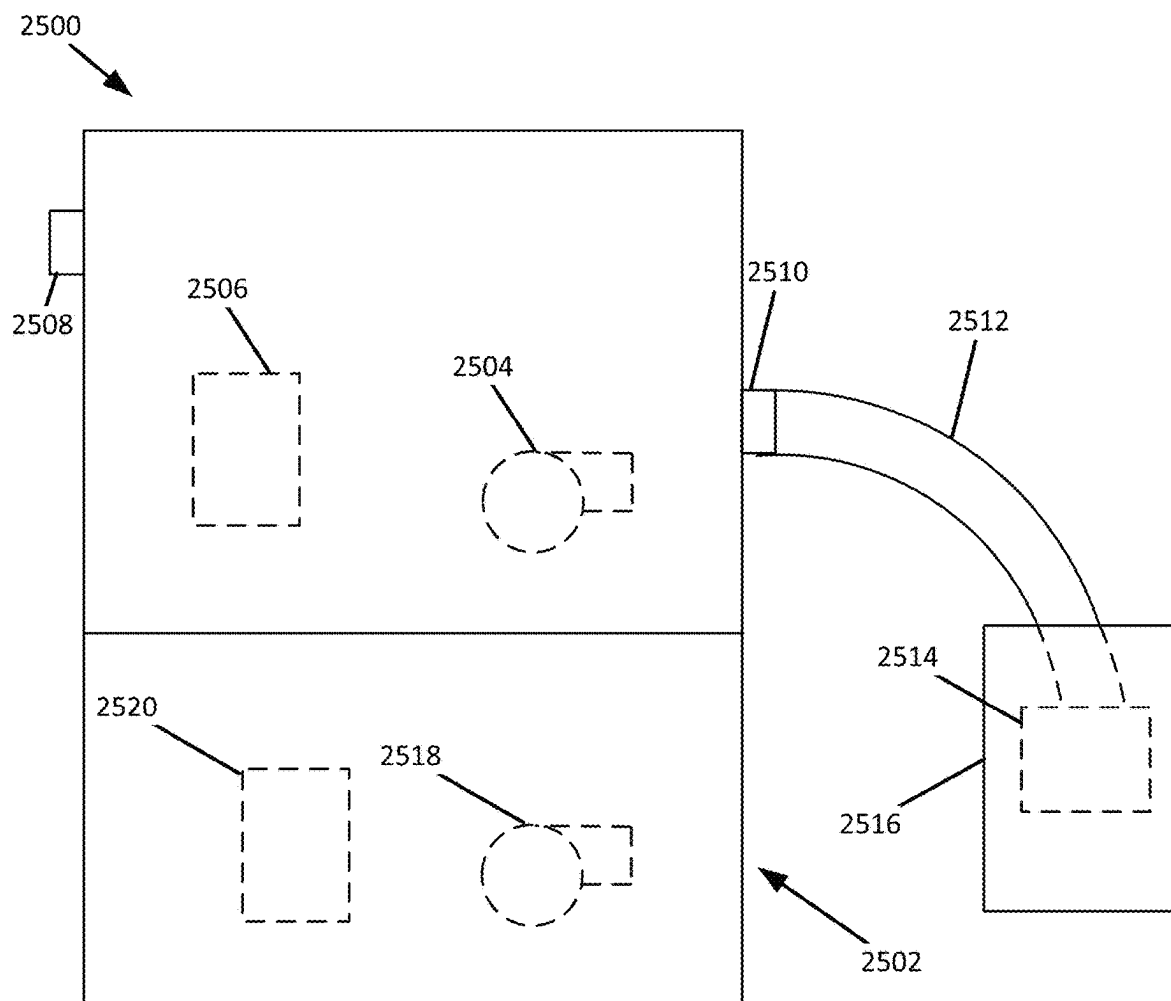


FIG. 25

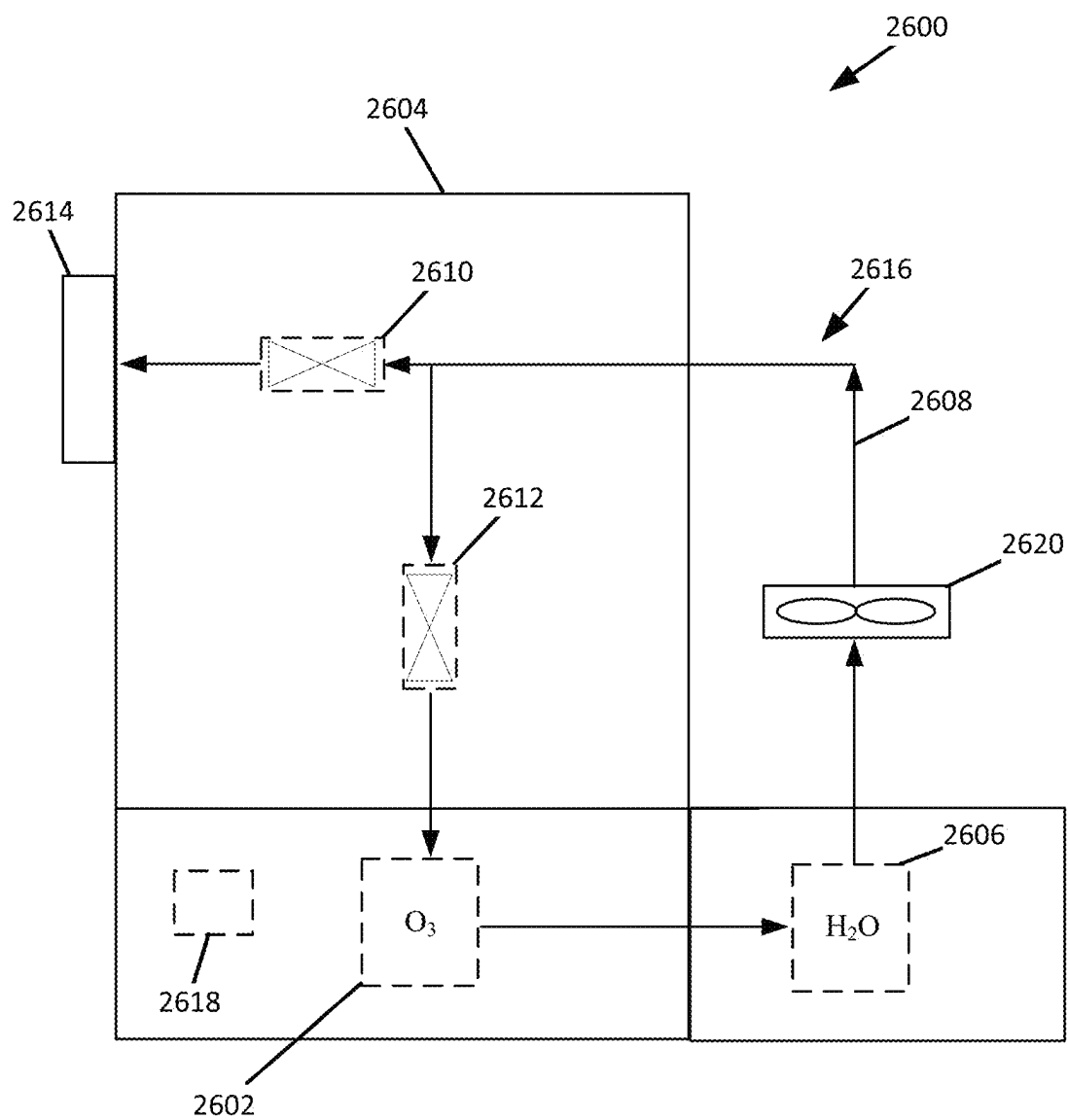


FIG. 26

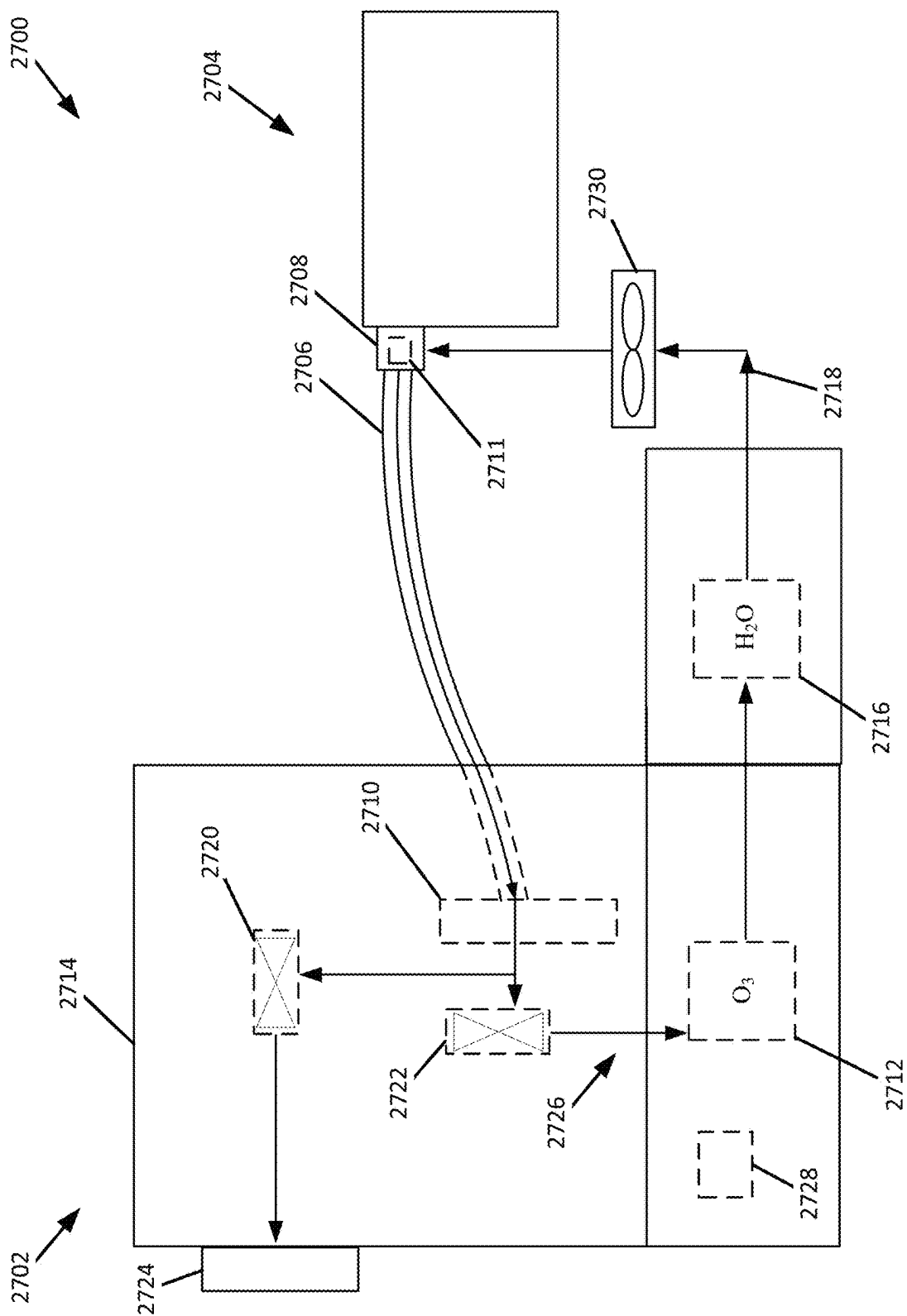


FIG. 27

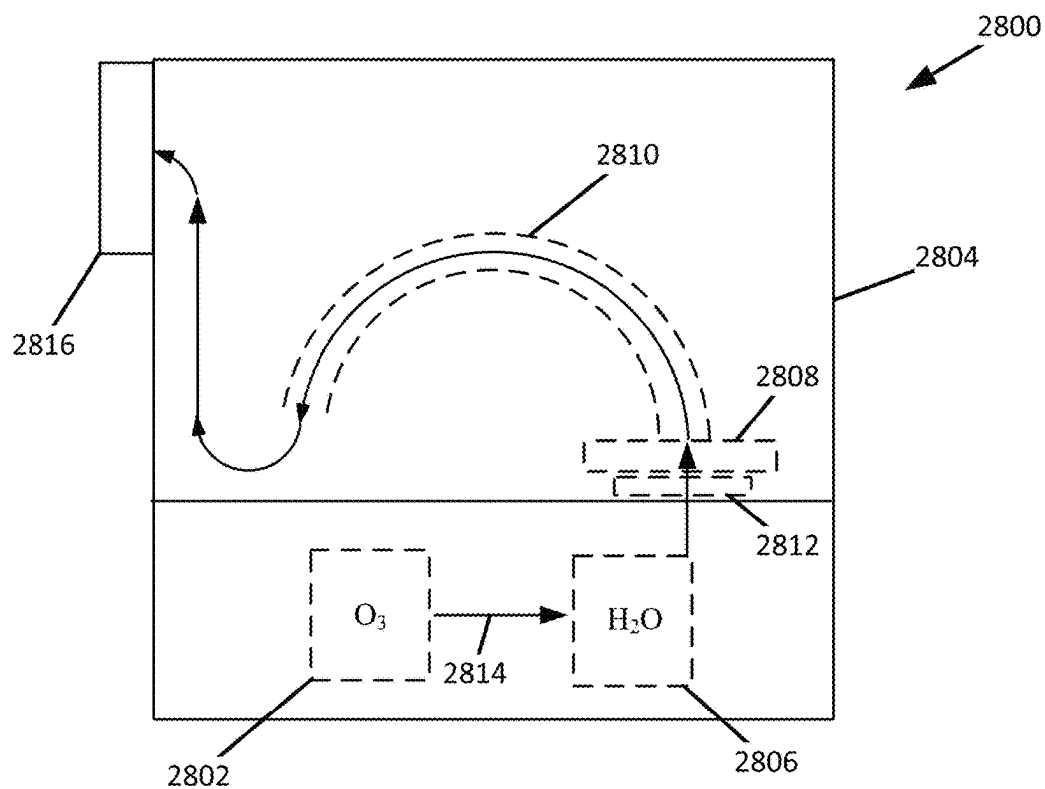


FIG. 28

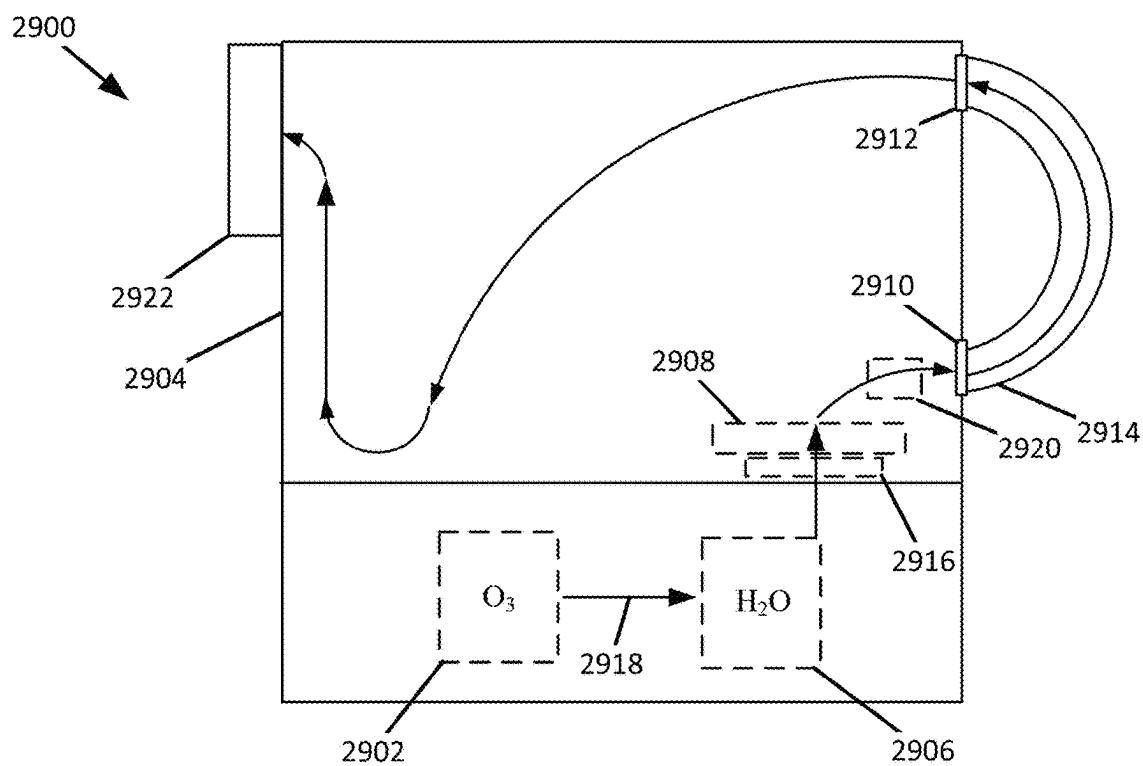


FIG. 29

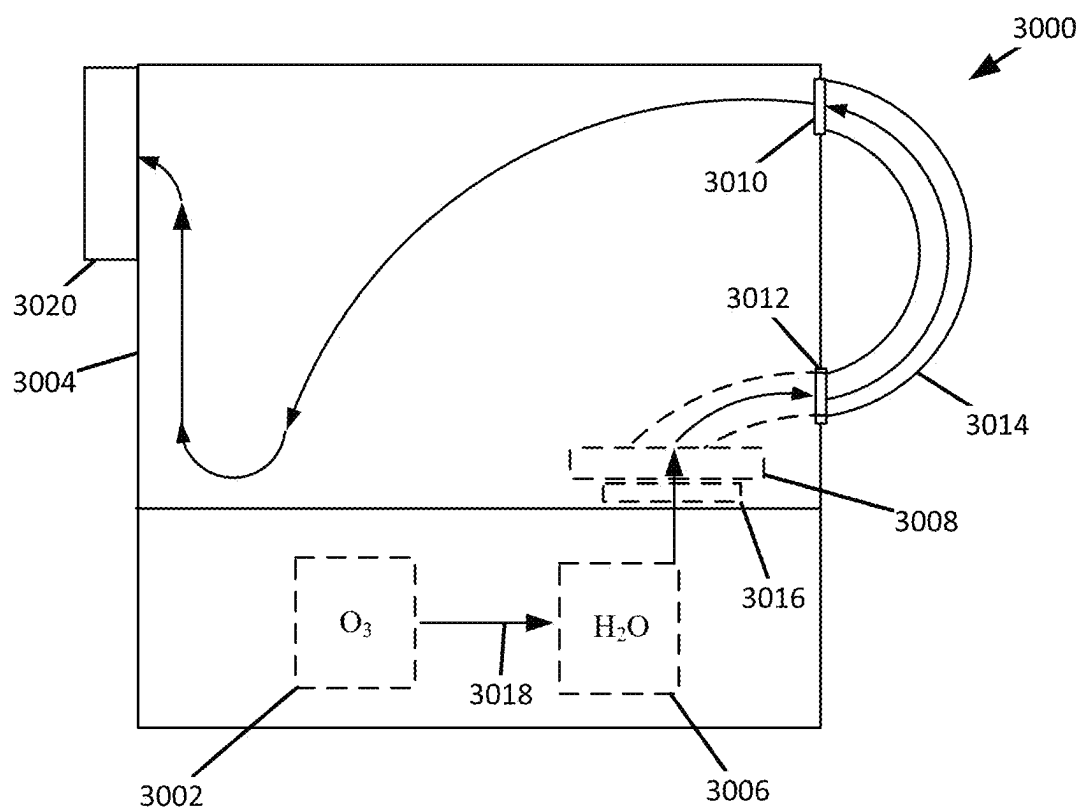


FIG. 30

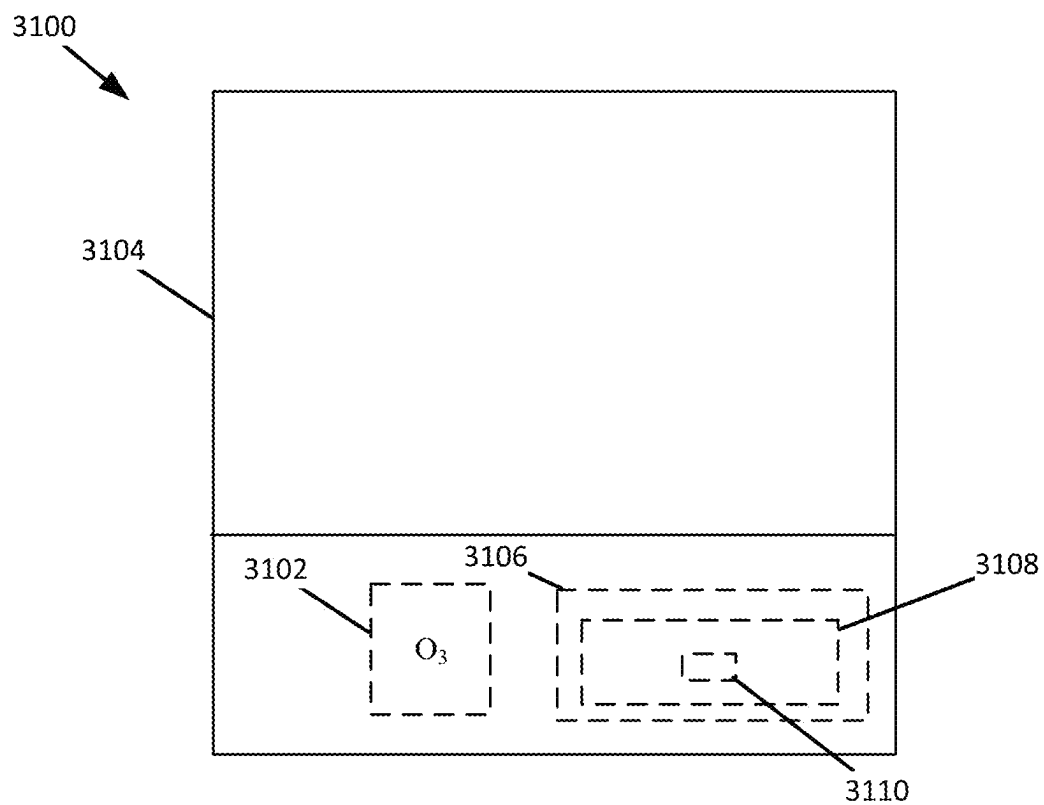


FIG. 31

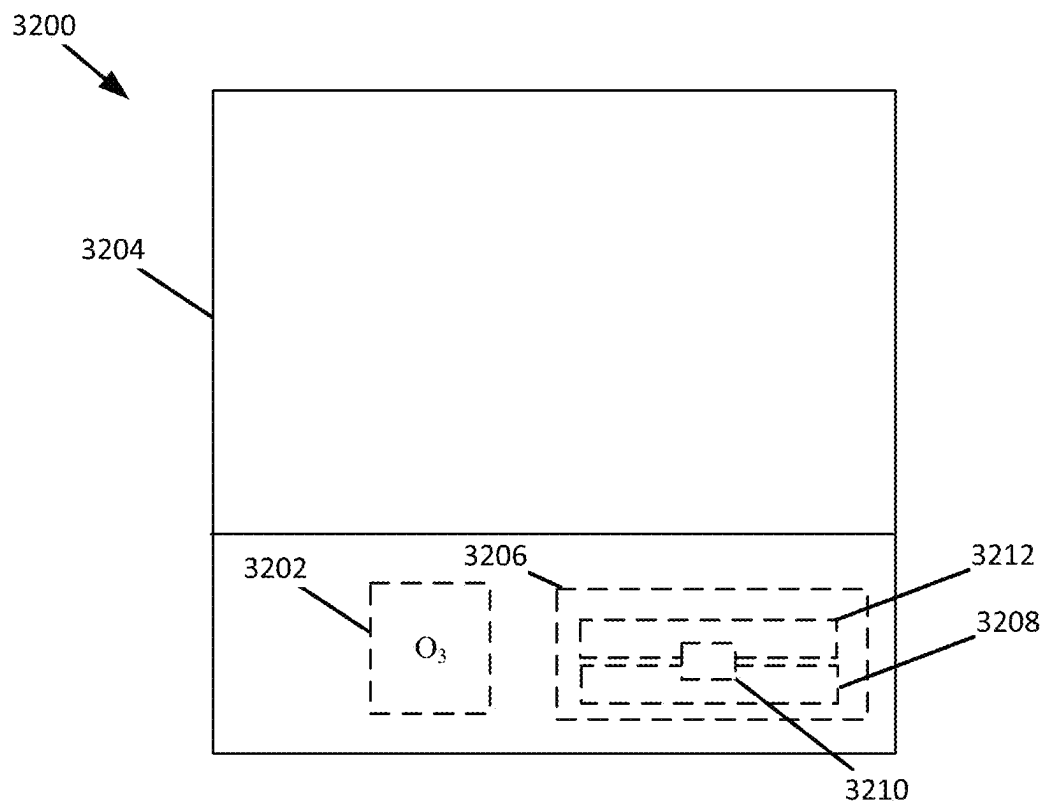


FIG. 32

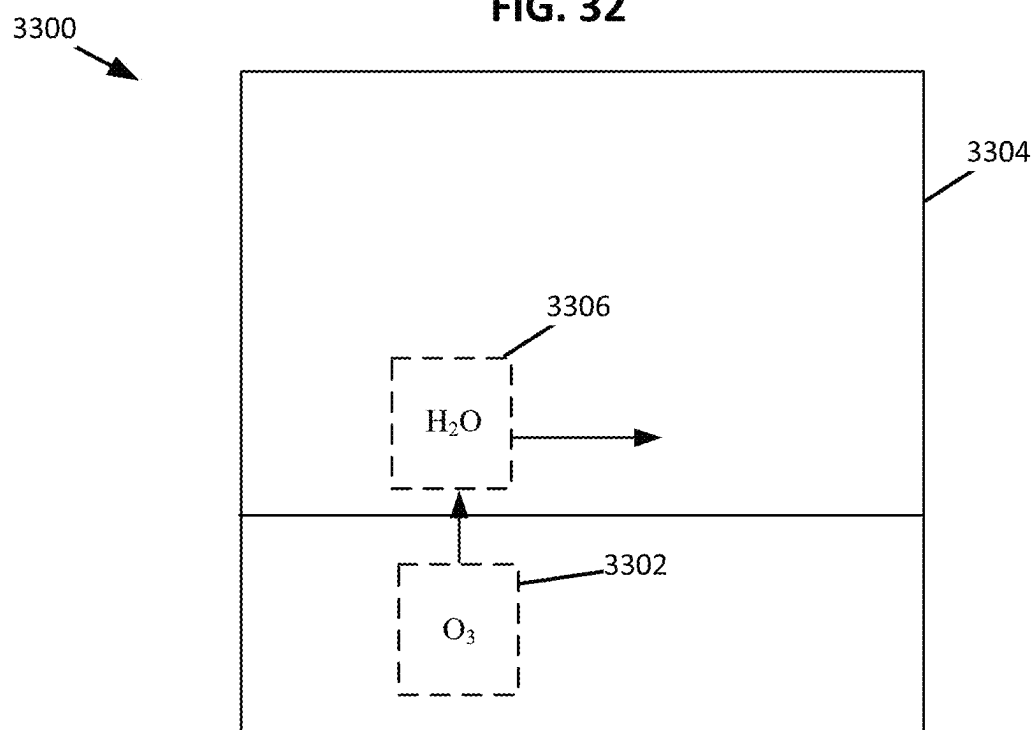


FIG. 33

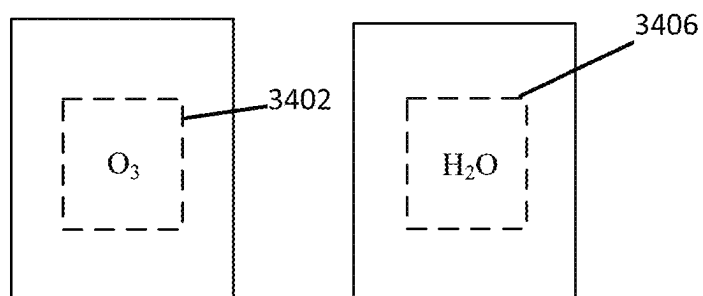


FIG. 34

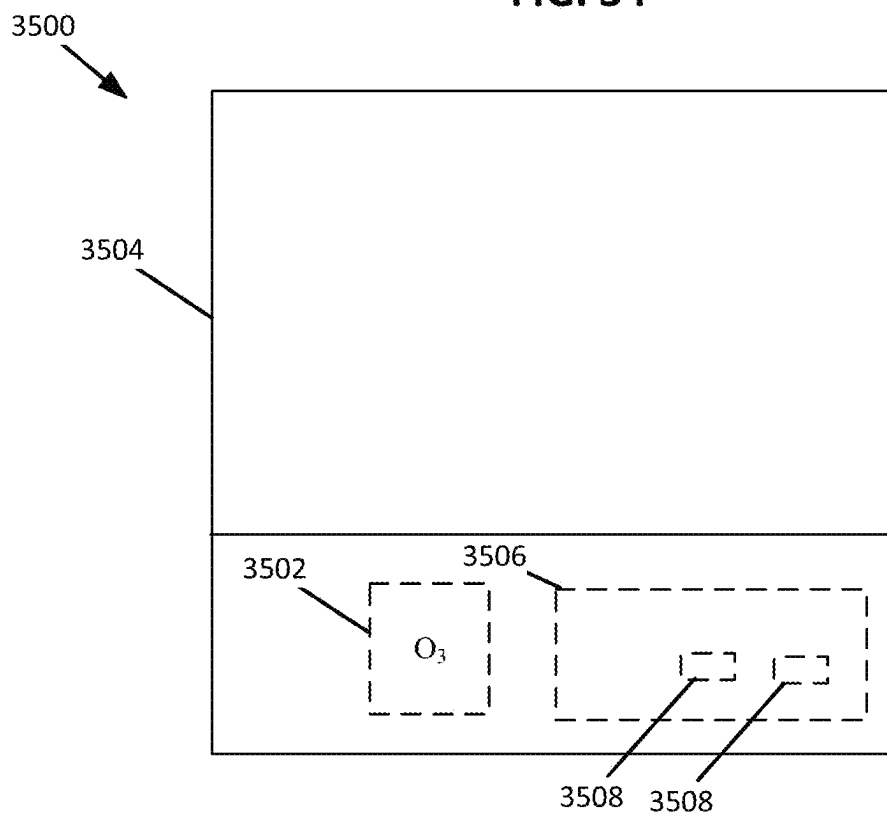


FIG. 35

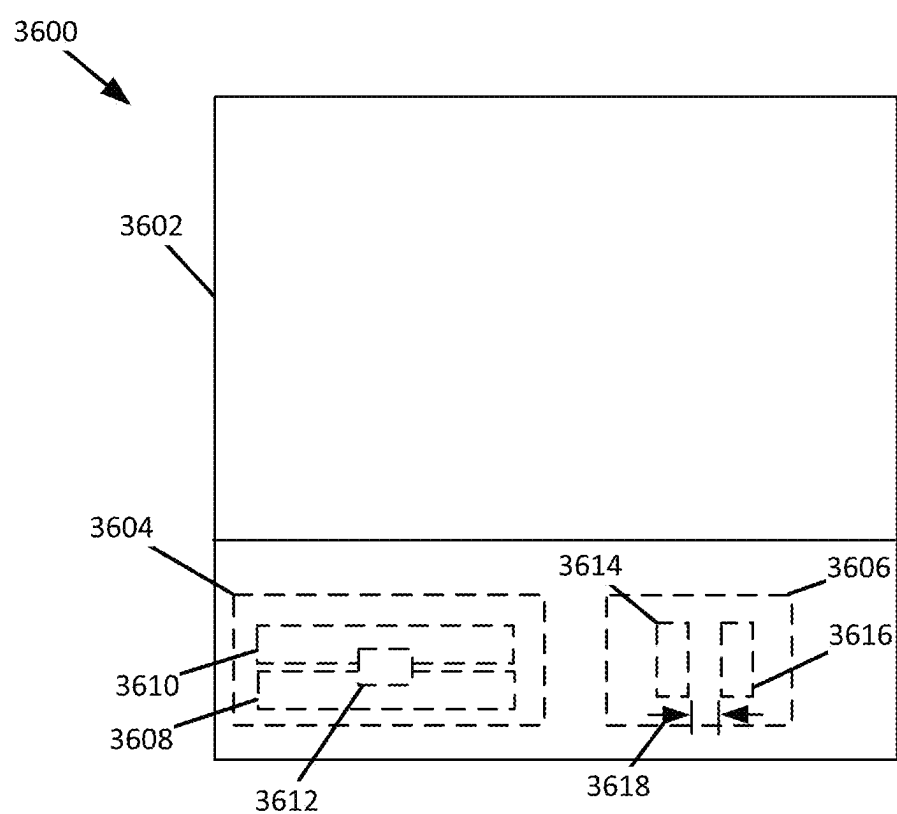


FIG. 36

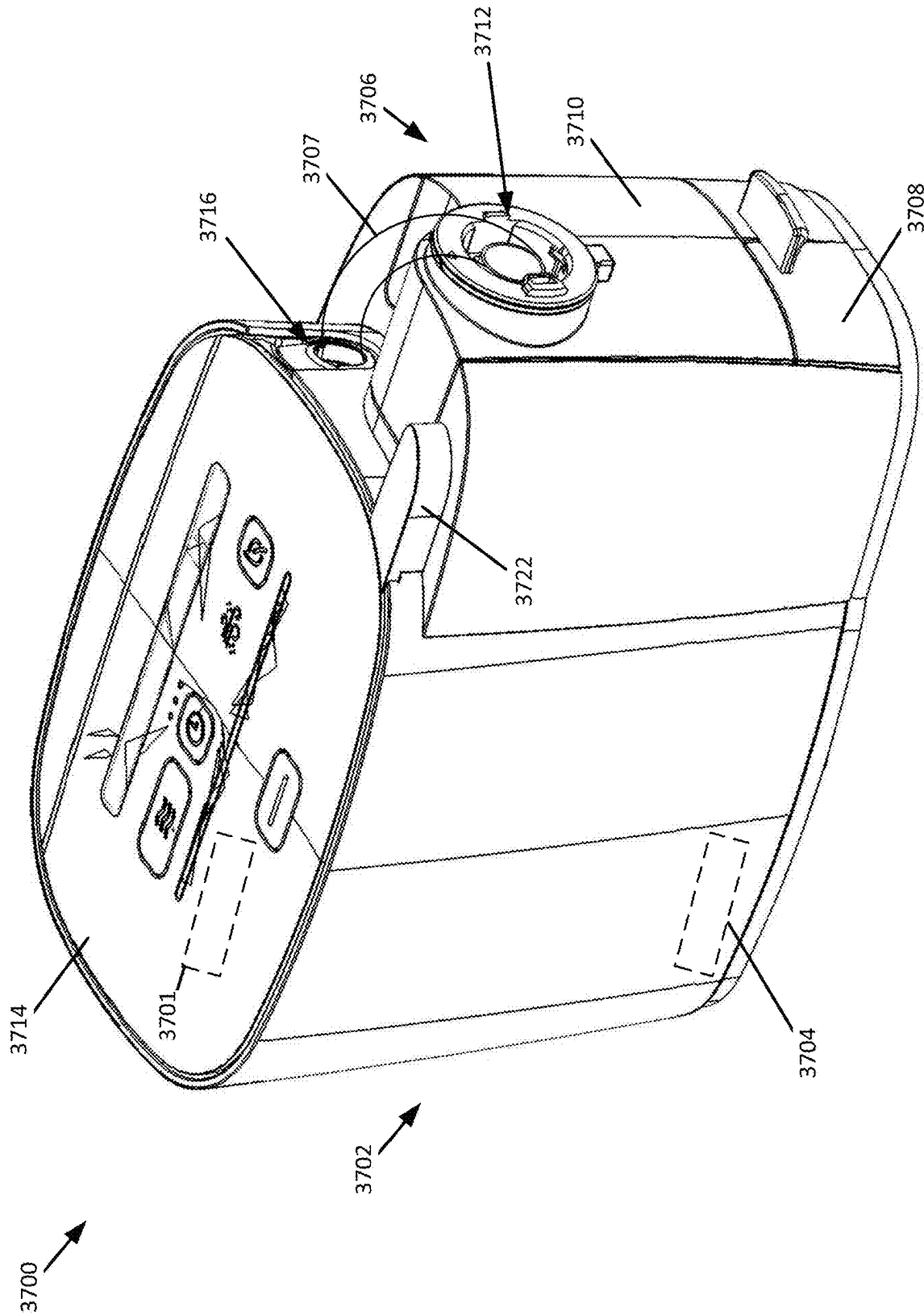


FIG. 37

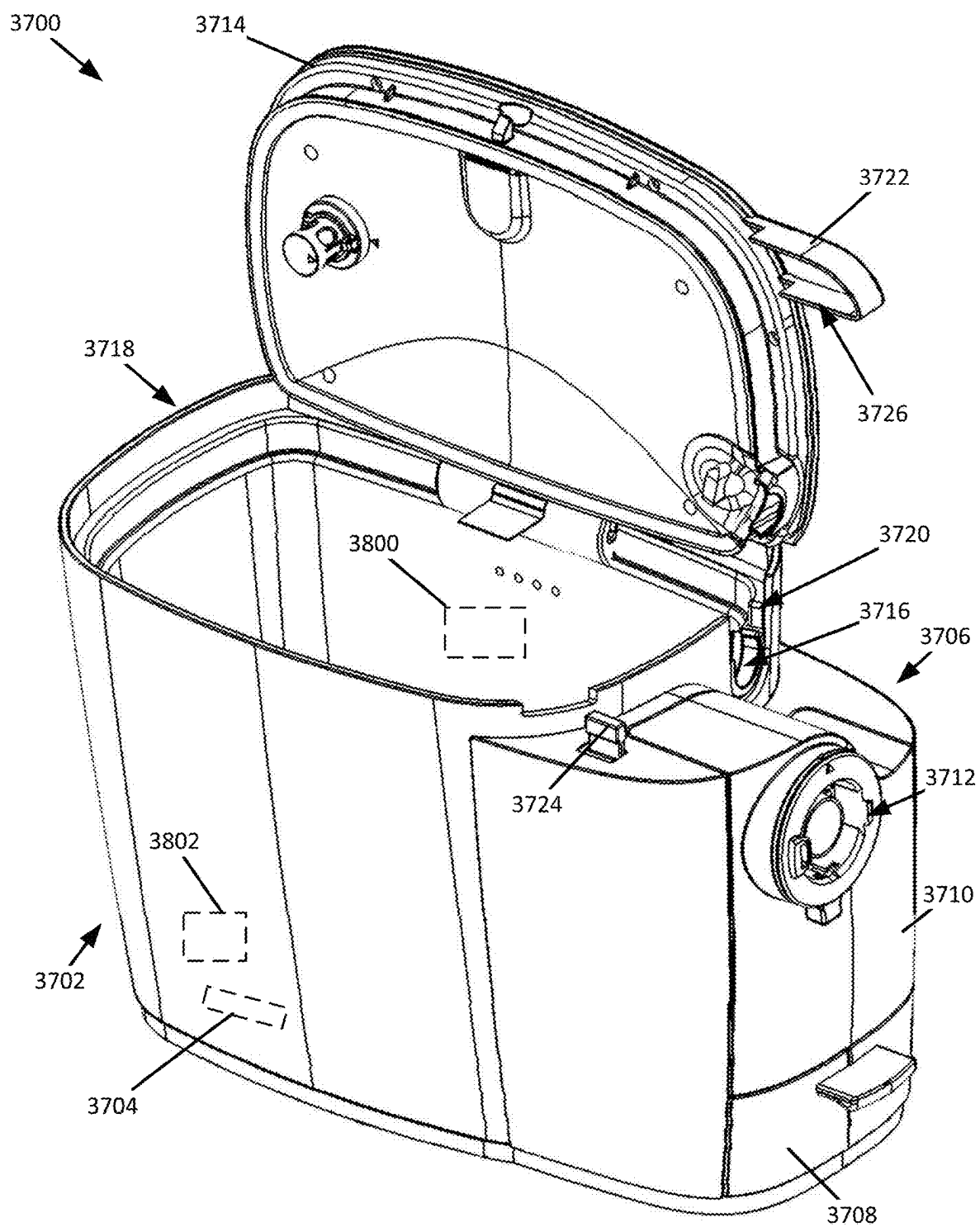


FIG. 38

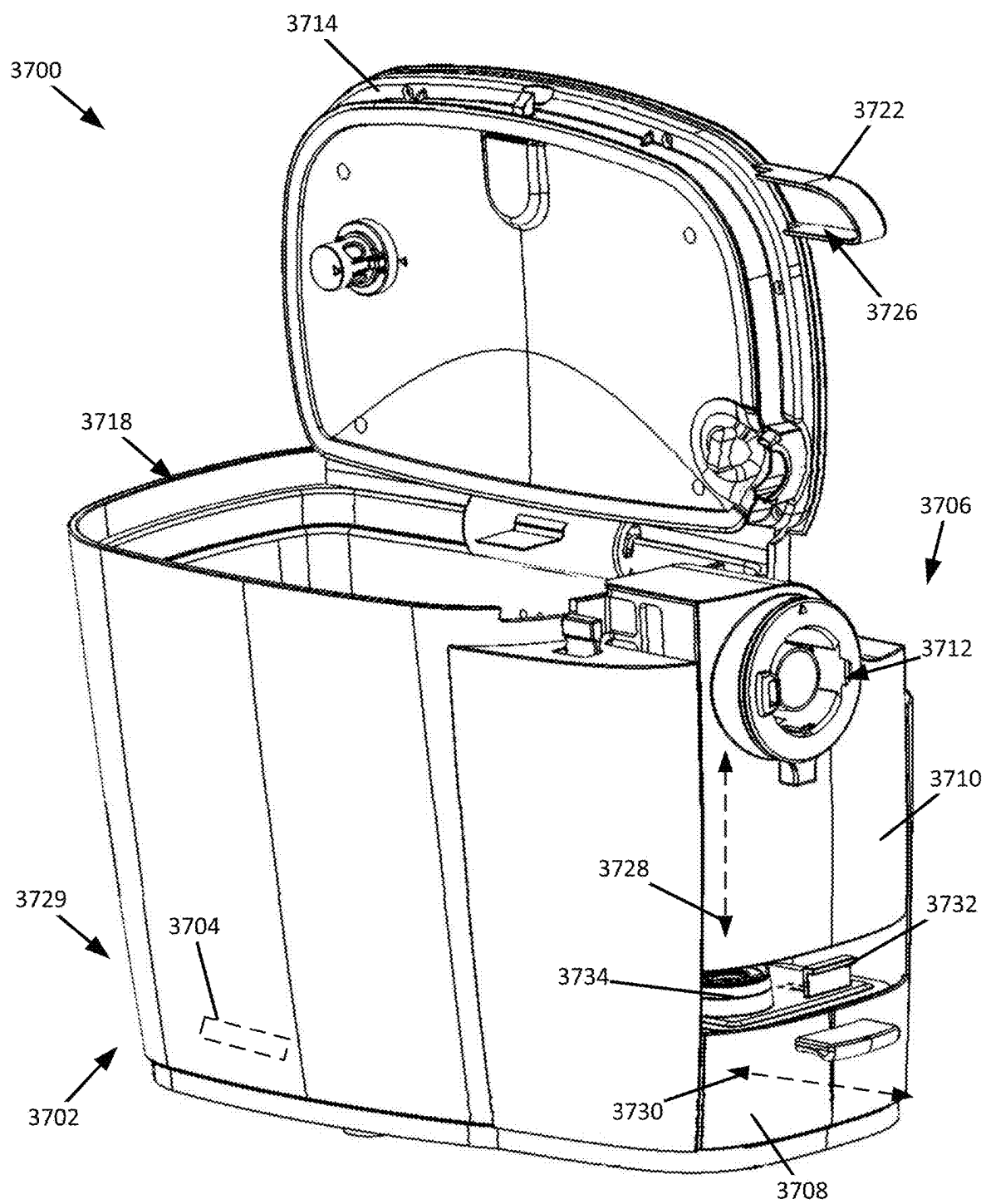
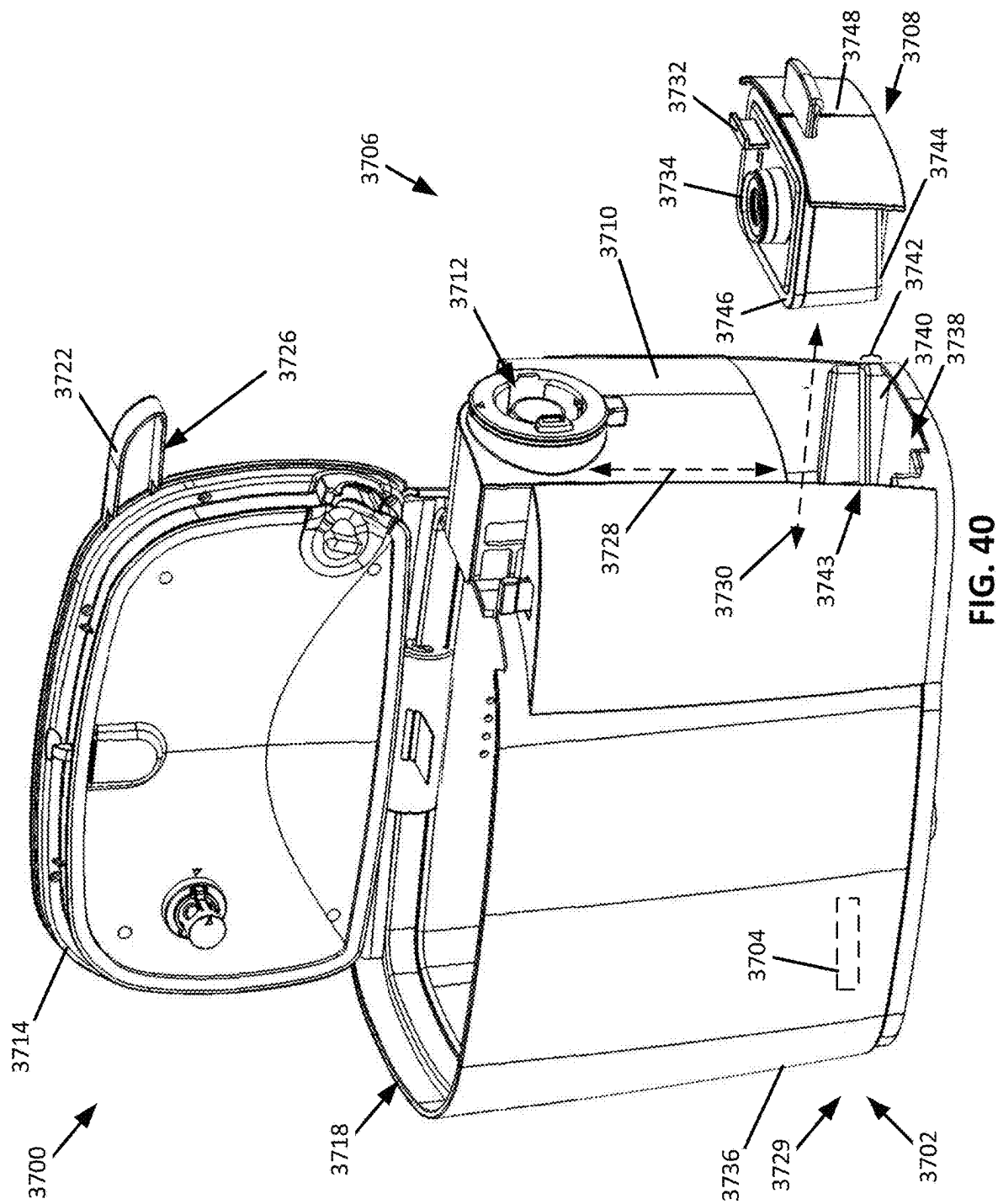


FIG. 39



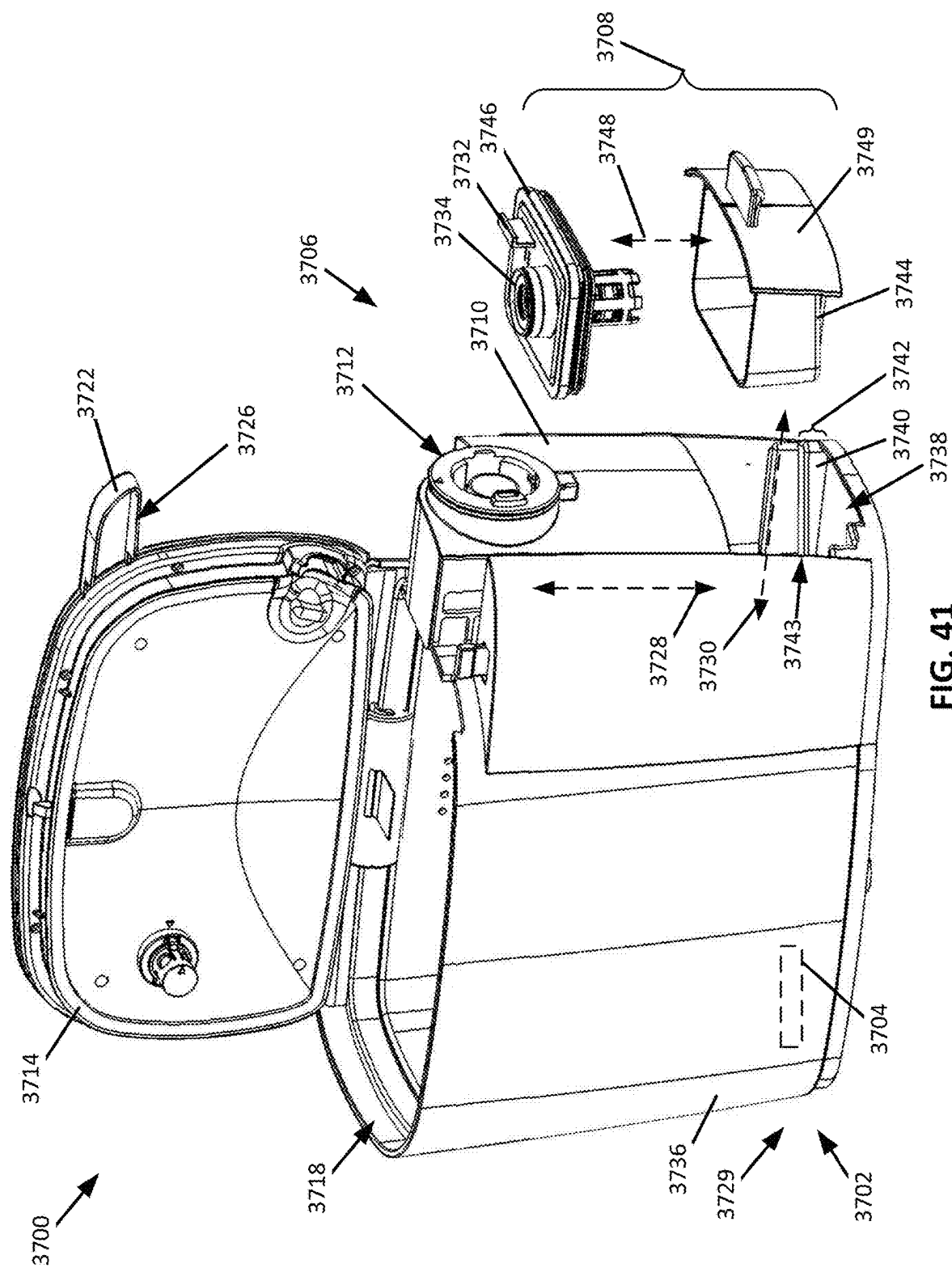


FIG. 41

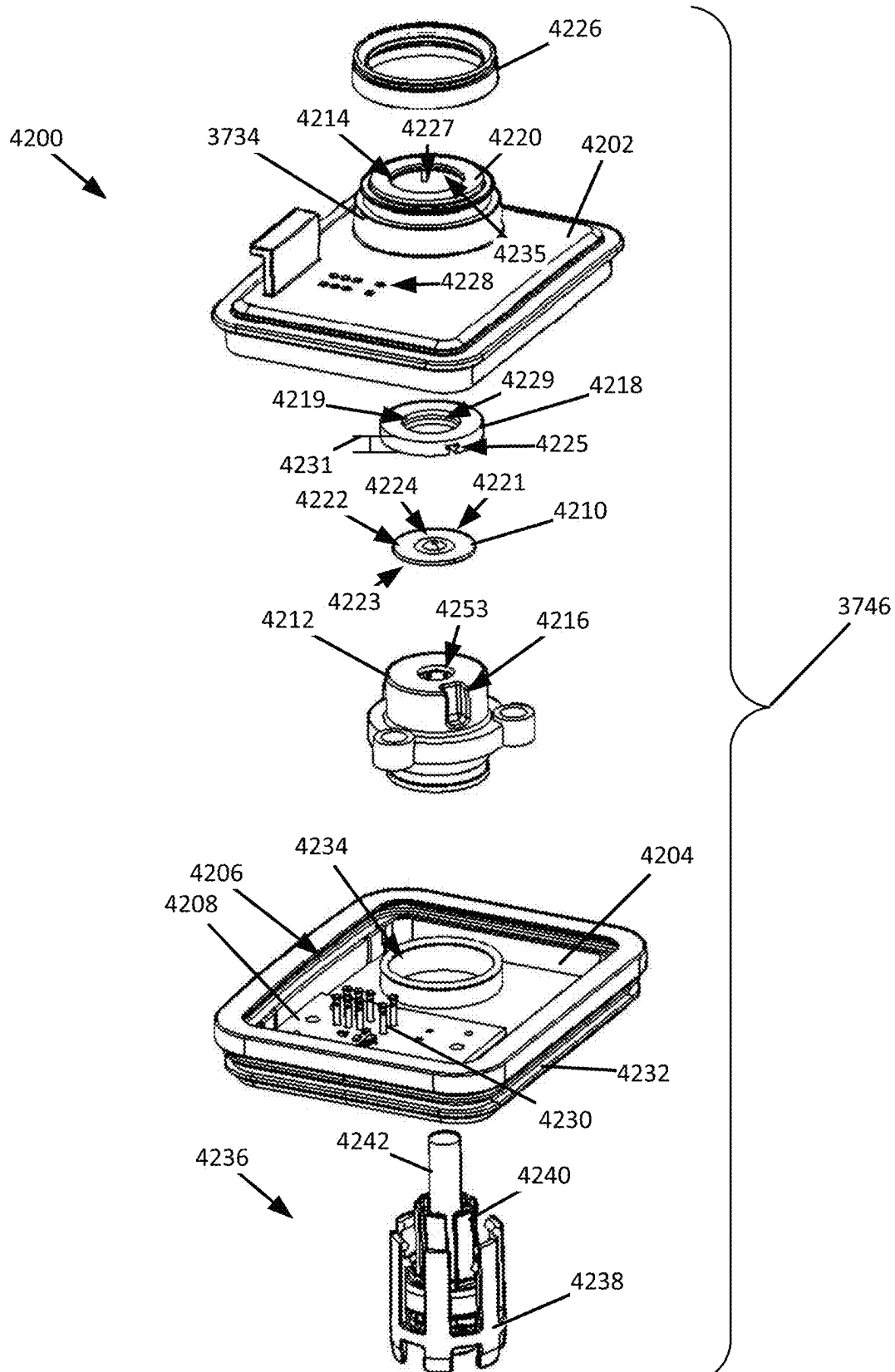


FIG. 42

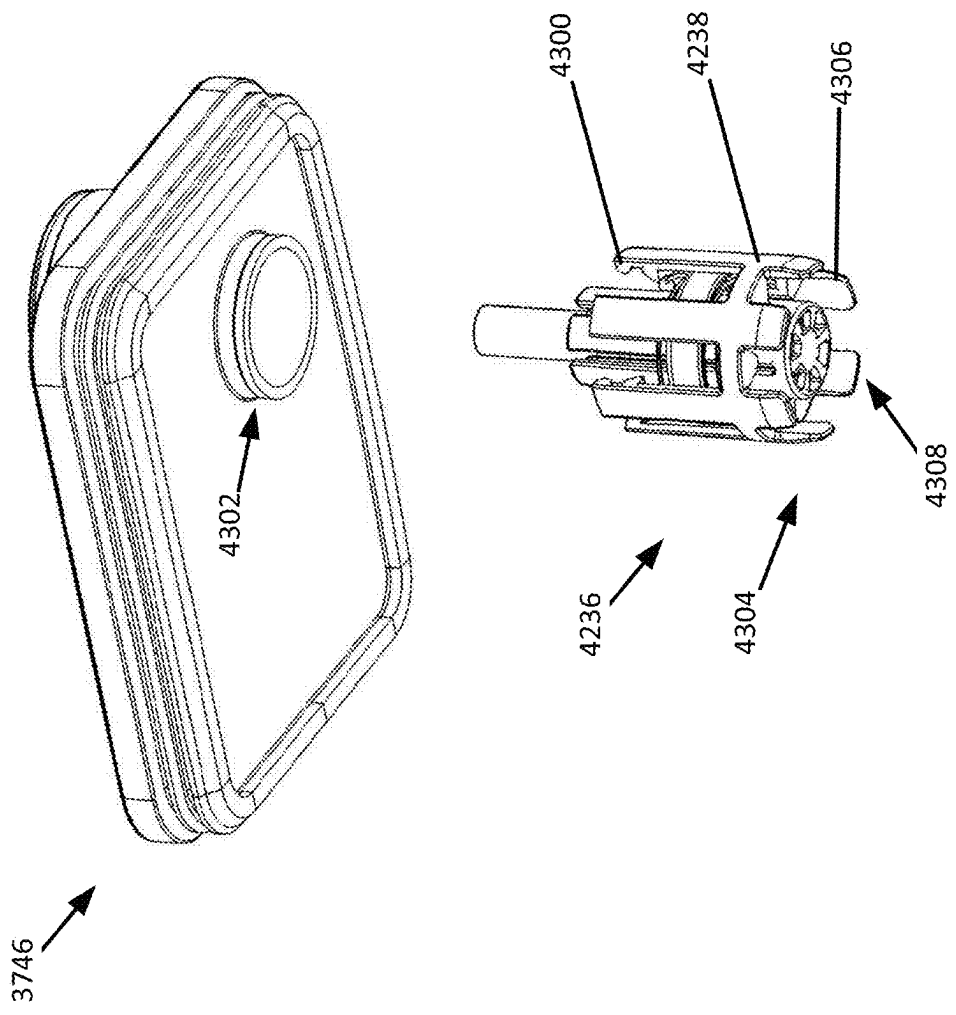


FIG. 43

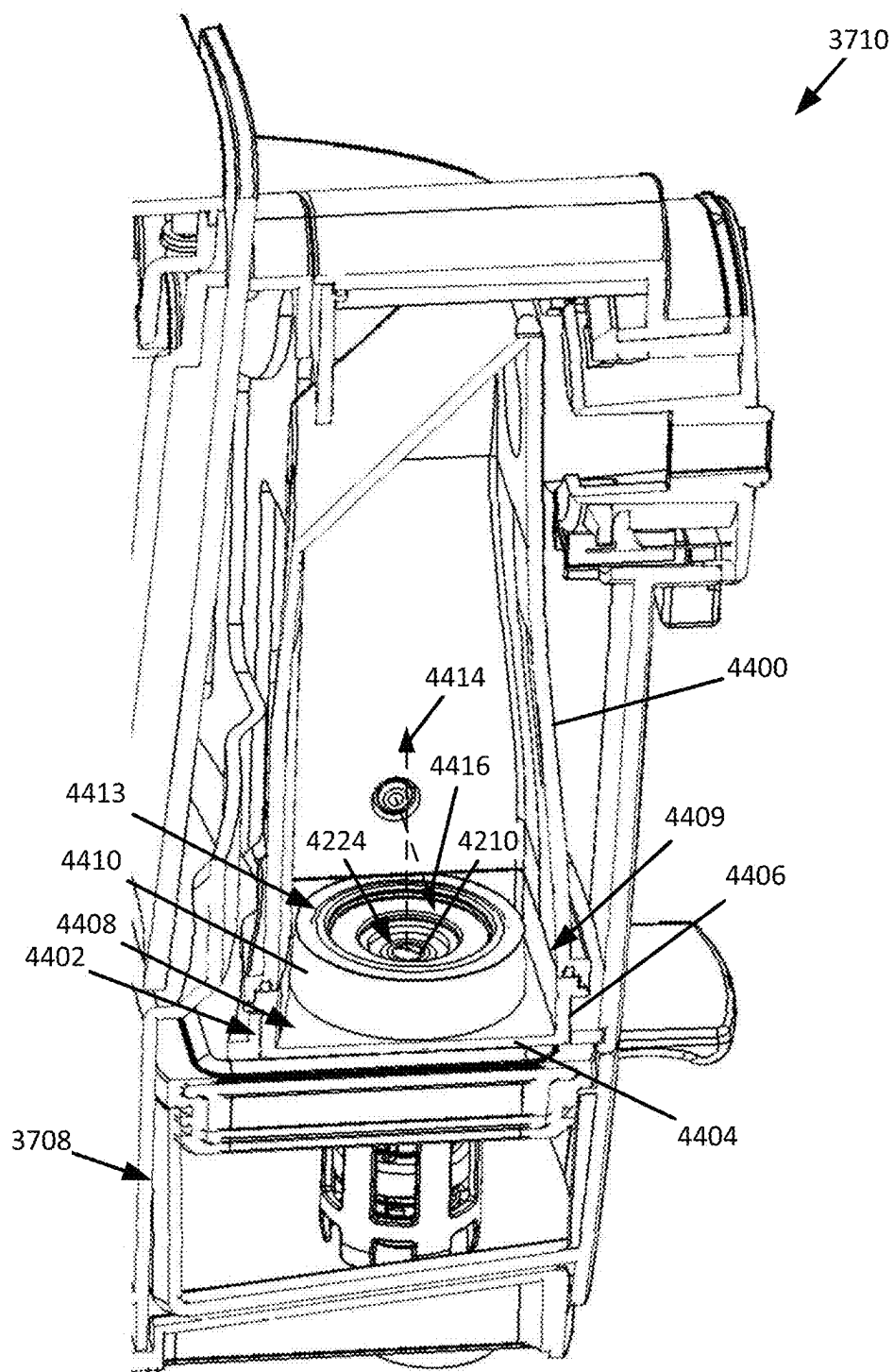


FIG. 44

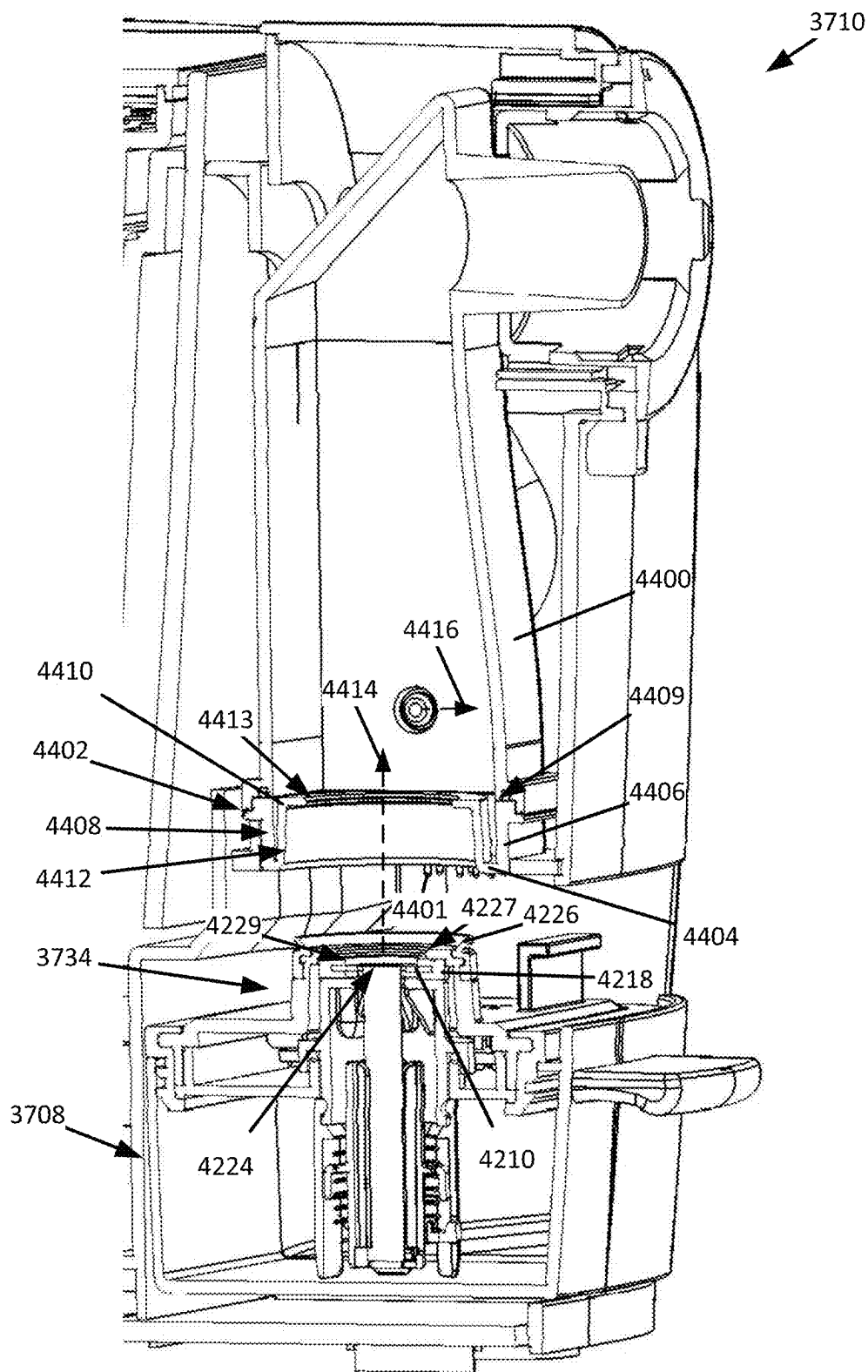


FIG. 45

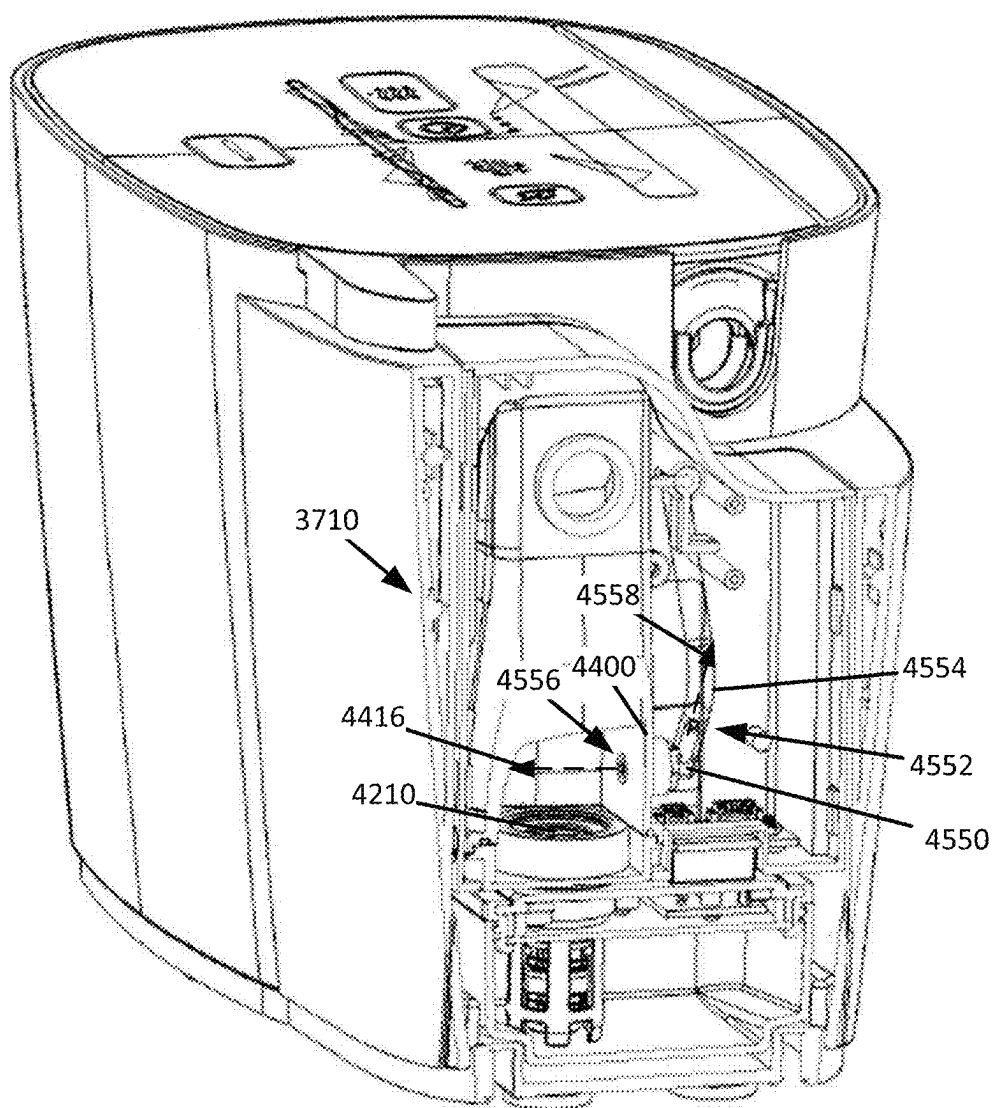


FIG. 45A

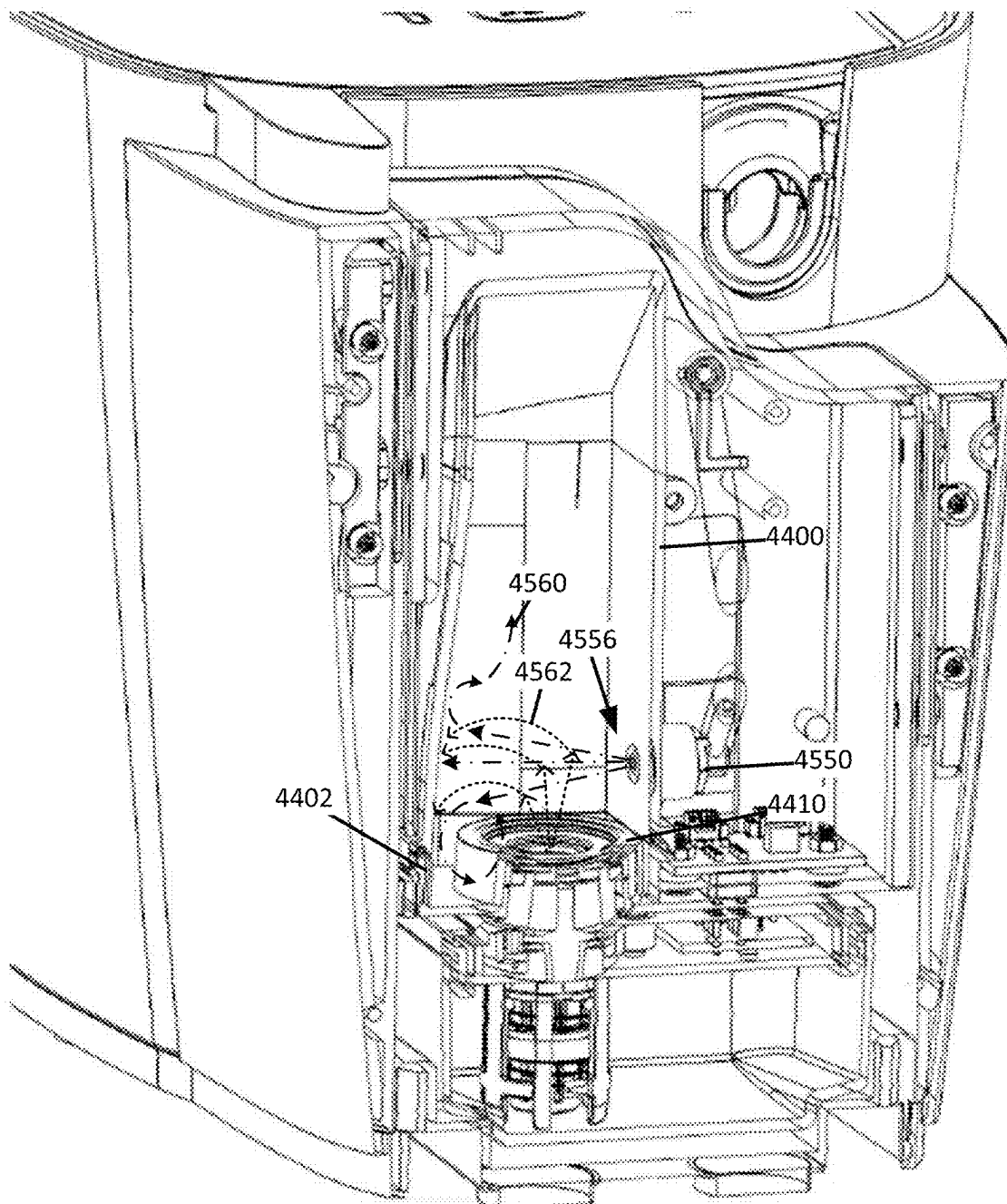


FIG. 45B

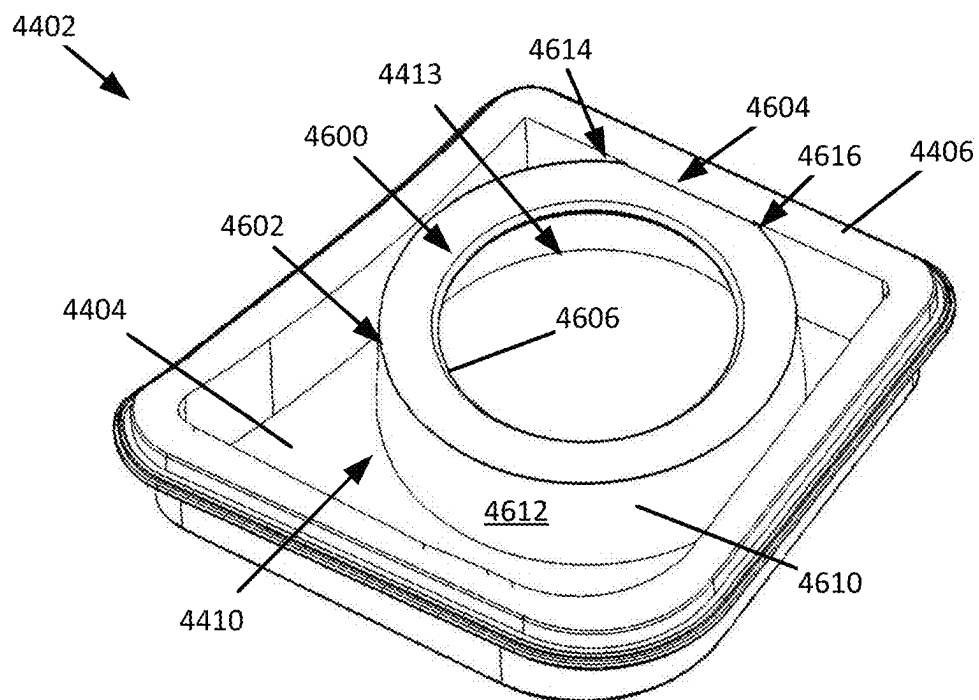


FIG. 46

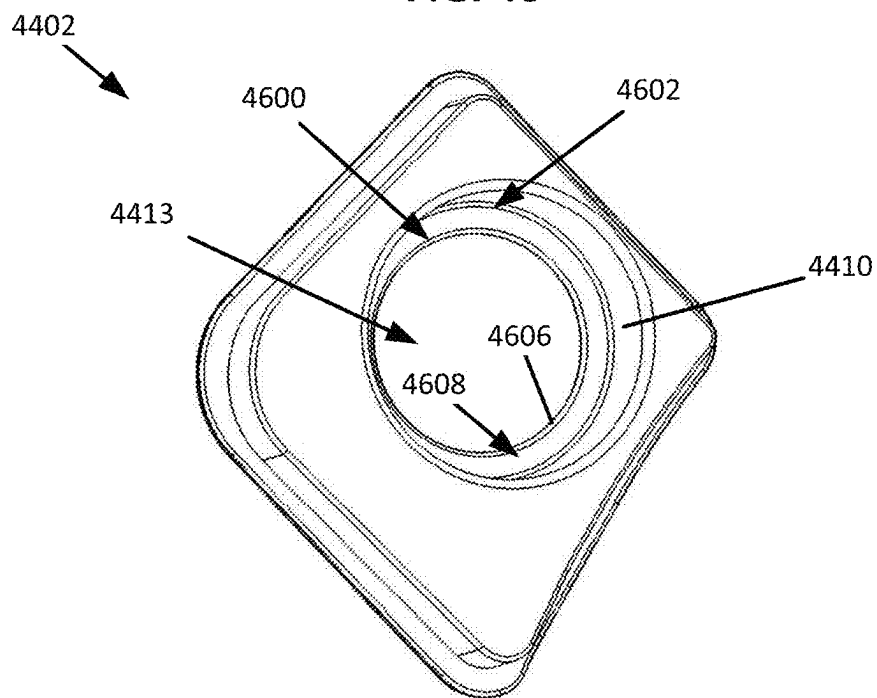


FIG. 47

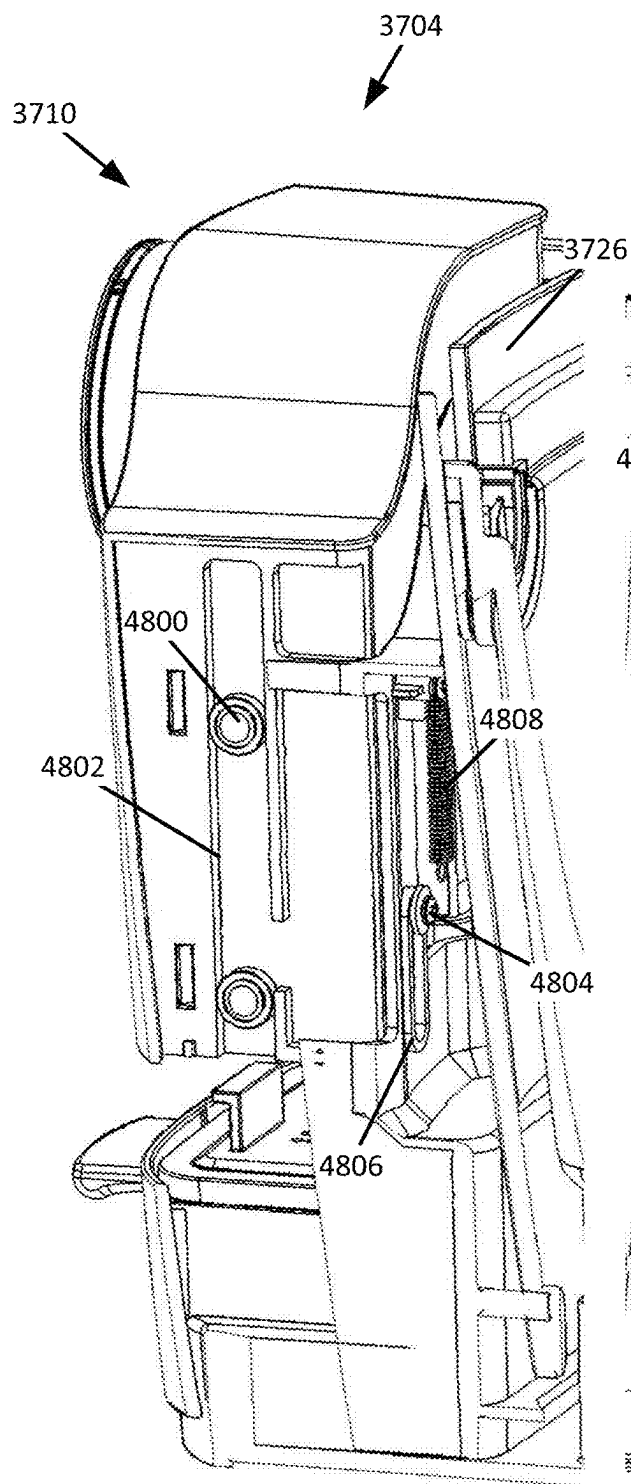


FIG. 48

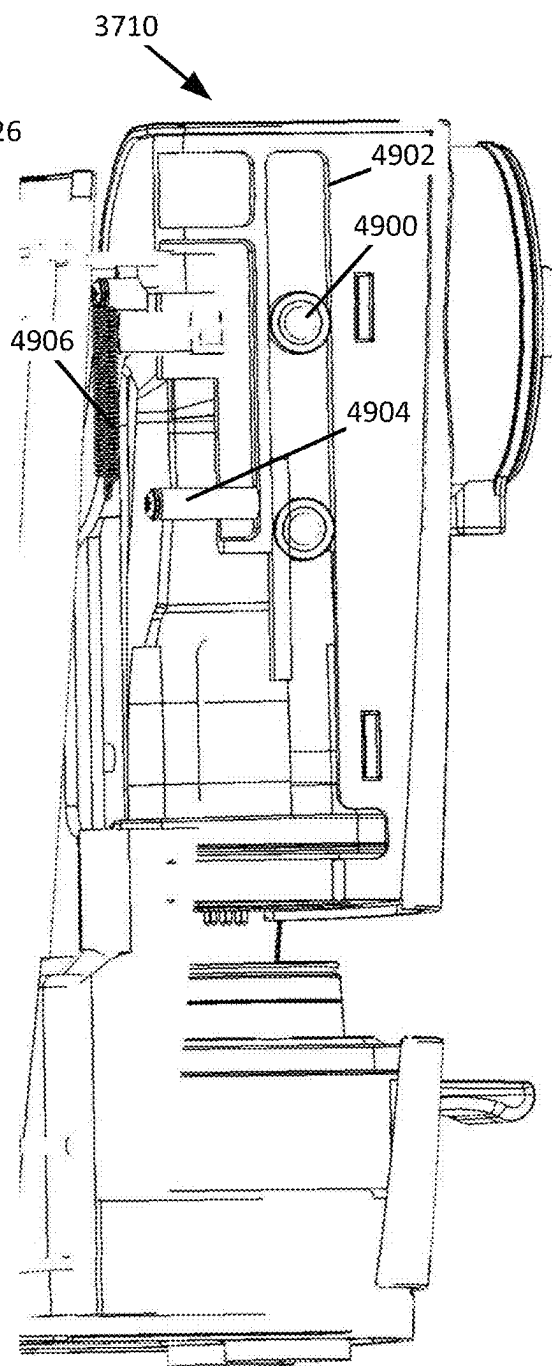


FIG. 49

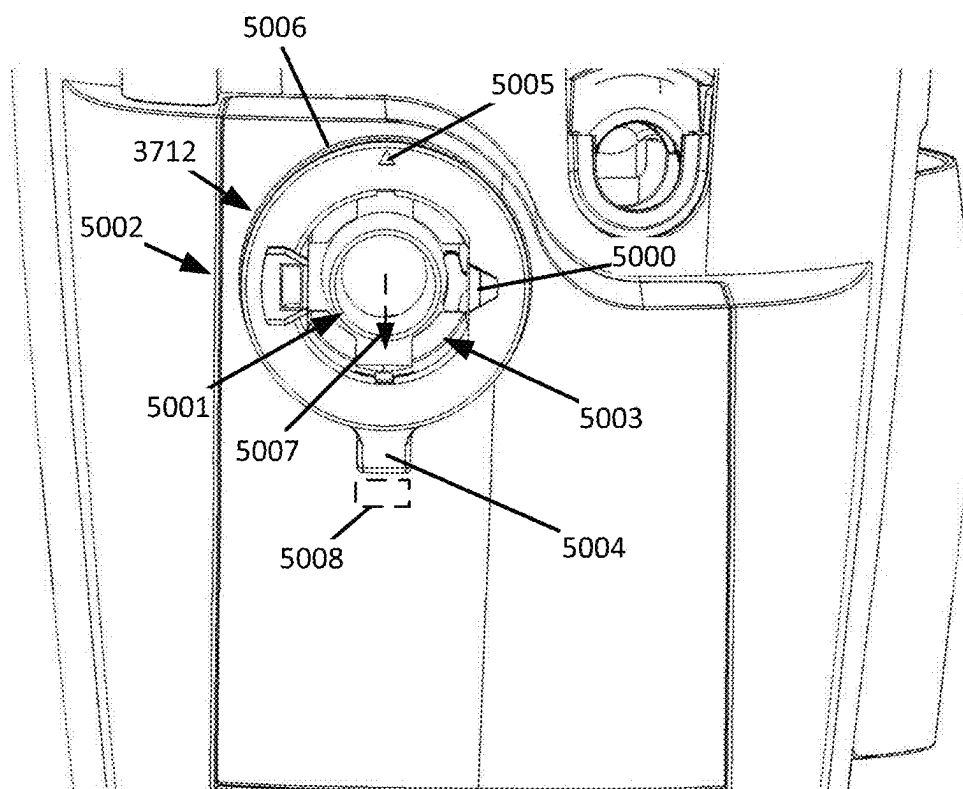


FIG. 50

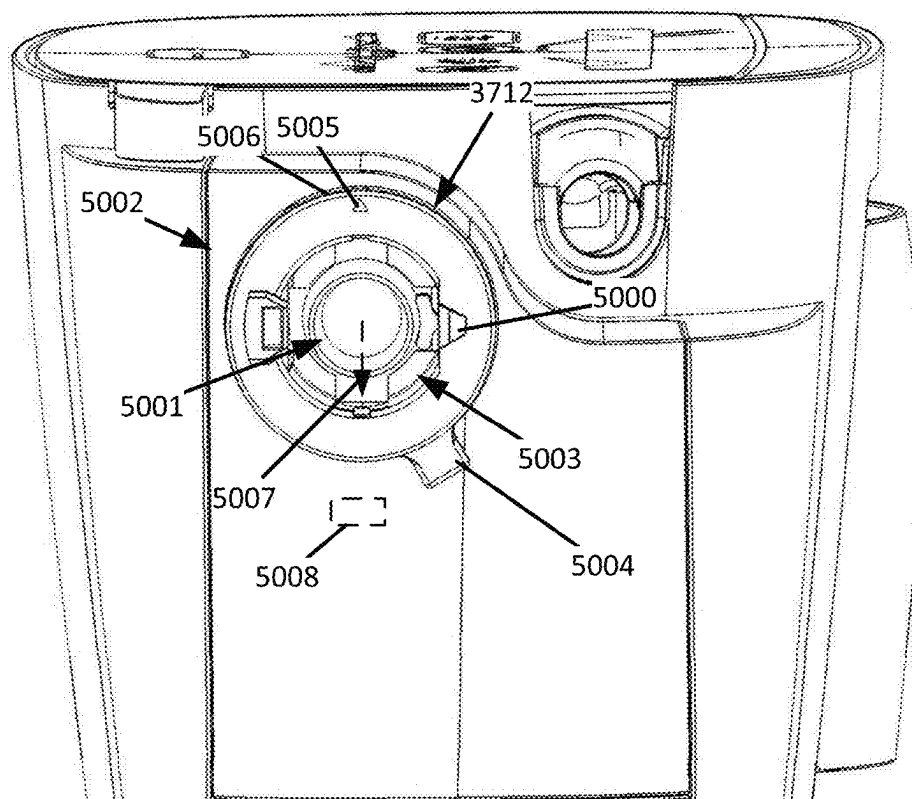


FIG. 51

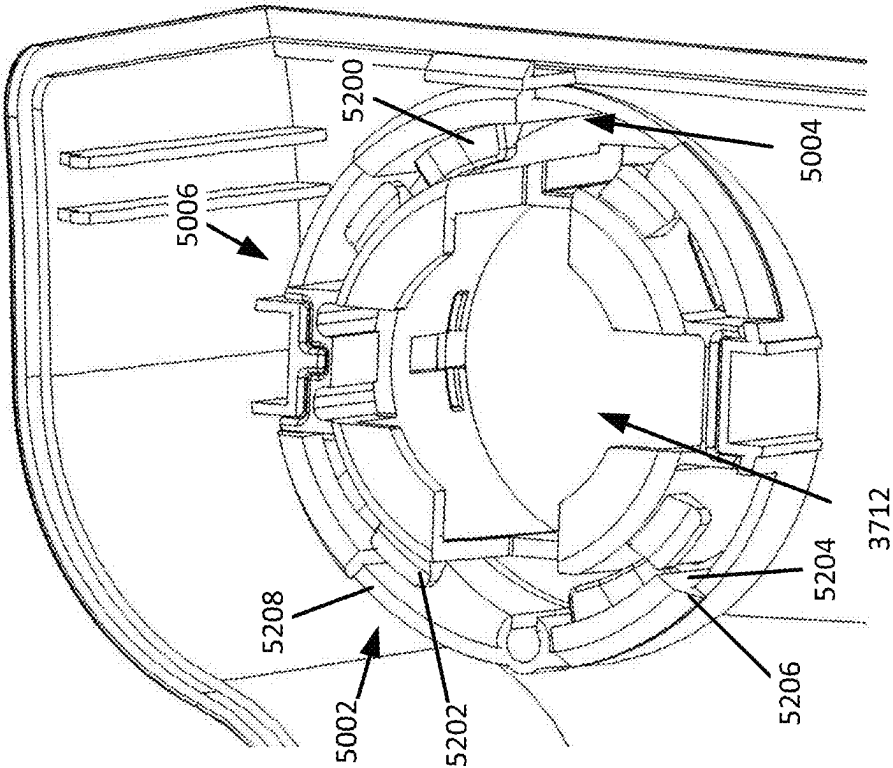


FIG. 53

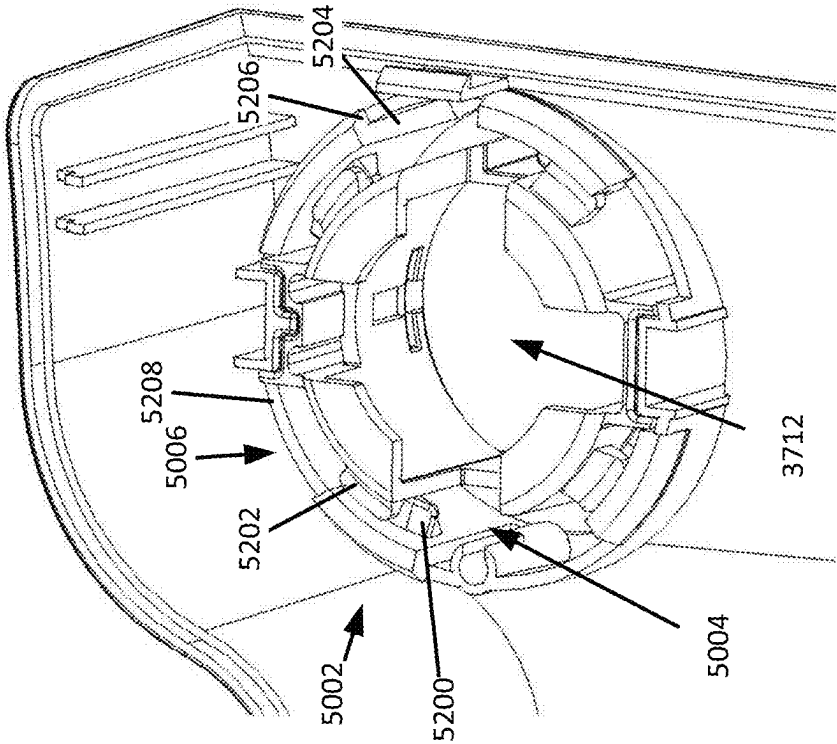


FIG. 52

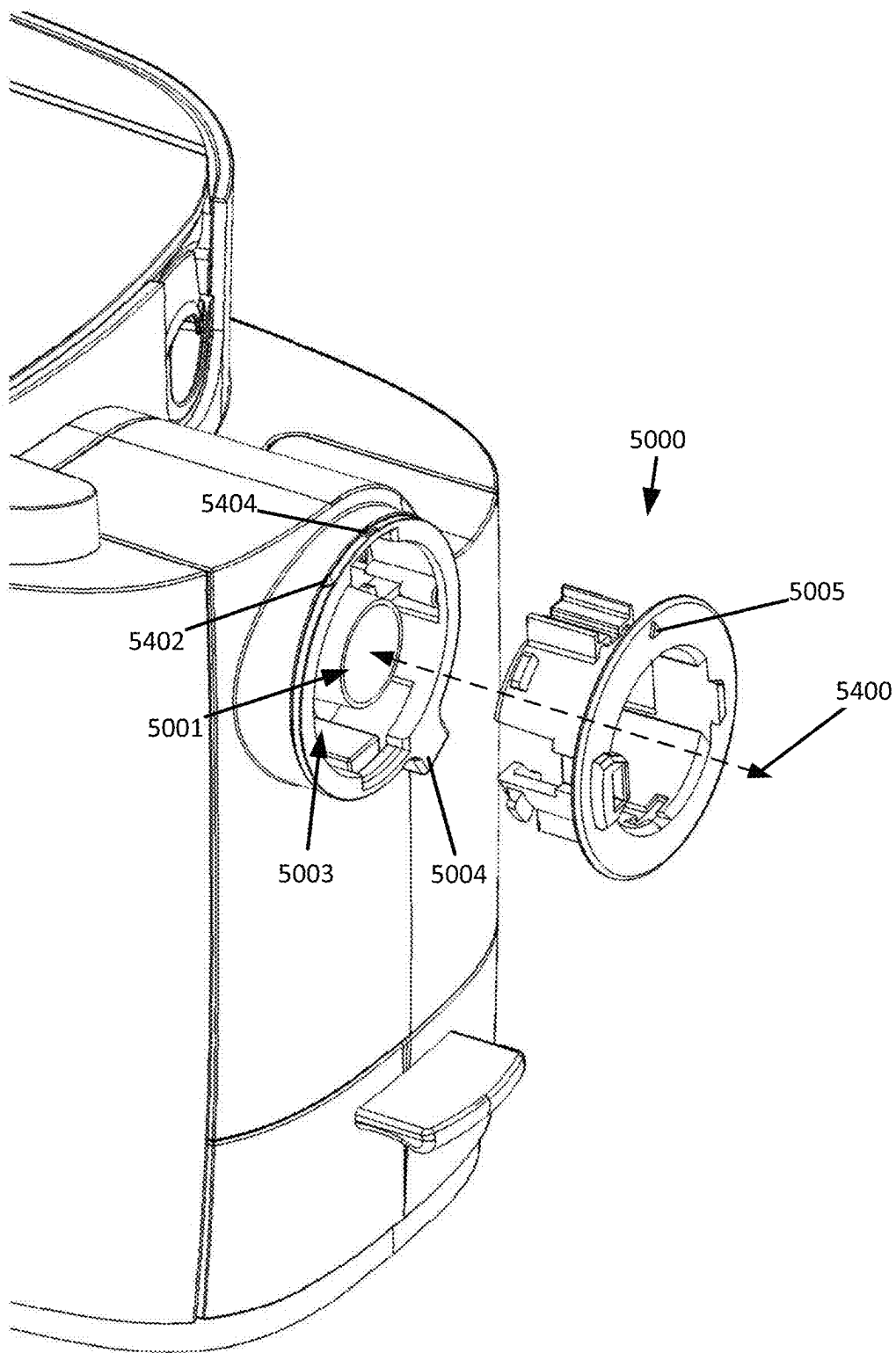


FIG. 54

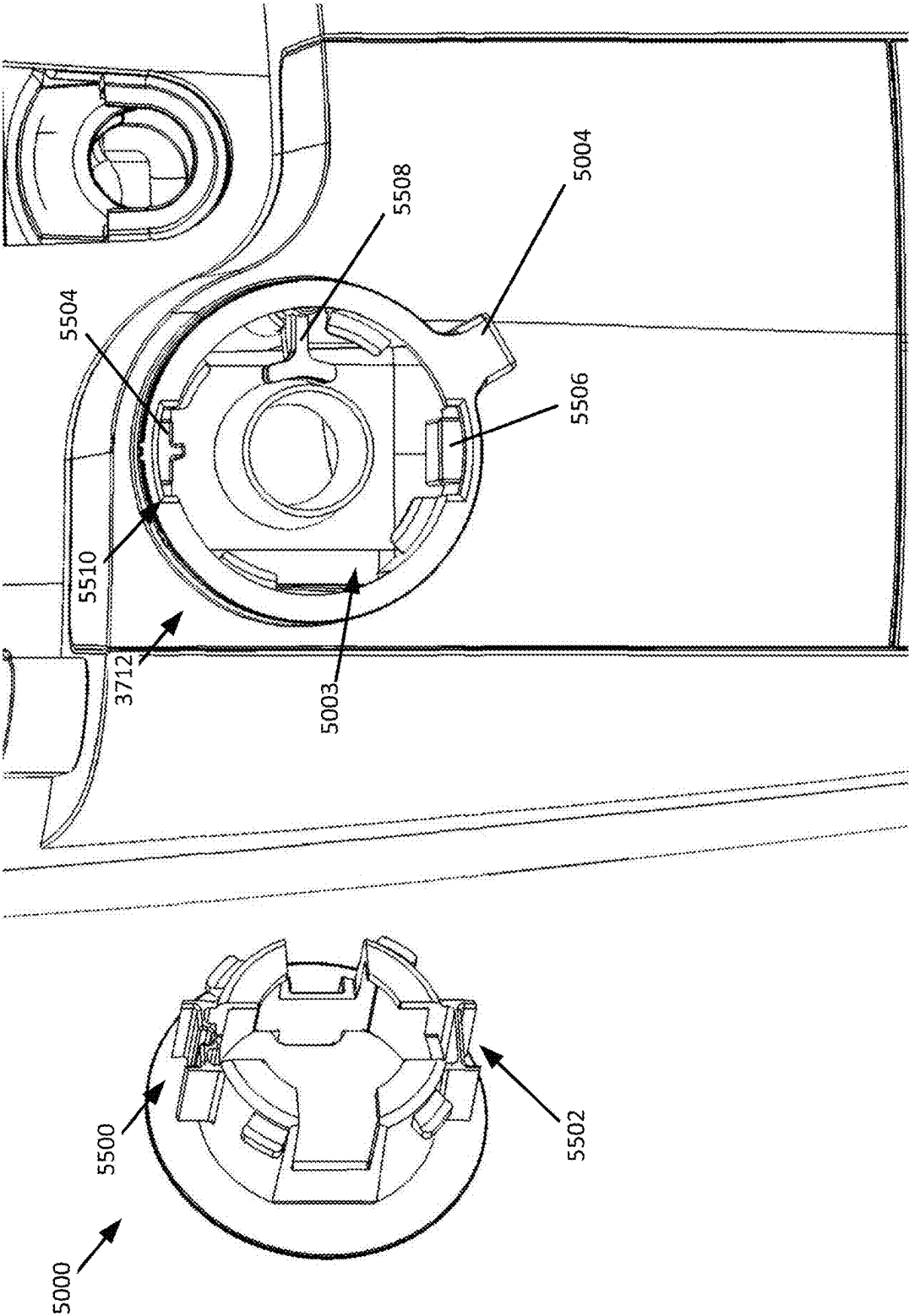


FIG. 55

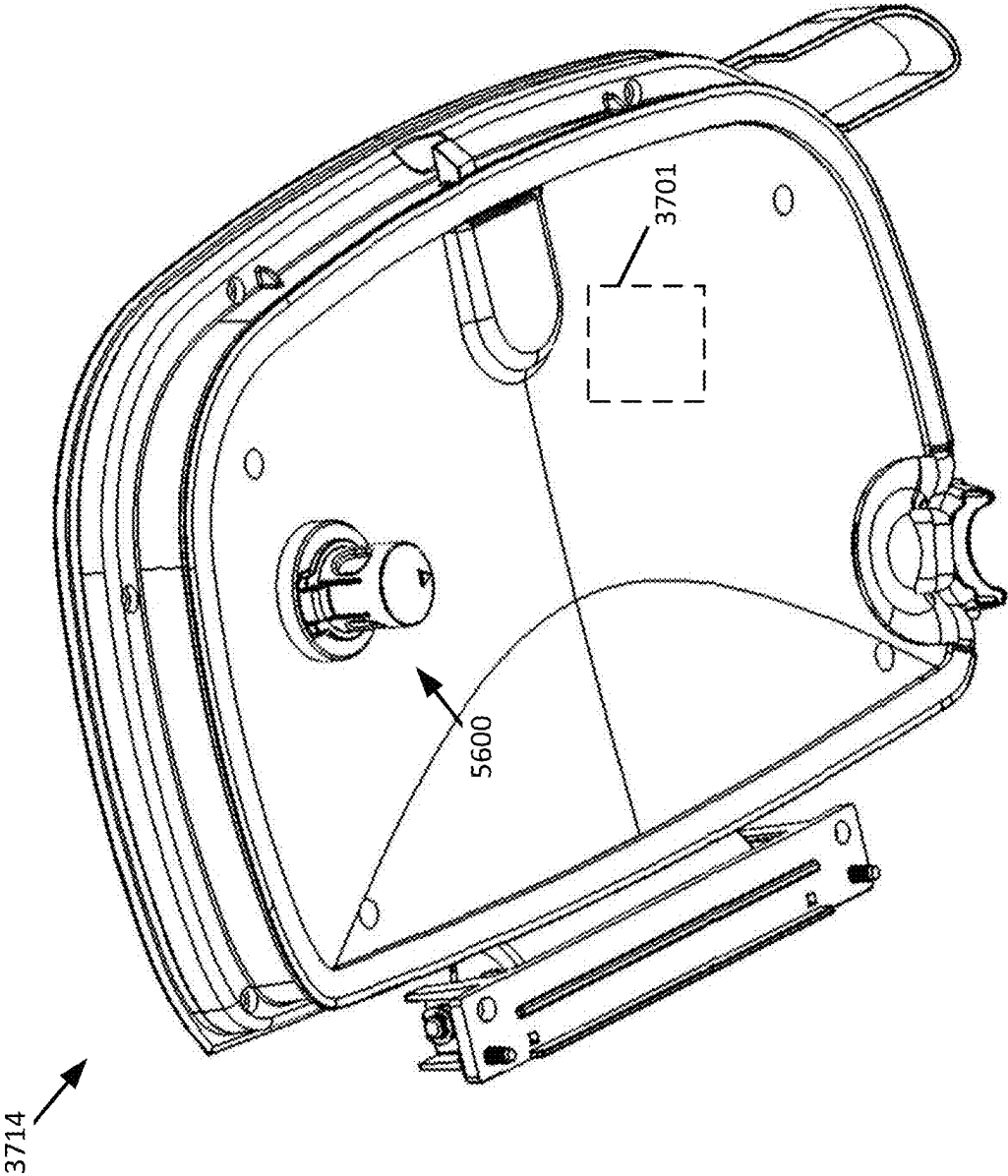


FIG. 56

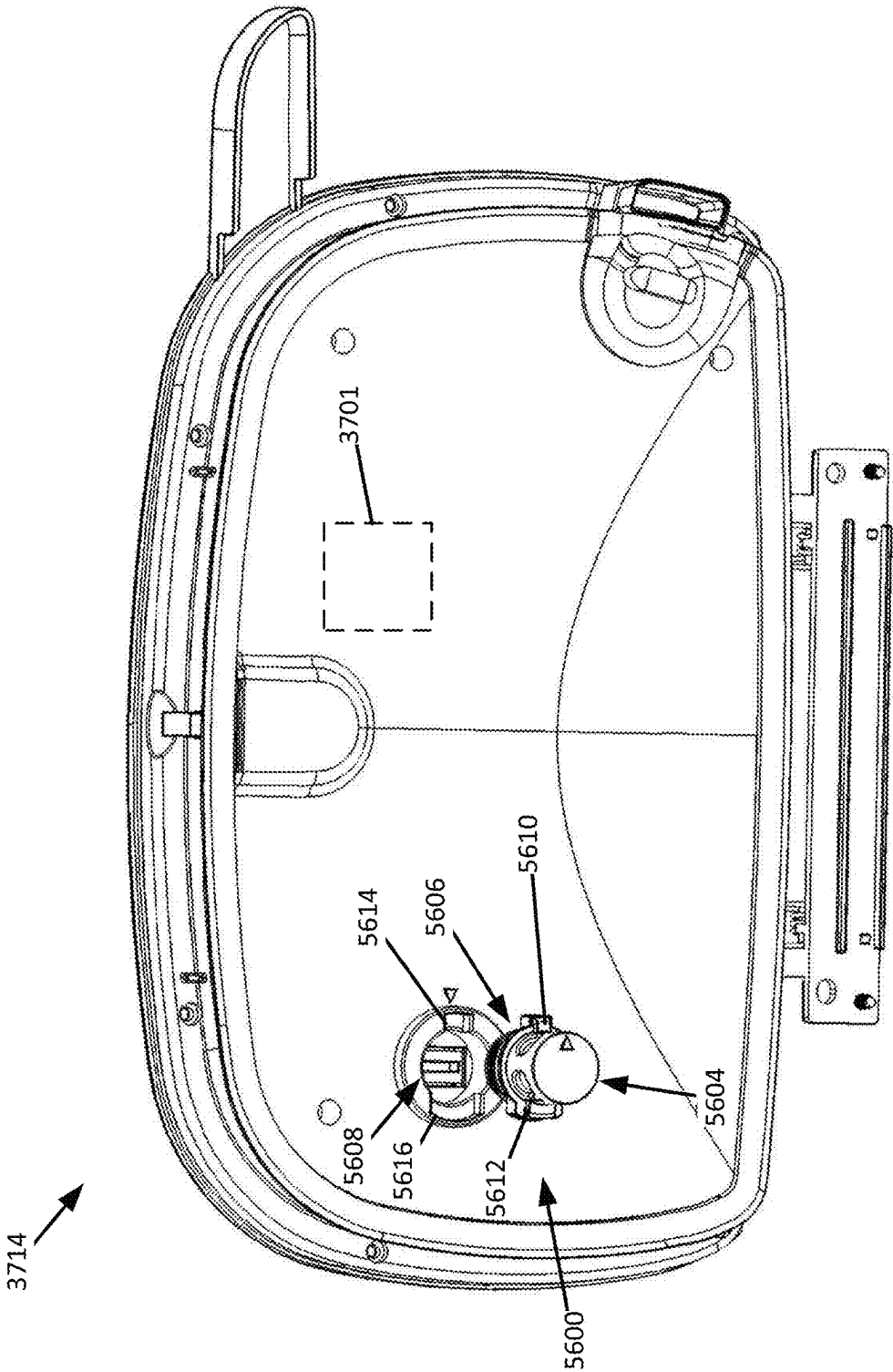


FIG. 57

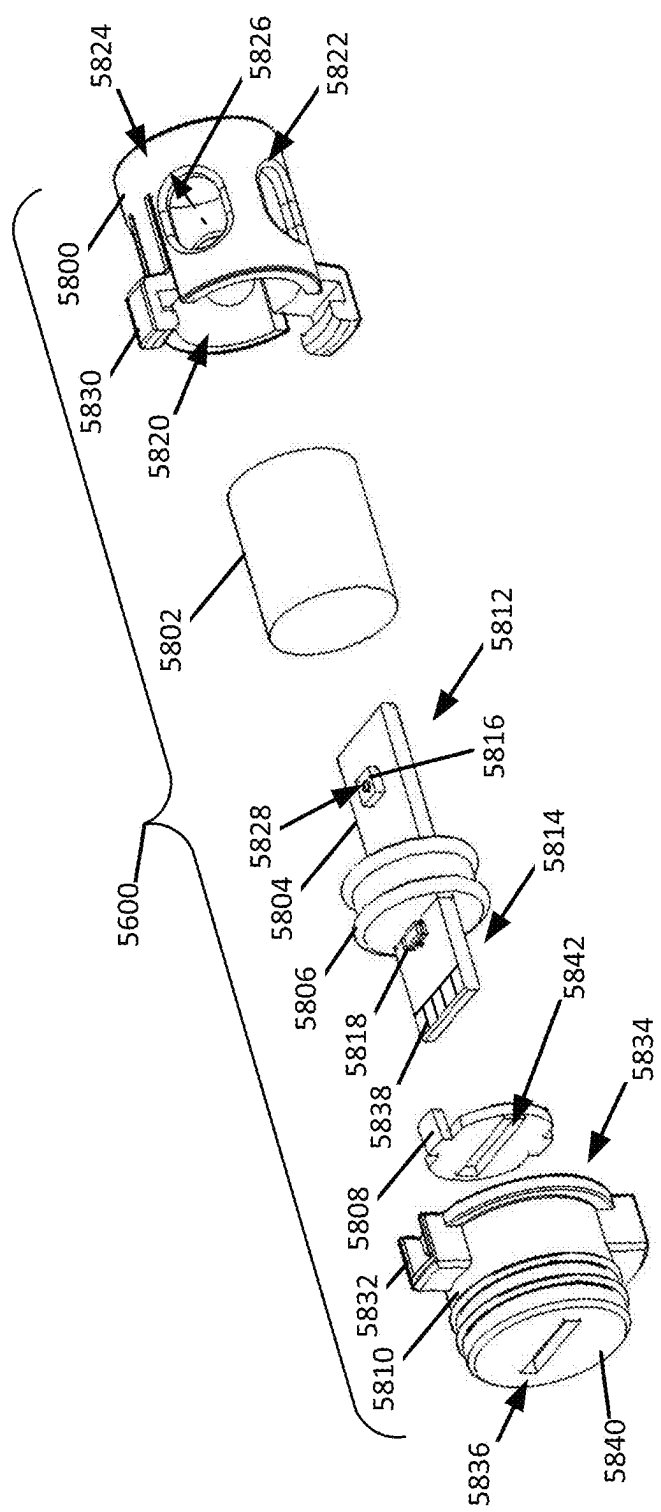


FIG. 58

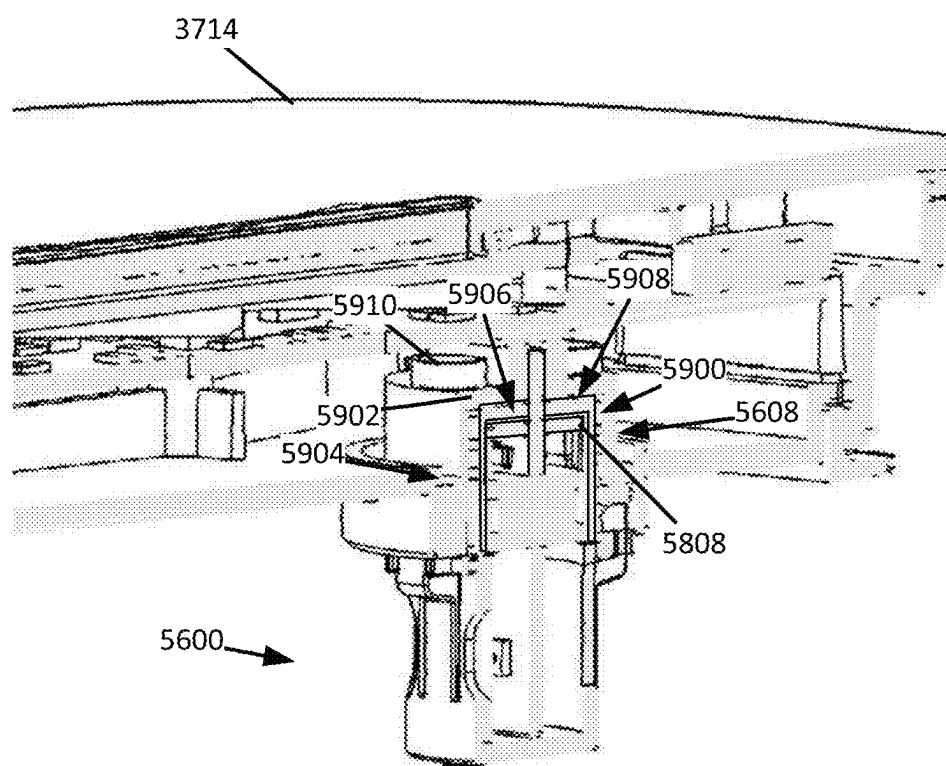


FIG. 59

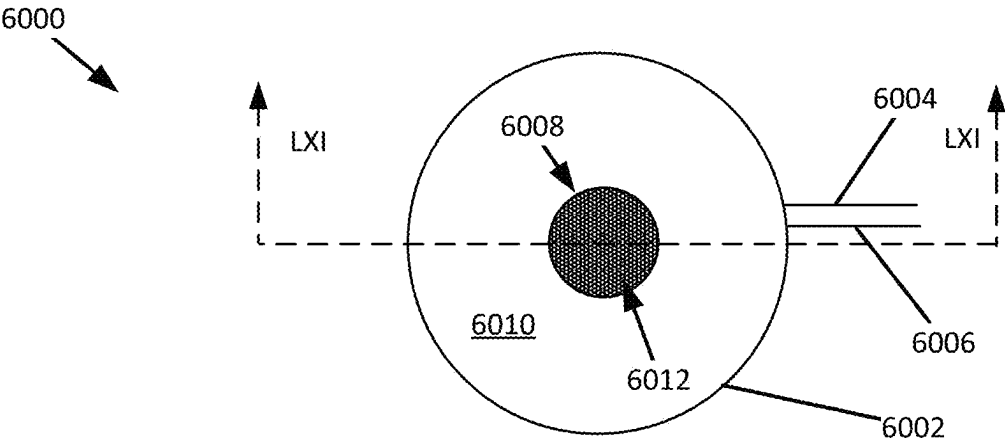


FIG. 60

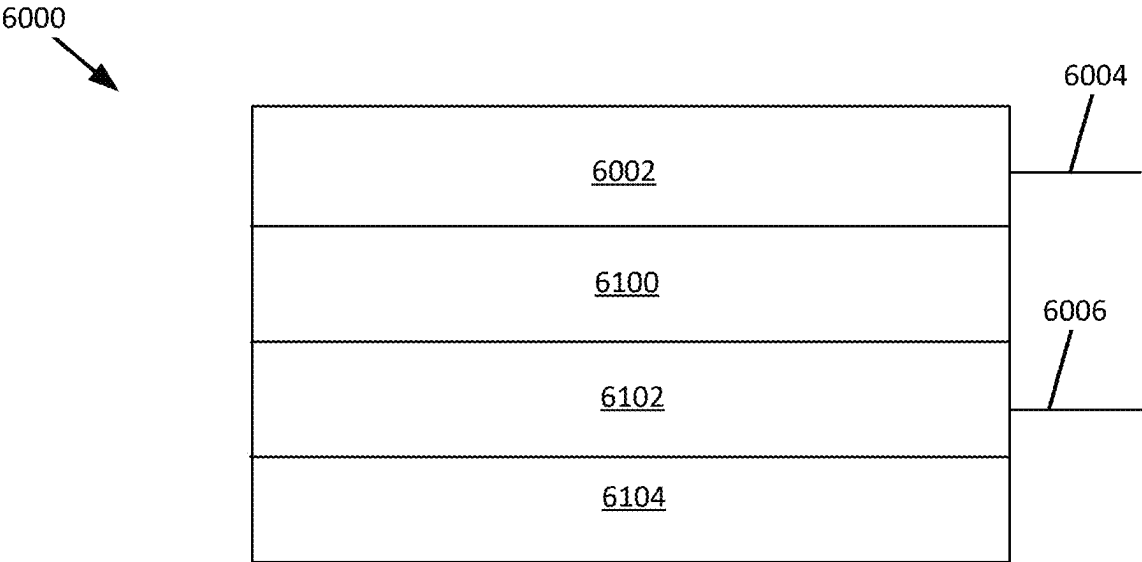


FIG. 61

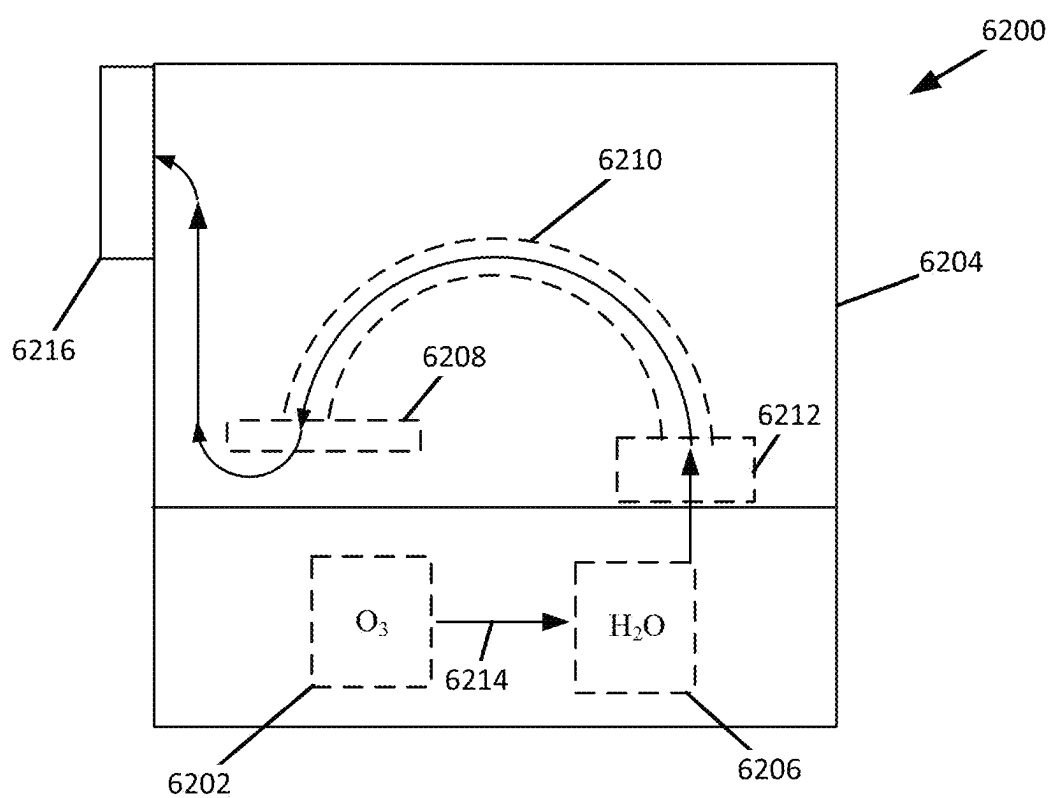


FIG. 62

DISINFECTION SYSTEMS AND METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. application Ser. No. 18/623,620, entitled Disinfection Systems and Methods, filed on Apr. 1, 2024, which is a continuation-in-part of U.S. application Ser. No. 18/368,598, entitled Disinfection Systems and Methods, filed on Sep. 15, 2023, which claims the benefit of U.S. Provisional Application Ser. No. 63/458,578 filed on Apr. 11, 2023, entitled Disinfection Systems and Methods, U.S. Provisional Application Ser. No. 63/458,527, filed on Apr. 11, 2023, entitled Disinfection Systems and Methods, U.S. Provisional Application Ser. No. 63/375,992 filed on Sep. 16, 2022, entitled Technologies for Sanitizing Medical Devices, and U.S. application Ser. No. 18/623,620 further claims the benefit of U.S. Provisional Application Ser. No. 63/458,578 filed on Apr. 11, 2023, entitled Disinfection Systems and Methods, U.S. Provisional Application Ser. No. 63/458,527, filed on Apr. 11, 2023, entitled Disinfection Systems and Methods, U.S. Provisional Application Ser. No. 63/609,991, filed on Dec. 14, 2023, entitled Piezoelectric Humidifier, and U.S. Provisional Application Ser. No. 63/613,867, filed on Dec. 22, 2023, entitled Disinfection Device, each of which are fully incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure is generally directed to disinfection systems and methods and more specifically to disinfection systems and methods using ozone.

BACKGROUND INFORMATION

[0003] Devices (e.g., medical devices, such as a mask, consumer devices, nebulizers etc.) are exposed to numerous environments in which pathogens (e.g., harmful bacteria and/or viruses) can come into contact with and reside on the devices. Without proper disinfection, further use of these devices after exposure can result in the spread of the pathogens. Accordingly, before reuse, devices may be disinfected in order to reduce the risk of spreading an infection to a user.

[0004] One example of a medical device is a mask, which may be coupled to one or more hoses (e.g., a continuous positive airway pressure mask and hose, a respirator mask, and/or any other type of mask). Masks are worn on a user's face and any hoses connected to the mask may come into direct contact with bodily fluids exhaled from a user's mouth and/or nose. These fluids may include pathogens. Accordingly, reuse of a mask, without proper disinfection, may increase a risk of infection.

[0005] Further, a hose coupled to the mask may include one or more crevices formed by a helical rib extending at least a substantial length of the hose and one or more pathogens may reside throughout the hose. The hose and/or the mask may include one or more regions (e.g., the crevices) which are difficult to disinfect. For example, a CPAP hose may include several crevices along the about 1.5 meter (m) to about 3 m length of the CPAP hose and/or the CPAP mask may have one or more difficult to disinfect places around the nasal passageways. The difficulty associated with disinfecting these regions may result in a build-up of one or more pathogens that may eventually pass to a user. Addi-

tionally, in some instances, a length of the CPAP hose may increase a difficulty in getting a disinfection fluid to traverse the entire length of the CPAP hose while achieving a desired disinfection performance, which may also lead to a build-up of one or more pathogens.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] These and other features advantages will be better understood by reading the following detailed description, taken together with the drawings wherein:

[0007] FIG. 1 shows a schematic block diagram of an example of a disinfection device, consistent with embodiments of the present disclosure.

[0008] FIG. 2 shows a schematic block diagram of an example of a disinfection device with humidity control, consistent with embodiments of the present disclosure.

[0009] FIG. 3 shows a schematic example of a disinfection device with humidity control configured to disinfect one or more components of a continuous positive air pressure (CPAP) machine, consistent with embodiments of the present disclosure.

[0010] FIG. 4 shows a schematic example of an operational timeline for the disinfection device of FIG. 3, consistent with embodiments of the present disclosure.

[0011] FIG. 5 shows a schematic example of an ozone generator, consistent with embodiments of the present disclosure.

[0012] FIG. 6 shows a schematic example of a humidifier, consistent with embodiments of the present disclosure.

[0013] FIG. 7 shows a perspective view of a disinfection device, consistent with embodiments of the present disclosure.

[0014] FIG. 7A shows a perspective view of the disinfection device of FIG. 7 having a CPAP hose coupled thereto, consistent with embodiments of the present disclosure.

[0015] FIG. 7B shows a cross-sectional perspective view of the disinfection device of FIG. 7A, consistent with embodiments of the present disclosure.

[0016] FIG. 8 shows a cross-sectional view of the disinfection device of FIG. 7 taken along the line VIII-VIII of FIG. 7, consistent with embodiments of the present disclosure.

[0017] FIG. 9 shows a cross-sectional view of the disinfection device of FIG. 7 taken along the line IX-IX of FIG. 7, consistent with embodiments of the present disclosure.

[0018] FIG. 9A is an exploded view of an example of an ozone reduction filter, consistent with embodiments of the present disclosure.

[0019] FIG. 10 shows a perspective view of a base region of the disinfection device of FIG. 7, consistent with embodiments of the present disclosure.

[0020] FIG. 11 shows a perspective view of a portion of a humidifier assembly of the disinfection device of FIG. 7, consistent with embodiments of the present disclosure.

[0021] FIG. 11A shows a perspective view of a liquid reservoir access, consistent with embodiments of the present disclosure.

[0022] FIG. 11B shows a perspective cross-sectional view of the liquid reservoir access of FIG. 11A, consistent with embodiments of the present disclosure.

[0023] FIG. 11C shows another cross-sectional perspective view of the liquid reservoir access of FIG. 11A, consistent with embodiments of the present disclosure.

[0024] FIG. 11D shows another cross-sectional perspective view of the liquid reservoir access of FIG. 11A, consistent with embodiments of the present disclosure.

[0025] FIG. 11E shows another cross-sectional perspective view of the liquid reservoir access of FIG. 11A, consistent with embodiments of the present disclosure.

[0026] FIG. 11F shows a bottom view of a platform of the liquid reservoir access of FIG. 11A, consistent with embodiments of the present disclosure.

[0027] FIG. 12 shows a cross-sectional view of the disinfection device of FIG. 7 taken along the line XII-XII of FIG. 7, consistent with embodiments of the present disclosure.

[0028] FIG. 12A shows a schematic example of an atomizer with a coating, consistent with embodiments of the present disclosure.

[0029] FIG. 12B shows an example of an insert configured to be coupled to (or formed from) a humidification chamber ozone inlet of the humidifier assembly of FIG. 11, consistent with embodiments of the present disclosure.

[0030] FIG. 12C shows a cross-sectional view of the insert of FIG. 12B, consistent with embodiments of the present disclosure.

[0031] FIG. 12D shows an example of the humidification chamber ozone inlet including the insert of FIG. 12B, consistent with embodiments of the present disclosure.

[0032] FIG. 12E shows a cross-sectional example of an atomizer having a recessed dimple configuration, consistent with embodiments of the present disclosure.

[0033] FIG. 13 shows a cross-sectional view of a portion of the humidifier assembly of the disinfection device of FIG. 7, consistent with embodiments of the present disclosure.

[0034] FIG. 14 shows a magnified view generally corresponding to region XIV-XIV of FIG. 13, consistent with embodiments of the present disclosure.

[0035] FIG. 14A shows a perspective view of an atomizer retainer, consistent with embodiments of the present disclosure.

[0036] FIG. 14B shows another perspective view of the atomizer retainer of FIG. 14A, consistent with embodiments of the present disclosure.

[0037] FIG. 15 shows a bottom view of a humidification chamber and a top view of a liquid reservoir of the disinfection device of FIG. 7, consistent with embodiments of the present disclosure.

[0038] FIG. 16 shows a perspective view of a wick assembly of the disinfection device of FIG. 7, consistent with embodiments of the present disclosure.

[0039] FIG. 17 shows a cross-sectional view of the wick assembly of FIG. 16 taken along the line XVII-XVII of FIG. 16, consistent with embodiments of the present disclosure.

[0040] FIG. 18 shows a cross-sectional view of the wick assembly of FIG. 16 coupled to the liquid reservoir of FIG. 15, consistent with embodiments of the present disclosure.

[0041] FIG. 19 shows a cross-sectional view of the disinfection device of FIG. 7 taken along the line XIX-XIX of FIG. 7, consistent with embodiments of the present disclosure.

[0042] FIG. 19A shows a schematic example of an operational timeline for the disinfection device of FIG. 7, consistent with embodiments of the present disclosure.

[0043] FIG. 19B shows one example of an efficacy outcome using one example of the disinfection device of FIG.

7 that is implementing one example of the operational timeline of FIG. 19A, consistent with embodiments of the present disclosure.

[0044] FIG. 20 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0045] FIG. 20A shows one example distribution of atomized water droplet sizes produced by an example of the piezo-electric atomizer of FIG. 20, consistent with embodiments of the present disclosure.

[0046] FIG. 21 shows a schematic example of an operational timeline for the disinfection device of FIG. 20, consistent with embodiments of the present disclosure.

[0047] FIG. 22 shows an example of a flow diagram, consistent with embodiments of the present disclosure.

[0048] FIG. 23 shows another example of a flow diagram, consistent with embodiments of the present disclosure.

[0049] FIG. 24 shows another example of a flow diagram, consistent with embodiments of the present disclosure.

[0050] FIG. 25 shows a schematic example a CPAP device having a disinfection device, consistent with embodiments of the present disclosure.

[0051] FIG. 26 shows a schematic example of a disinfection device with humidity control and a recirculation loop, consistent with embodiments of the present disclosure.

[0052] FIG. 27 shows a schematic example of a CPAP disinfection system, consistent with embodiments of the present disclosure.

[0053] FIG. 28 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0054] FIG. 29 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0055] FIG. 30 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0056] FIG. 31 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0057] FIG. 32 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0058] FIG. 33 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0059] FIG. 34 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0060] FIG. 35 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0061] FIG. 36 shows a schematic example of a disinfection device, consistent with embodiments of the present disclosure.

[0062] FIG. 37 shows a perspective view of a disinfection device, consistent with embodiments of the present disclosure.

[0063] FIG. 38 shows a perspective view of the disinfection device of FIG. 37 having a disinfection chamber lid in an open position, consistent with embodiments of the present disclosure.

[0064] FIG. 39 shows a perspective view of the disinfection device of FIG. 37 having a humidification chamber in a raised position, consistent with embodiments of the present disclosure.

[0065] FIG. 40 shows a perspective view of the disinfection device of FIG. 37 having a liquid reservoir removed from a reservoir receptacle, consistent with embodiments of the present disclosure.

[0066] FIG. 41 shows a perspective view of the disinfection device of FIG. 37, wherein the liquid reservoir has a reservoir lid removed therefrom, consistent with embodiments of the present disclosure.

[0067] FIG. 42 shows an exploded view of the reservoir lid of FIG. 41, consistent with embodiments of the present disclosure.

[0068] FIG. 43 shows a bottom perspective view of the reservoir lid of FIG. 41 having a wick assembly removed therefrom, consistent with embodiments of the present disclosure.

[0069] FIG. 44 shows a cross-sectional perspective view of the humidification chamber of the disinfection device of FIG. 37, consistent with embodiments of the present disclosure.

[0070] FIG. 45 shows another cross-sectional perspective view of the humidification chamber of the disinfection device of FIG. 37 in the raised position, consistent with embodiments of the present disclosure.

[0071] FIG. 45A shows a cross-sectional perspective view of the humidification chamber of the disinfection device of FIG. 37, consistent with embodiments of the present disclosure.

[0072] FIG. 45B shows a cross-sectional perspective view of the humidification chamber of the disinfection device of FIG. 37, consistent with embodiments of the present disclosure.

[0073] FIG. 46 shows a top perspective view of a chamber base of the humidification chamber of the disinfection device of FIG. 37, consistent with embodiments of the present disclosure.

[0074] FIG. 47 shows a bottom perspective view of the chamber base of FIG. 46, consistent with embodiments of the present disclosure.

[0075] FIG. 48 shows another cross-sectional perspective view of the humidification chamber of the disinfection device of FIG. 37 in the raised position, consistent with embodiments of the present disclosure.

[0076] FIG. 49 shows another cross-sectional perspective view of the humidification chamber of the disinfection device of FIG. 37 in the raised position, consistent with embodiments of the present disclosure.

[0077] FIG. 50 shows a perspective view of a humidification chamber outlet of the disinfection device of FIG. 37 having an adapter lock in a locked position, consistent with embodiments of the present disclosure.

[0078] FIG. 51 shows another shows a perspective view of a humidification chamber outlet of the disinfection device of FIG. 37 having the adapter lock in an unlocked position, consistent with embodiments of the present disclosure.

[0079] FIG. 52 shows a rear perspective view of the adapter lock of FIG. 50, consistent with embodiments of the present disclosure.

[0080] FIG. 53 shows a rear perspective view of the adapter lock of FIG. 51, consistent with embodiments of the present disclosure.

[0081] FIG. 54 shows a perspective view of the adapter lock of FIG. 50 in the unlocked position and a CPAP hose adapter removed from the humidification chamber outlet, consistent with embodiments of the present disclosure.

[0082] FIG. 55 shows another perspective view of the adapter lock of FIG. 50 in the unlocked position and the CPAP hose adapter removed from the humidification chamber outlet, consistent with embodiments of the present disclosure.

[0083] FIG. 56 shows a perspective bottom view of the disinfection chamber lid of the disinfection device of FIG. 37, consistent with embodiments of the present disclosure.

[0084] FIG. 57 shows another perspective bottom view of the disinfection chamber lid of the disinfection device of FIG. 37, having a humidity sensor assembly removed therefrom, consistent with embodiments of the present disclosure.

[0085] FIG. 58 shows an exploded view of the humidity sensor assembly of FIG. 57, consistent with embodiments of the present disclosure.

[0086] FIG. 59 shows a cross-sectional perspective view of the disinfection chamber of the disinfection device of FIG. 37 showing the humidity sensor assembly installed therein, consistent with embodiments of the present disclosure.

[0087] FIG. 60 shows a schematic top view of an example of a piezo-electric atomizer, consistent with embodiments of the present disclosure.

[0088] FIG. 61 shows a schematic cross-sectional view of the piezo-electric atomizer of FIG. 60 taken along the line LXI-LXI, consistent with embodiments of the present disclosure.

[0089] FIG. 62 shows a schematic view of a disinfection device, consistent with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0090] The present disclosure is generally directed to a disinfection device having humidity and/or disinfection performance control. The disinfection device includes an ozone generator, a humidifier, and a disinfection chamber. The ozone generator is configured to generate ozone such that generated ozone can become entrained within air passing through the ozone generator to create ozonated air. Ozonated air may pass through the humidifier and into the disinfection chamber. The ozone generator and the humidifier may be selectively enabled and disabled according to an operation profile. In one example, humidified ozonated air may pass from the humidifier and into the disinfection device via a medical device (e.g., a continuous positive air pressure (CPAP) hose and/or mask).

[0091] In some instances, the disinfection device may be configured to disinfect a CPAP hose and CPAP mask. For example, the disinfection device may be configured to achieve a disinfection performance of at least $4-\text{Log}_{10}$ (at least a 99.99% reduction) for a given one or more pathogens (e.g., viruses, bacteria, and/or any other pathogen) disposed on at least a portion of the CPAP hose and CPAP mask. In one example, the CPAP hose may have a length in a range of about 1.5 meters (m) to about 3 m, a diameter of about 3.8 centimeters (cm) to about 5 cm, and a connection point at a distal end at which the CPAP mask is coupled (e.g., removably coupled). As the CPAP mask is worn by a user, pathogens may be present in the greatest quantities at the CPAP mask and/or at a region of the CPAP hose proximate

to the connection point. As such, in some instances, the disinfection device may be configured such that disinfection performance at a region of the CPAP hose proximate to the connection point, at the CPAP mask (e.g., internal and/or external surfaces of the CPAP mask), and/or for an entire length of the CPAP hose is at least 4-Log_{10} .

[0092] In some instances, the humidifier may be configured to humidify the disinfection chamber prior to (or after) the generation of ozone. In these instances, ozone may not pass through the humidifier prior to entering the disinfection chamber. For example, the disinfection device may be configured to undergo a pre-humidification step in which a relative humidity within the disinfection chamber is achieved prior to the generation of ozone.

[0093] Addition of humidity to ozonated air may improve the disinfection performance of the ozonated air. Further, how the humidity is generated (e.g., using a bubbler system, a piezo-electric atomizer, a sprayer, a heat source, evaporation, and/or the like) may influence the disinfection performance of the ozonated air. Additionally, or alternatively, a size of water droplets generated by the humidifier that become entrained within the ozonated air may also influence the disinfection performance of the ozonated air. Further, the flow rate of the ozonated air, relative humidity of the environment, and/or exposure time to ozonated air may influence the disinfection performance.

[0094] One example operation profile may include a first period, a second period, and a third period. During the first period both the ozone generator and the humidifier may be operated continuously. During the second period, the humidifier may be disabled and the ozone generator may be pulsed (e.g., selectively enabled/disabled) according to a constant (or non-constant) pulse rate. During the third period, both the humidifier and the ozone generator may be disabled.

[0095] In some instances, the disinfection device, when operated according to an appropriate operation profile, may achieve at least a 4-Log_{10} kill rate for at least one pathogen (e.g., bacterial or viral) on an object within the disinfection chamber and/or within a flow path of the ozonated air (e.g., humidified ozonated air). In other words, the disinfection device may be configured to cause at least a 99.99% reduction of at least one pathogen. In other instances, the disinfection device, when operated according to an appropriate operation profile, may achieve at least a 6-Log_{10} kill rate for a pathogen on an object within the disinfection chamber and/or within a flow path of the ozonated air (e.g., humidified ozonated air). In other words, the disinfection device may be configured to cause at least a 99.9999% reduction of at least one pathogen. One example method for determining whether the disinfection device achieves a desired kill rate includes generating an inoculate having a sufficient quantity of one or more pathogens suspended therein to establish the desired reduction of the one or more pathogens was achieved and applying the inoculate to an object to be disinfected (e.g., a CPAP hose and/or a CPAP mask).

[0096] Although the technologies described herein can be used with many disinfection fluids, the present disclosure focuses on the use of ozone as a disinfecting gas. This is because ozone (O_3) gas is a relatively powerful disinfectant yet can be safely used within a consumer product. Because of its strong oxidizing properties, ozone can effectively kill or otherwise remove a wide range of organic and inorganic

contaminants such as yeasts, bacteria, molds, viruses, other pathogens, and/or pollutants with which it comes into contact (e.g., via oxidation). Naturally over time and/or as it oxidizes contaminants, ozone is chemically reduced to oxygen (O_2), which is safe for human consumption and for release into the environment. Ozone is also relatively easy to generate on site (and, thus, does not require the use of a storage tank), and leaves little or no chemical residue. For those and other reasons, ozone has been identified as a safe and effective disinfecting gas for use in the present disclosure. It should be understood, however, that the technologies described herein are not limited to the use of ozone and may be employed with a wide variety of disinfection fluids.

[0097] FIG. 1 shows a schematic block diagram of a disinfection device 100. The disinfection device 100 includes an ozone generator 102, a disinfection chamber 104, at least one pump 106 configured to cause air to pass into the ozone generator 102 and into the disinfection chamber 104, an ozone reduction filter 108 configured to breakdown ozone into oxygen, and a controller 110 configured to control operations of one or more of the ozone generator 102 and/or the pump 106.

[0098] The disinfection chamber 104 may include a disinfection chamber inlet 112, a disinfection chamber outlet 114, and define a volume (e.g., a selectively enclosable volume) that is configured to receive at least a portion of an object 116 (e.g., a medical device). The ozone reduction filter 108 can be positioned at (e.g., upstream of or downstream of) the disinfection chamber outlet 114. The ozone reduction filter 108 can be made of an ozone porous material that is configured to reduce ozone to oxygen. For example, the ozone reduction filter 108 may be a reticulated foam that is formed from, includes, and/or is coated with a material that reduces ozone to oxygen. Non-limiting examples of filter materials include activated carbon, magnesium oxide, and/or magnesium dioxide (either alone or in combination with activated carbon); however, other materials may be used.

[0099] In operation, the ozone generator 102 is caused to generate ozone and the pump 106 is configured to cause air to flow along a flow path 118. The flow path 118 passes through the ozone generator 102 into the disinfection chamber 104 and passes through the disinfection chamber outlet 114. The generated ozone is carried with the flow of air (which may generally be referred to as ozonated air) into the disinfection chamber 104. Once in the disinfection chamber 104, the ozone acts to disinfect the object 116 present within the disinfection chamber 104.

[0100] The rate of ozone generation and the flow rate of the air passing through the ozone generator 102 is controlled by the controller 110. For example, the controller 110 may cause the pump 106 and/or a fan 107 to generate a desired flow rate of air through the disinfection device 100. In this example, the pump 106 and the fan 107 may be operated at different speeds such that air flows through the disinfection device according to a flow rate. In this example, when the pump 106 generates a different (e.g., greater) flow rate than the fan 107, the disinfection performance may be improved. For example, the pump 106 generating a greater flow rate than the fan 107 may encourage a disinfection performance of at least 4-Log_{10} . In some instances, the controller 110 may be configured to control the disinfection device 100 such that a pressure and/or a vacuum is generated within the disinfection chamber 104. In other words, the controller 110

is configured to control the environment within the disinfection chamber 104 (e.g., the rate with which ozone is delivered to and/or removed from the disinfection chamber 104 may be used to adjust an amount of ozone within the disinfection chamber 104).

[0101] Varying the condition of the environment within the disinfection chamber 104 may influence an effectiveness of the disinfection of the object 116 within the disinfection chamber 104. For example, an amount of moisture within the air flow (humidity) may have an impact on disinfection effectiveness. In this example, humidified air may improve the disinfection effectiveness, potentially, allowing less ozone to be generated to achieve a desired amount of disinfection (e.g., a bacterial and/or viral kill rate). Generation of less ozone may, for example, improve the longevity of the ozone reduction filter 108.

[0102] FIG. 2 shows a schematic block diagram of a disinfection device 200 with humidity control. The disinfection device 200 includes an ozone generator 202, a disinfection chamber 204, a humidifier 206, at least one pump 208 configured to cause air to pass through the ozone generator 202 and the humidifier 206 and into the disinfection chamber 204, an ozone reduction filter 210 configured to breakdown ozone into oxygen, and a controller 212 configured to control operations of one or more of the ozone generator 202, the humidifier 206, and/or the pump 208. In some instances, the disinfection device 200 may further include a fan 209 downstream of the pump 208 and upstream of the ozone reduction filter 210. The controller 212 may be further configured to control operation of the fan 209. For example, the controller 212 may be configured to cause the fan 209 to operate at a different (e.g., slower) flow rate than the pump 208.

[0103] The disinfection chamber 204 may include a disinfection chamber inlet 214, a disinfection chamber outlet 216, and define a volume (e.g., a selectively enclosable volume) that is configured to receive at least a portion of an object 218 (e.g., a medical device). The ozone reduction filter 210 can be positioned at (e.g., upstream of or downstream of) the disinfection chamber outlet 216. The ozone reduction filter 210 can be an ozone porous material that is configured to reduce ozone to oxygen. For example, the ozone reduction filter 210 may be a reticulated foam that is formed from, includes, and/or is coated with a material that reduces ozone to oxygen. Non-limiting examples of filter materials include activated carbon, magnesium oxide, and/or magnesium dioxide (either alone or in combination with activated carbon); however, other materials may be used.

[0104] In operation, the pump 208 is configured to cause air to flow along a flow path 220. The flow path 220 extends through the ozone generator 202, the humidifier 206, the disinfection chamber 204, and the ozone reduction filter 210. The ozone generator 202 is configured to generate ozone as air passes through the ozone generator 202. When the ozone generator 202 is generating ozone, air exiting the ozone generator 202 includes ozone entrained therein (which may generally be referred to as ozonated air). The ozonated air enters the humidifier 206, wherein the humidifier 206 is configured to mix or combine the ozonated air with humidity. In some instances, ozone may become entrained within droplets of moisture generated by the humidifier 206. In some instances, the humidifier 206 may be configured to selectively humidify the ozonated air. For example, the humidifier 206 may be selectively operated such that ozo-

nated air can pass therethrough without humidification. Such a configuration, may allow the controller 212 to regulate the humidity within the disinfection chamber 204 independently of the amount of ozone. Additionally, or alternatively, the humidifier 206 may be positioned upstream of the ozone generator 202 such that the humidifier 206 adds humidity to non-ozonated air.

[0105] In some instances, when the disinfection device 200 includes the fan 209, the fan 209 may be configured to cooperate with the pump 208 to deliver ozonated air to the disinfection chamber 204. The pump 208 may be configured to push ozonated air into the disinfection chamber 204 (e.g., the pump 208 may be fluidly coupled to the disinfection chamber 204 at a location upstream of the disinfection chamber 204) and the fan 209 may be configured to pull ozonated air into the disinfection chamber 204 (e.g., the fan 209 may be fluidly coupled to the disinfection chamber 204 at a location downstream of the disinfection chamber 204). For example, the fan 209 may be configured to pull ozonated air into the disinfection chamber 204 at a first flow rate and the pump 208 may be configured to push ozonated air into the disinfection chamber 204 at a second flow rate, the second flow rate being different the first flow rate. In this example, during a disinfection cycle, the second flow rate may be greater than the first flow rate and, during a purge cycle (e.g., after completion of the disinfection cycle), the first flow rate may be greater than (or the same as) the second flow rate (e.g., in some instances, the second flow rate may be zero) in order to encourage the purging of ozonated air from the disinfection device 200. In some instances, the disinfection device may be configured such that the disinfection chamber 204 is under pressure and/or under vacuum.

[0106] FIG. 3 shows a schematic example of a disinfection device 300 with humidity control configured to disinfect one or more components of a continuous positive air pressure (CPAP) machine. The disinfection device 300 is an example of the disinfection device 200 of FIG. 2.

[0107] As shown, the disinfection device 300 includes a disinfection chamber 302 and an ozone and humidity generation assembly 304 external to the disinfection chamber 302 (e.g., the ozone and humidity generation assembly 304 may be coupled to an external surface of a sidewall 316 of the disinfection chamber 302). The ozone and humidity generation assembly 304 includes an ozone generator 306, a humidifier 308, and a pump 310 configured to urge air through the ozone generator 306 and the humidifier 308. The ozone generator 306, the humidifier 308, and the pump 310 are configured to cooperate together (e.g., to generate humidified ozonated air), forming the ozone and humidity generation assembly 304. In some instances, a chamber relative humidity sensor 305 may be configured to measure a relative humidity of the disinfection chamber 302. In these instances, an operation of the humidifier 308 may be controlled based, at least in part, on output from the chamber relative humidity sensor 305.

[0108] The ozone and humidity generation assembly 304 is configured to be removably coupled to a CPAP hose 312 at a first end 314 such that the CPAP hose 312 is fluidly coupled to the ozone generator 306, the humidifier 308, and the pump 310. The CPAP hose 312 is configured to fluidly couple the ozone and humidity generation assembly 304 to the disinfection chamber 302. For example, and as shown, the CPAP hose 312 is configured to pass through the sidewall 316 of the disinfection chamber 302. By way of

further example, the CPAP hose 312 may be configured to removably couple to a coupling that is fluidly coupled to the disinfection chamber 302. In some instances, a CPAP mask 318 may be disposed within the disinfection chamber 302. For example, the CPAP mask 318 may be coupled to the CPAP hose 312 at a second end 320, the second end 320 being opposite the first end 314. A length of the CPAP hose 312 extending between the first and second ends 314 and 320 may be, for example, in a range of about 1.5 meters (m) to 3 m. By way of further example, a length of the CPAP hose 312 extending between the first and second ends 314 and 320 may be about 1.8 m. An ozone reduction filter 322 may be disposed at an outlet 324 of the disinfection chamber 302. In some instances, a fan 328 may be provided to draw ozonated air from the disinfection chamber 302 and urge the ozonated air through the ozone reduction filter 322.

[0109] The humidifier 308, the pump 310, and/or the ozone generator 306 may be arranged in any configuration relative to the disinfection chamber 302 and/or may be disposed within separate or common housings. For example, one or more of the humidifier 308, the pump 310, and/or the ozone generator 306 may be arranged at least partially below the disinfection chamber 302. Additionally, or alternatively, and by way of further example, one or more of the humidifier 308, the pump 310, and/or the ozone generator 306 may be arranged at least partially along a side of and/or separate from the disinfection chamber 302. In some instances, one or more of the humidifier 308, the pump 310, and/or the ozone generator 306 may be movable relative to the disinfection chamber 302. For example, when the disinfection chamber 302 is in the form of a flexible bag, one or more of the humidifier 308, the pump 310, and/or the ozone generator 306 may be disposed within one or more external housings and be configured to fluidly couple to the disinfection chamber 302 via the CPAP hose 312.

[0110] In operation, a controller 326 is configured to selectively control each of the ozone generator 306, the humidifier 308, the pump 310, and/or the fan 328 to control an environment within the disinfection chamber 302. For example, the controller 326 may be configured to selectively urge air, ozonated air, or humidified ozonated air through the CPAP hose 312 and into the disinfection chamber 302 in order to disinfect the CPAP hose 312 and the CPAP mask 318. In this example, when the humidity has reached a desired level (e.g., estimated and/or measured), the humidifier 308 may be disabled and/or pulsed at a constant or non-constant pulse rate, and/or, when the ozone has reached a desired level (e.g., estimated and/or measured), the ozone generator 306 may be disabled and/or pulsed at a constant or non-constant pulse rate.

[0111] FIG. 4 shows a schematic example of an operational timeline 400 of the disinfection device 300 that corresponds to one example disinfection cycle. As shown, a first period (e.g., an initialization period) 402 extends between time T_0 and time T_1 , a second (e.g., an operation) period 404 extends between time T_1 and time T_2 , and a third (e.g., a purge) period 406 extends between time T_2 and time T_3 . Time T_3 may be greater than time T_2 , time T_2 may be greater than time T_1 , and time T_1 may be greater than time T_0 . The first period 402 may generally be referred to as a period in which the ozone and humidity generation assembly 304 is operated to achieve a desired initial condition (e.g., within a desired humidity and/or ozone range) within the hose 312, the mask 318, and/or the disinfection chamber 302

(the determination of whether the initial condition has been achieved may be estimated and/or measured). The second period 404 may generally be referred to as a period in which the ozone and humidity generation assembly 304 is operated to maintain a desired condition (e.g., within a desired humidity and/or ozone range that is estimated and/or measured). The third period 406 may generally be referred to as the period in which the ozone and humidity generation assembly 304 is operated to purge the ozone from the disinfection chamber 302 such that any residual ozone within the disinfection chamber 302 is removed and/or within an acceptable level.

[0112] In one example, during the first period 402, the ozone generator 306, the humidifier 308, the pump 310, and the fan 328 may be operated continuously or intermittently, during the second period 404, the ozone generator 306 may be operated continuously or intermittently, the humidifier 308 may be disabled, the pump 310 may be operated continuously or intermittently, and the fan 328 may be operated continuously or intermittently, and, during the third period 406, the ozone generator 306 and humidifier 308 may be disabled, the pump 310 may be operated continuously or intermittently, and the fan 328 may be operated continuously or intermittently. In another example, during the first period 402, one or more of the ozone generator 306, the humidifier 308, the pump 310, and/or the fan 328 may be operated continuously or intermittently, during the second period 404, one or more of the ozone generator 306, the humidifier 308, the pump 310, and/or the fan 328 may be disabled, and, during the third period 406, the ozone generator 306 and humidifier 308 may be disabled, the pump 310 may be operated continuously or intermittently, and the fan 328 may be operated continuously or intermittently.

[0113] A length of the first period 402, the second period 404, and the third period 406 may be predetermined or based, at least in part, on sensed conditions (e.g., within the disinfection chamber 302, the CPAP hose 312, and/or the CPAP mask 318) and/or estimated conditions (e.g., within the disinfection chamber 302, the CPAP hose 312, and/or the CPAP mask 318). Introduction of humidity during the first period 402 may result in one or more pathogens becoming more susceptible to breakdown by ozone.

[0114] While the terms, first, second, and third, are used to described the first period 402, the second period 404, and the third period 406, the terms first, second, and third are not intended to convey a specific order. For example, in some instances, the second period 404 and/or third period 406 may come before the first period 402. In some instances, there may be one or more additional periods that come before or after one or more of the first, second, and/or third periods 402, 404, and/or 406.

[0115] FIG. 5 is a schematic example of an ozone generator 500, which is an example of the ozone generator 202 of FIG. 2. As shown, the ozone generator 500 includes an ozone source 502, an ozone sensor 504, a humidity sensor 506 (e.g., a relative humidity sensor), and, in some instances, a temperature sensor 507. The ozone source 502, the ozone sensor 504, the humidity sensor 506, and the temperature sensor 507 may be communicatively coupled to the controller 212 (FIG. 2). For example, the controller 212 may be configured to control operation of the ozone source 502 based, at least in part, on outputs from the ozone sensor 504, the humidity sensor 506, and/or the temperature sensor 507. The ozone sensor 504 may be configured to measure a

quantity of ozone within air exiting the ozone sensor **504**. The humidity sensor **506** may be configured to measure a relative humidity of air entering the ozone source **502** (e.g., the relative humidity within a surrounding environment such as a room within which the ozone generator **500** is located). The temperature sensor **507** may be configured to measure a temperature of air entering the ozone source **502** (e.g., the temperature within a surrounding environment such as a room within which the ozone generator **500** is located). In one example, the controller **212** may operate the ozone source **502** at an initial operation level based, at least in part, on the relative humidity and measure the output of the ozone source **502** using the ozone sensor **504**, wherein the initial operation level corresponds to an estimated amount of ozone to be generated. The measured output of the ozone source **502** may be used to adjust an amount of ozone generated by the ozone source **502** until a desired ozone output is obtained. In some instances, the humidity sensor **506** and the temperature sensor **507** may be collectively referred to as a humidity/temperature sensor. In these instances, the humidity sensor **506** and the temperature sensor **507** may be part of the same sensor assembly or may be separate sensors.

[0116] As shown, in operation air is caused to flow along an ozone generator flow path **508** (which forms a portion of the flow path **220**, FIG. 2). The ozone generator flow path **508** extends through the ozone source **502** and the ozone sensor **504**, the ozone sensor **504** being downstream of the ozone source **502**. For example, the ozone sensor **504** may be downstream of the ozone source **502** and upstream of a humidifier (e.g., the humidifier **206** of FIG. 2) and/or a disinfection chamber (e.g., the disinfection chamber **204** of FIG. 2). By way of further example, the ozone sensor **504** may be downstream of the ozone source **502** and a humidifier (e.g., the humidifier **206** of FIG. 2) and upstream of a disinfection chamber (e.g., the disinfection chamber **204** of FIG. 2). By way of still further example, the ozone sensor **504** may be downstream of the ozone source **502** and a humidifier (e.g., the humidifier **206** of FIG. 2) and disposed within a disinfection chamber (e.g., the disinfection chamber **204** of FIG. 2).

[0117] FIG. 6 shows a schematic example of a humidifier **600**, which is an example of the humidifier **206** of FIG. 2. As shown, the humidifier **206** includes a liquid reservoir **602**, a humidification chamber **604**, and a humidity generator **606**. The humidity generator **606** is configured to urge liquid from the liquid reservoir **602** and to disperse the liquid into the humidification chamber **604**. For example, the humidity generator **606** may include a liquid collector **608** and a liquid disperser **610** (e.g., an atomizer, such as, a piezo-electric atomizer), the liquid collector **608** is configured to transfer fluid from the liquid reservoir **602** to the liquid disperser **610**. The liquid collector **608** may be an active collector (e.g., include a pump) or a passive collector (e.g., include a wick). In some instances, the humidity generator **606** may be configured to generate droplets having a droplet size (e.g., an approximate diameter or a maximum dimension) in a range of about 1.5 microns to about 4.5 microns (e.g., droplets having a droplet size of about 2.5 microns). In some instances, at least a portion of the humidity generator **606** (e.g., the liquid disperser **610**) may be at a bottom portion **620** (e.g., coupled at a bottom wall **622**) of the humidification chamber **604**.

[0118] The humidity generator **606** may include one or more of an atomizer (e.g., a piezo-electric atomizer, a nozzle

atomizer, an electrostatic atomizer, a centrifugal atomizer, and/or any other type of atomizer), a heat source (e.g., a boiler to generate steam), a passive evaporator, an ozone bubbler within a liquid bath, a mister, a sprayer, a humidity pack, and/or the like to generate humidity within the humidification chamber **604**. In some instances, one or more components of the humidity generator **606** may include a protective (e.g., hydrophobic) coating on at least a portion of the component.

[0119] The humidification chamber **604** includes a humidification chamber ozone inlet **612** through which air (e.g., ozonated air) enters the humidification chamber **604** and a humidification chamber outlet **614** through which air (e.g., humidified ozonated air or ozonated air) exits the humidification chamber **604** (e.g., to pass into the CPAP hose **312**). The humidification chamber ozone inlet **612** and the humidification chamber outlet **614** are arranged such that a humidifier flow path **616** (which forms a portion of the flow path **220**, FIG. 2) extends transverse to (e.g., perpendicular to) an emission axis **618** of the humidity generator **606**. In some instances, air flowing along the humidifier flow path **616** may be turbulent for at least a portion of the humidifier flow path **616** (the turbulence may improve mixing of ozone with atomized liquid). For example, a turbulent flow of air may be present at the humidification chamber ozone inlet **612** and may transition to a laminar flow (e.g., after passing through the humidification chamber outlet **614** and into the CPAP hose **312**). In some instances, the humidifier **600** may generally be described as being configured to encourage a mixing of ozone with humidity (e.g., within the humidification chamber **604**).

[0120] FIG. 7 shows an example of a disinfection device **700** with humidity control configured to disinfect one or more components of a continuous positive air pressure (CPAP) machine. The disinfection device **700** is an example of the disinfection device **300** of FIG. 3.

[0121] As shown, the disinfection device **700** includes a disinfection chamber **702**, an ozone generation assembly **704** (shown schematically in hidden lines), and a humidifier assembly **706**. The disinfection chamber **702** includes a CPAP hose passthrough **708** through which a CPAP hose **750** (see, FIG. 7A) can be inserted such that at least a portion of the CPAP hose **750** is received within the disinfection chamber **702**. In other words, the CPAP hose passthrough **708** is configured such that the CPAP hose **750** can pass therethrough. As shown in FIG. 7B, the CPAP hose **750** has a first hose end **756** and a second hose end **758**, wherein a CPAP hose adapter **752** is disposed at the first hose end **756** and a CPAP mask **754** is disposed at the second hose end **758** (e.g., removably coupled to a connection point at the second hose end **758**). As such, the CPAP hose **750** may generally be described as being fluidly coupled to the humidifier assembly **706** as the first hose end **756** and the humidifier assembly **706** may generally be described as being fluidly coupled to the disinfection chamber **702** via the CPAP hose **750** and CPAP mask **754**. As also shown, the CPAP hose **750** includes a helical rib **760** that extends at least a substantial length of the CPAP hose **750** and forms one or more crevices **762** (e.g., a helical groove that extends for at least a length of the helical rib **760**). In addition, the length of the CPAP hose **750** may pose one or more challenges to disinfection performance. As such, pathogens may tend to collect at the one or more crevices **762** and/or along a length of the CPAP hose **750**. The CPAP mask **754** may include one or more

features worn by a user (e.g., one or more nasal passages) that may also tend to collect one or more pathogens with use.

[0122] The CPAP hose passthrough 708 may be at least partially formed within a sidewall 710 of the disinfection chamber 702. In some instances, the CPAP hose passthrough 708 may include a passthrough open end 712. The passthrough open end 712 may be selectively closed by a disinfection chamber lid 714. In these instances, the CPAP hose passthrough 708 and the disinfection chamber lid 714 may cooperate to form a seal that extends about the CPAP hose 750. The seal may be configured to mitigate and/or prevent ozone from escaping from the disinfection chamber 702 at the CPAP hose passthrough 708. In some instances, the CPAP hose passthrough 708 may include a first hose sensor 716 (shown schematically in hidden lines) configured to detect a presence of the CPAP hose 750 within the CPAP hose passthrough 708. The first hose sensor 716 may be configured to prevent generation of ozone when the CPAP hose 750 is not received within the CPAP hose passthrough 708.

[0123] The humidifier assembly 706 includes a CPAP hose coupling 718 for removably coupling to the CPAP hose. As such, the CPAP hose fluidly couples the humidifier assembly 706 to the disinfection chamber 702. The CPAP hose coupling 718 may be configured to detect when a CPAP hose is coupled thereto. For example, the CPAP hose coupling 718 may include a second hose sensor 720 (shown schematically in hidden lines). The second hose sensor 720 may be configured to detect a presence of the CPAP hose 750 coupled thereto. In some instances, the CPAP hose 750 may be coupled to the CPAP hose coupling 718 using a CPAP hose adapter 752 (FIG. 7A) and the second hose sensor 720 may be configured to detect the presence of the CPAP hose 750 and/or the CPAP hose adapter 752. The second hose sensor 720 may be configured to prevent generation of ozone when the CPAP hose 750 is not coupled thereto.

[0124] The humidifier assembly 706 further includes a liquid reservoir access 722. The liquid reservoir access 722 is configured to allow a user to selectively replenish a liquid in the humidifier assembly 706. The liquid reservoir access 722 may be configured as a door, a drawer, and/or any other form of access. In some instances, the liquid reservoir access 722 may include a liquid level window 724 configured to enable a user to observe a quantity of liquid within a liquid reservoir.

[0125] FIG. 8 shows a cross-sectional view of the disinfection device 700 taken along the line VIII-VIII of FIG. 7 and FIG. 9 shows a cross-sectional view of the disinfection device 700 taken along the line IX-IX of FIG. 7. As shown, the ozone generation assembly 704 is disposed within a base region 800 of the disinfection device 700. At least a portion of the base region 800 is disposed beneath the disinfection chamber 702. For example, a base sidewall 801 that defines at least a portion of the disinfection chamber 702 may extend between the disinfection chamber 702 and the ozone generation assembly 704, separating the ozone generation assembly 704 from the disinfection chamber 702.

[0126] As also shown, at least a portion of the humidifier assembly 706 extends within the base region 800 and along at least a portion of the disinfection chamber 702 (e.g., an external surface of the sidewall 710 of the disinfection chamber 702). For example, the humidifier assembly 706 may include a liquid reservoir 802 and a humidification chamber 804, wherein at least a portion of the liquid

reservoir 802 extends within the base region 800 and at least a portion of the humidification chamber 804 extends along the disinfection chamber 702 and above the base region 800.

[0127] The liquid reservoir access 722 includes a movable platform 806, a carriage 808 configured to removably receive the liquid reservoir 802, one or more linkages 810 pivotally coupled to the movable platform 806 and the carriage 808, and an access door 812 including a handle 814 coupled to the movable platform 806. In use, a user exerts a force (e.g., a pulling or pushing force) on the handle 814 to transition the movable platform 806 from a use position to a refill position. When transitioning between the use and refill positions, the movable platform 806 may be caused to slide along an insertion axis 816. When the movable platform 806 transitions between the use and refill positions, the one or more linkages 810 pivot relative to the movable platform 806 and the carriage 808 such that the carriage 808 (and the liquid reservoir 802) are caused to move along the insertion axis 816 and a coupling axis 818. The insertion axis 816 extends transverse to (e.g., perpendicular to) the coupling axis 818. As shown, the insertion axis 816 extends generally (e.g., within 1°, 2°, 3°, 4°, or 5° of) parallel to a base 820 of the disinfection device 700. In some instances, the insertion axis 816 may be described as a generally horizontal axis. As shown, the coupling axis 818 extends transverse to (e.g., perpendicular to) the base 820 of the disinfection device 700. In some instances, the coupling axis 818 may be described as a generally vertical axis.

[0128] As also shown in FIG. 8, a flow path 822 extends from the ozone generation assembly 704, into the humidification chamber 804 via an inlet connector 824, out of the humidification chamber 804 via the CPAP hose coupling 718, into the CPAP hose 750 that extends through the CPAP hose passthrough 708, and exits the CPAP hose 750 within the disinfection chamber 702 (e.g., and passes through a CPAP mask 754, see, FIG. 7B, disposed within the disinfection chamber 702). As shown in FIG. 9, the flow path 822 extends from the disinfection chamber 702 through a disinfection chamber outlet 900, into an ozone reduction filter 902, through an upstream fan 904, and into the surrounding environment. The upstream fan 904 is configured to draw air and/or ozone through the ozone reduction filter 902. In some instances, the ozone reduction filter 902 may include a filter communication system 906 configured to communicatively couple with the disinfection device 700 (e.g., to provide firmware updates, updated disinfection profiles, and/or the like). In some instances, the filter communication system 906 may be a radio frequency identification (RFID) tag.

[0129] FIG. 9A shows an exploded view of the ozone reduction filter 902. As shown, the ozone reduction filter 902 includes a housing 920 having an upstream housing portion 922 and a downstream housing portion 924, wherein the upstream housing portion 922 and the downstream housing portion 924 define a filter cavity 926 configured to receive a filter medium 928 configured to reduce ozone. The upstream housing portion 922 includes an ozonated air inlet 930. The ozonated air inlet 930 may include a filter seal 932 for sealingly engaging with the disinfection chamber outlet 900 (FIG. 9). As shown, the filter communication system 906 is disposed within the filter cavity 926 at a location between the filter medium 928 and the upstream housing portion 922.

[0130] FIG. 10 shows a perspective view of the base region 800 of the disinfection device 700. As shown, the

base region **800** includes a pump **1000** fluidly coupled to the ozone generation assembly **704**. In some instances, a back flow preventer **1001** (e.g., a check-valve or Polytetrafluoroethylene (PTFE) filter) may be disposed downstream of the pump **1000** and/or downstream of the ozone generation assembly **704**, which may reduce or prevent water (e.g., ozonated water) from flowing into the pump **1000** and/or the ozone generation assembly **704**.

[0131] In operation, the pump **1000** is configured to cause air from a surrounding environment to flow through the ozone generation assembly **704** to create ozonated air. The environmental air has pre-existing conditions (e.g., relative humidity and/or any other pre-existing condition). The pre-existing conditions of the environmental air may influence the performance of the ozone generation assembly **704**. As such, the disinfection device **700** may, in some instances, include one or more environmental sensors (e.g., a relative humidity sensor) to sense one or more environmental conditions (e.g., a relative humidity) and to adjust the ozone generation assembly **704** based, at least in part, on at least one of the sensed environmental conditions. Operation of the ozone generation assembly **704** based, at least in part, on sensed environmental conditions may encourage more efficient and/or consistent operation of the ozone generation assembly **704**.

[0132] As shown, the ozone generation assembly **704** includes an ozone source **1002** (shown schematically in hidden lines), a relative humidity sensor **1004** (shown schematically in hidden lines), and an ozone sensor **1006**. The relative humidity sensor **1004** is configured to detect a relative humidity of the environmental air. The detected relative humidity is provided to a controller **1008**. The controller **1008** is configured to cause the ozone source **1002** to operate according to operation parameter(s) that are based, at least in part, on the detected relative humidity. The operation parameter(s) are configured to cause the ozone source **1002** to generate an estimated quantity of ozone. However, the estimated quantity of ozone to be generated may be different from an actual quantity of ozone generated. The ozone sensor **1006** is configured to detect the actual quantity of ozone generated by the ozone source **1002**. The detected quantity of ozone is provided to the controller **1008**. The controller **1008** compares the detected quantity of ozone to the estimated quantity of ozone and adjusts the operation parameter(s) of the ozone source **1002** if the detected quantity of ozone is greater than or less than the estimated quantity of ozone or outside of an acceptable range (e.g., within 1%, 2%, 3%, 4%, 5%, 10%, 15%, or 20% of the estimated quantity of ozone). In other words, output from the relative humidity sensor **1004** is used to set initial operation parameter(s) of the ozone source **1002** and the ozone sensor **1006** is used to adjust the initial operation parameter(s) to meet an acceptable ozone threshold or range. Such a configuration may allow the ozone source to more quickly reach a desired quantity of ozone production (e.g., steady state). The desired quantity of ozone, as detected by the ozone sensor **1006**, may be, for example, in a range of about 100 parts per million (ppm) to about 400 ppm. By way of further example, the desired quantity of ozone, as detected by the ozone sensor **1006**, may be, about (e.g., within 1%, 2%, 3%, 4%, 5%, 10%, 15%, or 20% of) 270 ppm.

[0133] As shown, the liquid reservoir access **722** further includes a liquid level sensor **1009** configured to detect a

level of liquid within the liquid reservoir **802**. When the level of liquid falls below a predetermined threshold (e.g., the liquid reservoir **802** is substantially empty or has insufficient liquid to complete a disinfection cycle) operation of the disinfection device **700** may be prevented. The liquid level sensor **1009** may include one or one or more arms **1010**. The one or more arms **1010** include a raised region **1012** configured to engage (e.g., contact) a portion of disinfection device **700** (e.g., one or more electrical contacts to form an electrical connection with the controller **1008**).

[0134] As also shown, the liquid reservoir access **722** may also include at least one wheel **1014** coupled to the carriage **808**. The at least one wheel **1014** is configured to engage a corresponding track **1016** extending from the base **820**. Engagement between the at least one wheel **1014** and the track **1016** may encourage the movement of the carriage along the coupling axis **818**.

[0135] FIG. **11** shows a perspective view of the carriage **808** urging the liquid reservoir **802** into engagement with the humidification chamber **804**. As shown, the liquid reservoir **802** includes an openable refill lid **1100** configured to be transitioned between an open position and a closed position. In some instances, the openable refill lid **1100** may be configured to be automatically opened as the movable platform **806** is transitioned from a use position to a refill position. Additionally, or alternatively, the openable refill lid **1100** may be opened by a user. As also shown, the movable platform **806** includes tracks **1102** including slots **1104** configured to slidably receive a guiding protrusion **1106** (FIG. **8**) of the disinfection device **700**.

[0136] FIGS. **11A-11E** show another example of a liquid reservoir access **1150**, which is an example of the liquid reservoir access **722**. The liquid reservoir access **1150** includes a carriage **1152**, a movable platform **1154**, and a plurality of linkages **1156** pivotally coupled to the carriage **1152** and the platform **1154**. The carriage **1152** is configured to receive a liquid reservoir **1158** which is an example of the liquid reservoir **802**. As shown, the liquid reservoir **1158** includes a keying protrusion **1160** configured to cooperate with a keying receptacle **1162** of the carriage **1152**. For example, the keying protrusion **1160** and the keying receptacle **1162** may be configured to cooperate to control an orientation of the liquid reservoir **1158** within the carriage **1152** (e.g., the liquid reservoir **1158** may be received within the carriage **1152** according to a single orientation). In this example, the keying protrusion **1160** and the keying receptacle **1162** may have corresponding shapes that are non-symmetrical along at least one axis.

[0137] The carriage **1152** further includes wheels **1164** configured to engage with corresponding tracks **1166**. As shown, as the carriage **1152** and the platform **1154** are moved along an insertion axis **1168**, the wheels **1164** cooperate with the tracks **1166** to move the carriage **1152** to move towards or away from the platform **1154** and along a coupling axis **1170**. As shown, as the carriage **1152** moves away from the platform **1154** a linkage angle ϵ , which opens in a direction of the tracks **1166**, increases to a fully inserted position (e.g., as shown in FIG. **11E**). When at the fully inserted position the linkage angle ϵ may be greater than or equal to 90° . Such a configuration may encourage the carriage **1152** and the platform **1154** to remain in the fully inserted position. When the linkage angle ϵ is less than 90° , the carriage **1152** and platform **1154** may be encouraged to

move along the insertion axis **1168** in a direction away from the fully inserted position (e.g., as a result of the weight of the liquid reservoir **1158**).

[0138] As shown, the plurality of linkages **1156** include a forward linkage **1156a** and a plurality of rearward linkages **1156b**. The forward linkage **1156a** may include a plurality of linkage arms **1172** connected together by a connection arm **1174**. The connection arm **1174** may increase a twisting resistance of the forward linkage **1156a**, which may improve alignment of the liquid reservoir **1158** relative to the humidification chamber **804**.

[0139] FIG. 11F shows a bottom view of the platform **1154**. As shown, the platform **1154** includes one or more stiffening structures **1176** extending along the platform **1154**. The one or more stiffening structures **1176** may increase torsional and/or linear stiffness.

[0140] FIG. 12 is a cross-sectional perspective view of a portion of the disinfection device **700** taken along the line XII-XII of FIG. 7 that illustrates the humidifier assembly **706**. As shown, the humidifier assembly **706** includes the liquid reservoir **802**, the humidification chamber **804**, and a humidity generator **1204**. The humidity generator **1204** is configured to draw liquid from the liquid reservoir **802** and disperse the liquid (e.g., as atomized droplets) into the humidification chamber **804**.

[0141] The humidity generator **1204** may include, for example, a wick assembly **1206** configured to cooperate with an atomizer **1208** to transfer liquid from the liquid reservoir **802** to the humidification chamber **804**. At least a portion of any ozone entering the humidification chamber **804** may become entrained within at least a portion of the liquid atomized by the atomizer **1208**. The wick assembly **1206** is configured to draw liquid from the liquid reservoir **802** through capillary action and deliver the liquid to the atomizer **1208**. The atomizer **1208** is configured to atomize delivered liquid into droplets that are dispersed within the humidification chamber **804**. The atomizer **1208** may be a piezo-electric (or ultrasonic) atomizer that is configured to generate droplets of liquid (e.g., water) in a given quantity and size. In some instances, the atomizer **1208** may be configured as a reverse piezo-electric atomizer (e.g., a stainless steel plate of the atomizer **1208** is more exposed to the humidification chamber **804** than a piezo-electric ceramic of the atomizer **1208**). For example, the atomizer **1208** may be piezo-electric atomizer having about (e.g., within 1%, 2%, 3%, 4%, 5%, 10%, or 15% of) 1,020 holes having a hole size of about (e.g., within 1%, 2%, 3%, 4%, 5%, 10%, or 15% of) 3.6 microns. By way of further example, the atomizer **1208** may be piezo-electric atomizer having about 1,320 holes having a hole size of about 4.3 microns. By way of further example, the atomizer **1208** may be piezo-electric atomizer having about 2,640 holes having a hole size of about 2.5 microns. The hole sizes of the piezo-electric atomizer may influence a droplet size that is formed and the quantity of holes may influence the quantity of droplets dispersed within the humidification chamber **804**. In some instances, the hole sizes may be configured such that the smallest droplet has a size that does not result in Brownian motion. In some instances, the droplet size, of the most prevalent droplets, may be in a range of, for example, about 1.0 microns to about 3 microns. By way of further example, the droplet size, of the most prevalent droplets, may be in range of about 1.0 microns to about 5 microns. By way of still further example, the droplet size, of

the most prevalent droplets, may be in range of about 1.5 microns to about 4.5 microns. By way of still further example, the droplet size, of the most prevalent droplets, may be about 2.5 microns. The piezo-electric atomizer may be an ultrasonic atomizer operated at about 110 kilohertz (kHz). In some instances, the piezo-electric atomizer **1208** may have a hole size of about 2.5 microns. In some instances, the disinfection device **700** may be configured such that the atomizer **1208** is caused to operate according to a self-clean cycle. In some instances, the controller **1008** may be configured to determine a status (e.g., functioning, damaged, and/or any other status) of the piezo-electric atomizer **1208** (e.g., by measuring a voltage across the piezo-electric atomizer **1208**).

[0142] The atomizer **1208** may further include a dimple **1210** configured to engage at least a portion of the wick assembly **1206**. The dimple **1210** can generally be described as a protruding dimple or a recessed dimple (see, e.g., FIG. 12E showing an atomizer **1280** having a recessed dimple **1282** engaging a wick **1284**), wherein a protruding dimple protrudes into the humidification chamber **804** (as shown). With reference to FIG. 12E, the wick **1284** may have a substantially planar dimple engaging surface **1286** configured to cooperate with (e.g., engaging) the recessed dimple **1282**. A more consistent engagement between the substantially planar dimple engaging surface **1286** may be obtained when using the recessed dimple **1282** when compared to the protruding dimple **1210**, which may improve a longevity of the wick assembly **1206**. For example, maintaining a substantially continuous force between the at least a portion of the wick **1284** (e.g., the substantially planar dimple engaging surface **1286**) and the atomizer **1280** in a range of about 0.02 Newtons (N) to about 0.5 N may result in improved performance. By way of further example, a substantially continuous force between the at least a portion of the wick **1284** (e.g., the substantially planar dimple engaging surface **1286**) and the atomizer **1280** in a range of about 0.05 N to about 0.25 N may result in improved performance.

[0143] The atomizer **1208** may be configured to operate for a period of at least about six months to at least about five years without failure within the environment of the disinfection device **700**. One example of the atomizer **1208** may include a protective coating that is applied to one or more surfaces of the atomizer **1208**. The protective coating may have a coating thickness in a range of, for example, about 2 microns to about 50 microns. By way of further example, the protective coating may have a coating thickness in a range of, for example, about 2 microns to about 12 microns. The coating thickness may be determined based, at least in part, on the properties of the coating and the ability of the coating to endure the high-frequency movement of the piezo-electric atomizer **1208** (e.g., the mechanical durability of the coating). As a thickness of the protective coating increases, performance of the atomizer **1208** may be begin to degrade (e.g., as a result of a thicker coating resisting vibration of the atomizer **1208**). An example protective coating may include a Parylene coating (e.g., Parylene N, Parylene C, Parylene D, or Parylene HT®). FIG. 12A shows a cross-sectional schematic example of a coated atomizer **1209** (which is an example of the atomizer **1208**) having a protective coating **1211** with a coating thickness **1213** that is applied to one or more surfaces **1207** (e.g., a humidification chamber facing surface) of the coated atomizer **1209**. Prior to application of the protective coating **1211**, the one or more

surfaces **1207** may be prepared (e.g., cleaned) to encourage a consistent adhesion of the protective coating **1211**.

[0144] Additionally, or alternatively, at least a portion of the atomizer **1208** may be made of one or more durable materials. Examples of durable materials may include stainless steel, silver, tungsten, lead zirconate titanate, silicone, and/or a ceramic material. A durable material and/or a protective coating may also be mechanically durable (e.g., in order to operate for a period of at least about six months to at least about five years without failure due to movement of the material when atomizing a liquid). The atomizer **1208** may also include a hydrophobic coating (e.g., the protective coating may be hydrophobic) and/or be at least partially made of a hydrophobic material.

[0145] In some instances, when the atomizer **1208** includes a protective coating, the surfaces of the atomizer **1208** to which the protective coating is applied may be pretreated in order to promote adhesion of the protective coating to the atomizer **1208**. For an atomizer **1208** having a stainless surface to which a protective coating (e.g., Parylene N) is to be applied, an adhesion promoter may be used. Additionally, or alternatively, polymer adhesion, conformal coatings, optical adhesion, sealers, and/or primers may be used to promote adhesion. In some instances, the atomizer **1208** may be configured to be user replaceable (e.g., as an atomizer module). Such a configuration may allow a user to replace the atomizer **1208** if the atomizer **1208** has experienced substantial degradation or mechanical failure.

[0146] The humidification chamber **804** and the atomizer **1208** may be configured to cooperate to generate a desired mix of atomized fluid and ozonated air to form humidified ozonated air. The resulting humidified ozonated air may be configured to have one or more disinfection properties that are effective against one or more pathogens (e.g., viruses, gram-positive bacteria, gram-negative bacteria, and/or any other pathogen).

[0147] The humidification chamber **804** includes a includes a bottom wall **1212**, a top wall **1214**, and one or more chamber sidewalls **1216** extending between the top and bottom wall **1212** and **1214**. The bottom wall **1212** may include a generator opening **1215** through which at least a portion of the humidity generator **1204** extends. For example, the atomizer **1208** may be coupled to the bottom wall **1212** on a chamber facing side **1217** of the bottom wall **1212** and the wick assembly **1206** may extend through the generator opening **1215** such that at least a portion of the wick assembly **1206** is on each side of the generator opening **1215**. As such, the wick assembly **1206** may be generally described as fluidly coupling the liquid reservoir **802** with the humidification chamber **804**.

[0148] The atomizer **1208** may be coupled to the bottom wall **1212** such that an emission axis **1218** of the atomizer **1208** extends in a direction of the top wall **1214**. In other words, the atomizer **1208** faces the top wall **1214** such that atomized liquid is directed toward the top wall **1214**. The top wall **1214** may be configured to define a surface that extends transverse to the bottom wall **1212**. In other words, at least a portion of the top wall **1214** may be sloped such that at least a portion of the top wall **1214** does not extend parallel to the bottom wall **1212**. Such a configuration may encourage condensation (e.g., as a result of the atomized liquid contacting the top wall **1214**) to flow along the top wall **1214** and down the one or more chamber sidewalls **1216**. Encour-

aging condensation to flow towards the one or more chamber sidewalls **1216** may discourage liquid from dripping onto the atomizer **1208**.

[0149] A slope angle θ (see, also, FIG. **13**) defined between the top wall **1214** and a plane parallel to the bottom wall **1212** (e.g., a horizontal plane) may be, for example, in a range of 30° to 60° . By way of further example, the slope angle θ may be in a range of 20° to 70° . By way of still further example, the slope angle θ may be about 45° .

[0150] The humidification chamber **804** may further include a humidification chamber ozone inlet **1220** that defines an injection axis **1222** that extends from the humidification chamber ozone inlet **1220** and across the humidification chamber **804**. The injection axis **1222** extends transverse to (e.g., perpendicular to) the emission axis **1218**. For example, the injection axis **1222** and the emission axis **1218** may intersect at about 90° .

[0151] The humidification chamber ozone inlet **1220** may be spaced apart from the atomizer **1208** by a vertical separation distance **1224** (e.g., as measured from a central point of the humidification chamber ozone inlet **1220**) and a horizontal separation distance **1226** (e.g., as measured from a central point of the atomizer **1208**). Adjusting the vertical and horizontal separation distances **1224** and **1226** such that the atomizer **1208** is proximate to the humidification chamber ozone inlet **1220** may encourage a uniform mixing of ozonated air with atomized liquid, which may improve disinfection performance of the resulting humidified ozonated air. The vertical separation distance **1224** may be, for example, in a range of 9 mm to 19 mm and the horizontal separation distance **1226** may be, for example, in a range of 20 mm to 80 mm. By way of further example, the vertical separation distance **1224** may be about 16 mm and the horizontal separation distance may be about 30 mm. In some instances, a larger humidification chamber **804**, a larger atomizer **1208**, and/or two or more atomizers **1208** may be used.

[0152] The humidification chamber ozone inlet **1220** may include or be coupled to the inlet connector **824**. As shown, the inlet connector **824** may be configured to change a direction of the flow path **822** (FIG. **8**). For example, the inlet connector **824** includes a connector inlet **1228** having a connector inlet axis **1230** and a connector outlet **1232** having a connector outlet axis **1234**, wherein the connector inlet axis **1230** extends transverse to (e.g., perpendicular to) the connector outlet axis **1234**. In some instances, the connector outlet axis **1234** may be colinear with the injection axis **1222**. The connector inlet axis **1230** may form a connector angle β with the connector outlet axis **1234**. The connector angle β may be configured such that the inlet connector **824** introduces turbulence into air flowing there-through (turbulence may encourage the mixing of ozone with atomized liquid). The connector angle β may be, for example, in a range of 50° to 120° . By way of further example the connector angle β may be about 90° . FIG. **22** shows one example of a flow diagram of the inlet connector **824**. FIG. **23** shows one example of a flow diagram of a linear inlet connector.

[0153] In some instances, the turbulent flow of air exiting the inlet connector **824** may encourage any condensation of atomized liquid to form within the humidification chamber **804** (e.g., as opposed to within the connected CPAP hose **750**). For example, the turbulent flow (e.g., the resultant

swirling) may encourage any condensation to occur on the top wall **1214** and/or the one or more chamber sidewalls **1216**.

[0154] FIG. 12B shows an example of an inlet insert **1250** configured to be coupled to (or formed in) the inlet connector **824**. FIG. 12C shows a cross-sectional view of the inlet insert **1250**. FIG. 12D an example of the inlet connector **824** that includes the inlet insert **1250**. FIG. 24 shows one example of a flow diagram of the inlet insert **1250**.

[0155] The inlet insert **1250** is configured to encourage generation of a turbulent flow and/or a fan-shaped dispersion pattern **1251** of air (which may include ozone) exiting therefrom. The fan-shaped dispersion pattern **1251** may have a pattern width **1253** that generally corresponds (e.g., about equal to) an emission pattern width **1255** of an atomizer emission **1257** from the atomizer **1208** (e.g., at a point of intersection between the dispersion pattern **1251** and the atomizer emission **1257**). Such a configuration may encourage mixing of the air with the generated humidity. The atomizer emission **1257** of the atomizer **1208** may be generally conical in shape. In these instances, the pattern width **1253** of the dispersion pattern **1251** may generally correspond to a diameter of the atomizer emission **1257** at a point of intersection between the dispersion pattern **1251** and the atomizer emission **1257**. In some instances, the inlet connector **824** may be angled such that the injection axis **1222** from the inlet connector **824** extends in a direction away from the atomizer **1208**. Such a configuration may increase an interaction area between the dispersion pattern **1251** and the atomizer emission **1257**. Additionally, or alternatively, one or more dimensions of the humidification chamber **804** (FIG. 8) may be increased to increase the interaction area between the dispersion pattern **1251** and the atomizer emission **1257**.

[0156] As shown, the inlet insert **1250** defines an insert passageway **1252** that extends from an insert inlet **1254** to an insert outlet **1256**. The insert passageway **1252** defines a passageway central axis **1258** that passes through both the insert inlet **1254** and the insert outlet **1256**. The insert passageway **1252** has a passageway width **1260** that increases from the insert inlet **1254** to a center portion **1262** of the insert passageway **1252** and decreases from the center portion **1262** to the insert outlet **1256**. The passageway width **1260** may be the same at the insert inlet **1254** and at the insert outlet **1256**. In this example, the insert passageway **1252** may have a shape that corresponds to the insert inlet **1254** that is revolved 180° about the passageway central axis **1258** while being moved linearly (e.g., for a distance of about 10 mm) along the passageway central axis **1258**. In other words, the insert passageway **1252** may have a twisted shape that corresponds to the shape of the insert inlet **1254**.

[0157] The insert inlet **1254** and the insert outlet **1256** may have the same dimensions. The insert inlet **1254** may have an insert width **1264** and an insert length **1266**, the insert length **1266** being greater than the insert width **1264**. Such a configuration may result in the generation of the fan-shaped dispersion pattern **1251**. The area of the insert inlet **1254** may be about the same as an area of an inlet connector passageway of the inlet connector **824**.

[0158] FIG. 13 shows a cross-sectional view of a portion of the humidifier assembly **706**. As shown, the humidification chamber ozone inlet **1220** faces in a different direction than a humidification chamber outlet **1300**. For example, the humidification chamber ozone inlet **1220** may be disposed in

a first sidewall **1302** and the humidification chamber outlet **1300** may be disposed in a second sidewall **1304**. The first and second sidewalls **1302** and **1304** may be immediately adjacent sidewalls. As also shown, the humidification chamber ozone inlet **1220** and the humidification chamber outlet **1300** may be vertically spaced apart by an inlet/outlet separation distance **1306** (e.g., as measured between a central point of the humidification chamber ozone inlet **1220** and the humidification chamber outlet **1300**). The inlet/outlet separation distance **1306** may be, for example, in a range of 30 mm to 100 mm. By way of further example, the inlet/outlet separation distance **1306** may be about 100 mm. As also shown, the top wall **1214** forms a slope angle λ with the second sidewall **1304** in a range of, for example, 30° to 60°. By way of further example, the slope angle λ may be in a range of 10° to 80°. By way of still further example, the slope angle λ may be about 45°.

[0159] As also shown, a cross-sectional width **1308** of the humidification chamber **804** may decrease in a direction of the humidification chamber outlet **1300**. For example, and as shown, at least one sidewall (e.g., the second sidewall **1304** that includes the humidification chamber outlet **1300**) may include a non-linear cross-section (e.g., include one or more arcuate or angular portions). In some instances, the humidification chamber **804** may have a chamber volume in a range of, for example 112 milliliters (mL). By way of further example, the chamber volume may be about 180 mL.

[0160] FIG. 14 is a magnified cross-sectional view generally corresponding to region XIV-XIV of FIG. 13 and FIG. 15 is an exploded view showing a bottom view of the humidification chamber **804** and a top view of the liquid reservoir **802**. As shown, the atomizer **1208** is coupled to the bottom wall **1212** of the humidification chamber **804** using a retainer **1400**. The retainer **1400** may be formed of, for example, silicone, plastic, metal, and/or any other suitable material. The retainer **1400** includes an atomizer opening **1402** (see, also, FIG. 14B) through which the atomizer **1208** may emit atomized water (e.g., distilled or demineralized water) and one or more retainer walls **1404** configured to engage (e.g., couple to) the bottom wall **1212** and/or stand-offs **1405** extending from the bottom wall **1212**. The one or more retainer walls **1404** may include one or more retainer wall drain openings **1406** (see, also, FIG. 14A showing a perspective view of the retainer **1400**) through which a drain path **1408** extends. From the one or more retainer wall drain openings **1406**, the drain path **1408** extends through one or more bottom wall drain openings **1410** defined in the bottom wall **1212** of the humidification chamber **804**. From the one or more bottom wall drain openings **1410**, the drain path **1408** extends through one or more reservoir drain openings **1412** defined within the liquid reservoir **802**. From the one or more reservoir drain openings **1412**, the drain path **1408** may extend along the wick assembly **1206** and into the liquid reservoir **802**. In operation, atomized water (which may contain ozone) that condenses on the top wall **1214** (instead of exiting the humidification chamber **804** via the humidification chamber outlet **1300**) and flows down the one or more chamber sidewalls **1216** may flow along the drain path **1408** and return to the liquid reservoir **802**. In some instances, the bottom wall **1212** may include a sloped region **1414** that encourages water to flow towards the retainer wall drain openings **1406**.

[0161] Liquid flowing along the drain path 1408 may have ozone entrained therein. The entrained ozone may disinfect the humidification chamber 804, the liquid reservoir 802, and/or the wick assembly 1206. Such a configuration may extend the time between cleanings of the humidifier assembly 706.

[0162] As also shown, the bottom wall 1212 includes a reservoir receptacle 1416 configured to receive at least a portion of a reservoir protrusion 1418. A portion of wick assembly 1206 extends within the reservoir protrusion 1418 such that a portion of wick assembly 1206 contacts the atomizer 1208 when the reservoir protrusion 1418 is received within the reservoir receptacle 1416.

[0163] The reservoir protrusion 1418 may include the one or more reservoir drain openings 1412 and one or more wick supports 1420 configured to support (e.g., contact) at least a portion of the wick assembly 1206 (e.g., a wick 1422) extending therein. The one or more wick supports 1420 may extend within a protrusion cavity 1419 defined by the reservoir protrusion 1418, wherein a portion of the wick assembly 1206 extends within the protrusion cavity 1419, and the one or more reservoir drain openings 1412 may be defined within the protrusion cavity 1419. When the wick 1422 is wetted, the structural rigidity of the wick 1422 may be reduced and the one or more wick supports 1420 may support the wetted wick 1422 in an upright position. In some instances, when the wick 1422 is dry it may have insufficient structural rigidity to remain in an upright position and be supported by the one or more wick supports 1420 in the upright position. The wick 1422 may be a soft or hard wick. A hard or a soft wick 1422 may each be effective provided that the structural integrity of the wick 1422 is sufficient to maintain the desired interaction between the wick 1422 and the dimple of the atomizer 1208. In some instances, for example, a substantially continuous force between the at least a portion of the wick 1422 and the atomizer 1208 may be in a range of about 0.02 N to about 0.5 N. By way of further example, a substantially continuous force between the at least a portion of the wick 1422 and the atomizer 1208 may be in a range of about 0.05 N to about 0.25 N. As shown, the drain path 1408 extends between the wick assembly 1206 and the reservoir protrusion 1418.

[0164] In some instances, a bottom region 1424 of the humidification chamber 804, which includes the bottom wall 1212, may be removable from a top region 1428 of the humidification chamber 804 (e.g., to facilitate cleaning of the humidification chamber 804 and/or replacement of the atomizer 1208). In these instances, a seal 1426 may extend between the bottom region 1424 and the top region 1428 of the humidification chamber 804.

[0165] FIG. 16 is a perspective view of the wick assembly 1206. As shown, the wick assembly 1206 includes the wick 1422 extending within a cartridge 1600 having a cartridge base 1602 slidably coupled to a cartridge body 1604. A biasing mechanism 1606 (e.g., a spring) extends within the cartridge body 1604 and urges the cartridge base 1602 into engagement (e.g., contact) with the wick 1422. Urging the cartridge base 1602 into engagement with the wick 1422 is configured to urge the wick 1422 into engagement with the atomizer 1208 (FIG. 12), encouraging a more efficient atomization of liquid. In some instances, the wick 1422 may have a shape that generally corresponds to a surface of the atomizer 1208 configured to be contacted by the wick 1422

such that a consistent contact between the atomizer 1208 and the wick 1422 may be maintained.

[0166] The cartridge base 1602 and/or the cartridge body 1604 include one or more liquid passthroughs 1608 and/or 1610. The liquid passthroughs 1608 and 1610 are configured to allow liquid to pass from the liquid reservoir 802 (FIG. 8) to the wick 1422.

[0167] FIG. 17 is a cross-sectional view of the wick assembly 1206 taken along the line XVII-XVII of FIG. 16. As shown, the cartridge base 1602 includes a base flange 1700 and one or more base wick supports 1702 configured to support the wick 1422. The base flange 1700 may be coupled to or formed from the cartridge base 1602. The biasing mechanism 1606 is configured to engage (e.g., contact) the base flange 1700. For example, the biasing mechanism 1606 may be a compression spring configured to urge the cartridge base 1602 in a direction of a cartridge open end 1704, wherein the wick 1422 extends through the cartridge open end 1704. In other words, the biasing mechanism 1606 cooperates with the cartridge base 1602 to urge the wick 1422 in a direction that extends outwardly from the cartridge open end 1704 (e.g., and into contact with the atomizer 1208 of FIG. 12). The cartridge body 1604 may include a locking receptacle 1706 that is configured to removably couple the wick assembly 1206 with the liquid reservoir 802 (FIG. 8). For example, the locking receptacle 1706 may be configured to engage a corresponding protrusion and/or seal of a wick coupler 1708 (FIG. 18) of the liquid reservoir 802. When coupled to the liquid reservoir 802 movement of the cartridge base 1602 within the cartridge body 1604 in a direction of the cartridge open end 1704 is restricted by the wick coupler 1708 of the liquid reservoir 802 engaging the base flange 1700 (see, e.g., FIG. 18).

[0168] The one or more base wick supports 1702 are configured to engage at least a portion of the wick 1422. When the wick 1422 becomes saturated with liquid, the structural rigidity of the wick 1422 may become compromised. The one or more base wick supports 1702 may be configured to retain the wick 1422 in an upright orientation when saturated with liquid (and/or if the wick 1422 is insufficiently rigid to maintain an upright orientation when dry). Such a configuration may encourage a more consistent contact between the atomizer 1208 and the wick 1422.

[0169] FIG. 19 is a cross-sectional view of the disinfection device 700 taken along the line XIX-XIX of FIG. 7. As shown, the disinfection chamber 702 includes a disinfection body 1900 defining a cavity 1902 that includes a disinfection receptacle 1904. The disinfection chamber lid 714 is pivotally coupled to the disinfection body 1900 and includes a disinfection chamber seal 1906 configured to sealingly engage with the disinfection receptacle 1904 when the disinfection chamber lid 714 is in a closed (e.g., operating) position. The disinfection chamber seal 1906 is configured to prevent and/or mitigate the leakage of ozone from the disinfection receptacle 1904. For example, the disinfection chamber seal 1906, the CPAP hose passthrough 708, and the ozone reduction filter 902 may be configured such that a total leakage of ozone from the disinfection device 700 is less than 0.05 ppm as measured 12 centimeters (cm) from the disinfection device 700. In some instances, the upstream fan 904 and/or pump 1000 (FIG. 10) may be operated to maintain ozone within the CPAP hose 750, the CPAP mask 754, and/or the disinfection receptacle 1904 within a pre-

determined range (e.g., by generating ozone pulses using the ozone generation assembly 704) and to maintain humidity within the CPAP hose 750, the CPAP mask 754, and/or the disinfection receptacle 1904 within a predetermined range (e.g., without generating humidity using the humidifier assembly 706). In other words, the upstream fan 904 and/or the pump 1000 may be operated such that a desired environment is maintained within the CPAP hose 750, the CPAP mask 754, and/or the disinfection receptacle 1904.

[0170] In some instances, when operating the disinfection device 700, a quantity of ozone within the CPAP hose 750, the CPAP mask 754, and/or the disinfection receptacle 1904 may be controlled. For example, the quantity of ozone within the disinfection receptacle 1904 may be controlled to be in a range of about 200 ppm to about 300 ppm. By way of further example, the quantity of ozone may be controlled to be in a range of about 100 ppm to about 400 ppm. By way of still further example, the quantity of ozone may be controlled to be in a range of about 120 ppm to about 200 ppm. By way of still further example, the quantity of ozone may be controlled to be about 285 ppm. By way of still further example, the quantity of ozone may be controlled to be about 160 ppm. In some instances, when operating the disinfection device 700, a humidity within the disinfection receptacle 1904 may be controlled. For example, the relative humidity within the disinfection receptacle 1904 may be controlled to be in a range of 50% to 99.9%. By way of further example, the relative humidity within the disinfection receptacle 1904 may be controlled to be in a range of 70% to 90%. By way of still further example, the relative humidity within the disinfection receptacle 1904 may be about 70%.

[0171] In addition, or in the alternative, to controlling the ozone generation assembly 704 to control a quantity of ozone within the CPAP hose 750, the CPAP mask 754, and/or the disinfection receptacle 1904 and/or controlling the humidifier assembly 706 to control a relative humidity within the CPAP hose 750, the CPAP mask 754, and/or the disinfection receptacle 1904, a speed of the upstream fan 904 and/or pump 1000 may be controlled. By controlling the speed of one or more of the upstream fan 904 and/or pump 1000 the rate at which ozone and/or humidity is delivered to (e.g., via the CPAP hose 750 and/or the CPAP mask 752) and removed from the disinfection receptacle 1904 may be controlled, which controls the quantity of ozone and/or the humidity within the disinfection receptacle 1904 and objects being disinfected (e.g., the CPAP hose 750 and/or the CPAP mask 752). For example, the pump 1000 may be operated to generate a flow rate in a range of about 1.1 standard liter per minute (SLPM) to 1.6 SLPM and the upstream fan 904 (FIG. 10) may be operated to generate a flow rate in a range of about 0.9 SLPM to about 1.6 SLPM. By way of further example, the upstream fan 904 may be operated to generate a flow rate of 1.3 SLPM and the pump 1000 of 1.3 SLPM. By way of still further example, the upstream fan 904 may be operated to generate a flow rate of 1.2 SLPM and the pump 1000 of 1.3 SLPM. In some instances, the disinfection device 700 may be configured such that the disinfection receptacle 1904 is under vacuum, under pressure, or alternates between being under vacuum and under pressure. Having, the upstream fan 904 operate at a lower flow rate than the pump 1000 may improve disinfection performance (e.g., encourage an at least 4-Log_{10} , at least 5-Log_{10} , or at least 6-Log_{10} disinfection performance). In some instances,

the flow rate may be configured such that, for a given quantity of entrained ozone, a desired level of disinfection (e.g., at least 4-Log_{10}) is obtained for one or more pathogens (e.g., one or more viruses, one or more gram-positive bacteria, one or more gram-negative bacteria, and/or any other pathogens).

[0172] FIG. 19A shows a schematic example of an operational timeline 1920 of the disinfection device 700 that corresponds to one example operation profile (e.g., a disinfection cycle) and is an example of the operational timeline 400 of FIG. 4. When the disinfection device 700 is operated according to the operational timeline 1920, an object (e.g., the CPAP mask 754 or CPAP hose 750) being disinfected by the disinfection device 700 may experience at least a 4-Log_{10} reduction in at least one pathogen. FIG. 19B shows one example of an efficacy outcome using one example of the disinfection device 700 that is implementing one example of the operational timeline 1920, wherein the efficacy achieved is greater than 4-Log_{10} for the listed pathogens.

[0173] As shown, a first period 1922 extends between time T_0 and time T_1 , a second period 1924 extends between time T_1 and time T_2 , and a third period 1926 extends between time T_2 and time T_3 , wherein time T_3 is greater than time T_2 , time T_2 is greater than time T_1 , and time T_1 is greater than time T_0 . A cycle time extending from T_0 to T_3 may be, for example, in a range of about (e.g., within 1%, 2%, 3%, 4%, or 5% of) 60 minutes to about 120 minutes. By way of further example, the cycle time may be about 90 minutes. The first period 1922 may be, for example, in a range of about 4 to about 6 minutes. By way of further example, the first period 1922 may be about 6 minutes. The second period 1924 may be, for example, in a range of about 60 to about 80 minutes. By way of further example, the second period 1924 may be about 72 minutes. The third period 1926 may be, for example, in a range of about 8 minutes to about 16 minutes. By way of further example, the third period 1926 may be about 12 minutes. The pump 1000 may be configured to operate at a pump flow rate and the upstream fan 904 may be configured to operate at a fan flow rate, the pump flow rate being different from the fan flow rate during at least one of the first period 1922, the second period 1924, and/or the third period 1926.

[0174] During the first period 1922, the ozone generation assembly 704 (FIG. 7) may be caused to generate ozone in a range of, for example, about (e.g., within 1%, 2%, 3%, 4%, 5%, or 10% of) 220 ppm to about 300 ppm, which may, in some instances, correspond to about 80 ppm to about 150 ppm of ozone within the disinfection chamber 702 (FIG. 7). By way of further example, during the first period 1922, the ozone generation assembly 704 may be caused to generate about 270 ppm of ozone. Depending on the travel time for ozone to travel from the ozone generation assembly 704 to the disinfection chamber 702, the ozone generation assembly 704 may be configured to generate a sufficient quantity of ozone such that the ozone within the CPAP hose 750, the CPAP mask 754, and/or the disinfection chamber 702 is in a range of about 80 ppm to about 150 ppm. Alternatively, in some instances, the ozone generation assembly 704 may be disabled during the first period 1922.

[0175] During the first period 1922, the humidifier assembly 706 (FIG. 7) may be caused to generate humidity that passes through the CPAP hose 750 and CPAP mask 754 to enter the disinfection chamber 702. For example, the

humidifier assembly **706** may be caused to generate humidity until a relative humidity (e.g., as measured or estimated) within the CPAP hose **750**, the CPAP mask **754**, and/or the disinfection chamber **702** is achieved. By way of still further example, the humidifier assembly **706** may be configured to generate humidity until a relative humidity within a range of about 74% to about 90% within the CPAP hose **750**, the CPAP mask **754**, and/or the disinfection chamber **702** is achieved. By way of still further example, the humidifier assembly **706** may be configured to generate humidity until a relative humidity is within a range of about 55% to about 99% within the CPAP hose **750**, the CPAP mask **754**, and/or the disinfection chamber **702** is achieved. Alternatively, in some instances, the humidifier assembly **706** may be disabled during the first period **1922**.

[0176] In some instances, during the first period **1922**, only one of the ozone generation assembly **704** or the humidifier assembly **706** is enabled. For example, only the humidifier assembly **706** may be enabled during the first period **1922** such that a relative humidity within the disinfection chamber **702** reaches a desired amount prior to the generation of ozone. In some instances, during the first period **1922**, the ozone generation assembly **704** and the humidifier assembly **706** may each be enabled. For example, the ozone generation assembly **704** and the humidifier assembly **706** may each be enabled during the first period **1922** without being enabled at the same time. By way of further example, the ozone generation assembly **704** and the humidifier assembly **706** may be both enabled at the same time during the first period **1922**. The ozone generation assembly **704** and/or the humidifier assembly **706** may be enabled for the entire first period **1922**.

[0177] During the first period **1922**, the pump **1000** (FIG. 10) may be operated according to an initialization pump flow rate and the upstream fan **904** may be operated according to an initialization fan flow rate. The initialization pump flow rate may be different from (e.g., greater than) the initialization fan flow rate. For example, a ratio of initialization fan flow rate to initialization pump flow rate (i.e., initialization fan flow rate divided by initialization pump flow rate) may be about 0.9. In some instances, the initialization pump flow rate may be in a range of about (within 1%, 2%, 3%, 4%, or 5% of) 1 SLPM to about 1.6 SLPM. In some instances, the initialization fan flow rate may be about 0.1 to 0.15 SLPM less than the initialization pump flow rate.

[0178] During the second period **1924**, the humidifier assembly **706** may be disabled and the ozone generation assembly **704** may be enabled for at least a portion of the second period **1924**. For example, the ozone generation assembly **704** may be cycled between being enabled and disabled (e.g., according to a fixed or varying duty cycle). In this example, the ozone generation assembly **704** may generally be described as being configured to generate ozone pulses at a pulse rate (or duty cycle). When generating ozone pulses, the duty cycle of the ozone generation assembly **704** may be such that a quantity of ozone within the disinfection chamber **702** (e.g., as measured or estimated) is in a range of about 80 ppm to about 150 ppm. One example pulse rate of the ozone generation assembly **704** may result in the ozone generation assembly **704** being enabled for about 10 seconds and disabled for about 65 seconds. In some instances, the duty cycle may be adjusted based, at least in part, on, for example, an output of an ozone sensor within

the disinfection chamber **702**. During the second period **1924**, the pump **1000** and/or upstream fan **904** may, in some instances, be enabled when the ozone generation assembly **704** is generating ozone and disabled when the ozone generation assembly **704** is not generating ozone.

[0179] During the second period **1924**, the pump **1000** may be operated according to an operational pump flow rate and the upstream fan **904** may be operated according to an operational fan flow rate. The operational pump flow rate may be different from (e.g., greater than) the operational fan flow rate. For example, the operational pump flow rate may be greater than the operational fan flow rate. Such a configuration, may encourage ozone to come into contact with one or more surfaces of the CPAP mask **754** and/or the CPAP hose **750**.

[0180] For example, a ratio of operational fan flow rate to operational pump flow rate (i.e., operational fan flow rate divided by operational pump flow rate) may be about 0.9. In some instances, operational pump flow rate may be in a range of about (within 1%, 2%, 3%, 4%, or 5% of) 1 SLPM to about 1.6 SLPM. In some instances, the operational fan flow rate may be about 0.1 to 0.15 SLPM less than the operational pump flow rate.

[0181] During the third period **1926** the humidifier assembly **706** and the ozone generation assembly **704** may be disabled. During the third period **1926**, the upstream fan **904** and/or the pump **1000** may be operated according to a purge fan/pump flow rate. The purge fan flow rate may be greater than about 90% (e.g., about 100%) of a maximum flow rate capable of being generated by the upstream fan **904** and/or the purge pump flow rate may be greater than about 90% (e.g., about 100%) of a maximum flow rate capable of being generated by the pump **1000**. The purge fan and purge pump flow rates may be configured to urge a substantial quantity of any remaining ozone through the ozone reduction filter **902** (FIG. 9) prior to expiration of the third period **1926**. In some instances, the purge fan flow rate may be greater than or equal to the purge pump flow rate. In some instances, the purge pump flow rate may be, for example, in a range of about (within 1%, 2%, 3%, 4%, or 5% of) 1 SLPM to about 1.6 SLPM and the purge fan flow rate may be in a range of about 10 SLPM to about 16 SLPM.

[0182] FIG. 20 is a schematic example of a disinfection device **2000**, which is an example of the disinfection device **700** of FIG. 7. The disinfection device **2000** is configured to achieve at least a 4-Log₁₀ (e.g., at least a 6-Log₁₀) kill rate of one or more bacteria on a CPAP mask **2002** and/or on a CPAP hose **2004**, wherein the one or more bacteria may include, but are not limited to, one or more of *Escherichia coli*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Staphylococcus haemolyticus*, *Streptococcus pyogenes*, *Staphylococcus hominis*, *Klebsiella pneumoniae*, and/or *Enterobacter cloacae*. In other words, the disinfection device **2000** is configured to cause at least a 99.9999% reduction of one or more bacteria in the CPAP mask **2002** and the CPAP hose **2004**.

[0183] As shown, the disinfection device **2000** includes a disinfection chamber **2006**, an ozone generator **2008**, a humidifier **2010**, a pump **2012**, an upstream fan **2014**, and an ozone reduction filter **2016**. The CPAP hose **2004** fluidly couples the pump **2012**, the ozone generator **2008**, and the humidifier **2010** to the disinfection chamber **2006**. As shown, the ozone generator **2008** is downstream of the pump **2012**, the humidifier **2010** is downstream of the ozone

generator **2008**, the CPAP hose **2004** is downstream of the humidifier **2010**, the disinfection chamber **2006** is downstream of the CPAP hose **2004**, the ozone reduction filter **2016** is downstream of the disinfection chamber **2006**, and the upstream fan **2014** is downstream of the ozone reduction filter **2016**. In some instances, the CPAP mask **2002** may be coupled to the CPAP hose **2004** and be disposed within the disinfection chamber **2006**.

[0184] The humidifier **2010**, the pump **2012**, and the ozone generator **2008** are external to the disinfection chamber **2006**. The CPAP mask **2002** and the CPAP hose **2004** fluidly couple the humidifier **2010**, the pump **2012**, and the ozone generator **2008** to the disinfection chamber **2006**. The humidifier **2010**, the pump **2012**, and/or the ozone generator **2008** may be arranged in any configuration relative to the disinfection chamber **2006**. For example, one or more of the humidifier **2010**, the pump **2012**, and/or the ozone generator **2008** may be arranged at least partially below the disinfection chamber **2006**. By way of further example, one or more of the humidifier **2010**, the pump **2012**, and/or the ozone generator **2008** may be arranged at least partially along a side of and/or separate from the disinfection chamber **2006**. In some instances one or more of the humidifier **2010**, the pump **2012**, and/or the ozone generator **2008** may be movable relative to the disinfection chamber **2006**.

[0185] The disinfection device **2000** further includes a controller **2018** that is communicatively coupled to one or more of the ozone generator **2008**, the humidifier **2010**, the pump **2012**, and/or the upstream fan **2014**. The controller **2018** is configured to operate one or more of the ozone generator **2008**, the humidifier **2010**, the pump **2012**, and/or the upstream fan **2014** according to one or more operation profiles (e.g., a disinfection cycle) that are configured to achieve a desired environment within the disinfection chamber **2006** (e.g., to achieve at least a 4-Log_{10} or at least a 6-Log_{10} kill rate).

[0186] The ozone generator **2008** includes a relative humidity sensor **2020** (e.g., positioned at an inlet side of the ozone generator **2008**), and an ozone sensor **2022** (e.g., positioned at an outlet side of the ozone generator **2008**). For example, the ozone sensor **2022** may be positioned downstream of an ozone source of the ozone generator **2008** and upstream of the humidifier **2010**. The controller **2018** is configured to operate the ozone generator **2008** according to an operation profile based, at least in part, on a detected ozone quantity as detected by the ozone sensor **2022** and/or a detected relative humidity as detected by the relative humidity sensor **2020**. For example, the controller **2018** may operate the ozone generator **2008** to generate a quantity of ozone in a range of about 200 ppm to about 300 ppm, as detected by the ozone sensor **2022**. In order to more quickly reach a desired ozone quantity, the controller **2018** may use the detected relative humidity to determine initial operation parameter(s) of the ozone generator **2008** (e.g., the controller **2018** generates an estimated quantity of ozone that is expected to be generated by the ozone generator **2008** for given operation parameter(s) at the detected relative humidity and causes the ozone generator **2008** to be operated accordingly).

[0187] The humidifier **2010** includes a liquid reservoir **2024**, a humidification chamber **2026**, and a piezo-electric atomizer **2028** configured to atomize liquid from the liquid reservoir **2024** into the humidification chamber **2026**. The controller **2018** is configured to operate the piezo-electric

atomizer **2028** according to the operation profile in order to generate a desired relative humidity. The piezo-electric atomizer **2028** may have about (e.g., within 1%, 2%, 3%, 4%, 5%, 10%, or 15% of) 2,640 holes having a hole size of about (e.g., within 1%, 2%, 3%, 4%, 5%, 10%, or 15% of) 2.5 microns with a theoretical maximum atomized volume of water per pulse of about 22.3 pico-liters (pL). Such a configuration for the piezo-electric atomizer **2028** may result in the most prevalent value of atomized water droplet size being in a range of, for example, about 2.5 microns to about 3 microns. FIG. 20A shows one example distribution of atomized water droplet sizes produced by an example of the piezo-electric atomizer **2028**, wherein droplet sizes were measured at an inlet **2029** of the CPAP hose **2004** (labeled IN in FIG. 20A) and an outlet **2031** of the CPAP hose **2004** (e.g., at the CPAP mask **2002**) (labeled OUT in FIG. 20A). As shown, droplets having a size in a range of 2 microns to 3 microns were the most prevalent and present in substantially similar numbers at both the inlet **2029** and the outlet **2031**. Droplet sizes within a range of 2 microns and 3 microns may mitigate a risk of condensation during a disinfection cycle (e.g., within the disinfection device **2000** and/or within the CPAP mask **2002** and/or CPAP hose **2004**), which may improve disinfection performance. Additionally, or alternatively, in some instances, one or more condensation reducers (e.g., mesh screen) may be included to reduce (e.g., prevent) condensation of droplets during a disinfection cycle (e.g., within the disinfection device **2000** and/or within the CPAP mask **2002** and/or CPAP hose **2004**).

[0188] The piezo-electric atomizer **2028** may be an atomizer with a recessed dimple, wherein a distal end of a wick engaging the atomizer **2028** may be substantially planar such that consistent contact between the wick and the atomizer **2028** is encouraged. When the piezo-electric atomizer **2028** is an atomizer with a protruding dimple, a distal end of a wick engaging the atomizer **2028** may have a shape that corresponds to that of a cavity created by the protruding dimple such that consistent contact between the atomizer **2028** and the wick is encouraged. The wick for an atomizer with a recessed dimple may be harder (or more rigid) than the wick for an atomizer with a protruding dimple, which may improve a longevity of the wick and/or atomizer.

[0189] Ozonated air from the ozone generator **2008** is injected into the humidification chamber **2026** in a direction transverse to (e.g., perpendicular to) an emission direction of the piezo-electric atomizer **2028**. The ozonated air may be injected such that the ozonated air has a turbulent flow (as opposed to a laminar flow) when entering the humidification chamber **2026**.

[0190] The controller **2018** operates the pump **2012** and upstream fan **2014** according to the operation profile in order to obtain a desired environment within the disinfection chamber **2006** (e.g., quantity of ozone and/or relative humidity). For example, the pump **2012** may be operated to generate a flow rate of about 1.3 SLPM and the upstream fan **2014** may be operated to generate a flow rate of about 1.2 SLPM. In some instances, the disinfection chamber **2006** may be under vacuum or under pressure.

[0191] FIG. 21 shows a schematic example of an operational timeline **2100** of the disinfection device **2000** that corresponds to one example operation profile (e.g., a disinfection cycle) and is an example of the operational timeline **400** of FIG. 4.

[0192] As shown, a first period **2102** extends between time T_0 and time T_1 , a second period **2104** extends between time T_1 and time T_2 , and a third period **2106** extends between time T_2 and time T_3 , wherein time T_3 is greater than time T_2 , time T_2 is greater than time T_1 , and time T_1 is greater than time T_0 . In other words, the second period **2104** occurs between the first period **2102** and the third period **2106**. The entire operation profile may have a duration in a range of, for example, 20 min to 120 min. By way of further example, the operation profile may have a duration of about 72 min.

[0193] The first period **2102** may generally be referred to as a period in which the ozone generator **2008** and/or the humidifier **2010** are operated to achieve a desired initial condition (e.g., a desired humidity and/or ozone range) within the disinfection chamber **2006**. The second period **2104** may generally be referred to as a period in which the ozone generator **2008** and/or the humidifier **2010** are operated to maintain a desired condition (e.g., a desired humidity and/or ozone range) within the disinfection chamber **2006**. The third period **2106** may generally be referred to as the period in which the ozone generator **2008** and/or the humidifier **2010** are operated to purge the ozone from the disinfection chamber **2006** such that any residual ozone within the disinfection chamber **2006** is removed and/or within an acceptable level. The conditions within the disinfection chamber **2006** may be measured (e.g., using one or more sensors) and/or estimated.

[0194] During the first period **2102**, the ozone generator **2008** generates ozone such that about 100 ppm to about 200 ppm of ozone is present in the disinfection chamber **2006** by the end of the first period **2102**, the humidifier **2010** humidifies the environment within the disinfection chamber **2006** to about 70% relative humidity by the end of the first period **2102**, the pump **2012** operates to generate a flow rate of about 1.3 SLPM, and the upstream fan **2014** operates to generate a flow rate of about 1.2 SLPM. The first period **2102** may have a duration in a range of, for example, about 1 minute to about 15 minutes. By way of further example, the first period **2102** may have a duration of about 6 minutes. During the first period **2102**, the ozone generator **2008** and the humidifier **2010** may operate continuously. In some instances, one or more of the ozone generator **2008** and/or the humidifier **2010** may operate intermittently during the first period **2102**. In these instances, the controller **2018** may be configured to determine whether to operate the ozone generator **2008** and/or the humidifier **2010** intermittently based, at least in part, on, for example, one or more sensed conditions within the disinfection chamber **2006**.

[0195] During the second period **2104**, the ozone generator **2008** operates intermittently (e.g., is pulsed according to a pulse rate), the humidifier **2010** is disabled, and the pump **2012** and upstream fan **2014** are operated to maintain a desired flow rate through the disinfection chamber **2006**. The ozone generator **2008** may be operated intermittently according to a pulse rate having a constant period or a non-constant period. For example, the ozone generator **2008** may be enabled for a period in a range of 3 seconds to 15 seconds and disabled for a period in a range of 30 seconds to 90 seconds, wherein the ozone generator **2008** is cycled between being enabled and disabled for the duration of the second period **2104**. By way of further example, the ozone generator **2008** may be enabled for a period of about 10 seconds and disabled for a period of about 65 seconds, wherein the ozone generator is cycled between being

enabled and disabled for the duration of the second period **2104**. The enabled period and the disabled period may be constant for the duration of the second period **2104** or at least one enabled period may have a different duration than at least one other enabled period and/or at least one disabled period may have a different duration than at least one other disabled period. Cycling of the ozone generator **2008** may be configured such that the quantity of ozone within the disinfection chamber **2006** is in a range of, for example, about 150 ppm to about 250 ppm. By way of further example, cycling of the ozone generator **2008** may be configured such that the quantity of ozone within the disinfection chamber **2006** is about 200 ppm.

[0196] During the second period, **2104** the pump **2012** operates to generate a flow rate of about 1.3 SLPM and the upstream fan **2014** operates to generate a flow rate of about 1.2 SLPM. The second period **2104** may have a duration in a range of, for example, about 15 minutes to about 120 minutes. By way of further example, the second period **2104** may have a duration of about 60 minutes. The flow rates of the pump **2012** and upstream fan **2014** may be such that the quantity of ozone and/or the relative humidity within the disinfection chamber **2006** stays within a predetermined acceptable range. One example of an acceptable range may include ozone within a range of about 100 ppm to about 285 ppm and/or relative humidity within a range of about 60% to about 99%.

[0197] During the third period **2106** the ozone generator **2008** and the humidifier **2010** are disabled and one or both of the pump **2012** and/or upstream fan **2014** are operated to urge at least a portion of the remaining ozone within the disinfection chamber **2006** through the ozone reduction filter **2016**. For example, the pump **2012** and/or upstream fan **2014** may be operated until a quantity of ozone is at or below a predetermined threshold. During the third period **2106**, the pump **2012** operates to generate a flow rate of about 1.3 SLPM and the upstream fan **2014** operates to generate a flow rate of about 1.2 SLPM. The third period **2106** may have a duration in a range of, for example, about 1 minute to about 20 minutes. By way of further example, the third period **2106** may have a duration of about 12 minutes.

[0198] The operation profile discussed in relation to FIG. 21 when carried out by the disinfection device **2000** of FIG. 20 results in the disinfection device **2000** achieving at least a 4-Log₁₀ (e.g., at least a 6-Log₁₀) kill rate. Through testing, it was determined that deviations in relative humidity (the quantity and how it is generated), quantity of ozone, and the flow rates of the pump **2012** and upstream fan **2014** from that described in relation to FIGS. 20 and 21 may result in a kill rate that was less than a 4-Log₁₀ kill rate.

[0199] FIG. 25 shows a schematic example a CPAP device **2500** having a disinfection device **2502**. The disinfection device **2502** may be integrated with the CPAP device **2500** or separate from the CPAP device **2500**. The CPAP device **2500** includes an air pump **2504**, a CPAP water reservoir **2506**, an environmental air inlet **2508** fluidly coupled to the air pump **2504**, an air outlet **2510** fluidly coupled to the air inlet **2508**, a CPAP hose **2512** fluidly coupled to the air outlet **2510**, and a CPAP mask **2514** fluidly coupled to the CPAP hose **2512**.

[0200] The CPAP device **2500** may have an operation mode and a disinfection mode. In the operation mode, the air pump **2504** urges air from the environment and through the

CPAP hose **2512**. The CPAP water reservoir **2506** is configured to add humidity to air flowing through the CPAP device **2500**.

[0201] In the disinfection mode, the CPAP device **2500** may enable the disinfection device **2502** causing humidified ozonated air to flow through at least the CPAP hose **2512** and the CPAP mask **2514**. In some instances, the humidified ozonated air may be caused to additionally flow through the CPAP water reservoir **2506**. The humidified ozonated air may be generated and caused to flow in a manner consistent with any of the examples discussed herein. As such, for the sake of brevity, a detailed discussion of the generation and flowing of humidified ozonated air in the context of FIG. **25** is omitted.

[0202] When the disinfection mode is enabled, the CPAP device **2500** may be configured to be transitioned to a closed system. For example, the environmental air inlet **2508** may be closed (e.g., sealed) to substantially prevent the escape of ozone from the environmental air inlet and the CPAP mask **2514** may be disposed within a disinfection chamber **2516**. The disinfection chamber **2516** may be coupled to the CPAP device **2500** or separate from the CPAP device **2500**. For example, the disinfection chamber **2516** may be a disposable (or reusable) flexible bag that is ozone impermeable.

[0203] In some instances, any water within the CPAP water reservoir **2506** may be emptied prior to commencing the disinfection mode. For example, the disinfection device **2502** may include a water pump **2518** configured to transfer water from the CPAP water reservoir **2506**. In this example, the water pump **2518** may transfer water to a disinfection water reservoir **2520** of the disinfection device **2502** such that the water can be used for humidifying ozonated air during the disinfection mode. In some instances, upon completion of the disinfection mode, the water in the disinfection water reservoir **2520** may be returned to the CPAP water reservoir **2506**. As ozone may have become entrained within the water when in the disinfection water reservoir **2520** at least a portion of any pathogens in the water may be killed. In some instances, the CPAP water reservoir **2506** may be used as the disinfection water reservoir **2520**.

[0204] FIG. **26** is a schematic example of a disinfection device **2600** with humidity control and a recirculation loop. The disinfection device **2600** is an example of the disinfection device **200** of FIG. **2**.

[0205] The disinfection device **2600** includes an ozone generator **2602**, a disinfection chamber **2604**, and a humidifier **2606**. As shown, a flow path **2608** extends through the ozone generator **2602** through the humidifier **2606** and into the disinfection chamber **2604**. The disinfection chamber **2604** may include one or more valves (e.g., a first valve **2610** and/or a second valve **2612**). In some instances, the one or more valves may be two-way valves, three-valves, four-way valves, and/or any other valve configuration. For example, and as shown, the first valve **2610** is a two-way valve and the second valve **2612** is a two-way valve. In this example, the first valve **2610** is configured to selectively fluidly couple the flow path **2608** to a disinfection chamber outlet **2614** (which may include an ozone reduction filter) and the second valve **2612** may selectively fluidly couple the flow path **2608** to an inlet side of the ozone generator **2602** (selectively creating a recirculation loop **2616**). A controller **2618** may be configured to selectively actuate the first and second valves **2610** and **2612** such that the recirculation loop **2616** can be selectively created during a disinfection cycle.

The controller **2618** may further control a fan **2620** such that ozonated air flows along the flow path **2608**.

[0206] FIG. **27** is a schematic example of a CPAP disinfection system **2700** having a disinfection device **2702**, a CPAP machine **2704**, a CPAP hose **2706** coupled to the CPAP machine **2704** via a disinfection adapter **2708**, and a CPAP mask **2710** coupled to the CPAP hose **2706**. The disinfection device **2702** is an example of the disinfection device **2600** of FIG. **26**.

[0207] The disinfection adapter **2708** is configured to fluidly couple the CPAP hose **2706** and CPAP mask **2710** to the disinfection device **2702** and to fluidly couple the CPAP hose **2706** and the CPAP mask **2710** to the CPAP machine **2704**. As such, a user may not have to remove the CPAP hose **2706** from the CPAP machine **2704** when disinfecting the CPAP hose **2706** and/or the CPAP mask **2710**. In some instances, the disinfection adapter **2708** may include adapter valving **2711** configured to selectively alter a flow path through the disinfection adapter **2708** (e.g., between a user mode in which the CPAP machine **2704** is fluidly coupled with the CPAP hose **2706** through the disinfection adapter **2708** and a disinfection mode in which the disinfection device **2702** is fluidly coupled with the CPAP hose **2706** through the disinfection adapter **2708**). Such a configuration may substantially prevent ozone generated by the disinfection device **2702** from entering the CPAP machine **2704**. In some instances, the CPAP machine **2704** may be operated during a disinfection cycle to substantially prevent ozone from entering the CPAP machine **2704**.

[0208] The disinfection device **2702** includes an ozone generator **2712**, a disinfection chamber **2714** configured to receive at least a portion of the CPAP hose **2706** and/or the CPAP mask **2710**, and a humidifier **2716**. As shown, a flow path **2718** extends through the ozone generator **2712** through the humidifier **2716** and into the disinfection chamber **2714** via the CPAP hose **2706** and the CPAP mask **2710**. The disinfection chamber **2714** may include one or more valves (e.g., a first valve **2720** and/or a second valve **2722**). In some instances, the one or more valves may be two-way valves, three-valves, four-way valves, and/or any other valve configuration. For example, and as shown, the first valve **2720** is a two-way valve and the second valve **2722** is a two-way valve. In this example, the first valve **2720** is configured to selectively fluidly couple the flow path **2718** to a disinfection chamber outlet **2724** (which may include an ozone reduction filter) and the second valve **2722** may selectively fluidly couple the flow path **2718** to an inlet side of the ozone generator **2712** (selectively creating a recirculation loop **2726**). A controller **2728** may be configured to selectively actuate the first and second valves **2720** and **2722** such that the recirculation loop **2726** can be selectively created during a disinfection cycle. The controller **2728** may further control a fan **2730** such that ozonated air flows along the flow path **2718**.

[0209] FIG. **28** shows a schematic example of a disinfection device **2800**, which is an example of the disinfection device **200** of FIG. **2**.

[0210] The disinfection device **2800** includes an ozone generator **2802**, a disinfection chamber **2804**, and a humidifier **2806**. The disinfection chamber **2804** is configured to receive a CPAP mask **2808** and/or a CPAP hose **2810**. For example, the disinfection chamber **2804** may include a mask mount **2812**. The mask mount **2812** may be fluidly coupled to the ozone generator **2802** and the humidifier **2806** and

configured to removably couple to the CPAP mask **2808**. As shown, the mask mount **2812** is downstream of the ozone generator **2802** and the humidifier **2806** and upstream of the CPAP hose **2810**. As such, ozonated air (e.g., humidified ozonated air) comes into contact with the CPAP mask **2808** prior to coming into contact with the CPAP hose **2810**. Such a configuration may improve disinfection at the CPAP mask **2808**.

[0211] As shown, a flow path **2814** extends through the ozone generator **2802** through the humidifier **2806** and into the disinfection chamber **2804** via the mask mount **2812**, the CPAP mask **2808**, and the CPAP hose **2810**. Once in the disinfection chamber **2804**, the ozonated air may be recirculated or pass through a disinfection chamber outlet **2816**.

[0212] FIG. 29 shows a schematic example of a disinfection device **2900**, which is an example of the disinfection device **200** of FIG. 2.

[0213] The disinfection device **2900** includes an ozone generator **2902**, a disinfection chamber **2904**, and a humidifier **2906**. The disinfection chamber **2904** is configured to receive a CPAP mask **2908**. A first hose coupling **2910** and a second hose coupling **2912** are fluidly coupled to the disinfection chamber **2904** such that a CPAP hose **2914** that is external to the disinfection chamber **2904** can be fluidly coupled to the disinfection chamber **2904**. The first and the second hose couplings **2910** and **2912** may be configured to detect whether a CPAP hose **2914** is coupled thereto such that the disinfection device **2900** is prevented from carrying out a disinfection cycle when the CPAP hose **2914** is not coupled to both the first and the second hose coupling **2910** and **2912**.

[0214] The disinfection chamber **2904** may include a mask mount **2916**. The mask mount **2916** may be fluidly coupled to the ozone generator **2902** and the humidifier **2906** and configured to removably couple to the CPAP mask **2908**. As shown, the mask mount **2916** is downstream of the ozone generator **2902** and the humidifier **2906** and upstream of the CPAP hose **2914**. As such, ozonated air (e.g., humidified ozonated air) comes into contact with the CPAP mask **2908** prior to coming into contact with the CPAP hose **2914**. Such a configuration may improve disinfection at the CPAP mask **2908**.

[0215] As shown, a flow path **2918** extends through the ozone generator **2902** through the humidifier **2906** and into the disinfection chamber **2904** via the mask mount **2916** and CPAP mask **2908**. Once in the disinfection chamber **2904**, the ozonated air may be urged to pass through the CPAP hose **2914** (e.g., using a fan **2920**). After passing through the CPAP hose **2914**, the ozonated air may be recirculated or pass through a disinfection chamber outlet **2922**.

[0216] FIG. 30 shows a schematic example of a disinfection device **3000**, which is an example of the disinfection device **200** of FIG. 2.

[0217] The disinfection device **3000** includes an ozone generator **3002**, a disinfection chamber **3004**, and a humidifier **3006**. The disinfection chamber **3004** is configured to receive a CPAP mask **3008**. A hose coupling **3010** and a hose pass through **3012** are fluidly coupled to the disinfection chamber **3004** such that at least a portion of a CPAP hose **3014** is external to the disinfection chamber **3004** and at least a portion of the CPAP hose **3014** is disposed within the disinfection chamber **3004**, while being fluidly coupled to the disinfection chamber **3004**. The hose coupling **3010** and the hose passthrough **3012** may be configured to detect

whether a CPAP hose **3014** is coupled thereto such that the disinfection device **3000** is prevented from carrying out a disinfection cycle when the CPAP hose **3014** is not coupled to both the hose coupling **3010** and the hose passthrough **3012**.

[0218] The disinfection chamber **3004** may include a mask mount **3016**. The mask mount **3016** may be fluidly coupled to the ozone generator **3002** and the humidifier **3006** and configured to removably couple to the CPAP mask **3008**. As shown, the mask mount **3016** is downstream of the ozone generator **3002** and the humidifier **3006** and upstream of the CPAP hose **3014**. As such, ozonated air (e.g., humidified ozonated air) comes into contact with the CPAP mask **3008** prior to coming into contact with the CPAP hose **3014**. Such a configuration may improve disinfection at the CPAP mask **3008**.

[0219] As shown, a flow path **3018** extends through the ozone generator **3002** through the humidifier **3006** and into the disinfection chamber **3004** via the mask mount **3016**, the CPAP mask **3008**, and the CPAP hose **3014**. After passing through the CPAP hose **3014**, the ozonated air may be recirculated or pass through a disinfection chamber outlet **3020**.

[0220] FIG. 31 shows a schematic example of a disinfection device **3100**, which is an example of the disinfection device **200** of FIG. 2.

[0221] The disinfection device **3100** includes an ozone generator **3102**, a disinfection chamber **3104**, and a humidifier **3106**. The humidifier **3106** includes a water chamber **3108** and a bubbler **3110** within the water chamber **3108** that is configured to bubble ozonated air through water within the water chamber **3108**. Bubbling of ozonated air through water may result in the addition of humidity to the ozonated air. The humidified ozonated air may enter the disinfection chamber **3104**.

[0222] FIG. 32 shows a schematic example of a disinfection device **3200**, which is an example of the disinfection device **200** of FIG. 2.

[0223] The disinfection device **3200** includes an ozone generator **3202**, a disinfection chamber **3204**, and a humidifier **3206**. The humidifier **3206** includes a water chamber **3208**, an atomizer nozzle **3210** fluidly coupled to the water chamber **3208**, and a humidification chamber **3212**. The atomizer nozzle **3210** is configured to emit atomized water into the humidification chamber **3212** in a direction transverse to (e.g., perpendicular to) a flow of ozonated air entering the humidification chamber **3212** to humidify the ozonated air. The humidified ozonated air may enter the disinfection chamber **3204**.

[0224] FIG. 33 shows a schematic example of a disinfection device **3300**, which is an example of the disinfection device **200** of FIG. 2. As shown, the disinfection device **3300** includes an ozone generator **3302**, a disinfection chamber **3304**, and a humidifier **3306**. At least a portion of the humidifier **3306** is disposed within the disinfection chamber **3304**. As such, the humidifier **3306** may be configured to emit humidified air and/or humidified ozonated air directly into the disinfection chamber **3304**.

[0225] FIG. 34 shows a schematic example of a disinfection device **3400**, which is an example of the disinfection device **200** of FIG. 2. As shown, the disinfection device **3400** includes an ozone generator **3402**, a disinfection chamber **3404**, and a humidifier **3406**. As shown, the humidifier **3406** may be separate from one or more of the

disinfection chamber **3404** and/or the ozone generator **3402**. Such a configuration may allow, for example, the humidifier **3406** to be moved independently of one or more of the disinfection chamber **3404** and/or the ozone generator **3402**. In some instances, such a configuration may allow the humidifier **3406** to be used as a module for a disinfection device **3400** which does not have a humidifier.

[0226] In some instances, the ozone generator **3402** and the humidifier **3406** may be coupled together and separate from the disinfection chamber **3404**. Such a configuration may allow the ozone generator **3402** and the humidifier **3406** to be moved independently of the disinfection chamber **3404**. In these instances, the disinfection chamber **3404** may be collapsible or compressible (e.g., a flexible bag).

[0227] FIG. 35 shows a schematic example of a disinfection device **3500**, which is an example of the disinfection device **200** of FIG. 2. As shown, the disinfection device **3500** includes an ozone generator **3502**, a disinfection chamber **3504**, and a humidifier **3506**. As shown, the humidifier **3506** includes a plurality of humidity generators **3508** (e.g., a plurality of atomizers and associated wicks).

[0228] FIG. 36 shows a schematic example of a disinfection device **3600**. As shown, the disinfection device includes a disinfection chamber **3602**, a humidifier **3604**, and an electrolytic cell **3606**. The humidifier **3604** includes a liquid reservoir **3608**, a humidification chamber **3610**, and a humidity generator **3612** configured to generate humidity within the humidification chamber **3610**. The electrolytic cell **3606** is configured to generate hypochlorous acid and to emit the hypochlorous acid into the humidification chamber **3610** to mix with humidity generated by the humidity generator **3612**. For example, the humidity generator **3612** may be configured to emit atomized water into the humidification chamber **3610** and the electrolytic cell **3606** may be configured to emit atomized hypochlorous acid into the humidification chamber **3610** such that the atomized water and the atomized hypochlorous acid mix. The electrolytic cell **3606** may be upstream or downstream of the humidity generator **3612** and/or the humidification chamber **3610**. The mixing of hypochlorous acid with humidity may be accomplished in a similar manner as discussed herein in relation to ozone. As such, in some instances, one or more of the ozone based embodiments disclosed herein may use the electrolytic cell **3606** instead of (or in addition to) an ozone generator.

[0229] As shown, the electrolytic cell **3606** includes an anode **3614** and a cathode **3616** spaced apart by an anode-cathode separation distance **3618**. The anode-cathode separation distance **3618** may be, for example, in a range of about 1 millimeter (mm) to about 10 mm. By way of further example, the anode-cathode separation distance **3618** may be in a range of about 2 mm to about 4 mm. A close proximity of the anode **3614** to the cathode **3616** may encourage generation of hypochlorous acid.

[0230] In response to a current running between the anode **3614** and the cathode **3616** in the presence of water, the electrolytic cell **3606** may generate hypochlorous acid. In some instances, use of chlorinated water may improve the generation of hypochlorous acid.

[0231] The mixing of hypochlorous acid with humidity may be accomplished in a similar manner as discussed herein in relation to ozone. As such, in some instances, one or more of the ozone based embodiments disclosed herein may use the electrolytic cell **3606** instead of (or in addition

to) an ozone generator. As such, ozone and hypochlorous acid may be generally referred to herein as a disinfection fluid which may be generated by a disinfection fluid generator (e.g., an electrolytic cell, such that the electrolytic cell **3606**, or an ozone generator, such as the ozone generator **202**). In other words, the embodiments disclosed in relation to, for example, FIGS. 1-61 utilize at least one disinfection fluid to achieve a desired level of disinfection performance.

[0232] FIGS. 37 to 41 show another example of a disinfection device **3700** with humidity control configured to disinfect equipment (e.g., one or more components of a continuous positive air pressure (CPAP) machine). The disinfection device **3700** may be an example of the disinfection device **300** of FIG. 3. As should be appreciated one or more features of the disinfection device **700** may also be incorporated into the disinfection device **3700**. For example, the disinfection device **3700** may utilize similar disinfection cycles as described in relation to the disinfection device **700**. By way of further example, ozonated air and humidified ozonated air may be generated and delivered in the disinfection device **3700** using similar liquid dispersers, pumps, fans, ozone generators, ozone sensors, and/or the like as described in relation to disinfection device **700**. As such, in an interest of brevity, some of the features common to both the disinfection device **700** and the disinfection device **3700** may not be redescribed in the context of the disinfection device **3700**.

[0233] As shown, the disinfection device **3700** includes a disinfection chamber **3702**, an ozone generator **3704** (shown schematically in hidden lines), and a humidifier assembly **3706**. A conduit **3707** (shown schematically), such as a CPAP hose (e.g., the CPAP hose **750** of FIG. 7A), may be configured to fluidly couple the humidifier assembly **3706** with the disinfection chamber **3702**. Such a configuration allows humidified ozonated air to pass through the conduit **3707** (e.g., CPAP hose) when entering the disinfection chamber **3702**, disinfecting the conduit **3707** (e.g., CPAP hose).

[0234] In some instances, a CPAP hose may be entirely disposed within the disinfection chamber **3702** and configured to fluidly couple to the conduit **3707** via a connector within the disinfection chamber **3702**. Additionally, or alternatively, one or more pieces of equipment may be disposed within the disinfection chamber **3702** (e.g., a CPAP mask may be disposed within the disinfection chamber **3702** and coupled to the CPAP hose). While a CPAP hose is provided as an example of a conduit, other conduits may be used. For example, the conduit **3707** may include any type of rigid or flexible tubing. The disinfection device **3700** further includes a device controller **3701** configured to control one or more functions of the disinfection device (e.g., humidity generation, ozone generation, and/or any other function).

[0235] The humidifier assembly **3706** is configured to encourage a mixing of ozonated air with humidity. The humidifier assembly **3706** includes a liquid reservoir **3708** and a humidification chamber **3710** fluidly connected to the ozone generator **3704** and the liquid reservoir **3708**. The humidifier assembly **3706** includes a humidification chamber outlet **3712** configured to be fluidly coupled to the disinfection chamber **3702** via the conduit (e.g., the CPAP hose). In operation, fluid (e.g., water) stored within the liquid reservoir **3708** is dispersed into the humidification chamber **3710** to mix with ozonated air entering the humidification chamber **3710** to form humidified ozonated air.

[0236] The disinfection chamber 3702 includes a disinfection chamber lid 3714 and a conduit (e.g., CPAP hose) passthrough 3716 configured to receive a portion of the conduit 3707 (e.g., CPAP hose). The disinfection chamber lid 3714 is configured to transition between an open position (FIG. 38) and a closed position (FIG. 37) to selectively close a chamber open end 3718 of the disinfection chamber 3702 and to selectively close a passthrough open end 3720 of the conduit passthrough 3716. When the chamber lid 3714 is in the open position, the conduit 3707 (e.g., CPAP hose) may be inserted into the conduit passthrough 3716 such that a portion of the conduit 3707 (e.g., CPAP hose) is disposed within the disinfection chamber 3702 and a portion of the conduit 3707 (e.g., CPAP hose) is disposed outside the disinfection chamber 3702. When the conduit 3707 (e.g., CPAP hose) is received within the conduit passthrough 3716 and the chamber lid 3714 is in the closed position, each of the chamber lid 3714 and the conduit passthrough 3716 form a seal with a respective portion of the conduit 3707 (e.g., CPAP hose) such that a seal is formed about an entire perimeter of the conduit 3707 (e.g., CPAP hose). While a conduit passthrough is shown, other configurations are possible. For example, instead of (or in addition to) the conduit passthrough 3716, the humidifier assembly 3706 may be fluidly coupled to the disinfection chamber via a port, wherein the port is configured to fluidly couple to the conduit 3707 and/or equipment disposed entirely within the disinfection chamber 3702. The port may be fluidly coupled to the humidifier assembly 3706 by one or more internal supply lines (e.g., internal to the disinfection device 3700).

[0237] The disinfection chamber lid 3714 further includes a toggle shroud 3722. The toggle shroud 3722 is configured to move (e.g., pivot) with the disinfection chamber lid 3714 between the open and closed positions. The toggle shroud 3722 is configured to selectively enclose a humidification chamber toggle 3724. When the disinfection chamber lid 3714 is in the closed position, the toggle shroud 3722 at least partially (e.g., entirely) encloses the humidification chamber toggle 3724 to prevent actuation of the humidification chamber toggle 3724. When the disinfection chamber lid 3714 is in the open position, the toggle shroud 3722 exposes the humidification chamber toggle 3724 such that the humidification chamber toggle 3724 can be actuated. As shown in FIG. 38, the toggle shroud 3722 defines a toggle cavity 3726 configured to receive at least a portion of the humidification chamber toggle 3722.

[0238] Actuation of the humidification chamber toggle 3724 causes the humidification chamber 3710 to at least partially disengage the liquid reservoir 3708. In other words, the humidification chamber 3710 may be generally described as being configured to selectively engage the liquid reservoir 3708 to form a fluid coupling between the humidification chamber 3710 and the liquid reservoir 3708. For example, actuation of the humidification chamber toggle 3724 may cause the humidification chamber 3710 to transition from a lowered position (FIG. 38) to a raised position (FIG. 39). In this example, a user may exert a force on the humidification chamber 3710 to transition the humidification chamber 3710 from the raised position to the lowered position. When the humidification chamber 3710 is in raised position, the liquid reservoir 3708 may be removable from the disinfection device 3700 (e.g., for the purposes of refilling the liquid reservoir 3708).

[0239] For example, and as shown in FIG. 39, actuation of the humidification chamber toggle 3724 may cause the humidification chamber 3710 to move along a chamber axis 3728 from the lowered position to the raised position. The chamber axis 3728 intersects a base region 3729 of the disinfection device 3700 and extends in a direction of the disinfection chamber lid 3714. For example, the chamber axis 3728 may be a substantially vertical axis.

[0240] When the humidification chamber 3710 is in the raised position, the liquid reservoir 3708 is removable along a removal axis 3730. When the humidification chamber 3710 is in the lowered position, removal of the liquid reservoir 3708 is prevented as a result of one or more portions of the liquid reservoir 3708 engaging the humidification chamber 3710. For example, the liquid reservoir 3708 may include one or more of a catch 3732 and/or a liquid dispersion protrusion 3734 that is configured to interfere with a portion of the humidification chamber 3710 if removal of the liquid reservoir 3708 is attempted when the humidification chamber 3710 is in the lowered position. As shown, the removal axis 3730 may extend transverse to the chamber axis 3728, forming a perpendicular or non-perpendicular angle with the chamber axis 3728.

[0241] As shown in FIG. 40, at least a portion of a device housing 3736 of the disinfection device 3700 and/or at least a portion of the humidification chamber 3710 may collectively define reservoir receptacle 3738 for selectively receiving the liquid reservoir 3708 therein. The disinfection chamber 3702 (FIG. 37), the humidifier assembly 3706 (FIG. 37), and/or the ozone generator 3704 (FIG. 37) may be coupled to and/or define a portion of (e.g., a portion of an outer surface of) the device housing 3736. The device housing 3736 may be, for example, a single monolithic body or a plurality of bodies connected together that define at least a portion of an open or enclosed cavity (e.g., a cavity configured to receive one or more components, walls, or supports).

[0242] The reservoir receptacle 3738 may include one or more reservoir ramps 3740 along which the liquid reservoir 3708 slides when being received within or removed from the reservoir receptacle 3738. The reservoir ramps 3740 may be angled such that the reservoir ramps 3740 taper inwardly from an insertion end 3743 of the reservoir receptacle 3738. In other words, a ramp height 3742 of the reservoir ramps 3740 decreases with increasing distance from the insertion end 3743 of the reservoir receptacle 3738.

[0243] As also shown, the liquid reservoir 3708 includes corresponding ramp guides 3744. The ramp guides 3744 are configured to slidably engage the reservoir ramps 3740. The ramp guides 3744 may include a taper, wherein the taper of the ramp guides 3744 is opposite a taper of the reservoir ramps 3740. For example, when the reservoir ramps 3740 taper inwardly, the ramp guides 3744 may taper outwardly. Such a configuration, may discourage (e.g., prevent) the inadvertent removal of the liquid reservoir 3708 when the humidification chamber 3710 is in the raised position (e.g., as a result of a user moving the disinfection device 3700). Additionally, or alternatively, such a configuration may encourage insertion of the liquid reservoir 3708 (e.g., gravity may naturally encourage the liquid reservoir 3708 to slide along the reservoir ramps 3740 into place).

[0244] The liquid reservoir 3708 may further include a removable reservoir lid 3746. As shown in FIG. 41, the removable reservoir lid 3746 is removable along a reservoir lid axis 3748. For example, a user may apply a force on the

catch 3732 to remove the reservoir lid 3746. Once the reservoir lid 3746 is removed, a user may clean and/or refill a liquid container 3749 of the liquid reservoir 3708. The reservoir lid 3746 may be poka-yoked to prevent operation of the disinfection device 3700 with an improperly installed reservoir lid 3746. For example, one or more components on the reservoir lid 3746 may be positioned such that the reservoir lid 3746 is non-symmetrical. Such a non-symmetrical configuration is configured to prevent components on the reservoir lid 3746 from properly mating with corresponding components (e.g., one or more electrical connectors) on the humidity chamber 3710 when the humidity chamber 3710 is in the lowered position. By way of further example, a shape of the reservoir lid 3746 may be non-symmetric over at least one central axis (e.g., such that the reservoir lid 3746 is wider along a first side than along a second, opposing, side).

[0245] FIG. 42 shows a perspective exploded view of the removable reservoir lid 3746. As shown, the reservoir lid 3746 includes a lid housing 4200 having an upper housing 4202 and a lower housing 4204. The upper and lower housing 4202 and 4204 are configured to couple together to form a lid cavity 4206 therebetween. The lid cavity 4206 is configured to receive a lid controller 4208 electrically coupled to a piezo-electric atomizer 4210. The piezo-electric atomizer 4210 is configured to disperse liquid from the liquid reservoir 3708 into the humidity chamber 3710. A particle size of the dispersed liquid may be configured such that a quantity of a most prevalent particle size is substantially the same (e.g., similar to as discussed in relation to FIG. 20A) at an inlet side of the conduit 3707 (FIG. 37) fluidly coupled to the humidity chamber 3710 and at an outlet side of the conduit 3707 disposed within the disinfection chamber 3702 (FIG. 37). The dispersed liquid may have a particle size distribution, wherein a most prevalent particle size is, for example, in a range of 0.2 microns to 10 microns. By way of further example, the most prevalent particle size may be in a range of 0.3 to 3 microns. By way of still further example, the most prevalent particle size may be in a range of 0.3 microns to 1 microns. By way of still further example, the most prevalent particle size may be in a range of 0.3 microns to 0.5 microns. By way of still further example, the most prevalent particle size may be about 0.5 microns. By way of still further example, the most prevalent particle size may be about 0.3 microns. By way of still further example, the most prevalent particle size may be about 1 micron. While the reservoir lid 3746 is shown as including the piezo-electric atomizer 4210, other liquid dispersers may be used. For example, the reservoir lid 3746 may include a spray nozzle fluidly coupled to a pump.

[0246] The piezo-electric atomizer 4210 is coupled to an atomizer support 4212 extending from the lower housing 4204 and into a protrusion cavity 4214 defined by the liquid dispersion protrusion 3734. In other words, the liquid dispersion protrusion 3734 includes the piezo-electric atomizer 4210 when the reservoir lid 3746 is assembled. The atomizer support 4212 includes a wire channel 4216 through which one or more wires of the piezo-electric atomizer 4210 may be routed. At least a portion of the atomizer gasket 4218 may be disposed between the piezo-electric atomizer 4210 and an emission end wall 4220 of the liquid dispersion protrusion 3734. The emission end wall 4220 may have a substantially continuous engagement surface 4235 (e.g., that extends along at least 90%, at least 95%, at least 96%, at least 97%,

at least 98%, at least 99%, or 100% of an outer perimeter of the piezo-electric atomizer 4210) that forms a seal with the atomizer gasket 4218.

[0247] The atomizer gasket 4218 may sealingly engage the emission end wall 4220 and a peripheral region 4222 of the piezo-electric atomizer 4210. The peripheral region 4222 surrounds an emission region 4224 of the piezo-electric atomizer 4210. The sealing engagement between the emission end wall 4220 and the peripheral region 4222 may reduce (e.g., prevent) ingress of liquid dispersed by the piezo-electric atomizer 4210 (which may include ozone entrained therein) from reentering the liquid reservoir 3708. Such a configuration may prolong a useable life of one or more of the electronic components within the lid cavity 4206 (e.g., the piezo-electric atomizer 4210). In some instances, the atomizer gasket 4218 may include a groove 4219 extending along an inner perimeter, the groove 4219 being configured to receive at least a portion of the peripheral region 4222 of the piezo-electric atomizer 4210. In these instances, the atomizer gasket 4218 may extend along both an emission side 4221 and a supply side 4223 of the piezo-electric atomizer 4210, the emission side 4221 being opposite the supply side 4223. For example, the atomizer gasket 4218 may extend substantially continuously around the entire outer perimeter of the piezo-electric atomizer 4210 (e.g., enclosing at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% of an outer perimeter of the piezo-electric atomizer 4210). In some instances, the atomizer gasket 4218 may extend continuously along the emission side 4221 of the piezo-electric atomizer 4210 and substantially continuously along the supply side 4223 of the piezo-electric atomizer 4210 (e.g., to accommodate a wire passthrough 4225).

[0248] The emission side 4221, in one example, may include a stainless steel alloy and the supply side 4223 may include a ceramic material. In another example, the emission side 4221 may include a ceramic material and the supply side 4223 may include a stainless steel alloy. In these examples, the material of the emission side 4221 and the supply side 4223 may be selected based, at least in part, on an environment in which the emission and supply sides 4221 and 4223 may be exposed during use.

[0249] The groove 4219 may be connected with the wire passthrough 4225 through which one or more wires of the piezo-electric atomizer 4210 may be routed. The wire passthrough 4225 may be aligned with and configured to cooperate with the wire channel 4216 of the atomizer support 4212. In some instances, a protective coating (e.g., a Parylene coating) may be disposed on the supply side 4223 and the emission side 4221 within the peripheral region 4222 (but not within the emission region 4224). In other words, the protective coating may have an annular shape. In these instances, the protective coating may also extend over a peripheral edge extending between the emission and supply sides 4221 and 4223 of the piezo-electric atomizer 4210.

[0250] When received within the groove 4219, the piezo-electric atomizer 4210 may be exposed to a compression force as a result of the atomizer gasket 4218 forming a seal with the emission end wall 4220 of the liquid dispersion protrusion 3734. The compression force experienced by the piezo-electric atomizer 4210 may be a function of a compression ratio, wherein the compression ratio is a change in gasket thickness 4231 of the atomizer gasket 4218 between

the compressed and uncompressed states divided by the uncompressed gasket thickness **4231**. The compression ratio may be, for example, in a range of 0.01 to 0.3. By way of further example, the compression ratio may be less than or equal to 0.06 but greater than 0, less than or equal to 0.05 but greater than 0, less than or equal to 0.04 but greater than 0, less than or equal to 0.03 but greater than 0, less than or equal to 0.02 but greater than 0, less than or equal to 0.01 but greater than 0. By way of still further example, the compression ratio may be about 0.02. By way of still further example, the compression ratio may be in a range of 0.1 to 0.26. By way of still further example, the compression ratio may be in a range of 0.21 to 0.26. By way of still further example, the compression ratio may be in a range of 0.1 to 0.2.

[0251] When the reservoir lid **3746** is assembled, the atomizer gasket **4218** and the atomizer support **4212** may extend substantially continuously along a perimeter of the emission region **4224** of the piezo-electric atomizer **4210**. Such a configuration may result in the formation of a substantially continuous seal and/or an application of a substantially consistent clamping force within the peripheral region **4222**.

[0252] The emission end wall **4220** includes a first atomizer emission opening **4227** and the atomizer gasket **4218** includes a second atomizer opening **4229**. The first and second atomizer openings **4227** and **4229** are sized such that at least the emission region **4224** of the piezo-electric atomizer **4210** is exposed. In other words, the first and second atomizer openings **4227** and **4229** are sized and aligned to enable the piezo-electric atomizer **4210** to disperse fluid therethrough.

[0253] As shown, the upper housing **4202** of the reservoir lid **3746** includes the liquid dispersion protrusion **3734** extending therefrom. The liquid dispersion protrusion **3734** is configured to selectively fluidly couple the piezo-electric atomizer **4210** with the humidification chamber **3710** (FIG. 37). In some instances, a protrusion seal **4226** may extend around perimeter of the liquid dispersion protrusion **3734**. The protrusion seal **4226** is configured to sealingly engage with the humidification chamber **3710** to mitigate (e.g., prevent) ozone passing through the humidification chamber **3710** from leaking from the humidification chamber **3710** at an interface between the humidification chamber **3710** and the liquid dispersion protrusion **3734**.

[0254] The upper housing **4202** of the reservoir lid **3746** may further include a plurality of electrical connection passthroughs **4228** within which lid electrical connectors **4230** of the lid controller **4208** extend. The lid electrical connectors **4230** are configured to electrically couple the lid controller **4208** with the disinfection device **3700**. For example, the lid electrical connectors **4230** may be configured to receive one or more of power and/or operating instructions from the disinfection device **3700** (e.g., the device controller **3701** and/or a power supply of the disinfection device **3700**). The lid electrical connectors **4230** may form part of a pogo-pin connection. Additionally, or alternatively, the reservoir lid **3746** may include one or more wireless communication and/or power transfer interfaces (e.g., a near field communication interface, an inductive power transfer interface, and/or any other wireless interface).

[0255] The lower housing **4204** of the reservoir lid **3746** may include a lid gasket **4232** and a wick passthrough **4234**.

The lid gasket **4232** is configured to sealingly engage with the liquid container **3749** (FIG. 37). The wick passthrough **4234** is configured to receive a portion of a wick assembly **4236**. In some instances, the wick passthrough **4234** may receive at least a portion of the atomizer support **4212**, wherein the atomizer support **4212** includes a wick opening **4253** through which a portion of the wick assembly **4236** (e.g., a wick **4242**) extends.

[0256] The wick assembly **4236** may include a wick cage **4238**, a wick support **4240**, and the wick **4242**. In operation, the wick cage **4238** may be configured to releasably engage with a portion of the lower housing **4204** (e.g., using a snap fit connection). The wick support **4240** is coupled to the wick cage **4238** such that the wick support **4240** is located within the wick cage **4238**. The wick support **4240** is configured to removably receive the wick **4242** therein. When the wick assembly **4236** is coupled to the lower housing **4204**, the wick **4242** directly contacts the piezo-electric atomizer **4210**. The wick **4242** is configured to passively draw liquid (e.g., water) from the liquid container **3749** to the piezo-electric atomizer **4210** such that the piezo-electric atomizer **4210** disperses the liquid into the humidification chamber **3710**.

[0257] Having the removable reservoir lid **3746** include the piezo-electric atomizer **4210** and the lid controller **4208** may allow a user to easily replace the piezo-electric atomizer **4210** and the lid controller **4208** should one or more of the piezo-electric atomizer **4210** and/or the lid controller **4208** become damaged. In these instances, the lid controller **4208** may include one or more anti-counterfeit mechanisms (e.g., software and/or hardware based) configured to prevent the use of unauthorized reservoir lids (e.g., by preventing operation of the disinfection device **3700** when an unauthorized reservoir lid is detected). The one or more anti-counterfeit mechanisms may include, for example, one or more of a radio frequency identification system, a near field communication system, and/or the like. In some instances, at least one of the one or more anti-counterfeit mechanisms may be separate from (e.g., electrically isolated from) the lid controller **4208**.

[0258] FIG. 43 shows a bottom perspective view of the removable reservoir lid **3746** having the wick assembly **4236** removed therefrom. As shown, the wick cage **4238** includes cage protrusions **4300** configured to cooperate with a corresponding passthrough groove **4302** to form a snap-fit connection. As also shown, the wick cage **4238** defines a castellated base **4304** having a plurality of spaced apart castellations **4306**. An end surface **4308** of the castellations may engage with a bottom portion of the liquid container **3749** (FIG. 37).

[0259] FIG. 44 shows a cross-sectional perspective view of the humidification chamber **3710** in the lowered position and engaging the liquid reservoir **3708** and FIG. 45 shows a cross-sectional perspective view of the humidification chamber **3710** in the raised position and out of engagement with the liquid reservoir **3708**.

[0260] As shown, the humidification chamber **3710** includes a chamber housing **4400**, a chamber base **4402** coupled to the chamber housing **4400**, and one or more chamber electrical connectors **4401** (FIG. 45) configured to electrically couple to the lid electrical connectors **4230** (FIG. 42) when the humidification chamber **3710** is in the lowered position. In other words, the chamber electrical connectors **4401** are configured to selectively electrically couple to the

lid electrical connectors **4230**. The one or more chamber electrical connectors **4401** are configured to receive one or more of power and/or operating instructions from the disinfection device **3700** (e.g., the device controller **3701** and/or a power supply of the disinfection device **3700**). The one or more chamber electrical connectors **4401** may form part of a pogo pin connection.

[0261] The chamber base **4402** may include a bottom wall **4404** and one or more base sidewalls **4406** extending from the bottom wall **4404** to define a base cavity **4408** having an open end **4409** opposite the bottom wall **4404**. A reservoir receptacle **4410** extends from the bottom wall **4404** and into the base cavity **4408**. The reservoir receptacle **4410** defines a reservoir cavity **4412** (FIG. 45) for selectively receiving at least a portion of the liquid dispersion protrusion **3734** of the liquid reservoir **3708**. The reservoir cavity **4412** is configured to sealingly engage with at least a portion of the protrusion seal **4226**.

[0262] The reservoir receptacle **4410** includes an atomizer opening **4413**. The atomizer opening **4413** is configured to align with the first atomizer emission opening **4227** of the emission end wall **4220** and the second atomizer emission opening **4229** of the atomizer gasket **4218**. As such, an emission axis **4414** of the piezo-electric atomizer **4210** extends, unobstructed, from the emission region **4224** of the piezo-electric atomizer **4210** and into the humidification chamber **3710**. In other words, the atomizer opening **4413** is aligned relative to piezo-electric atomizer **4210** such that the emission axis **4414** extends, unobstructed, into the humidification chamber **3710**. The emission axis **4414** extends transverse to the ozone injection axis **4416** at any angle between 0° and 180° (e.g., a perpendicular angle) to encourage a mixing of ozonated air with dispersed liquid. In other words, the emission axis **4414** and the injection axis **4416** are oriented relative to each other to encourage mixing of ozonated air with dispersed liquid.

[0263] The atomizer opening **4413** may be the only opening in the one or more base sidewalls **4406** and/or the bottom wall **4404** of the chamber base **4402**. Such a configuration may mitigate (e.g., prevent) a backflow of liquid collected within the chamber base **4402** and into the liquid reservoir **3708**. In other words, when in use, the piezo-electric atomizer **4210** may be generally described as providing a one-way fluid coupling from the liquid reservoir **3708** and into humidification chamber **3710** such that liquid dispersed into the humidification chamber **3710** does not return to the liquid reservoir **3708**. Additionally, or alternatively, such a configuration may encourage mixing of ozonated air with dispersed liquid from the piezo-electric atomizer **4210**.

[0264] As shown in FIG. 45A, ozonated air enters the humidification chamber **3710** via an inlet connector **4550** having connector inlet **4552** configured to engage a tube **4554** (shown as transparent) and connector outlet **4556** from which the ozone injection axis **4416** extends. The inlet connector **4550** is coupled to the humidification chamber **3710** (e.g., coupled to the chamber housing **4400**). The inlet connector **4550** is fluidly coupled to the ozone generator **3704** (FIG. 37). The connector outlet **4556** may be vertically spaced apart from the piezo-electric atomizer **4210**. As shown, the inlet connector **4550** may be configured to change a flow direction of ozonated air passing there-through. For example, a connector inlet axis **4558** may extend transverse to (e.g., perpendicular to) the ozone injection axis **4416**. Changing a flow direction of air within the

inlet connector **4550** may encourage generation of turbulent air. The inlet connector **4550** may be similar to the inlet connector **824** of FIG. 8.

[0265] As shown in FIG. 45B, ozonated air **4560** entering the humidification chamber **3710** along the ozone injection axis **4416** (FIG. 44) intersects liquid **4562** dispersed from the piezo-electric atomizer **4210** along the emission axis **4414** (FIG. 44). The intersection of the dispersed liquid **4562** with the ozonated air **4560** may encourage the ozonated air **4560** to mix with the dispersed liquid **4562** to form humidified ozonated air. As also shown, ozonated air **4560**, along with any liquid entrained therein, may intersect the chamber housing **4400** at a side of the chamber housing **4400** opposite the connector outlet **4556** of the inlet connector **4550**. Intersection with the chamber housing **4400** may cause the humidified ozonated air to extend along one or more sidewalls of the chamber housing **4400** while changing a flow direction of the humidified ozonated air. A portion of the humidified ozonated air may extend along a portion of the chamber base **4402**, wherein the reservoir receptacle **4410** may interfere with the flow. The direction changes and/or flow interference may encourage generation of a turbulent flow of ozonated humidified air, which may encourage further mixing of the ozonated air with the dispersed liquid **4562** (increasing a relative humidity of the ozonated humidified air).

[0266] While the piezo-electric atomizer **4210** is shown as being located in the chamber base **4402** and the inlet connector **4550** is shown as being located in the chamber housing **4400** (e.g., a sidewall), other configurations are possible. For example, the piezo-electric atomizer **4210** may be disposed in a sidewall of the chamber housing **4400** and the inlet connector **4550** may be disposed within the chamber base **4402**.

[0267] FIGS. 46 and 47 show perspective views of the chamber base **4402**. As shown, the reservoir receptacle **4410** has an inner perimeter **4600** that defines the atomizer opening **4413** and an outer perimeter **4602** spaced apart from the inner perimeter **4600** in an outward direction. The inner perimeter **4600** may have a shape that generally corresponds to a shape of the piezo-electric atomizer **4210** (e.g., a circular shape). The outer perimeter **4602** may be the same or different in shape from the inner perimeter **4600**. For example, as shown, the outer perimeter **4602** has a shape different from that of the inner perimeter **4600**. In the illustrated example, the outer perimeter **4602** has a truncated circular shape. A flat surface of the truncated circular shape may be defined at an intersection region **4604** of the outer perimeter **4602** of the reservoir receptacle **4410** with at least a portion of at least one of the one or more base sidewalls **4406** of the chamber base **4402**. The intersection region **4604** may be generally described as a region wherein a portion of the reservoir receptacle **4410** intersects with at least one of the one or more base sidewalls **4406**. The intersection region **4604** may correspond to (e.g., overlie) a position of the wire passthrough **4225** of the atomizer gasket **4218**. Additionally, or alternatively, the intersection region **4604** may be disposed beneath (e.g., a portion of the intersection region **4604** may be directly beneath) the connector outlet **4556** (FIG. 45A).

[0268] As also shown, a flange **4606** may extend from the inner perimeter **4600** towards the outer perimeter **4602**, the flange **4606** defining a mating surface **4608**. The mating surface **4608** is configured to engage (e.g., directly contact)

the protrusion seal **4226** (FIG. 42) to form a seal at an interface between the mating surface **4608** and the protrusion seal **4226**. The mating surface **4608** may be a substantially continuous (e.g., continuous) surface. The flange **4606** may extend substantially continuously (e.g., continuously) around the atomizer opening **4413** and define the inner perimeter **4600**.

[0269] The reservoir receptacle **4410** includes one or more receptacle sidewalls **4610** extending between the flange **4606** and the bottom wall **4404** of the chamber base **4402**. The one or more receptacle sidewalls **4610** may define a substantially continuous (e.g., continuous) exterior surface **4612** that extends at least partially around at least a portion of the liquid dispersion protrusion **3734** (FIG. 39). For example, the substantially continuous exterior surface **4612** extends substantially around (e.g., around at least 70%, around at least 75%, around at least 85%, around at least 90%, around at least 95%, around at least 97%, around at least 99%, or around 100%) at least a portion of the liquid dispersion protrusion **3734**. In some instances, the exterior surface **4612** may extend substantially continuously (e.g., without interruptions, such as openings therein) from a first end **4614** of the intersection region **4604** to a second end **4616** of the intersection region **4604**. As such, there may be no gaps between the exterior surface **4612** and the one or more base sidewalls **4406** of the chamber base **4402** through which air (e.g., ozonated air) can pass at a location of the intersection region **4604**.

[0270] The configuration of the chamber base **4402** (e.g., the absence of air outlets in the chamber base **4402**, the shape and position of the reservoir receptacle **4410**, and/or the like) may be configured to encourage the generation of a turbulent flow of ozonated air entering the humidification chamber **3710** and/or to encourage the mixing of ozonated air with liquid dispersed by the piezo-electric atomizer **4210** into the humidification chamber **3710**. Generation of turbulence may encourage mixing of air with ozone to become humidified ozonated air, creating a disinfection medium. With one or more turbulence features, humidified ozonated air may also be encouraged to traverse an entire length of the conduit **3707** (FIG. 37), which may encourage disinfection of the conduit **3707**. Additionally, or alternatively, use of a fan **3800** (shown schematically in FIG. 38) to pull the humidified ozonated air through the conduit **3707** may encourage the humidified ozonated air to traverse an entire length of the conduit **3707**. In some instances, the fan **3800** may cooperate with a pump **3802** (shown schematically in FIG. 38), the pump **3802** being configured to generate a pushing flow through the conduit **3707**.

[0271] FIG. 48 shows a cross-sectional perspective view of the disinfection device **3700** showing the humidification chamber **3710** in the raised position. As shown, the humidification chamber **3710** includes a first plurality of bearings **4800** configured to move within a first guide track **4802** and an extension limiter **4804** configured to move within a first retention track **4806**. The first guide track **4802** and the first retention track **4806** may be part of the humidifier assembly **3706** (FIG. 37) and may be coupled (directly or indirectly) to, for example, the device housing **3736** (FIG. 40). The extension limiter **4804** may be configured to cooperate with (e.g., be coupled to) a first biasing mechanism **4808** (e.g., a tension spring). The first biasing mechanism **4808** is coupled to a fixed portion of the humidifier assembly **3706**, the fixed portion being fixed relative to the humidification chamber

3710. The first biasing mechanism **4808** is configured to urge the humidification chamber **3710** towards the raised position. As such, when a user actuates the humidification chamber toggle **3724** (FIG. 38), the humidification chamber **3710** may automatically transition from the lowered position to the raised position. When transitioning between the lowered and raised positions, the bearings **4800** move within the first guide track **4802**.

[0272] FIG. 49 shows an opposing cross-sectional perspective view, relative to FIG. 48, of the disinfection device **3700** showing the humidification chamber **3710** in the raised position. As shown, the humidification chamber **3710** includes a second plurality of bearings **4900** configured to move within a second guide track **4902** and a lift protrusion **4904** configured to cooperate with a second biasing mechanism **4906** (e.g., a tension spring) to urge the humidification chamber **3710** towards the raised position. The second plurality of bearings **4900** and/or the second guide track **4902** may be vertically offset from the first plurality of bearings **4800** and the first guide track **4802**.

[0273] FIGS. 50 and 51 show the humidification chamber outlet **3712** having a removable conduit adapter such as CPAP hose adapter **5000** disposed therein. A removable CPAP hose adapter **5000** may allow a user to couple multiple different CPAP hoses (e.g., from different brands or having different designs) to the humidification chamber outlet **3712**. Additionally, or alternatively, one or more additional non-CPAP hose adapters may be configured to cooperate with the humidification chamber outlet **3712** such that one or more conduits may be coupled thereto. For example, a non-CPAP hose adapter may be configured to couple to a beverage conduit (e.g., for carbonated beverages, water, and/or the like), air conduits (e.g., for oxygen delivery in a medical setting), medical tubing (e.g., intravenous lines), and/or any other form of conduit.

[0274] As shown, the humidification chamber outlet **3712** includes an outlet opening **5001** through which humidified ozonated air exits the humidification chamber **3710** and a mounting region **5003** extending around the outlet opening **5001**. The mounting region **5003** is configured to receive the removable CPAP hose adapter **5000** such that a CPAP hose can be removably coupled to the CPAP hose adapter **5000** and fluidly coupled to the outlet opening **5001**. In other words, the mounting region **5003** is configured to selectively receive the removable CPAP hose adapter **5000**. The outlet opening **5001** of the humidification chamber outlet **3712** is fluidly coupled to the humidification chamber **3710**. The humidification chamber outlet **3712** includes an adapter lock **5002** configured to selectively couple the CPAP hose adapter **5000** with the humidification chamber outlet **3712** (e.g., with the mounting region **5003**).

[0275] As shown, the adapter lock **5002** includes a rotatable locking body **5004** and a fixed body **5006**. The fixed body **5006** may form part of or be coupled to the device housing **3736** (FIG. 40). The rotatable locking body **5004** is configured to rotate relative to the fixed body **5006** between a locked position (FIG. 50) and an unlocked position (FIG. 51). In some instances, the CPAP hose adapter **5000** may include a locking indicator **5005** configured to align with a corresponding locking indicator on the rotatable locking body **5004**. For example, when the rotatable locking body **5004** is in the locked position, the locking indicator **5005** may align with a locked indicator on the rotatable locking body **5004** and, when in the unlocked position, the locking

indicator **5005** may align with a corresponding unlocked indicator on the rotatable locking body **5004**. The rotating locking body **5004** may be configured to rotate about a locking axis **5007**. The locking axis **5007** may pass through a center of the outlet opening **5001** of the humidification chamber outlet **3712**. For example, the locking axis **5007** may be a horizontal axis that passes through the center of the outlet opening **5001**.

[0276] When the rotatable locking body **5004** is in locked position, the adapter lock **5002** is configured to prevent removal of the CPAP hose adapter **5000** from the humidification chamber outlet **3712**. When the rotatable locking body **5004** is in the unlocked position, the adapter lock **5002** is configured to allow removal of the CPAP hose adapter **5000** from the humidification chamber outlet **3712**. In some instances, the disinfection device **3700** may include one or more lock sensors (shown schematically in hidden lines) communicatively coupled with the device controller **3701**. The one or more lock sensors **5008** may be configured to sense one or more of a position of the rotatable locking body **5004** (e.g., locked position or unlocked position), a presence of the removable CPAP hose adapter **5000** within the humidification chamber outlet **3712**, and/or a presence of a CPAP hose coupled to the CPAP hose adapter **5000**. Based, at least in part, on an output of the one or more lock sensors **5008** one or more operations of the disinfection device **3700** may be enabled or disabled.

[0277] FIG. 52 shows a rear view of the of the adapter lock **5002** with the locking body **5004** in the locked position and engaging the removable CPAP hose adapter **5000**. FIG. 53 shows a rear view of the adapter lock **5002** with the locking body **5004** in the unlocked position.

[0278] As shown, the locking body **5004** includes a plurality of locking tabs **5200** that rotate with the locking body **5004**. The locking tabs **5200** are configured to cooperate with corresponding retention tabs **5202** of the CPAP hose adapter **5000** to selectively retain the CPAP hose adapter **5000** in the humidification chamber outlet **3712**. When transitioning between the locked and unlocked positions, the locking tabs **5200** are configured to move into and out of engagement with the retention tabs **5202**. As the locking tabs **5200** are rotated into engagement with the retention tabs **5202**, the locking tabs **5200** and the retention tabs **5202** may be configured to interfere with each other such that the locking tabs **5200** exert a retaining force on the retention tabs **5202**. For example, one or more of the locking tabs **5200** and/or the retention tabs **5202** may be cammed at an interfacing surface. Application of a retaining force may mitigate (e.g., prevent) a leak at an interface between the removable CPAP hose adapter **5000** and the humidification chamber outlet **3712**.

[0279] As also shown, the locking body **5004** includes a plurality of guide arms **5204** from which the locking tabs **5200** extend. The guide arms **5204** further include a runner **5206** extending from the guide arms **5204** in a direction opposite that of the locking tabs **5200**. The runner **5206** is configured to slidably engage a rail **5208** of the fixed body **5006**. The runner **5206** and the rail **5208** are configured to cooperate to rotatably couple the locking body **5004** to the fixed body **5006**. In some instances, the rail **5208** and the runner **5206** may be configured to form a snap fit connection between the locking body **5004** and the fixed body **5006**.

[0280] FIG. 54 shows a perspective view of the locking body **5004** in the unlocked position and the CPAP hose

adapter **5000** removed from the humidification chamber outlet **3712**. As shown, the CPAP hose adapter **5000** is configured to be removed from (and inserted into) the mounting region **5003** along an adapter removal axis **5400**. The adapter removal axis **5400** may be substantially parallel to the locking axis **5007**. In some instances, the adapter removal axis **5400** and the locking axis **5007** may be coincident. As shown, the mounting region **5003** of the humidification chamber outlet **3712** extends completely around the outlet opening **5001**.

[0281] The locking body **5004** includes a locked indicator **5402** and an unlocked indicator **5404**. Rotation of the locking body **5004** to the locked position will cause the locked indicator **5402** to align with the locking indicator **5005** of the CPAP hose adapter **5000**. Rotation of the locking body **5004** to the unlocked position will cause the unlocked indicator to align with the locking indicator **5005** of the CPAP hose adapter **5000**.

[0282] FIG. 55 shows a perspective view of the locking body **5004** in the unlocked position with the CPAP hose adapter **5000** removed from the humidification chamber outlet **3712**, wherein the CPAP hose adapter **5000** is rotated relative to the humidification chamber outlet **3712**. As shown, the CPAP hose adapter **5000** is poka-yoked to discourage (e.g., prevent) improper insertion of the CPAP hose adapter **5000** into the mounting region **5003** of the humidification chamber outlet **3712**. For example, the CPAP hose adapter **5000** may include a first key hole **5500** having a first key hole profile and a second key hole **5502** having a second key hole profile, the first key hole profile being different from the second key hole profile. The first key hole **5500** is configured to receive a corresponding first key **5504** extending within the mounting region **5003** and the second key hole **5502** is configured to receive a corresponding second key **5506** extending within the mounting region **5003**. The first key **5504** has a first key profile that corresponds to the first key hole profile and the second key **5506** has a second key profile that corresponds to the second key hole profile, the first key profile being different from the second key profile. While, in the example shown, the first key hole **5500** is shown being opposite the second key hole **5502** along a central axis of the CPAP hose adapter **5000** and the first key **5504** is shown as being opposite the second key **5506** along a central plane of the mounting region **5003**, other configurations are possible.

[0283] As also shown in FIG. 55, the lock sensor **5008** may include a switch **5508** that is configured to be depressed when the CPAP hose adapter **5000** is installed in the mounting region **5003**. In some instances, the switch **5508** may be a two stage switch, wherein the first stage is actuated when the CPAP hose adapter **5000** is installed and the second stage is actuated when a CPAP hose is coupled to the CPAP hose adapter **5000**. Such a configuration may allow for the generation of context specific alerts (e.g., hose adapter missing alert, CPAP hose missing alert, and/or the like). Additionally, or alternatively, the switch **5508** may be actuated in response to the CPAP hose adapter **5000** being inserted and the locking body **5004** being in the locked position.

[0284] As also shown, in some instances, the locking body **5004** may include cutouts **5510** that correspond to the first and second keys **5504** and **5506** such that the portion of the CPAP hose adapter **5000** defining the first and second key holes **5500** and **5502** may pass therethrough.

[0285] FIG. 56 shows a perspective view of the disinfection chamber lid 3714 having a humidity sensor assembly 5600 removably coupled thereto and FIG. 57 shows a perspective view of the disinfection chamber lid 3714 having the humidity sensor assembly 5600 decoupled therefrom. The humidity sensor assembly 5600 is configured to measure a relative humidity within the disinfection chamber 3702. The measured relative humidity may be used to control operation of one or more components of the disinfection device 3700 (FIG. 37). For example, the humidity sensor assembly 5600 may be communicatively coupled to the device controller 3701, wherein the device controller 3701 may control an output of one or more of the ozone generator 3704 (FIG. 37) and/or the humidifier assembly 3706 (FIG. 37). In this example, the device controller 3701 may be configured to cause the ozone generator 3704 to generate more or less ozone and/or the humidifier assembly 3706 to generate more or less humidity based, at least in part, on the relative humidity detected by the humidity sensor assembly 5600. Additionally, or alternatively, and by way of further example, the humidity sensor assembly 5600 may be communicatively coupled to the device controller 3701, wherein the device controller 3701 is configured to stop a disinfection cycle based, at least in part, on a sensed relative humidity. In this example, the disinfection cycle may be stopped in response to the detected relative humidity being less than a threshold humidity level (e.g., greater than or equal to 50% relative humidity, greater than or equal to 60% relative humidity, greater than or equal to 70% relative humidity, greater than or equal to 75% relative humidity, greater than or equal to 80% relative humidity, greater than or equal to 90% relative humidity, or greater than or equal to 95% relative humidity) after a predetermined time (e.g., after 1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 7 minutes, 8 minutes, 9 minutes, or 10 minutes of commencement of humidity generation). Additionally, or alternatively, if the humidity level is less than the threshold humidity level, the device controller 3701 may generate (e.g., display) a user alert. The user alert may include one or more of an instruction to check a water level in the liquid reservoir 3708, an indication that a disinfection procedure was unsuccessful (e.g., that the cycle was not completed or, if the cycle was completed, that the disinfection may not have been successful), to check a status of the wick assembly 4236, and/or the like.

[0286] The humidity sensor assembly 5600 may be removably coupled to the disinfection chamber lid 3714 (e.g., for replacement in the event of damage). As such, the humidity sensor assembly 5600 may include a sensing end 5604 that is exposed to an environment within the disinfection chamber 3702 and a communication end 5606 configured to communicatively couple with a communication socket 5608 disposed within the disinfection chamber lid 3714. The communication socket 5608 is configured to communicatively couple the humidity sensor assembly 5600 with the device controller 3701. If the humidity sensor assembly 5600 is not detected (or is determined to be not operating properly), the device controller 3701 may prevent operations of one or more of the ozone generator 3704 and/or the humidifier assembly 3706.

[0287] As shown, the humidity sensor assembly 5600 may be poka-yoked to discourage (e.g., prevent) improper insertion of the humidity sensor assembly 5600 into the communication socket 5608. For example, the humidity sensor

assembly 5600 may have a first sensor key 5610 having a first key profile and a second sensor key 5612 having a second key profile, the first key profile being different from the second key profile. In this example, the communication socket 5608 may include a first socket key hole 5614 having a first key hole profile and a second socket key hole 5616 having a second key hole profile, the first key hole profile being different from (e.g., wider than) the second key hole profile. The first key profile may generally correspond to the first key hole profile and the second key profile may correspond to the second key hole profile.

[0288] FIG. 58 shows an exploded view of the humidity sensor assembly 5600. As shown, the humidity sensor assembly 5600 includes a cage 5800, a mesh (e.g., a screen) 5802, a printed circuit board (PCB) 5804, a PCB gasket 5806, an end cap plate 5808, and an end cap 5810. The PCB 5804 includes a first PCB portion 5812 and a second PCB portion 5814, the first and second PCB portions 5812 and 5814 being on opposite sides of the PCB gasket 5806. The first PCB portion 5812 includes a humidity sensor 5816 and the second PCB portion 5814 includes an anti-counterfeit identifier 5818. The anti-counterfeit identifier 5818 is configured to prevent counterfeit humidity sensor assemblies from being used by a user. In some instances, the anti-counterfeit identifier 5818 may include calibration data associated with the humidity sensor 5816. At least a portion of the PCB 5804 and/or one or more components on the PCB 5804 (e.g., electrical traces, the anti-counterfeit identifier 5818, and/or any other component) may have a coating (e.g., a hydrophobic coating) applied thereto to mitigate potentially adverse effects of humidity on the PCB 5804 and/or components on the PCB 5804.

[0289] The cage 5800 defines a cage cavity 5820 configured to receive at least a portion of the first PCB portion 5812. In other words, the cage 5800 is configured to extend around (e.g., enclose) the humidity sensor 5816. The cage 5800 further includes a plurality of cage openings 5822 that extend from an exterior surface 5824 of the cage 5800 and into the cage cavity 5820. The plurality of cage openings 5822 are spaced apart from each other such that no cage opening 5822 directly overlies the humidity sensor 5816. In other words, the cage openings 5822 are positioned such that no cage opening central axis 5826 intersects a sensing surface 5828 of humidity sensor 5816 at a perpendicular angle, the cage opening central axis 5826 extending perpendicular to a plane within which the cage opening 5822 extends. Such a configuration may create a tortuous path for humidified ozonated air to reach the humidity sensor 5816, potentially creating more reliable results and/or extending a useful life of the humidity sensor 5816.

[0290] The mesh 5802 is disposed within the cage cavity 5820 and configured to extend around the first PCB portion 5812. The mesh 5802 is disposed between the first PCB portion 5812 and the cage openings 5822. The mesh 5802 may be configured such that large droplets of water are discouraged from reaching the humidity sensor 5816. As such, a pore size of the mesh 5802 may be configured to capture at least a portion of water droplets greater than a certain size.

[0291] The cage 5800 includes plurality of snap fit arms 5830. The plurality of snap fit arms 5830 are configured to engage corresponding snap fit connections 5832 on the end cap 5810. The plurality of snap fit arms 5830 and/or the snap

fit connections **5832** may form at least a portion of the first and second sensor keys **5610** and **5612**.

[0292] The end cap **5810** defines an end cap cavity **5834** configured to receive at least a portion of the second PCB portion **5814** and the PCB gasket **5806**. The PCB gasket **5806** is configured to sealingly engage with a sidewall of the end cap cavity **5834**. The seal formed between the end cap cavity **5834** and the PCB gasket **5806** is configured to mitigate (e.g., prevent) an ingress of humidified ozonated air into the end cap cavity **5834**. The PCB gasket **5806** may include one or more of a silicone and/or rubber grommet.

[0293] The end cap **5810** further includes a communication opening **5836** from which a communication connector **5838** of the second PCB portion **5814** extends. The communication connector **5838** is configured to communicate data generated by one or more of the humidity sensor **5816** and/or the anti-counterfeit identifier **5818** to the device controller **3701** (FIG. 56) via the communication socket **5608** (FIG. 56).

[0294] The end cap plate **5808** is disposed within the end cap cavity **5834**. The end cap plate **5808** may be disposed between an end wall **5840** of the end cap **5810** and the PCB gasket **5806**. The end cap plate **5808** may include a plate passthrough **5842** configured to align with the communication opening **5836** such that a portion of the second PCB portion **5814** extends through both the plate passthrough **5842** and the communication opening **5836** to expose the communication connector **5838**. The end cap plate **5808** may form part of a magnetic connection with the communication socket **5608**. For example, the end cap plate **5808** may be a ferromagnetic material configured to cooperate with a magnetic material within the communication socket **5608**. By way of further example, the end cap plate **5808** may be a magnetic material configured to cooperate with a ferromagnetic material within the communication socket **5608**.

[0295] FIG. 59 shows a cross-sectional perspective view of a portion of the disinfection chamber lid **3714** having the humidity sensor assembly **5600** installed in the communication socket **5608**. As shown, the communication socket **5608** defines a socket cavity **5900** having a cavity end wall **5902** and a cavity open end **5904** opposite the cavity end wall **5902**. As shown, the cavity end wall **5902** includes an interior surface **5906** that extends within the socket cavity **5900** and an exterior surface **5908** opposite the interior surface **5906**. The exterior surface **5908** may include (e.g., be coupled with) one or more magnets **5910**. For example, a plurality of magnets **5910** may be disposed on opposing sides of the PCB **5804**. The one or more magnets **5910** are configured to cooperate with the end cap plate **5808** (which includes a magnetic material) to encourage a secure connection of the humidity sensor assembly **5600** within the communication socket **5608**.

[0296] FIG. 60 shows a schematic top view of example of a piezo-electric atomizer **6000**, which may be an example of any one of the piezo-electric atomizers disclosed herein. As shown, the piezo-electric atomizer **6000** includes an atomization plate **6002** and a plurality of electrical conductors **6004** and **6006**. The atomization plate **6002** includes an emission region **6008** and peripheral region **6010** enclosing the emission region **6008**. The emission region **6008** includes a plurality of holes **6012** through which water can pass. The emission region **6008** may be raised or recessed relative to the peripheral region **6010**.

[0297] FIG. 61 shows a schematic cross-sectional view of the piezo-electric atomizer **6000** taken along the line LXI-LXI of FIG. 60. As shown, the piezo-electric atomizer **6000** further includes a piezo-electric layer **6100** and a conductive layer **6102**, wherein the piezo-electric layer **6100** is disposed between the atomization plate **6002** and the conductive layer **6102**. In some instances, the atomization plate **6002** and the conductive layer **6102** may be directly coupled to (e.g., in direct contact with) the piezo-electric layer **6100**. The atomization plate **6002** and the conductive layer **6102** are electrically coupled (e.g., directly electrically coupled) to the piezo-electric layer **6100**. The first electrical conductor **6004** may be electrically coupled (e.g., directly electrically coupled) to the atomization plate **6002** and the second electrical conductor **6006** may be electrically coupled (e.g., directly electrically coupled) to the conductive layer **6102**. The piezo-electric atomizer **6000** may also include a ceramic layer **6104**, wherein the conductive layer **6102** is disposed between the ceramic layer **6104** and the piezo-electric layer **6100**. In some instances, the ceramic layer may be directly coupled to (e.g., in direct contact with) the conductive layer **6102**.

[0298] The conductive layer **6102** may include an electrically conductive corrosion resistant material. The electrically conductive corrosion resistant material may be selected based, at least in part, on an environment within which the piezo-electric atomizer **6000** is to be used. For example, the piezo-electric atomizer **6000** may be used within an environment containing ozone. Examples of an electrically conductive corrosion resistant material may include carbon nanotubes, gold, platinum, stainless steel, and/or the like.

[0299] FIG. 62 shows a schematic example of a disinfection device **6200**, which is an example of the disinfection device **200** of FIG. 2.

[0300] The disinfection device **6200** includes an ozone generator **6202**, a disinfection chamber **6204**, and a humidifier **6206**. The disinfection chamber **6204** is configured to receive equipment **6208** (e.g., a CPAP mask) and/or a conduit **6210** (e.g., a CPAP hose). For example, the disinfection chamber **6204** may include a conduit port **6212**. The conduit port **6212** may be fluidly coupled to the ozone generator **6202** and the humidifier **6206** and configured to removably couple to the conduit **6210**. The conduit port **6212** is downstream of the ozone generator **6202** and the humidifier **6206**. The conduit port **6212** may be coupled to any sidewall defining the disinfection chamber **6204** (e.g., a horizontally extending and/or a vertically extending sidewall).

[0301] As shown, a flow path **6214** extends through the ozone generator **6202** through the humidifier **6206** and into the disinfection chamber **6204** via the conduit port **6212**, the conduit **6210**, and the equipment **6208**. Once in the disinfection chamber **6204**, the ozonated air may be recirculated or pass through a disinfection chamber outlet **6216**.

[0302] The disinfection fluid described herein may be used as a disinfectant for a variety of home or industrial products including but not limited to: dentistry and orthodontic appliances and equipment, surgical tools, ventilators, respirators, nebulizers, hospital equipment, rooms, toys, daycare appliances, baby products (e.g. bottles, pumps, pump equipment, pacifiers), kitchen equipment, fruits and vegetables, sports equipment and apparel and other uses where the systems and methods described herein may readily be incorporated into

a disinfection device and improve disinfection of an object over existing disinfection methods.

[0303] An example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, a disinfection fluid generator (e.g., an ozone generator and/or an electrolytic cell) configured to generate a disinfection fluid, and a humidifier fluidly coupled to the disinfection chamber and the disinfection fluid generator, the humidifier being downstream of the disinfection fluid generator and upstream of the disinfection chamber.

[0304] In some instances, the humidifier may include a piezo-electric atomizer configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the piezo-electric atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the piezo-electric atomizer may include a protective coating. In some instances, the humidifier may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis and the humidifier further includes an atomizer having an emission axis, the emission axis extending transverse to the injection axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may include a liquid reservoir, a wick assembly removably coupled to the liquid reservoir, and a piezo-electric atomizer configured to engage the wick assembly. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of engagement with the piezo-electric atomizer. In some instances, the humidifier may include a humidification chamber and an atomizer, the humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, wherein the atomizer is coupled to the bottom wall. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall.

[0305] An example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being downstream of the ozone generator and upstream of the disinfection chamber.

[0306] In some instances, the humidifier may include a liquid disperser such as a piezo-electric atomizer configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the piezo-electric atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the piezo-electric atomizer may include a protective coating. In some instances, the humidifier may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis and the humidifier further

includes an atomizer having an emission axis, the emission axis extending transverse to the injection axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may include a liquid reservoir, a wick assembly removably coupled to the liquid reservoir, and a piezo-electric atomizer configured to engage the wick assembly. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of engagement with the piezo-electric atomizer. In some instances, the humidifier may include a humidification chamber and an atomizer, the humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, wherein the atomizer is coupled to the bottom wall. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall.

[0307] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier including a liquid reservoir, a humidification chamber, and a liquid disperser, such as a piezo-electric atomizer or a nozzle, configured to emit atomized droplets into the humidification chamber along an emission axis.

[0308] In some instances, the liquid disperser may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the liquid disperser may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the liquid disperser may include a protective coating. In some instances, the humidifier may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis that extends transverse to the emission axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may include a wick assembly removably coupled to the liquid reservoir and configured to engage a piezo-electric atomizer. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move a wick assembly into and out of engagement with the liquid disperser. In some instances, the humidification chamber may include a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, wherein the liquid disperser is coupled to the bottom wall. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the

bottom wall. In some instances, liquid collected on the bottom wall may be returned to the liquid reservoir.

[0309] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber having a disinfection chamber outlet, an ozone reduction filter fluidly coupled to the disinfection chamber outlet, an ozone generator, and a humidifier fluidly coupled to the disinfection chamber and the ozone generator. The humidifier may include a liquid reservoir, a humidification chamber having an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector, and a piezo-electric atomizer configured to emit atomized droplets into the humidification chamber along an emission axis.

[0310] In some instances, the inlet connector may include a connector inlet axis and a connector outlet axis, the connector inlet axis extending transverse to the connector outlet axis. In some instances, the connector inlet axis may be perpendicular to the connector outlet axis. In some instances, the piezo-electric atomizer may include a protective coating. In some instances, the piezo-electric atomizer may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns.

[0311] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier fluidly coupled to the disinfection chamber and the ozone generator such that the humidifier is downstream of the ozone generator and upstream of the disinfection chamber, the humidifier including a piezo-electric atomizer.

[0312] In some instances, the piezo-electric atomizer may include a protective coating. In some instances, the protective coating may have a thickness in a range of about 2 microns to about 12 microns. In some instances, the protective coating may be a Parylene coating. In some instances, the protective coating may be hydrophobic. In some instances, the piezo-electric atomizer may include a hydrophobic coating. In some instances, the piezo-electric atomizer may be formed from one or more durable materials. In some instances, the piezo-electric atomizer may include a protective coating. In some instances, the humidifier may include a piezo-electric atomizer configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the piezo-electric atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the piezo-electric atomizer may include a protective coating. In some instances, the humidifier may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis that extends transverse to an emission axis of the piezo-electric atomizer. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may include a liquid reservoir, a wick assembly removably coupled to the liquid reservoir and configured to engage the piezo-electric atomizer. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of

engagement with the piezo-electric atomizer. In some instances, the humidifier may include a humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, wherein the atomizer is coupled to the bottom wall. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall.

[0313] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier including a liquid reservoir, a humidification chamber, and a piezo-electric atomizer fluidly coupled to the liquid reservoir and configured to emit atomized droplets into the humidification chamber along an emission axis.

[0314] In some instances, the piezo-electric atomizer may include a protective coating. In some instances, the protective coating may have a thickness in a range of about 2 microns to about 12 microns. In some instances, the protective coating may be a Parylene coating. In some instances, the protective coating may be hydrophobic. In some instances, the piezo-electric atomizer may include a hydrophobic coating. In some instances, the piezo-electric atomizer may be formed from one or more durable materials. In some instances, the piezo-electric atomizer may include a protective coating. In some instances, the piezo-electric atomizer may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the piezo-electric atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the piezo-electric atomizer may include a protective coating.

[0315] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator fluidly coupled to the disinfection chamber, a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being downstream of the ozone generator and upstream of the disinfection chamber, and a controller configured to control operation of the ozone generator and the humidifier according to a disinfection cycle.

[0316] In some instances, the disinfection cycle may include a first period, a second period, and a third period, wherein during the first period the controller causes both the ozone generator and the humidifier to operate concurrently. In some instances, the humidifier may be disabled during the second period. In some instances, during the second period, the controller may selectively enable and disable the ozone generator, causing the ozone generator to generate ozone pulses. In some instances, the humidifier and the ozone generator may be disabled during the third period. In some instances, to generate the ozone pulses, the controller may alternate between enabling the ozone generator for about 10 seconds and disabling the ozone generator for about 65 seconds. In some instances, the controller may be configured to cause the ozone generator to generate the ozone pulses at a pulse rate, wherein the pulse rate is configured such that a quantity of ozone within the disinfection chamber is in a range of about 80 parts-per-million (ppm) to about 150 ppm. In some instances, the first period may be about 6 minutes, the second period may be about 72 minutes, and the third

period may be about 12 minutes. In some instances, during the first period, the controller may operate the humidifier such that a relative humidity within the disinfection chamber is in a range of about 65% to about 99%. In some instances, the first period may occur before the second period and the second period may occur before the third period. In some instances, the second period may be longer than both the third period and the first period. In some instances, the first period may be shorter than the third period.

[0317] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator fluidly coupled to the disinfection chamber, a pump fluidly coupled to the ozone generator and upstream of the ozone generator, a fan fluidly coupled to the ozone generator and downstream of the ozone generator, a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being downstream of the ozone generator and upstream of the disinfection chamber, and a controller configured to control operation of the ozone generator, the humidifier, the pump, and the fan according to a disinfection cycle, the disinfection cycle including a first period, a second period, and a third period.

[0318] In some instances, the controller may cause the fan to operate at a fan flow rate and the pump to operate at a pump flow rate, the pump flow rate being different from the fan flow rate during at least one of the first period, the second period, and/or the third period. In some instances, the fan flow rate may be less than the pump flow rate during the first period and the second period. In some instances, a ratio of the fan flow rate to the pump flow rate during the first period and the second period may be about 0.9. In some instances, the fan flow rate may be greater than the pump flow rate during the third period. In some instances, the pump flow rate may be in a range of about 1 standard liter per minute (SLPM) to about 1.6 SLPM during the first period and the second period. In some instances, the fan flow rate may be about 0.1 SLPM to about 0.15 SLPM less than the pump flow rate during the first period and the second period. In some instances, during the first period the controller may cause both the ozone generator and the humidifier to operate concurrently. In some instances, the humidifier may be disabled during the second period. In some instances, during the second period, the controller may selectively enable and disable the ozone generator, causing the ozone generator to generate ozone pulses. In some instances, the humidifier and the ozone generator may be disabled during the third period. In some instances, to generate the ozone pulses, the controller may alternate between enabling the ozone generator for about 10 seconds and disabling the ozone generator for about 65 seconds. In some instances, the controller may be configured to cause the ozone generator to generate the ozone pulses at a pulse rate, wherein the pulse rate is configured such that a quantity of ozone within the disinfection chamber is in a range of about 80 parts-per-million (ppm) to about 150 ppm. In some instances, the first period may be about 6 minutes, the second period may be about 72 minutes, and the third period may be about 12 minutes. In some instances, during the first period, the controller may operate the humidifier such that a relative humidity within the disinfection chamber is in a range of about 65% to about 99%. In some instances, the first period may occur before the second period and the second period may occur before the third period. In some instances, the second period may be

longer than both the third period and the first period. In some instances, the first period may be shorter than the third period.

[0319] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier including a humidification chamber and a liquid reservoir.

[0320] In some instances, the humidification chamber may include a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall. In some instances, liquid collected on the bottom wall may be returned to the liquid reservoir. In some instances, the humidifier may further include a liquid disperser, such as a piezo-electric atomizer or a nozzle, coupled to the bottom wall or a sidewall and configured to emit atomized droplets along an emission axis that extends in a direction of the top wall. In some instances, the liquid disperser may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the liquid disperser may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the liquid disperser may include a protective coating. In some instances, the humidification chamber may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis and the humidifier further includes an atomizer having an emission axis, the emission axis extending transverse to the injection axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may further include a wick assembly removably coupled to the liquid reservoir and a piezo-electric atomizer configured to engage the wick assembly. In some instances, the humidifier may further include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of engagement with the piezo-electric atomizer.

[0321] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator fluidly coupled to the disinfection chamber, a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being downstream of the ozone generator and upstream of the disinfection chamber, and a controller configured to control operation of the ozone generator and the humidifier, wherein the controller is configured to cause both the ozone generator and the humidifier to operate concurrently for a first period of time and to cause the ozone generator to operate independent of the humidifier for a second period of time.

[0322] In some instances, the second period of time may be greater than the first period of time. In some instances, the humidifier may include an inlet connector configured to

introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis and the humidifier further includes an atomizer having an emission axis, the emission axis extending transverse to the injection axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may include a liquid reservoir, a wick assembly removably coupled to the liquid reservoir, and a piezo-electric atomizer configured to engage the wick assembly. In some instances, the humidifier may include a liquid disperser such as a nozzle. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of engagement with the piezo-electric atomizer. In some instances, the humidifier may further include a humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, wherein the atomizer is coupled to the bottom wall. In some instances, the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall. In some instances, liquid collected on the bottom wall may be returned to the liquid reservoir.

[0323] An example of a disinfection device for cleaning one or more components of a continuous positive airway pressure (CPAP) machine, consistent with the present disclosure, may include a disinfection chamber, an ozone generator fluidly coupled to the disinfection chamber, a humidifier fluidly coupled to the ozone generator, the humidifier being downstream of the ozone generator, a CPAP hose fluidly coupling the humidifier to the disinfection chamber, and a controller configured to control operation of the ozone generator and the humidifier.

[0324] In some instances, the controller may be configured to cause both the ozone generator and the humidifier to operate concurrently for a first period of time and to cause the ozone generator to operate independent of the humidifier for a second period of time, the second period of time being greater than the first period of time. In some instances, the humidifier may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis and the humidifier further includes an atomizer having an emission axis, the emission axis extending transverse to the injection axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may include a liquid reservoir, a wick assembly removably coupled to the liquid reservoir, and a piezo-electric atomizer configured to engage the wick assembly. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of engagement with the

piezo-electric atomizer. In some instances, the humidifier may further include a humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, wherein the atomizer is coupled to the bottom wall. In some instances, the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall. In some instances, liquid collected on the bottom wall may be returned to the liquid reservoir.

[0325] An example of a humidifier for use with a disinfection device, consistent with the present disclosure, may include an atomizer (and/or any other liquid disperser), a liquid reservoir, and a humidification chamber, the atomizer configured to emit atomized liquid from the liquid reservoir into the humidification chamber, the humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, at least a portion of the top wall is sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall.

[0326] In some instances, liquid collected on the bottom wall may be returned to the liquid reservoir. In some instances, the atomizer (or other liquid disperser) may be coupled to the bottom wall of the humidification chamber. In some instances, the atomizer may be configured to emit atomized droplets along an emission axis that extends in a direction of the top wall. In some instances, the humidification chamber may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis, the injection axis extending transverse to the emission axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the atomizer may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the atomizer may be a piezo-electric atomizer. In some instances, the atomizer may include a protective coating. In some instances, the humidifier may further include a wick assembly removably coupled to the liquid reservoir and the atomizer is a piezo-electric atomizer that is configured to engage the wick assembly. In some instances, the humidifier may further include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis.

[0327] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator configured to generate ozone, and a humidifier configured to generate humidity and configured to be fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being configured to encourage a mixing of generated ozone with generated humidity.

[0328] In some instances, the disinfection chamber may be fluidly coupled to the humidifier via a continuous positive air pressure (CPAP) hose. In some instances, a first hose end of the CPAP hose may be coupled to the humidifier and a

second hose end of the CPAP hose may be coupled to a CPAP mask, the disinfection chamber being configured to receive the CPAP mask. In some instances, the humidifier may include a liquid reservoir, a humidification chamber, and a humidity generator, the humidity generator configured to urge liquid from the liquid reservoir into the humidification chamber. In some instances, the humidity generator may include an atomizer. In some instances, the atomizer may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the ozone generator may be configured to generate a quantity of ozone in a range of about 200 parts-per-million (ppm) to about 300 ppm. In some instances, the humidifier may be downstream of the ozone generator and may include an inlet connector configured to introduce a turbulent flow to air passing therethrough. In some instances, the mixing of generated ozone with generated humidity may result in at least a portion of the generated ozone becoming entrained within droplets of moisture generated by the humidifier. In some instances, a pump may be fluidly coupled to the disinfection chamber at location upstream of the disinfection chamber and a fan may be fluidly coupled to the disinfection chamber at a location downstream of the disinfection chamber. In some instances, for at least a portion of a disinfection cycle, the fan may be operated at a fan flow rate and the pump may be operated at a pump flow rate, the fan flow rate may be less than the pump flow rate.

[0329] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber configured to receive a continuous positive air pressure (CPAP) mask and at least a portion of a CPAP hose, an ozone generator configured to generate ozone, and a humidifier configured to generate humidity and configured to be fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being configured to encourage a mixing of generated ozone with generated humidity.

[0330] In some instances, the disinfection chamber may be fluidly coupled to the humidifier via the CPAP hose. In some instances, a first hose end of the CPAP hose may be coupled to the humidifier and a second hose end of the CPAP hose may be coupled to the CPAP mask. In some instances, the humidifier may include a liquid reservoir, a humidification chamber, and a humidity generator, the humidity generator configured to urge liquid from the liquid reservoir into the humidification chamber. In some instances, the humidity generator may include an atomizer. In some instances, the atomizer may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the ozone generator may be configured to generate a quantity of ozone in a range of about 200 parts-per-million (ppm) to about 300 ppm. In some instances, the humidifier may be downstream of the ozone generator and may include an inlet connector configured to introduce a turbulent flow to air passing therethrough. In some instances, the mixing of generated ozone with generated humidity may result in at least a portion of the generated ozone becoming entrained within droplets of moisture generated by the humidifier. In some instances, a pump may be fluidly coupled to the disinfection chamber at a location upstream of the disinfection chamber and a fan

may be fluidly coupled to the disinfection chamber at a location downstream of the disinfection chamber. In some instances, for at least a portion of a disinfection cycle, the fan may be operated at a fan flow rate and the pump may be operated at a pump flow rate, the fan flow rate may be less than the pump flow rate.

[0331] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber configured to receive a continuous positive air pressure (CPAP) mask, the disinfection chamber including a hose passthrough configured such that a CPAP hose can pass therethrough, an ozone generator configured to generate ozone, a humidifier configured to generate humidity and configured to be fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being configured to encourage a mixing of generated ozone with generated humidity such that at least a portion of the generated ozone becomes entrained within droplets of moisture generated by the humidifier, wherein a first end of the CPAP hose is coupled to the humidifier and a second end of the CPAP hose is coupled to the CPAP mask, a pump fluidly coupled to the disinfection chamber at location upstream of the disinfection chamber, and a fan fluidly coupled to the disinfection chamber at a location downstream of the disinfection chamber, wherein, for at least a portion of a disinfection cycle, the fan is operated at a fan flow rate and the pump is operated at a pump flow rate, the fan flow rate being less than the pump flow rate.

[0332] In some instances, the humidifier may include a liquid reservoir, a humidification chamber, and an atomizer. In some instances, the atomizer may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the ozone generator may be configured to generate a quantity of ozone in a range of about 200 parts-per-million (ppm) to about 300 ppm. In some instances, the humidifier may be downstream of the ozone generator and includes an inlet connector configured to introduce a turbulent flow to air passing therethrough.

[0333] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier configured to generate humidity and configured to be fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being configured to encourage a mixing of generated ozone with generated humidity.

[0334] In some instances, the humidifier may include a humidity generator configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the humidity generator may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the humidity generator may include a protective coating. In some instances, the humidifier may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis and the humidifier may further include an atomizer having an emission axis, the emission axis extending transverse to the injection axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may include a liquid reservoir, a

wick assembly removably coupled to the liquid reservoir, and a piezo-electric atomizer configured to engage the wick assembly. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of engagement with the piezo-electric atomizer. In some instances, the humidifier may include a humidification chamber and an atomizer, the humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, wherein the atomizer is coupled to the bottom wall. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall.

[0335] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier fluidly coupled to the ozone generator and configured to be fluidly coupled to the disinfection chamber via a conduit (e.g., a continuous positive air pressure (CPAP) hose), the humidifier including a liquid reservoir, a humidification chamber, and a humidity generator.

[0336] In some instances, the humidity generator may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the humidity generator may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the humidity generator may include a protective coating. In some instances, the humidifier may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis that extends transverse to an emission axis of the humidity generator. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may further include a wick assembly removably coupled to the liquid reservoir. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move a wick assembly into and out of engagement with a liquid disperser of the humidity generator. In some instances, the humidification chamber may include a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, wherein at least a portion of the humidity generator is coupled to the bottom wall. In some instances, the humidity generator may be received within a holder (e.g., extending from the bottom wall), wherein the holder is configured to encourage turbulent flow within the humidification chamber. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more

sidewalls to collect on the bottom wall. In some instances, liquid collected on the bottom wall may be returned to the liquid reservoir.

[0337] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber having a disinfection chamber outlet, an ozone reduction filter fluidly coupled to the disinfection chamber outlet, an ozone generator, and a humidifier fluidly coupled to the disinfection chamber and the ozone generator. The humidifier may include a liquid reservoir, a humidification chamber having an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector, and a humidity generator configured to emit droplets into the humidification chamber along an emission axis.

[0338] In some instances, the inlet connector may include a connector inlet axis and a connector outlet axis, the connector inlet axis extending transverse to the connector outlet axis. In some instances, the connector inlet axis may be perpendicular to the connector outlet axis. In some instances, the humidity generator may include a protective coating. In some instances, humidity generator may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns.

[0339] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier including a humidity generator having a protective coating.

[0340] In some instances, the humidity generator may include a piezo-electric atomizer. In some instances, the protective coating may have a thickness in a range of about 2 microns to about 12 microns. In some instances, the protective coating may be a Parylene coating. In some instances, the protective coating may be hydrophobic. In some instances, the humidifier may include a piezo-electric atomizer configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the piezo-electric atomizer may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the piezo-electric atomizer may include the protective coating. In some instances, the humidifier may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, the piezo-electric atomizer may be received within a holder, wherein the holder is configured to cooperate with the inlet connector to encourage generation of humidified ozonated air and/or to encourage generation of turbulent flow. In some instances, air may exit the inlet connector along an injection axis that extends transverse to an emission axis of the humidity generator. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may include a liquid reservoir, a wick assembly removably coupled to the liquid reservoir. In some instances, the humidifier may include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of engagement with a liquid disperser of the humidity generator. In some instances, the

humidifier may include a humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall and wherein at least a portion of the humidity generator is coupled to the bottom wall. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall.

[0341] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier fluidly coupled to the ozone generator and configured to be fluidly coupled to disinfection chamber via a continuous positive air pressure (CPAP) hose, the humidifier including a liquid reservoir, a humidification chamber, and humidity generator fluidly coupled to the liquid reservoir and configured to emit droplets into the humidification chamber along an emission axis.

[0342] In some instances, the humidity generator may include a protective coating. In some instances, the protective coating may have a thickness in a range of about 2 microns to about 12 microns. In some instances, the protective coating may be a Parylene coating. In some instances, the protective coating may be hydrophobic. In some instances, the humidity generator may include a piezo-electric atomizer having a protective coating. In some instances, humidity generator may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the humidity generator may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the humidity generator may include a piezo-electric atomizer having a protective coating. In some instances, the disinfection chamber may be configured to receive a CPAP mask coupled to the CPAP hose.

[0343] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator fluidly coupled to the disinfection chamber, a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being upstream of the disinfection chamber, and a controller configured to control operation of the ozone generator and the humidifier according to a disinfection cycle.

[0344] In some instances, the disinfection cycle may include a first period, a second period, and a third period, wherein during the first period the controller causes both the ozone generator and the humidifier to operate concurrently. In some instances, the humidifier may be disabled during the second period. In some instances, during the second period, the controller may selectively enable and disable the ozone generator, causing the ozone generator to generate ozone pulses. In some instances, the humidifier and the ozone generator may be disabled during the third period. In some instances, to generate the ozone pulses, the controller may alternate between enabling the ozone generator for about 10 seconds and disabling the ozone generator for about 65 seconds. In some instances, the controller may be configured to cause the ozone generator to generate the ozone pulses at a pulse rate, wherein the pulse rate is configured such that a quantity of ozone within the disinfection chamber is in a range of about 80 parts-per-million (ppm) to about 150 ppm. In some instances, the first period may be about 6 minutes, the second period may be about 72 minutes, and the third

period may be about 12 minutes. In some instances, during the first period, the controller may operate the humidifier such that a relative humidity within the disinfection chamber is in a range of about 65% to about 99%. In some instances, the first period may occur before the second period and the second period may occur before the third period. In some instances, the second period may be longer than both the third period and the first period. In some instances, the first period may be shorter than the third period.

[0345] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator fluidly coupled to the disinfection chamber, a pump fluidly coupled to the ozone generator and upstream of the ozone generator, a fan fluidly coupled to the ozone generator and downstream of the ozone generator, a humidifier fluidly coupled to the disinfection chamber and the ozone generator, the humidifier being upstream of the disinfection chamber, and a controller configured to control operation of the ozone generator, the humidifier, the pump, and the fan according to a disinfection cycle, the disinfection cycle including a first period, a second period, and a third period.

[0346] In some instances, the controller may cause the fan to operate at a fan flow rate and the pump to operate at a pump flow rate, the pump flow rate being different from the fan flow rate during at least one of the first period, the second period, and/or the third period. In some instances, the fan flow rate may be less than the pump flow rate during the first period and the second period. In some instances, a ratio of the fan flow rate to the pump flow rate during the first period and the second period may be about 0.9. In some instances, the fan flow rate may be greater than or equal to the pump flow rate during the third period. In some instances, the pump flow rate may be in a range of about 1 standard liter per minute (SLPM) to about 1.6 SLPM during the first period and the second period. In some instances, the fan flow rate may be about 0.1 SLPM to about 0.15 SLPM less than the pump flow rate during the first period and the second period. In some instances, during the first period the controller may cause both the ozone generator and the humidifier to operate concurrently. In some instances, the humidifier may be disabled during the second period. In some instances, during the second period, the controller may selectively enable and disable the ozone generator, causing the ozone generator to generate ozone pulses. In some instances, the humidifier and the ozone generator may be disabled during the third period. In some instances, to generate the ozone pulses, the controller may alternate between enabling the ozone generator for about 10 seconds and disabling the ozone generator for about 65 seconds. In some instances, the controller may be configured to cause the ozone generator to generate the ozone pulses at a pulse rate, wherein the pulse rate is configured such that a quantity of ozone within the disinfection chamber is in a range of about 80 parts-per-million (ppm) to about 150 ppm. In some instances, the first period may be about 6 minutes, the second period may be about 72 minutes, and the third period may be about 12 minutes. In some instances, during the first period, the controller may operate the humidifier such that a relative humidity within the disinfection chamber is in a range of about 65% to about 99%. In some instances, the first period may occur before the second period and the second period may occur before the third period. In some instances, the second period may be longer than both the

third period and the first period. In some instances, the first period may be shorter than the third period.

[0347] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, an ozone generator, and a humidifier being configured to be fluidly coupled to the disinfection chamber via a continuous positive air pressure (CPAP) hose and configured to be fluidly coupled to the ozone generator, the humidifier including a humidification chamber and a liquid reservoir, the humidifier being configured to encourage a mixing of generated ozone with generated humidity.

[0348] In some instances, the humidification chamber may include a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall. In some instances, at least a portion of the top wall may be sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall. In some instances, liquid collected on the bottom wall may be returned to the liquid reservoir. In some instances, the humidifier further may include a liquid disperser coupled to the bottom wall and configured to emit droplets along an emission axis that extends in a direction of the top wall. In some instances, the liquid disperser may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the liquid disperser may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the liquid disperser may include a protective coating. In some instances, the humidification chamber may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis and the humidifier further includes a liquid disperser having an emission axis, the emission axis extending transverse to the injection axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the humidifier may further include a wick assembly removably coupled to the liquid reservoir and a liquid disperser configured to engage the wick assembly. In some instances, the humidifier may further include a liquid reservoir access that includes a carriage for receiving the liquid reservoir, the carriage being configured to move along an insertion axis and a coupling axis, the coupling axis extending transverse to the insertion axis. In some instances, movement of the carriage along the coupling axis may move the wick assembly into and out of engagement with the liquid disperser.

[0349] Another example of a humidifier for use with a disinfection device, consistent with the present disclosure, may include a liquid disperser, a liquid reservoir, and a humidification chamber, the liquid disperser configured to emit liquid from the liquid reservoir into the humidification chamber, the humidification chamber having a bottom wall, a top wall, and one or more sidewalls extending between the bottom wall and the top wall, at least a portion of the top wall is sloped to direct liquid condensed thereon in a direction of at least one of the one or more sidewalls such that the liquid flows down the at least one of the one or more sidewalls to collect on the bottom wall.

[0350] In some instances, liquid collected on the bottom wall may be returned to the liquid reservoir. In some instances, the liquid disperser may be coupled to the bottom wall of the humidification chamber. In some instances, the

liquid disperser may be configured to emit droplets along an emission axis that extends in a direction of the top wall. In some instances, the humidification chamber may include an inlet connector configured to introduce a turbulent flow to air passing therethrough and an outlet downstream of the inlet connector. In some instances, air may exit the inlet connector along an injection axis, the injection axis extending transverse to the emission axis. In some instances, the emission axis may be substantially perpendicular to the injection axis. In some instances, the liquid disperser may be configured to generate droplets having a droplet size in a range of about 1.5 microns to about 4.5 microns. In some instances, the liquid disperser may be configured to generate droplets having a droplet size of about 2.5 microns. In some instances, the liquid disperser may be a piezo-electric atomizer.

[0351] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, a device controller, an ozone generator configured to generate ozonated air, and a humidifier assembly configured to encourage a mixing of the ozonated air with humidity. The humidifier assembly may include a humidification chamber that includes a humidification chamber outlet, a chamber housing, a chamber base coupled to the chamber housing, and chamber electrical connectors electrically coupled to the device controller and a liquid reservoir that includes a liquid container and a reservoir lid removably coupled to the liquid container, the reservoir lid including a piezo-electric atomizer and lid electrical connectors configured to selectively electrically couple with the chamber electrical connectors.

[0352] In some instances, disinfection device may further include a humidification chamber toggle, wherein actuation of the humidification chamber toggle causes the humidification chamber to at least partially disengage the liquid reservoir. In some instances, actuation of the humidification chamber toggle may cause the humidification chamber to move from a lowered position to a raised position and, wherein, when in the humidification chamber is in the raised position, the liquid reservoir is removable from the disinfection device. In some instances, the disinfection chamber may include a disinfection chamber lid configured to transition between an open position and a closed position. In some instances, the disinfection chamber lid may include a toggle shroud, the toggle shroud being configured to at least partially enclose the humidification chamber toggle when the disinfection chamber lid is in the closed position. In some instances, the chamber base may include a bottom wall, one or more sidewalls extending from the bottom wall to define a base cavity having an open end, and a reservoir receptacle that extends from the bottom wall and into the base cavity, the reservoir receptacle defining a reservoir cavity and an atomizer opening. In some instances, the reservoir lid may include a liquid dispersion protrusion and, wherein, the reservoir cavity is configured to selectively receive at least a portion of the liquid dispersion protrusion to selectively fluidly couple the piezo-electric atomizer with the humidification chamber. In some instances, the disinfection device may further include a protrusion seal that extends around a perimeter of the liquid dispersion protrusion, at least a portion of the protrusion seal is configured to sealingly engage with the reservoir cavity. In some instances, disinfection device may further include an atomizer gasket, the atomizer gasket including a groove extend-

ing along an inner perimeter of the atomizer gasket, a portion of the piezo-electric atomizer being received within the groove. In some instances, the atomizer gasket may extend substantially continuously around an entire outer perimeter of the piezo-electric atomizer. In some instances, the atomizer gasket may extend continuously along an emission side of the piezo-electric atomizer and substantially continuously along a supply side of the piezo-electric atomizer. In some instances, the humidification chamber outlet may include an outlet opening fluidly coupled to the humidification chamber and a mounting region extending around the outlet opening, the mounting region configured to selectively receive a hose adapter. In some instances, the humidification chamber outlet may further include an adapter lock configured to selectively couple the hose adapter in the mounting region. In some instances, the adapter lock may include a rotatable locking body configured to rotate between a locked position and an unlocked position. In some instances, the disinfection device may further include one or more lock sensors configured to detect one or more of a position of the rotatable locking body, a presence of the hose adapter, or a presence of a hose coupled to the hose adapter, the one or more lock sensors being communicatively coupled to the device controller.

[0353] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, a device controller, an ozone generator configured to generate ozonated air, and a humidifier assembly configured to encourage a mixing of the ozonated air with humidity. The humidifier assembly may include a humidification chamber that includes a humidification chamber outlet having an outlet opening through which humidified ozonated air exits the humidification chamber and a mounting region extending around the outlet opening, a liquid reservoir that includes a liquid container, a liquid disperser, and a reservoir lid removably coupled to the liquid container, and a hose adapter selectively coupled to the mounting region.

[0354] In some instances, the disinfection device may further include a humidification chamber toggle, wherein actuation of the humidification chamber toggle causes the humidification chamber to at least partially disengage the liquid reservoir. In some instances, actuation of the humidification chamber toggle may cause the humidification chamber to move from a lowered position to a raised position and, wherein, when in the humidification chamber is in the raised position, the liquid reservoir is removable from the disinfection device. In some instances, the disinfection chamber may include a disinfection chamber lid configured to transition between an open position and a closed position. In some instances, the disinfection chamber lid may include a toggle shroud, the toggle shroud being configured to at least partially enclose the humidification chamber toggle when the disinfection chamber lid is in the closed position. In some instances, the humidification chamber may further include a chamber housing, a chamber base coupled to the chamber housing, and chamber electrical connectors electrically coupled to the device controller. In some instances, the liquid disperser may be a piezo-electric atomizer and the reservoir lid may include lid electrical connectors configured to selectively electrically couple with the chamber electrical connectors. In some instances, the chamber base may include a bottom wall, one or more sidewalls extending from the bottom wall to define a base cavity having an open end, and a reservoir receptacle that extends from the bottom wall

and into the base cavity, the reservoir receptacle defining a reservoir cavity and an atomizer opening. In some instances, the reservoir lid may include a liquid dispersion protrusion and, wherein, the reservoir cavity is configured to selectively receive at least a portion of the liquid dispersion protrusion to selectively fluidly couple the piezo-electric atomizer with the humidification chamber. In some instances, the disinfection device may further include a protrusion seal that extends around a perimeter of the liquid dispersion protrusion, at least a portion of the protrusion seal is configured to sealingly engage with the reservoir cavity. In some instances, the disinfection device may further include an atomizer gasket, the atomizer gasket including a groove extending along an inner perimeter of the atomizer gasket, a portion of the piezo-electric atomizer being received within the groove. In some instances, the atomizer gasket may extend substantially continuously around an entire outer perimeter of the piezo-electric atomizer. In some instances, the atomizer gasket may extend continuously along an emission side of the piezo-electric atomizer and substantially continuously along a supply side of the piezo-electric atomizer. In some instances, the humidification chamber outlet may further include an adapter lock configured to selectively couple the hose adapter in the mounting region, the adapter lock includes a rotatable locking body configured to rotate between a locked position and an unlocked position. In some instances, the disinfection device may further include one or more lock sensors configured to detect one or more of a position of the rotatable locking body, a presence of the hose adapter, or a presence of a hose coupled to the hose adapter, the one or more lock sensors being communicatively coupled to the device controller.

[0355] Another example of a disinfection device, consistent with the present disclosure, may include a disinfection chamber, a device controller, an ozone generator configured to generate ozonated air, and a humidifier assembly configured to encourage a mixing of the ozonated air with humidity, the humidifier assembly including a liquid reservoir having a liquid dispersion protrusion and a humidification chamber. The humidification chamber may include a humidification chamber outlet, a chamber housing, an inlet connector coupled to the chamber housing and fluidly coupled to the ozone generator, and a chamber base coupled to the chamber housing. The chamber base may include a bottom wall, one or more base sidewalls extending from the bottom wall to define a base cavity having an open end opposite the bottom wall, and a reservoir receptacle extending from the bottom wall and into the base cavity, the reservoir receptacle defining a reservoir cavity for receiving at least a portion of the liquid dispersion protrusion.

[0356] In some instances, the liquid dispersion protrusion may include a piezo-electric atomizer and the reservoir includes an atomizer opening aligned relative to the piezo-electric atomizer such that an emission axis of the piezo-electric atomizer extends, unobstructed, into the humidification chamber. In some instances, the reservoir receptacle may include one or more receptacle sidewalls that define an exterior surface that extends at least partially around at least a portion of the liquid dispersion protrusion. In some instances, the reservoir receptacle may intersect with at least one of the one or more base sidewalls to form an intersection region. In some instances, the exterior surface may extend substantially continuously from a first end of the intersection region to a second end of the intersection region. In some

instances, the reservoir receptacle and the inlet connector may be configured to cooperate to encourage generation of turbulent flow within the humidification chamber.

[0357] While the principles of the invention have been described herein, it is to be understood by those skilled in the art that this description is made only by way of example and not as a limitation as to the scope of the invention. Other embodiments are contemplated within the scope of the present invention in addition to the exemplary embodiments shown and described herein. Modifications and substitutions by one of ordinary skill in the art are considered to be within the scope of the present invention, which is not to be limited except by the claims.

What is claimed is:

1-30. (canceled)

31. A disinfection device comprising:

a disinfection chamber;

an ozone generator configured to generate ozonated air; and

a humidifier assembly configured to cooperate with the ozone generator to generate humidified ozonated air, the humidifier assembly including:

a humidification chamber configured to move along a chamber axis between a lowered position and a raised position;

a liquid reservoir including a liquid container and a reservoir lid removably coupled to the liquid container, wherein:

when the humidification chamber is in the raised position, the liquid reservoir is removable from the humidifier assembly; and

when the humidification chamber is in the lowered position, the liquid reservoir is prevented from being removed from the humidifier assembly; and

a liquid disperser configured to disperse liquid from the liquid reservoir and into the humidification chamber to generate the humidified ozonated air.

32. The disinfection device of claim **31**, wherein the liquid disperser includes a piezo-electric atomizer and a wick, the piezo-electric atomizer configured to engage the wick and the wick is configured to extend into the liquid container.

33. The disinfection device of claim **32**, wherein, the humidification chamber includes a reservoir receptacle configured to receive at least a portion of the liquid disperser when the humidification chamber is in the lowered position.

34. The disinfection device of claim **33**, wherein the piezo-electric atomizer has an emission axis that extends, unobstructed, into the humidification chamber when the humidification chamber is in the lowered position.

35. The disinfection device of claim **31** further comprising a housing, the disinfection chamber, the ozone generator, and the humidifier assembly being coupled to the housing, wherein the humidification chamber moves relative to the housing when transitioning between the lowered position and the raised position.

36. The disinfection device of claim **31**, wherein the humidification assembly further includes a humidification chamber toggle, actuation of the humidification chamber toggle being configured to cause the humidification chamber to transition from the lowered position to the raised position.

37. The disinfection device of claim **36**, wherein the humidification chamber is biased towards the raised position.

38. The disinfection device of claim **31**, wherein the humidification chamber includes a humidification chamber outlet configured to selectively fluidly couple a continuous positive air pressure (CPAP) hose to the humidification chamber.

39. The disinfection device of claim **38**, wherein the humidification chamber outlet includes an adapter lock configured to selectively couple to a CPAP hose adapter, the CPAP hose adapter being configured to couple to the CPAP hose.

40. A disinfection device comprising:

a disinfection chamber;

an ozone generator configured to generate ozonated air; and

a humidifier assembly configured to cooperate with the ozone generator to generate humidified ozonated air, the humidifier assembly including:

a humidification chamber;

a liquid container configured to be removable from the humidifier assembly;

a piezo-electric atomizer configured to disperse liquid from the liquid container into the humidification chamber; and

a wick configured to deliver liquid from the liquid container to the piezo-electric atomizer, wherein, when the liquid container is removed from the humidifier assembly, the wick and the humidification chamber move relative to each other.

41. The disinfection device of claim **40**, wherein the wick and the humidification chamber move relative to each other along an emission axis of the piezo-electric atomizer.

42. The disinfection device of claim **40**, wherein the humidification chamber is configured to transition between a lowered position and a raised position and, when the humidification chamber is in the raised position, the container is removable from the humidifier assembly and, when the humidification chamber is in the lowered position, the piezo-electric atomizer is configured to disperse liquid from the liquid container into the humidification chamber.

43. The disinfection device of claim **42**, wherein the humidification assembly further includes a humidification chamber toggle, actuation of the humidification chamber toggle being configured to cause the humidification chamber to transition from the lowered position to the raised position.

44. The disinfection device of claim **40**, wherein the humidification chamber includes a humidification chamber outlet configured to selectively fluidly couple a continuous positive air pressure (CPAP) hose to the humidification chamber.

45. The disinfection device of claim **44**, wherein the humidification chamber outlet includes an adapter lock configured to selectively couple to a CPAP hose adapter, the CPAP hose adapter being configured to couple to the CPAP hose.

46. A disinfection device comprising:

a disinfection chamber;

an ozone generator configured to generate ozonated air; and

a humidifier assembly configured to cooperate with the ozone generator to generate humidified ozonated air, the humidifier assembly including:

a humidification chamber;

a liquid container configured to be removable from the humidifier assembly;

- a piezo-electric atomizer configured to disperse liquid from the liquid container into the humidification chamber; and
- a wick configured to deliver liquid from the liquid container to the piezo-electric atomizer, wherein the wick and the humidification chamber are movable relative to each other along an emission axis of the piezo-electric atomizer.

47. The disinfection device of claim **46**, wherein the humidification chamber is configured to transition between a lowered position and a raised position and, when the humidification chamber is in the raised position, the liquid container is removable from the humidifier assembly and, when the humidification chamber is in the lowered position, the piezo-electric atomizer is configured to disperse liquid from the liquid container into the humidification chamber.

48. The disinfection device of claim **47**, wherein the humidification assembly further includes a humidification chamber toggle, actuation of the humidification chamber toggle being configured to cause the humidification chamber to transition from the lowered position to the raised position.

49. The disinfection device of claim **46**, wherein the humidification chamber includes a humidification chamber outlet configured to selectively fluidly couple a continuous positive air pressure (CPAP) hose to the humidification chamber.

50. The disinfection device of claim **49**, wherein the humidification chamber outlet includes an adapter lock configured to selectively couple to a CPAP hose adapter, the CPAP hose adapter being configured to couple to the CPAP hose.

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